University of Douala

National Advanced School of Engineering of Douala (ENSPD)

Laboratory of Computer Engineering, Data Science and Artificial Intelligence

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Design of a Telemonitoring System for Patients with Chronic Diseases (Hypertension, Diabetes)

Using Consultation Data, Home Measurements and Possibly Mobile Sensors

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Master of Science Thesis

Data Science and Artificial Intelligence

Presented by

NSAME AYUNI Rodrick

Thesis supervised by

Dr. Malong Main supervisor, ENSPD

Dr. Noulapeu Co-supervisor, ENSPD

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Chapter 1

General Introduction

1.1 Background and Motivation

Hypertension is the leading modifiable risk factor for premature death worldwide. The first global report of the World Health Organization (WHO) devoted to the condition establishes that the number of adults aged 30–79 years living with hypertension doubled between 1990 and 2019, reaching 1.3 billion, that about 78% of them live in low- and middle-income countries, and that elevated systolic blood pressure is responsible for an estimated 10.8 million deaths each year (**who2023hypertension**). Sub-Saharan Africa combines a high and increasing prevalence with persistently poor detection, treatment and control (**gafane2024**): in the WHO African Region, only 12% of hypertensive adults have their blood pressure controlled.

Diabetes compounds this burden. In Africa, more than one person with diabetes in two is also hypertensive, and the highest sub-regional estimate concerns Central Africa, the region in which the present work is deployed (**hinneh2023**). The two conditions share their risk factors, aggravate each other, and call for the same kind of care: repeated measurement and treatment adjustment sustained over decades.

At the same time, the technological environment of the region is changing. Mobile telephony is now widespread and health technology is among the leading digital investment sectors in sub-Saharan Africa (**segun2025**), and a controlled study in Cameroonian district hospitals has shown that remote clinician support is feasible in primary care (**bediang2021**). A care model that fails at follow-up, and an environment in which remote follow-up has become conceivable: these two observations motivate the present work.

1.2 Problem Statement

Care for hypertension and diabetes in Cameroon is episodic. Blood pressure and blood glucose are measured at consultations that occur a few times a year, recorded on paper

or in fragmented registers, and rarely assembled into a history that a clinician can read as a trend. Yet diagnosis and treatment adjustment are longitudinal acts: the WHO protocol requires measurements repeated over several visits and monthly follow-up until the blood pressure target is reached (**who2021guideline**; **who2023hypertension**). The consequences are loss to follow-up, which often affects half of patients or more in resource-limited settings, and therapeutic inertia, the failure to intensify treatment when blood pressure remains uncontrolled.

Telemonitoring could close this gap, but three obstacles stand in the way. First, the deployment environment imposes technological, organisational, financial, individual, cultural and regulatory barriers (**dodoo2022**), and the comparative evidence on telemedicine adoption in the region comes from South Africa, Kenya and Nigeria rather than from Central Africa (**agbeyangi2025**). Second, the clinical evidence that telemonitoring improves blood pressure and glycaemic control comes almost entirely from high-income settings. Third, a system that processes real patient data through a mobile and cloud pipeline must earn the trust of patients in the confidentiality and security of their data (**alhammad2024**).

The research question addressed in this thesis is therefore the following: *how can consultation data, home measurements and mobile sensor data be integrated into a telemonitoring system that detects the deterioration of patients with hypertension and diabetes early enough to support a clinical decision, under the infrastructural, organisational and regulatory constraints of the Cameroonian context?*

1.3 Objectives

The general objective is to design a telemonitoring system for patients with hypertension and type 2 diabetes that combines consultation data, home measurements and, where available, mobile sensor data, and that supports the clinician through risk assessment and alert prioritisation. The specific objectives are:

1. to synthesise the evidence on the burden of the two conditions, the effectiveness of telemonitoring, the enabling technologies, the predictive methods and the deployment constraints (Chapter 2), and to derive from it the requirements of the system (Chapter 3);
2. to construct a multi-source dataset with a documented schema, preprocessing pipeline and labelling rule, and to develop and evaluate predictive risk models on it (Chapter 4);
3. to design the architecture of the system: acquisition, cloud backend, patient application and clinician dashboard (Chapter 5).

1.4 Research Hypotheses

Three hypotheses, set out in the research framing document of this work, guide the experimental phase. They are confronted with the literature in Section 3.4.7.

H1. A deep temporal model obtains better recall on high and critical risk states than robust classical models, on locally collected telemonitoring data.

H2. Short-horizon forecasting of physiological parameters allows degradation events to be anticipated with a clinically useful lead time.

H3. Given the limited size and irregularity of the local data, the gain of the deep model over classical models may remain modest.

1.5 Organisation of the Thesis

Chapter 2 reviews the literature in six themes, from the burden of disease to data protection, and states the research gap. Chapter 3 derives from this review the clinical, acquisition, architectural, modelling and decision-support requirements of the system. Chapter 4 describes data collection, preprocessing, feature engineering and predictive modelling. Chapter 5 presents the architecture and software design of the system. The references cited throughout are listed in the bibliography at the end of the document.

Chapter 2

Literature Review

Scope and Organisation of the Review

This chapter reviews the literature on which the design of the proposed system rests. It draws on nineteen peer-reviewed sources, each checked against PubMed, PubMed Central, Crossref or the publisher's record, complemented by three institutional documents of the WHO (**who2023hypertension**; **who2021guideline**; **whoHEARTS**) and by the TELEMED-CAM study (**kingue2013**), the earliest Cameroonian evaluation of hypertension telemanagement. The 2023 guidelines of the European Society of Hypertension (ESH) (**mancia2023**), from which the blood pressure thresholds of the risk levels are taken, are cited wherever those thresholds are applied.

The sources are organised in six themes that follow the logic of the argument, from the problem to the response (Table 2.1): the global burden of hypertension and diabetes (Section 2.1); hypertension care and digital health in sub-Saharan Africa (Section 2.2); the clinical evidence for mobile health and telemonitoring (Section 2.3); sensor and Internet of Things (IoT) technologies (Section 2.4); machine learning for risk prediction (Section 2.5); and data privacy and ethics (Section 2.6). Within each theme, sub-themes go from the problem to the response, and the studies of each sub-theme are presented in chronological order of publication, so that the progression of the evidence is visible. Each study is presented once, in the theme to which it contributes most, and is only recalled elsewhere. Section 2.7 synthesises the findings and states the research gap.

2.1 Global Burden of Hypertension and Diabetes

2.1.1 The global burden of hypertension

The WHO global report on hypertension (**who2023hypertension**) is the reference framing of the problem. For population surveillance it defines hypertension as a systolic blood pressure (SBP) $\geq$ 140 mmHg, a diastolic blood pressure (DBP) $\geq$ 90 mmHg, or the use of

Table 2.1: Sources of the review, by theme and in chronological order within each theme.

Theme	Source	Design and setting
1. Burden of disease	**hinneh2023<empty citation>**	systematic review and meta-analysis, Africa
	who2023hypertension<empty citation>	institutional report
2. Care and digital health in sub-Saharan Africa	**kingue2013<empty citation>**	interventional pilot, rural Cameroon
	bediang2021<empty citation>	controlled multicentre study, Cameroon
	dodoo2022<empty citation>	review of 66 empirical studies, sub-Saharan Africa
	gafane2024<empty citation>	narrative review, sub-Saharan Africa
	agbeyangi2025<empty citation>	systematic review, South Africa, Kenya, Nigeria
	segun2025<empty citation>	perspective, sub-Saharan Africa
3. Mobile health and clinical evidence	**alessa2019<empty citation>**	review and content analysis of applications, United Kingdom
	mao2020<empty citation>	meta-analysis of 51 randomised trials
	zhou2024<empty citation>	meta-analysis of randomised trials
	bandeira2024<empty citation>	pooled meta-analysis of randomised trials
	mihevc2025<empty citation>	multicentre randomised trial, Slovenia
4. Sensors and IoT	**alhaddad2022<empty citation>**	review of non-invasive glucose sensing
	tai2024<empty citation>	IoT architecture and deep learning technique
	efendi2025<empty citation>	IoT and cloud implementation with device validation
	liu2025review<empty citation>	review of IoT sensing and machine learning
	liu2025ppg<empty citation>	smartphone photoplethysmography study
5. Machine learning	**chatrati2022<empty citation>**	glucose monitoring system with machine learning
	mendoza2025<empty citation>	cross-sectional predictive study, Mexico
6. Privacy and ethics	**alhammad2024<empty citation>**	systematic review of patient perspectives

antihypertensive medication, in adults aged 30–79 years. This threshold is used throughout this thesis for epidemiological figures; the lower 130/80 mmHg classification of the ESH (**mancia2023**), used later to define the risk levels, explains part of the discrepancies between publications.

The report establishes that the number of hypertensive adults rose from 650 million in 1990 to 1.3 billion in 2019, while the age-standardised prevalence barely changed, from 32% to 33%: the increase in absolute numbers is mostly explained by population growth and ageing rather than by a rise in individual risk. Elevated systolic blood pressure is the leading risk factor for premature death among the eighty-seven factors compared, ahead of smoking and hyperglycaemia, with 10.8 million deaths and 235 million disability-adjusted life years each year. Progress towards the global target is insufficient: the prevalence of uncontrolled hypertension fell only from 29% in 2010 to 26% in 2019, and no WHO region is on track. The report notes, however, that the target would be met if about half of the adults already diagnosed were effectively treated. Among the recommended responses, it endorses team-based care, community and home blood pressure measurement, and telemonitoring.

As an institutional synthesis, the report relies on modelled estimates and on references that it does not detail; it does not address patient-level monitoring technology. It is used here for the global framing and for its normative positions, not as primary evidence.

2.1.2 The failure of the care cascade

The WHO structures the care of hypertension as a cascade of nested stages measured on the same denominator, all hypertensive adults (Table 2.2).

Table 2.2: Hypertension care cascade by WHO region, adults aged 30–79 years, 2019 (adapted from **who2023hypertension**, Table 2).

Indicator	**African Region**	**World**	**Americas**	**Europe**
Age-standardised prevalence	36%	33%	35%	37%
Diagnosis coverage	43%	54%	70%	66%
Treatment coverage	27%	42%	60%	53%
Effective treatment (control)	12%	21%	36%	26%

Two observations follow. First, the European Region has a higher prevalence than the African Region but a control rate more than twice as high: the African deficit is organisational rather than epidemiological. Second, the conditional losses derived from these figures are spread along the whole chain: in the African Region, 57% of hypertensive adults are undiagnosed, 37% of the diagnosed are untreated, and 56% of the treated are not controlled. The largest conditional loss lies at the last link, among patients already in care. A screening device acts only on the first link; a longitudinal follow-up system acts on the second and third.

2.1.3 The hypertension–diabetes comorbidity

The association between hypertension and type 2 diabetes rests on shared risk factors—physical inactivity, unbalanced diet and obesity—and on a direct pathophysiological link: diabetes stiffens small vessels, which raises peripheral resistance and hence blood pressure, while hypertension accelerates the vascular damage of diabetes.

hinneh2023<empty citation> produced the first systematic quantification of this comorbidity in Africa. Their meta-analysis, conducted according to PRISMA on PubMed, EMBASE and HINARI up to March 2023, pooled forty-one cross-sectional studies totalling 144 433 participants. The pooled prevalence of hypertension among people with diabetes is 58.1% (95% CI 52.0–63.2), and 60.8% among people with type 2 diabetes alone. Older age, overweight and obesity, longer duration of diabetes, urban residence and male sex are associated with a higher risk. The authors also note that patients with both conditions paradoxically have lower chances of receiving appropriate care before complications occur.

The sub-regional estimates place Central Africa highest (Table 2.3 and Figure 2.1).

Table 2.3: Sub-regional pooled prevalence of hypertension among people with diabetes in Africa (**hinneh2023**).

Sub-region	Pooled prevalence	95% CI	Studies
Central Africa	77.6%	53.0–91.4	2
Southern Africa	69.1%	59.8–77.1	10
Eastern Africa	53.0%	45.8–59.1	15
Western Africa	51.5%	41.8–61.1	10

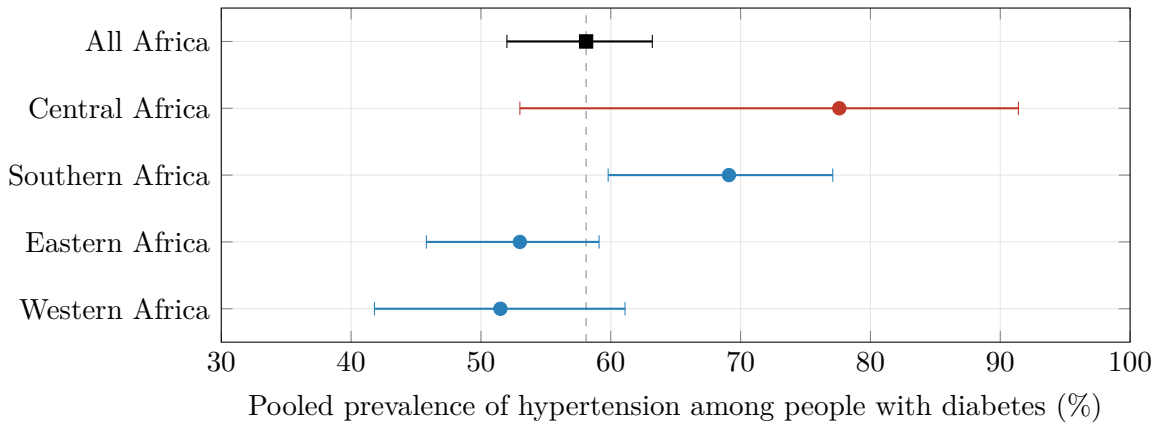


Figure 2.1: Point estimates and 95% confidence intervals of Table 2.3; the dashed line marks the pooled African estimate. The Central African interval spans 38 points.

Reliability of the Central African estimate. The value of 77.6% must be reported with its confidence interval, 53.0–91.4%, and it cannot by itself carry the justification of this work. It rests on two small Cameroonian hospital series (109 and 210 patients),

while three other Central African studies listed in the article (1 832 patients, prevalences of 35.0–50.4%) do not appear in this subgroup. Heterogeneity is extreme ($I^2 \approx 98\%$), the included studies are mostly hospital-based and use varying diagnostic thresholds, and the restriction to English-language publications under-represents francophone Central Africa, as the authors themselves acknowledge. The figure also measures hypertension among people with diabetes, not the reverse.

The case for monitoring blood pressure and blood glucose jointly therefore rests on the robust continental estimate of 58.1%, on the pathophysiological mechanism, and on the position of the WHO (**who2023hypertension**): uncontrolled hypertension, not hyperglycaemia, is among the risk factors most strongly associated with premature death in people with diabetes; Member States have adopted a target of 80% of people with diagnosed diabetes having good blood pressure control; and the HEARTS package developed for hypertension is explicitly applicable to diabetes and other long-term conditions.

2.1.4 Comparability of estimates

The prevalences quoted in this chapter are not directly comparable. They differ in reference population (general population, people with diabetes or hospital attendees), in diagnostic threshold (140/90 or 130/80 mmHg), in age structure, in the number of measurement visits—a single-visit measurement overestimates prevalence—and in recruitment mode, opportunistic screening and self-selection producing samples systematically different from probability sampling. None of the available sources is nationally representative for Cameroon. The nature of each estimate is therefore stated where it is used.

2.2 Hypertension Care and Digital Health in Sub-Saharan Africa

2.2.1 Burden, determinants and care gaps

gafane2024<empty citation> synthesise the hypertension literature of sub-Saharan Africa published between January 2017 and June 2023, drawn from MEDLINE, PubMed, CINAHL, African Journals Online and grey literature. Most countries of the region reported prevalence data over the period, confirming a high burden with poor detection and control. The authors relate the rise of hypertension to the shift towards a westernised lifestyle—salt, sugar and fat, physical inactivity, obesity, tobacco and alcohol—together with socio-economic and psychosocial factors, and they identify human immunodeficiency virus infection as a regional specificity; among the mechanisms still under investigation they list sodium handling, endothelial function and circadian profiles such as nocturnal dipping. They distinguish individual obstacles to care (health literacy, cost, distance)

from system obstacles (medicine availability, staffing and training, uneven political will). Community-based task shifting improves linkage to care and medication adherence, but its effect on blood pressure over time is inconsistent.

Their inventory of national surveys is the most recent available for Central Africa (Table 2.4). Dispersion is wide, from 13.5% to 42.1%, and reflects the diversity of instruments as much as real differences; three countries of the sub-region have no data for the period. In Cameroon, the May Measurement Month campaign of 2019 found a prevalence of 20.8%, with only 11.2% of hypertensive participants controlled.

Table 2.4: Most recent hypertension surveys in Central Africa (adapted from **gafane2024**, Table 1).

Country	**Instrument**	**Year**	n	**Prev.**	**Aware.**	**Treat.**	**Control**
Burundi	hospital screening	2018	353	16.7%	–	–	–
Cameroon	May Measurement Month	2019	30 167	20.8%	29.9%	24.0%	11.2%
Central African Rep.	WHO STEPS	2017	3 265	42.1%	30.2%	–	–
Comoros	national survey	2019	2 602	38.4%	47.9%	45.4%	30.2%
DR Congo	May Measurement Month	2019	29 857	25.5%	33.1%	23.2%	11.9%
Congo (Rep.)	May Measurement Month	2019	3 157	33.5%	42.6%	37.3%	23.3%
Gabon	retrospective study	2013–15	2 104	13.5%	–	–	–
Chad, Eq. Guinea, São Tomé	no data	–	–	–	–	–	–

The review is narrative rather than pooled, the surveys it gathers are heterogeneous, and some figures are inconsistent: the same pair of values (48% and 34%) is presented once as prevalence and once as diagnosis rate. It is used here for the regional profile and the care cascade, not for pooled prevalence.

2.2.2 Why the episodic care model fails

The WHO documents the mechanisms behind the failure of the cascade (**who2023hypertension**). *Loss to follow-up* often affects half of patients or more in resource-limited settings; the report therefore holds that screening is useful only if linked to effective follow-up, and it recommends opportunistic detection in health facilities together with retention of identified patients rather than population screening. *Therapeutic inertia*, the tendency not to intensify treatment despite uncontrolled blood pressure, explains why the last link of the cascade is the weakest; a simple standardised protocol helps reduce it. Lifestyle

modification alone lowers systolic pressure by only about 1.3 mmHg in primary-care trials, so almost all diagnosed patients need medication, and its timely adjustment. *Consultation time* is often below five minutes in primary care, which requires digital tools to be fast and to replace paper rather than add to it. Finally, *paper records* are duplicated, lost and unavailable during the visit, whereas a secure longitudinal record is described as essential for hypertension, diabetes and other long-term conditions. The core indicators of the HEARTS programme, two of which are temporal, define what such a record must produce (Section 3.1.3a).

The expected follow-up frequency is set by Recommendation 7 of the WHO guideline on the pharmacological treatment of hypertension: monthly follow-up after initiating or changing treatment until the target is reached, then every three to six months once blood pressure is controlled (**who2021guideline**). The HEARTS treatment-protocol module adds that the return interval should usually be four weeks or less during titration (**whoHEARTS**). The gap between this requirement and the one or two contacts a year that an episodic model delivers is quantified in Section 3.1.2a.

The programmes cited by the WHO point in the same direction. In Sri Lanka, 86% of enrolled patients return every three months and 57% are controlled; in the Philippines, India and Cuba, part of the follow-up was moved out of the facility, towards the home or the community. These are associations observed in programmes that changed several components at once, not causal estimates. The Indian programme also shows that retention depends on medicine supply: stock-outs soon after launch discouraged patients from returning, so that a telemonitoring system deployed during stock-outs would probably not improve control, whatever its technical performance.

2.2.3 Telemedicine evidence from Cameroon

Two controlled evaluations of cardiovascular telemedicine have been conducted in Cameroonian primary care.

TELEMED-CAM (2008). **kingue2013<empty citation>** evaluated a hypertension management package in thirty rural health centres supported by a telemedicine centre at Yaoundé General Hospital. The package combined a standardised protocol restricted to generic drugs, lifestyle counselling, training of general practitioners and nurses, and a telecare channel through which providers reported patient data daily by text or voice message and received telephone feedback. Among 268 participants followed for 24 weeks, attendance at scheduled visits was higher in the intervention arm at every grade of hypertension ($p \leq 0.04$), and systolic pressure fell from the second visit. The benefit on control was confined to stage III hypertension (no improvement in 16.7% against 33.3%, $p = 0.04$); for stages I and II the proportions at target were similar (65.2% against 70%, $p = 0.20$).

Allocation was not strictly random, clustering was ignored, the arms differed at baseline (systolic pressure 169 against 161 mmHg, diabetes 9.1% against 43.6%), and telecare was only one component of the package, so its own effect cannot be isolated. The study also quantifies the cost of systematic remote review: about six calls per provider per day, of about five minutes each.

Tele-expertise in district hospitals (2016–2017). **bediang2021<empty citation>** compared two district hospitals equipped with a twelve-lead electrocardiograph and an internet tele-expertise platform (Mbouda and Akonolinga) with two matched control hospitals (Foumbot and Sa'a). Among 171 patients, access to an electrocardiogram with a cardiologist's opinion reached 98.9% against 33.3% (relative risk 2.97), the expert modified the local provider's diagnosis in 46.2% of cases against 17.9%, and patient satisfaction was higher. The study was not randomised and the groups differed at baseline; its primary outcome measures the availability of the service rather than its clinical effect. The reported blood pressure benefit (40.9% against 17.9% of patients with a favourable evolution) counts the 52 control patients who never returned as not improved; restricted to the patients actually reviewed at three months, the proportions become 41.3% against 53.8%. The specialist's opinion also took 10.3 days on average against a protocol maximum of 72 hours, no faster than in the control centres (11.2 days, $p = 0.82$): the bottleneck was expert availability, not transmission.

Table 2.5: The two Cameroonian evaluations compared.

	kingue2013<empty citation>	**bediang2021<empty citation>**
Period	2008	2016–2017
Function evaluated	telemanagement of treatment	tele-expertise for diagnosis
Design	prospective, partly randomised by centre	controlled, matched centres, not randomised
Participants	268	171
Technology	GSM handset, SMS and voicemail	portable 12-lead ECG, web platform
Principal result	control improved in stage III only	access to ECG with expertise, 98.9% vs 33.3%
Retention result	visit adherence higher at every grade	attendance at 3 months, 98.9% vs 33.3%
Main weakness	clustering ignored, package effect	non-comparable groups, differential attrition

Read together (Table 2.5), the two studies support three conclusions. Retention is the most robust result, observed by different mechanisms eight years apart. An effect on blood pressure control is not established in Cameroon. And the value added lies in the clinical decision—a revised diagnosis, a supported prescriber—rather than in a tool placed in the patient's hands. Both studies also show that systematic expert review of every case does not scale where specialists are scarce.

2.2.4 Digital health landscape and barriers to adoption

dodoo2022<empty citation> reviewed sixty-six empirical studies and government reports on telemedicine in sub-Saharan Africa, drawn from eight databases under adapted PRISMA guidelines. Progress is markedly uneven between countries, and the barriers fall into six dimensions (Table 2.6). Two findings bear directly on design: most projects lost momentum once external funding ended, and health professionals facing a heavy workload perceive telemedicine as an additional task. The authors also report that store-and-forward services suffered from low user satisfaction for lack of real-time interaction. Their inventory lists for Cameroon telecardiology, wound-care and tele-education services, including the Telemed-cam project, and they record forty-one countries with national digital health strategies. Francophone countries, Cameroon included, are under-represented in the literature they review.

Table 2.6: Six dimensions of barriers to telemedicine in sub-Saharan Africa (**dodoo2022**).

Dimension	**Definition given by the authors**
Technological	information systems and connectivity
Organisational	stakeholder engagement, sustainability of systems
Individual	users' experience, knowledge and awareness
Financial	financial resources and operating cost
Cultural	social constraints and traditional beliefs
Legal and regulatory	policies, legislation, confidentiality and privacy protection

agbeyangi2025<empty citation> extend this picture with a PRISMA-guided systematic review of fifty-three studies published between 2014 and 2024, comparing South Africa, Kenya and Nigeria. South Africa leads adoption, including specialist teleconsultation and chronic disease management; Kenya shows strong integration of mobile health; Nigeria is advancing remote diagnosis under infrastructural constraints. Regulatory gaps, digital literacy and infrastructure remain the persistent barriers. Central Africa is absent from the comparison.

segun2025<empty citation> assess the digital health landscape of the region from the perspective of cardiovascular care. More than 80% of the population holds a mobile telephone subscription, mobile internet penetration stood at 25% in 2023, and smartphone adoption is projected to reach 88% by 2030; yet about 550 million people lack reliable access to electricity and the internet, and up to 18% of countries had no registered cardiologist in 2020. The collapse of the Rwandan platform Babyl in 2023, after about 2.5 million users had registered, illustrates that adoption is not sustainability. Artificial intelligence remains largely untested in the region, with few validation studies in African populations and a documented risk that externally trained algorithms perform poorly there. As a perspective article without a declared search strategy, this source is used for contextual

orders of magnitude, not as pooled evidence.

Taken together, these sources describe a population that is reachable by mobile networks and a specialist who is not. A system that forwards every reading to a clinician reproduces the scarcity it aims to relieve; a system that sorts readings and orders the clinician's attention addresses it. They also impose offline operation, a simple interface for both patients and providers, and a documented operating cost.

2.3 Mobile Health Self-Management and Clinical Evidence

2.3.1 What self-management applications offer

alessa2019<empty citation> analysed the 186 hypertension self-management applications available in English on the two main application stores in January 2018, coding thirty of them in depth. Most applications offered a narrow set of functions: 58.1% offered a single function, only 17.7% offered three or more, and none offered all eight identified. Self-monitoring was present in 53.2% of applications, feedback in 28.0% and reminders in 21.0%; most of the applications that tracked blood pressure required the patient to type in the value displayed by the monitor. None of the thirty applications analysed in depth had been scientifically evaluated or certified as a medical device, and none reported the involvement of patients in its design. Three applications claimed to measure blood pressure with the smartphone camera without any evidence of validity. Of the thirty, only twenty had a privacy policy, and fourteen of these declared sharing data with third parties. In terms of behaviour change techniques, coded with the taxonomy of Michie and colleagues, the applications contained a median of nine techniques, with self-monitoring, prompts and cues, and action planning present in all thirty. As a content analysis of a market snapshot, the study describes what applications contain, not whether they work.

2.3.2 Pooled clinical effects

mao2020<empty citation> pooled fifty-one randomised trials published between 2008 and 2019, with 13 054 participants with diabetes or hypertension. Mobile health interventions reduced glycated haemoglobin (HbA1c) by 0.39 points (−0.50 to −0.29), fasting glucose by 0.52 mmol/L, systolic pressure by 2.99 mmHg (−4.19 to −1.80) and diastolic pressure by 1.14 mmHg. Two subgroup results are decisive for this work. Interventions combining mobile health with professional management reduced systolic pressure by 6.17 mmHg (−8.83 to −3.50), whereas mobile health alone produced a non-significant 2.16 mmHg. And the effect on systolic pressure was present in developed countries (−5.72 mmHg) but absent in developing countries (+0.25 mmHg, −3.10 to 3.59), whose trials used mobile

health without professional management; the glycaemic effect, by contrast, was larger in developing countries (−0.52 against −0.35 points). Only one of the fifty-one trials was conducted in sub-Saharan Africa. Heterogeneity was moderate to high (I^2 of 57–67%) and most trials were short.

zhou2024<empty citation> focused on mobile application self-management for adults with hypertension: twenty randomised trials published from 2017 onwards, sixteen of them meta-analysed, with 4 168 participants from nine countries, none in sub-Saharan Africa. Systolic pressure fell by 5.78 mmHg (−7.97 to −3.59; low certainty, $I^2 = 82\%$) and diastolic pressure by 3.28 mmHg (moderate certainty). The subgroup analyses link the effect to the design of the intervention (Table 2.7): interventions with at least eleven behaviour change techniques and those addressing several risk factors were markedly more effective, and three-month interventions outperformed longer ones, which the authors attribute to the weakening of habit after about twelve weeks. Fifteen of the twenty trials were at high risk of bias, and interventions based mainly on text messages or calls—the type most often evaluated in Africa—were excluded.

Table 2.7: Subgroup analyses on systolic pressure (**zhou2024**). BCT: behaviour change technique.

Factor	**Modality**	**Effect on SBP (95% CI)**	*P*
Risk factors addressed	multiple	−6.50 mmHg (−9.00 to −3.99)	.007
	single	−1.54 mmHg (−4.15 to 1.06)	
Number of techniques	≥ 11 BCT	−9.68 mmHg (−13.49 to −5.87)	<.001
	< 11 BCT	−2.88 mmHg (−3.90 to −1.86)	
Duration	3 months	−8.87 mmHg (−10.90 to −6.83)	.09
	≥ 6 months	−5.76 mmHg (−8.74 to −2.77)	

bandeira2024<empty citation> updated the synthesis of randomised trials of mobile health interventions on blood pressure. They confirm statistically and clinically significant reductions across diverse populations and settings, and identify telemonitoring with clinician feedback as the most effective modality. Heterogeneity between interventions, short follow-up in several trials and the scarcity of data from low- and middle-income countries limit the generalisation of these results.[1]

2.3.3 A randomised trial in the target population

mihevc2025<empty citation> conducted the trial most directly relevant to this work: a multicentre, open-label randomised trial in three Slovenian primary-care centres between

[1] [**To verify**] Extract the pooled systolic and diastolic estimates and the subgroup estimate for telemonitoring with clinician feedback from **bandeira2024<empty citation>**, and add them to Table 2.8.

2021 and 2023, enrolling 128 patients aged 65 years or more with both hypertension and type 2 diabetes. In addition to usual care, the intervention group measured blood pressure and blood glucose at home with Bluetooth devices that transmitted automatically through a smartphone to a cloud platform; a telemedicine centre coordinated by a general practitioner reviewed the values according to a three-level protocol (green, yellow, red) detailed in Section 3.1.4c. After twelve months, systolic pressure was 6.9 mmHg lower than in the control group (−11 to −2.7), after a difference of 10.5 mmHg at six months, and HbA1c was 0.5 points lower.[2]

Three further results matter. Telemonitoring had no effect on other cardiovascular risk factors, on diabetes-related quality of life or on lifestyle: the authors attribute the clinical effect to improved communication, medication adherence, early detection of exacerbations and treatment adjustment based on accurate data, not to behaviour change. Seven of the 64 patients of the intervention arm (10.9%) withdrew in the first month because they could not use the equipment, despite one hour of individual training. And about a third of the systolic effect was lost between six and twelve months. The trial was conducted in a high-income setting, among older adults, without blinding; its applicability to low-resource settings requires adaptation.

2.3.4 Synthesis of the clinical evidence

Table 2.8 and Figure 2.2 gather the available estimates in chronological order.

Table 2.8: Estimates of the effect of mobile health and telemonitoring on systolic blood pressure and glycaemic control.

Source	Design	Effect on SBP (95% CI)	Glycaemic effect
mao2020<empty citation>	80 RCTs,	on SBP	−2.99 mmHg (−4.19 to −1.80)	HbA1c −0.39
overall			
with professional management	subgroup	−6.17 mmHg (−8.83 to −3.50)	–
mobile health alone	subgroup	−2.16 mmHg (−5.07 to 0.75), n.s.	–
developing countries	subgroup	+0.25 mmHg (−3.10 to 3.59), n.s.	HbA1c −0.52
zhou2024<empty citation>	16 RCTs	−5.78 mmHg (−7.97 to −3.59), low certainty	not assessed
mihevc2025<empty citation>	RCT, 117	analysed	−6.9 mmHg (−11 to −2.7)	HbA1c −0.5

Four conclusions emerge. First, mobile health and telemonitoring produce a modest but clinically relevant effect, of the order of −3 to −7 mmHg on systolic pressure and −0.4 to −0.5 points on HbA1c; these are intermediate outcomes, and none

[2] [**To verify**] The four HbA1c means reported by **mihevc2025<empty citation>** (7.2 to 6.7 and 7.1 to 6.7) imply a difference in change of −0.1, not the reported −0.5 (−0.8 to −0.2). Check Table 3 of the article before citing the glycaemic result.

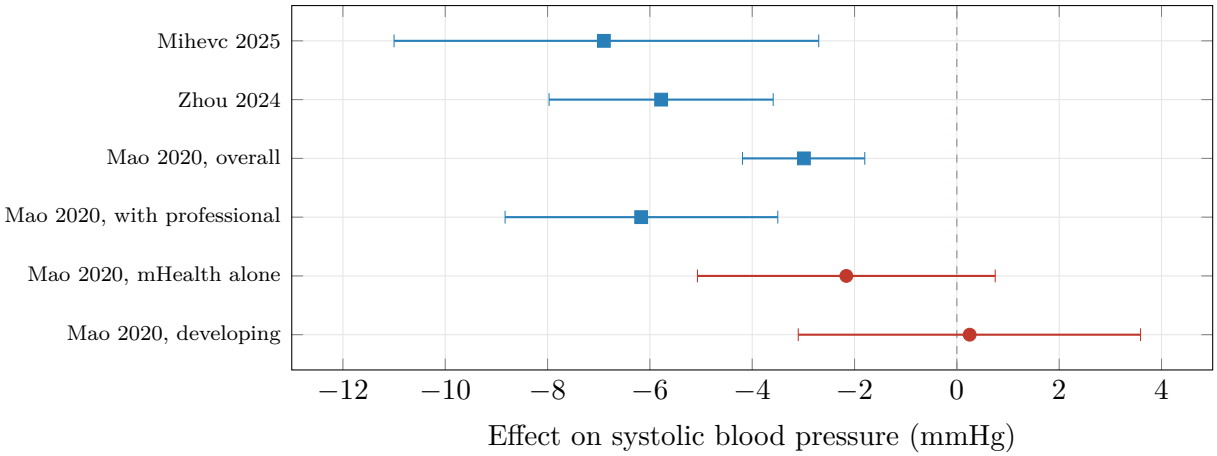


Figure 2.2: Estimates of Table 2.8. Squares mark intervals excluding the null, circles those including it. The two non-significant estimates describe interventions without professional management or conducted in developing countries.

of the sources measures morbidity or mortality. Second, the effect depends on professional oversight: **mao2020<empty citation>**, **bandeira2024<empty citation>** and **mihevc2025<empty citation>** converge on this point by independent routes, and so do the Cameroonian studies of Section 2.2.3. Third, the evidence from sub-Saharan Africa is almost absent, and the absence of effect observed in developing countries is plausibly a design gap—no clinician at the end of the chain—rather than a context effect. Fourth, two problems remain open: the erosion of the effect over time, and the inability of a fraction of patients to use the equipment. Telemonitoring must therefore be built as a decision-support system linked to a clinician, not as a self-tracking tool handed to the patient.

2.4 Sensor and IoT Technologies for Chronic Disease Monitoring

2.4.1 The normative position on measuring devices

The WHO sets the frame within which any sensor must be assessed (**who2023hypertension**). Manual devices should be replaced by independently validated automated electronic devices, and automated upper-arm cuff devices are recommended. Cuffless devices, including popular connected wristbands, are not recommended for the management of hypertension, because of unresolved questions about their accuracy and their ability to track blood pressure change; as of 2023, no alternative to the validated upper-arm cuff is accurate enough for diagnosis and management. Only about one automated device in five on the market has evidence of validation.

2.4.2 Non-invasive sensing of glucose and blood pressure

alhaddad2022<empty citation> reviewed the non-invasive sensing modalities proposed for blood glucose—electrocardiography, electromagnetic sensing, bioimpedance and photoplethysmography (PPG)—and the machine learning models applied to them. PPG- and ECG-based sensing combined with deep learning shows promise for detecting glucose trends and hypoglycaemia, but not for clinical-grade absolute measurement: many studies have small samples, inter-individual variation limits generalisation, and accuracy remains below the standards recognised by the Food and Drug Administration.

liu2025ppg<empty citation> estimated blood pressure from the PPG signal recorded by a smartphone camera alone, without additional hardware. Combining waveform feature engineering with machine learning, and using SHAP values to trace each prediction back to interpretable physiological features, they report a strong correlation with reference devices and assess agreement by Bland–Altman analysis on 127 participants. The sample is modest and single-site, accuracy remains below that of clinical cuff devices, and generalisation across skin tones, devices and lighting conditions is not established.

These two studies, set against the WHO position and the unvalidated camera applications found by **alessa2019<empty citation>**, define a consistent position. Blood glucose must come from a glucometer and blood pressure from a validated upper-arm cuff, because these are the measurements on which clinical decisions rest. Smartphone PPG is a plausible low-cost source of heart rate and an exploratory source of blood pressure trends, to be evaluated rather than relied upon (Section 3.2.4c).

2.4.3 IoT architectures for remote monitoring

tai2024<empty citation> address the difficulty of applying conventional machine learning to noisy, heterogeneous and high-dimensional IoT health data. They propose a human-centric deep learning technique for remote chronic disease monitoring, embedded in a standardised three-phase architecture: data acquisition, data processing and data visualisation. The architecture supports accurate and interpretable predictions in their evaluation, which was conducted in controlled conditions and does not address the connectivity constraints of low-resource settings.[3]

efendi2025<empty citation> describe and validate a complete IoT system for monitoring elderly people. Its three layers are a physical layer (a MAX30100 optical sensor for heart rate and oxygen saturation and a MAX30205 temperature sensor, read by an ESP32 microcontroller), a network layer (MQTT transmission to a Firebase real-time database) and an application layer (mobile application and website accessed over HTTP). Firebase was chosen for real-time synchronisation, built-in authentication, and

[3] **[To verify]** Only the bibliographic record and abstract-level summary of **tai2024<empty citation>** were used. Check the evaluation dataset and reported performance against the full text.

local persistence with automatic merging when connectivity returns. Against a digital oximeter and thermometer, six participants measured for ten minutes at rest gave a mean absolute percentage error of 0.90%, and XGBoost was the best of five classifiers of the user's state as stable or not stable. The system measures heart rate, oxygen saturation and temperature, not blood pressure or blood glucose; its validation sample is small, the reference instruments are not identified as validated, and its classifier is examined critically in Section 2.5.

liu2025review<empty citation> survey the use of IoT mobile sensing and machine learning for cardiovascular disease, cancer, chronic respiratory disease and diabetes. They classify devices into portable, wearable, intelligent clinical, remote, implantable and ingestible categories, describe a seven-layer platform from device access to decision support, and identify the weaknesses of traditional monitoring: manual processes, information silos, lack of real-time feedback and of early-warning mechanisms. They also note that electronic medical records require cleaning of duplicates, unit errors and values carried over identically between visits. Although presented as a systematic review, the article reports no protocol, flow diagram or number of included studies, so it is treated here as a narrative survey. Controlled settings dominate the studies it covers, and deployment in low- and middle-income countries is poorly represented.

The architectures converge on a common structure: acquisition, transmission and storage, processing, and presentation. None of them has been validated on real patients in routine care in a low-resource setting, and none acquires blood pressure and blood glucose through its own sensors. The detailed implications for acquisition and architecture are developed in Sections 3.2 and 3.3.

2.5 Machine Learning for Clinical Risk Prediction

chatrati2022<empty citation> proposed a smart home monitoring system that analyses home blood pressure and glucose readings and notifies the healthcare provider when an abnormality is detected, combining conditional rules for hypertension with a learned classifier for diabetes. It is the closest published precedent of a system combining both measurements with machine learning. As described, however, it classifies single readings rather than series, and it has not been evaluated on patients in routine care.[4]

liu2025review<empty citation> report that artificial neural networks, support vector machines, random forests, gradient boosting and deep learning are the dominant algorithm families for chronic disease monitoring. The accuracies they collect cluster between 95% and 100%, whereas the only recurrent network applied to a genuine clinical prediction task, the prediction of readmission in lupus, reaches an area under the curve

[4] [**To verify**] Only the abstract of **chatrati2022<empty citation>** was consulted. Check the dataset, the rule thresholds, the reported performance and whether readings or windows are modelled.

(AUC) of 0.70. They name the black-box nature of these models and their dependence on large labelled datasets as their main limitations.

The classifier comparison of **efendi2025<empty citation>** illustrates why such figures must be read with care. XGBoost reached an accuracy of 0.973 and the four other models exceeded 0.92. However, the label *stable* is defined by the same normal ranges that are used to discretise the three input features, so the label is a deterministic function of the inputs: the models recover a rule rather than predict a state. Some reported metrics are also mutually inconsistent, and the training configuration described (epochs, sigmoid activation, Adam optimiser) does not apply to tree-based models. The result does not show that XGBoost outperforms other algorithms (Section 3.4.3).

mendoza2025<empty citation> provide the counter-example of a genuine prediction task. Using 1 308 hypertensive adults from the nationally representative Mexican survey ENSANUT 2022, they predicted uncontrolled blood pressure (130/80 mmHg threshold, prevalence 30.3%) from seven predictors selected by LASSO: sex, age, socio-economic status, years since diagnosis, history of diabetes, source of medical care and adherence to a nutritional plan. Random forest reached an AUC of 0.75 (0.72–0.77), logistic regression 0.61 and XGBoost 0.54 (0.49–0.60), the last being indistinguishable from chance. Variable importance and SHAP values ranked age and years since diagnosis first, and the data and code are public. The best model nevertheless detects only 51% of uncontrolled cases (specificity 0.98); the logistic regression row is computed with a different positive class from the other two, and the discussion attributes to random forest a sensitivity that is in fact its specificity (Section 3.4.4). The design is cross-sectional and the predictors are demographic and administrative rather than physiological.

Three lessons follow. Reported performance is governed by how the task is constructed—label integrity and informativeness of the features—far more than by the choice of algorithm. Ensemble tree models, random forest and XGBoost, are plausible candidates, and interpretability through feature importance and SHAP is now expected practice, as **mendoza2025<empty citation>** and **liu2025ppg<empty citation>** both show. And no source forecasts a future physiological state from a patient's own series of blood pressure and blood glucose readings, nor evaluates such a model in a low-resource setting. The methodological consequences are developed in Section 3.4.

2.6 Data Privacy, Security and Ethics in Digital Health

alhammad2024<empty citation> conducted the first systematic review of patients' perspectives on the confidentiality, privacy and security of the data collected by mobile health applications: thirty-three studies published between 2014 and 2022, identified in PubMed, Ovid and ScienceDirect under PRISMA guidance. Patients' concerns about their data are a consistent and significant barrier to adoption. Encryption, secure authentication

and regular risk assessment are recommended as baseline safeguards, and transparency about how data are handled improves trust. Most included studies come from high-income countries and measure self-reported perceptions rather than actual incidents.

These findings converge with earlier observations of this review: most commercial applications either lack a privacy policy or declare sharing data with third parties (**alessa2019**), the legal and regulatory dimension is one of the six barriers to telemedicine in the region (**dodoo2022**), and the reference implementation of **efendi2025<empty citation>** stores its data in a cloud region outside the country of use, which raises the question of data residency. Since the present work processes real, hospital-sourced patient data through a mobile and cloud pipeline, data protection is treated as a first-class design constraint: anonymisation before modelling, encrypted transmission, role-based authentication, and explicit identification of the applicable Cameroonian legal framework (Sections 3.3.5 and 5.1.6).

2.7 Synthesis and Research Gap

Table 2.9 summarises, for each theme, what the literature establishes, what it leaves open, and the consequence for this work.

Table 2.9: Synthesis of the literature review.

Theme	Established	Not established	Consequence for this work
Burden	high burden; cascade fails at every stage, most at control; 58% of people with diabetes are hypertensive	reliable Central African and Cameroonian estimates	longitudinal follow-up of diagnosed patients, joint BP and glucose monitoring
Sub-Saharan context	loss to follow-up, inertia, paper records; six barrier dimensions; specialist scarcity	BP effect of telemedicine in Cameroon; evidence from Central Africa	offline-first, low workload, triage of clinician attention
Clinical evidence	modest effect on BP and HbA1c, conditional on professional oversight	effectiveness in sub-Saharan Africa; long-term effect	clinician-in-the-loop decision support
Sensors and IoT	validated cuff and glucometer required; layered architectures with offline storage	validated cuffless BP; validation on real patients in low-resource settings	instruments for BP and glucose, smartphone sensing exploratory
Machine learning	ensemble trees plausible; interpretability expected; task construction drives performance	forecasting from individual BP and glucose series	leakage-free forecasting formulation, honest metrics
Privacy	data concerns hinder adoption; baseline safeguards known	evidence from African patients	anonymisation, encryption, declared data residency

The research gap can therefore be stated precisely. The need for longitudinal follow-up is established, and telemonitoring linked to a clinician improves blood pressure and

glycaemic control in high-income settings, including in older patients with both conditions (**mihevc2025**). The only validated protocol for this population, however, relies on fixed thresholds applied uniformly to all patients, triggers on the level of a reading rather than on its trend, and uses no learned model. No evaluated system combines consultation data, home measurements and mobile sensor data for hypertension and diabetes; none applies machine learning to the longitudinal series of the individual patient to anticipate deterioration and to order the clinician's attention; and none has been evaluated in Central Africa, under the connectivity, workload, sustainability and data-protection constraints documented in this chapter. The present work addresses this gap. Chapter 3 derives from the review the requirements that such a system must meet.

Part I

Methodology, Modelling and System Architecture Design

Chapter 3

Design Requirements Derived from the Literature

Chapter 2 established what is known and what is missing. This chapter turns those findings into requirements for the proposed system. It proceeds from the clinical requirements of measurement and follow-up (Section 3.1) to the acquisition of the data (Section 3.2), its transport and storage (Section 3.3), its modelling (Section 3.4) and the generation of alerts (Section 3.5). Each section closes with a specification table that states every requirement together with its basis. Beyond the sources of the review, two reference architectures for chronic disease telemonitoring are used here as engineering references (**yazdani2018**; **mucchi2021**), and the research framing document of this work is confronted with the evidence wherever it states a design choice.

3.1 Clinical Requirements for Measurement and Follow-up

The review showed that the episodic care model fails because it produces point-in-time information, whereas diagnosis, treatment adjustment and programme monitoring all require repeated measurements organised as a series (Section 2.2.2). This section translates that finding into measurement and follow-up requirements.

3.1.1 Diagnosis and the variability of blood pressure

3.1.1a What diagnosis requires

The diagnostic protocol recommended by the WHO is demanding, and its detail is worth reporting, because the insufficiency of the episodic model follows from it.

Measurement must be performed by trained staff, in a standardised way, with a cuff of appropriate size, the patient seated comfortably, back supported, bladder empty and legs

uncrossed. Two readings must be taken at the first encounter, and the result of the second guides the decision whether to schedule a confirmation visit. What follows depends on the level observed (Table 3.1).

Table 3.1: Recommended action by confirmed pressure level at the first encounter (**who2023hypertension**).

Confirmed pressure on both readings of the day	**Recommended action**
SBP $\geq$ 160 or DBP $\geq$ 100 mmHg	start treatment the same day
SBP 140–159 or DBP 90–99 mmHg	confirm at an additional visit, usually one to four weeks later

Diagnosis is made only if, at both visits, systolic pressure is $\geq$ 140 mmHg and/or diastolic pressure is $\geq$ 90 mmHg.

Diagnosis is therefore, by construction, a longitudinal act: it requires two measurement points separated by one to four weeks (the recheck window already noted in Section 2.1.2). A care model that does not guarantee the patient's return for the second visit does not produce imperfect diagnoses; it produces none.

3.1.1b What variability imposes

Blood pressure is a highly dynamic vital sign, influenced by time of day, activity, stress and the measurement environment. The WHO observes accordingly that, in population screening, many people with elevated pressure no longer meet the diagnostic criteria when assessed more carefully at follow-up (**who2023hypertension**).

This variability can be formalised. For patient i measured at time t,

$$b_{i,t} = \mu_i + \varepsilon_{i,t}, \tag{3.1}$$

where μ_i is the patient's usual pressure, the only clinically relevant quantity, and $\varepsilon_{i,t}$ gathers circadian variability, variability due to activity and stress, the effect of the measurement environment and instrument error. A single measurement estimates μ_i with a variance equal to that of ε; n sufficiently spaced measurements, whose errors are independent, reduce it by a factor n:

$$\mathrm{Var}(\hat{\mu}_i) = \frac{\sigma_\varepsilon^2}{n}. \tag{3.2}$$

Independence is an assumption: two readings taken in the same sitting share their context, so the gain from the second is smaller than the formula suggests, which is one more reason why the protocol also asks for a second visit.

This is the logic of the two-reading, two-visit protocol (Figure 3.1). It is also, and this is the central argument of this thesis, the logic that favours home measurement: where a

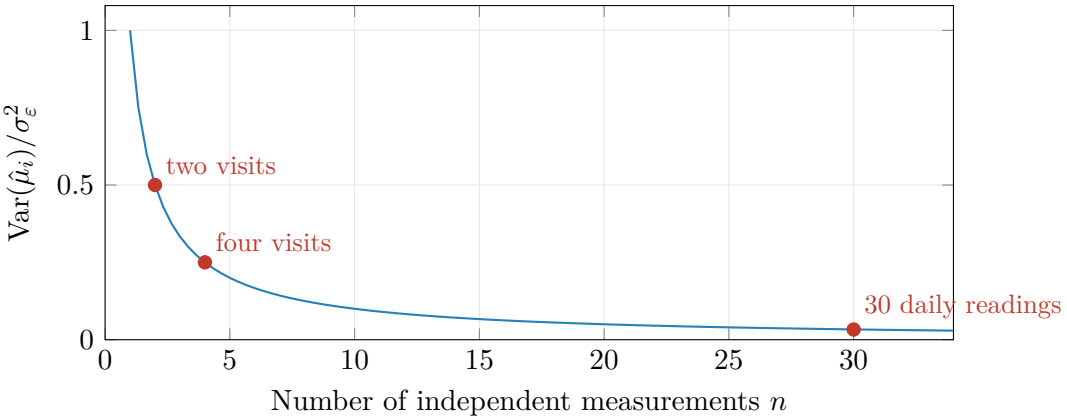


Figure 3.1: Relative variance of the estimate of the usual pressure μ_i as a function of the number of independent measurements. Errors are assumed independent.

consultation provides at best two to four points per year, home follow-up produces several dozen, which reduces uncertainty about μ_i and additionally allows σ_ε itself, the patient's blood pressure variability, to be estimated, an independent risk factor.

3.1.2 Follow-up frequency

3.1.2a What follow-up requires

Recommendation 7 of the WHO 2021 guideline on the pharmacological treatment of hypertension (**who2021guideline**) sets the expected follow-up frequency:

- monthly follow-up after initiating or changing antihypertensive treatment, until the patient reaches the blood pressure target;
- follow-up every three to six months for patients whose pressure is controlled.

The therapeutic protocol module of the HEARTS technical package (**whoHEARTS**) further specifies that a simple protocol must set the return-visit interval, "usually four weeks or less"[†], to allow rapid titration of doses and drugs until control is reached.

Let us confront this requirement with the data of Section 2.1. In the African Region, treatment coverage is 27% and control 12%, from which we derived that 56% of treated patients do not reach the target (Section 2.1.2). According to recommendation 7, these patients should be seen monthly until they do.

The gap can be quantified (Table 3.2). An uncontrolled patient should receive about twelve contacts per year, a controlled patient two to four. For an active list in which a share $\pi_u = 0.56$ of patients is uncontrolled, the theoretical need is

$$\bar{n} = \pi_u \times 12 + (1-\pi_u) \times n_c \approx 7.6 \text{ to } 8.5 \text{ contacts per patient per year}, \qquad n_c \in [2, 4]. \tag{3.3}$$

If the episodic model delivers one to two contacts per year, it delivers between about

Table 3.2: Follow-up contacts per patient per year required by WHO recommendation 7.

Patient profile	**Contacts per year**
Uncontrolled: monthly follow-up until the target is reached	12
Controlled: follow-up every three to six months	2 to 4
Active list with 56% uncontrolled (Section 2.1.2)	7.6 to 8.5

an eighth and about a quarter of the recommended frequency.[1] It is this cumulative gap, and not the duration of each consultation taken in isolation, that constitutes the structural deficit.

3.1.3 From act data to series data

The longitudinal record that the WHO describes as essential (Section 2.2.2) matters because clinical information only acquires its meaning in a series.

This distinction between act data and series data is central to the system design (Figure 3.2). A pressure of 152/94 mmHg does not have the same meaning depending on whether it follows a stable series around 135/85 or is part of a progressive drift observed over three months. The first case calls for a check, the second for therapeutic intensification. The paper register cannot distinguish the two situations; from an informational viewpoint, this is the reason why therapeutic inertia persists.

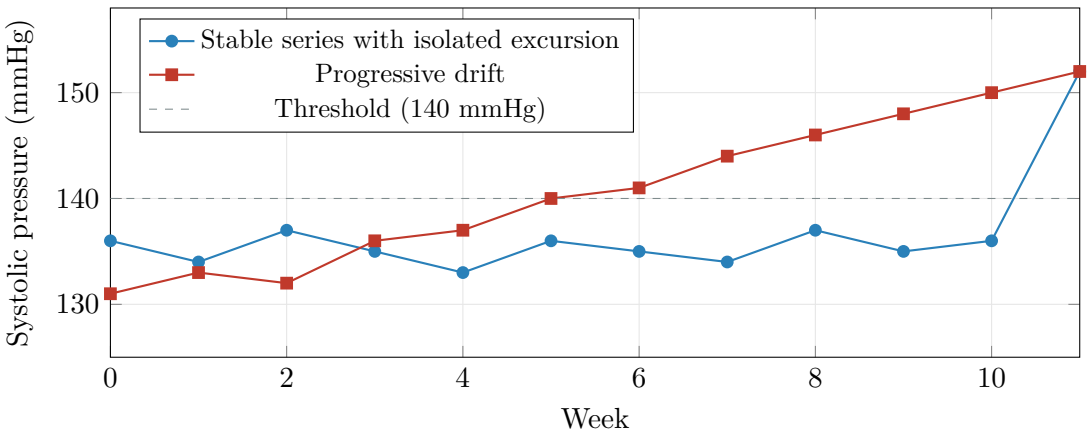


Figure 3.2: Same last value, different meaning. Schematic illustration with synthetic values, not patient data: a single reading of 152 mmHg is a check-up case in one series and an intensification case in the other.

A further requirement follows. Information silos are not only a consequence of paper: they re-form in a digital environment as soon as each device produces its own format

[1] [**To verify**] The figure of one to two delivered contacts per year has no source in the draft, where it was introduced as "available data for the region". Cite a regional or Cameroonian source, or present it explicitly as an illustrative scenario. The draft's "one fifth to one quarter" assumed half of the list uncontrolled; with the 56% derived in Section 2.1.2 the range is about 12–26%.

(**liu2025review**). The present work must therefore define a single, documented data schema, rather than aggregating heterogeneous flows after the fact.

liu2025review<empty citation> note that electronic medical records themselves require prior cleaning, the typical artefacts being duplicates, unit errors and values carried over identically from one visit to the next. This must be kept for the experimental part: digitisation does not remove the data-quality problem, it displaces it. Any model trained on real consultation data must include an explicit step to detect these three artefacts, and the identically carried-over value, which falsely simulates clinical stability, is the most dangerous of the three for a trend-detection model.

3.1.3a What the WHO expects from an information system

The report does not stop at the finding: it defines the expected outputs of a hypertension information system, as a short list of core HEARTS indicators. This list is a directly usable functional specification (Table 3.3).

Table 3.3: Core HEARTS programme indicators (**who2023hypertension**).

Indicator	**Definition**	**Temporal**
Programme coverage	number of hypertensive patients registered in the catchment area	no
Blood pressure control	proportion of registered patients with BP $< 140/90$ mmHg over the last three months	yes
Retention in care	proportion of registered patients with no visit in the last three months, and in the last twelve months	yes
Availability	availability of essential antihypertensive medicines and BP measurement devices	no

Two observations are in order. First, two of the four core indicators are temporal: they can only be computed from a dated history, and a paper register cannot produce them except at the cost of an impracticable manual review. Second, the control indicator is itself defined over a three-month sliding window, which confirms that the quantity of interest is a property of the series and not of a single measurement.

The WHO specifies that when programme managers receive these indicators monthly, they can identify lagging facilities and implement targeted improvements. It cites the DHIS2 Hypertension Tracker as an example of a user-friendly mobile application. Among the six maturity domains conditioning the national scale-up of a programme, it also includes the presence of a digital information system for hypertension management, with, as a clinical quality factor, a missed-visit rate below 25%, defined as failure to return to a scheduled visit within the preceding three or six months. The other five domains are clinical capacity, service delivery, clinical quality, medicine supply and political commitment.

3.1.4 Self-measurement protocols

3.1.4a Blood pressure self-measurement

The WHO, whose protocol we reported in Section 3.1.1a, sets the conditions of measurement: trained staff or patient, cuff of appropriate size, seated position with back supported, empty bladder, legs uncrossed, two readings of which the second is retained. It further establishes that only independently validated automated electronic upper-arm cuff devices are recommended, and that no cuffless alternative was recommended in 2023 (Section 2.4.1).

mihevc2025<empty citation> operationalise this protocol in a real telemonitoring device, and their description constitutes the most precise specification at our disposal: patients took two readings in the morning and in the evening, after five minutes of rest in a seated position, with back and arm supported. A one-minute interval was programmed by the device between the two readings, and the second was retained as valid.

Two elements of this description deserve to be retained for the design. First, the interval between the two readings is imposed by the device and not left to the patient's initiative, which removes a major source of non-compliance with the protocol. Second, the validity rule—retain the second reading—is applied automatically, and not by a computation demanded of the patient.

For establishing the reference value at the assessment time points, the authors apply a stricter rule still: the second morning reading and the second evening reading are averaged over seven days. This is the direct application of the variance-reduction logic set out in Section 3.1.1b: with fourteen retained measurements, the standard error of the estimator of the usual pressure is divided by $\sqrt{14} \approx 3.7$ relative to an isolated measurement.

3.1.4b Capillary blood glucose self-monitoring

mihevc2025<empty citation> describe a glycaemic protocol articulated in two regimes.

In the *routine* regime, the patient performs two measurements: one in the morning while fasting and one ninety minutes after lunch.

In the *verification* regime, triggered by an abnormal value, the patient performs a one-day glycaemic profile comprising six measurements: fasting, 90 minutes after breakfast, before lunch, 90 minutes after lunch, before dinner, and 90 minutes after dinner.

The therapeutic targets retained, adjusted for an elderly population, are a fasting glucose below 7 mmol/L and a postprandial glucose below 10 mmol/L. For Cameroonian practice, which usually expresses blood glucose in grams per litre, these thresholds correspond to about 1.26 g/L and 1.80 g/L, the conversion factor being 1 mmol/L $\approx$ 0.18 g/L.

The blood pressure target retained is a home pressure below 135/85 mmHg—and not 140/90 mmHg—in accordance with the convention whereby self-measurement thresholds are five points below consultation thresholds, the absence of a white-coat effect lowering

the values measured at home.

3.1.4c The decision architecture of the joint protocol

The most directly exploitable contribution of **mihevc2025<empty citation>** is their three-level classification system, applied symmetrically to both conditions (Table 3.4).

Table 3.4: Three-level classification of **mihevc2025<empty citation>**.

Code	Interpretation	Action triggered
Green	normal values	no change to the follow-up protocol
Yellow	abnormal values	intensified surveillance: seven-day blood pressure profile, or one-day six-measurement glycaemic profile
Red	critically abnormal values	immediate teleconsultation with the general practitioner, further assessment or treatment modification

The operational thresholds, as presented in Figure 1 of the article, are reproduced in Table 3.5. We have reorganised them by decision level and by meal timing, because the original presentation leaves the reading of the glycaemic intervals ambiguous.

Table 3.5: Operational thresholds of the two protocols (**mihevc2025**), reorganised by decision level. BP in mmHg, glucose in mmol/L.

Level	Blood pressure (current reading)	Blood glucose (current reading)
Red (low)	SBP $<$ 110 or DBP $<$ 60	$<$ 4 (any timing)
Green	110–134 / 60–84	4–7 before a meal; 7–10 after a meal
Yellow	135–174 / 85–104	7.1–10 before a meal; 10.1–14 after a meal
Red (high)	SBP $\geq$ 175 or DBP $\geq$ 105	$>$ 10 before a meal; $>$ 14 after a meal
On the profile mean	$<$ 110/60 or $\geq$ 135/85 $\Rightarrow$ teleconsultation	$>$ 7 before a meal or $>$ 10 after a meal $\Rightarrow$ teleconsultation

This architecture possesses three remarkable properties, which it is appropriate to state explicitly because they directly orient the design of the proposed system.

It is bidirectional. Low thresholds trigger as much as high ones. For diabetes this is an obvious clinical necessity—hypoglycaemia is an acute risk—but the device applies the same logic to blood pressure, hypotension in the treated elderly patient being a risk of falls. A system detecting only elevated values would be clinically incomplete.

It is two-stage. An isolated abnormal value does not trigger a clinical action but an additional collection of information. Only in the light of the profile mean is the teleconsultation triggered. This is exactly the operational answer to the variance problem set out in Section 3.1.1b: the device distinguishes noise from trend by increasing n before deciding, rather than by lowering the threshold (Figure 3.3).

It is economical in human attention. Only the red class mobilises a physician. The yellow class mobilises the patient. In a context where, as **dodoo2022<empty citation>**

establish, professionals perceive telemonitoring as an additional workload (Section 2.2.4), this distribution is a condition of adoption and not a refinement.

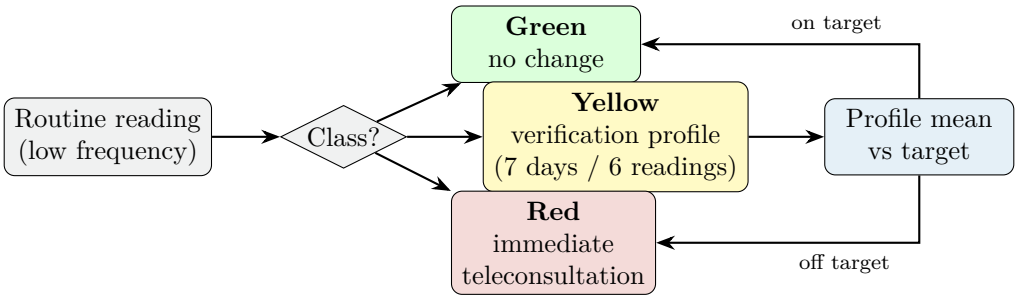


Figure 3.3: The two-stage triggering architecture of **mihevc2025<empty citation>**. An isolated abnormal value increases the number of measurements instead of mobilising a clinician; only the mean of the verification profile reaches a clinical decision.

3.1.4d The question of frequency

The sources diverge on the frequency of the routine regime, and this divergence calls for clarification before any reuse.

The abstract and the body of the article of **mihevc2025<empty citation>** indicate that participants measured their blood pressure twice a week and their blood glucose once a week. Figure 1 of the same article, by contrast, carries the labels "weekly blood pressure measurements" and "monthly blood glucose measurements". These two statements are incompatible. We retain the textual description, confirmed by the abstract, and we signal the figure as erroneous, without being able to settle the matter definitively without recourse to the separately published protocol.[2]

The frequency trade-off must in any event be posed explicitly, because it sets two contrary requirements against each other. Variance reduction argues for maximal frequency, since $\mathrm{Var}(\hat{\mu}) = \sigma_\varepsilon^2/n$. Adherence argues for minimal frequency, every measurement demanded being an occasion for abandonment. The device of **mihevc2025<empty citation>** resolves this tension by modulation: a low routine frequency, sufficient for surveillance, and a high frequency triggered only when a signal justifies it. This is a solution the proposed system should adopt, because it optimises both criteria simultaneously instead of sacrificing one of them.

One residual ambiguity of the threshold table must be noted, because any implementation will have to resolve it: the blood pressure rules are stated on two variables, and the source does not say what happens when the two disagree—a systolic value in the yellow band together with a diastolic value in the low red band, for instance. The safe

[2] **[To verify]** Resolve this frequency divergence against the published protocol of **mihevc2025<empty citation>** (ISRCTN31471852) before reusing either value as a design parameter.

convention, which we adopt, is that the most severe of the two classifications prevails, red before yellow and yellow before green. This is our decision, not a statement of the source.

3.1.5 Consequences of the metrological constraint

The WHO position on measuring devices (Section 2.4.1) has three consequences for the design.

Three consequences must be explicitly assumed:

1. The "mobile sensors" component cannot rely on cuffless devices for blood pressure measurement. These can provide context variables—physical activity, heart rate, sleep—useful as model covariates, but blood pressure must come from a validated automated upper-arm monitor. This restriction does not amputate the project: it moves it from the ambition of unconstrained continuous measurement to that of reliable repeated measurement, which is what the clinical protocol actually requires.
2. The list of admissible devices must be restricted to devices on a recognised validation register, and this criterion must be part of the system specification.
3. The cost of the device, higher for validated devices, must be integrated into the economic feasibility analysis alongside the cost of treatment, which the WHO puts at 5 to 10 US dollars per patient per year for quality-assured medicines, amlodipine and hydrochlorothiazide generally being available at 0.01 dollar per tablet or less, but with prices varying more than tenfold between countries and mark-ups recorded in Kenya exceeding 340% for amlodipine and 280% for hydrochlorothiazide.

3.1.6 Specification

3.2 Multi-Source Acquisition: IoT/IoMT, Biomedical Sensors, Mobile Entry

Section 3.1 established what the system must measure and how often. This section addresses the first link of the technical chain, the acquisition of the data, drawing on the sensor and IoT literature reviewed in Section 2.4, on the reference implementation of **efendi2025<empty citation>**, and on the acquisition design stated in the research framing document.

3.2.1 Three acquisition pipelines, three different kinds of data

The system draws on three sources that differ not only in their technology but, more importantly for what follows, in their statistical properties. Treating them as a single undifferentiated input stream would be the first modelling error, and the distinction must be established before any architecture is described (Figure 3.4).

Table 3.6: Clinical requirements established in Section 3.1 and their basis.

Requirement	Basis
Validated automated upper-arm monitor, never cuffless measurement	**who2023hypertension<empty citation>**; **alessa2019<empty citation>**
Automatic transmission, no manual entry	**alessa2019<empty citation>**: 73 of 84 applications require entry; **mihevc2025<empty citation>**: Bluetooth to a concentrator
Measurement protocol imposed by the device: 2 readings, 1-minute interval, second retained	**mihevc2025<empty citation>**
Reference value = seven-day mean of the second morning and evening readings	**mihevc2025<empty citation>**
Three-level classification, bidirectional (high and low thresholds)	**mihevc2025<empty citation>**
Two-stage architecture: anomaly → verification profile → clinical decision	**mihevc2025<empty citation>**
Most severe of the systolic and diastolic classifications prevails	our decision; the source is silent (Section 3.1.4c)
Low routine frequency, triggered intensification	**mihevc2025<empty citation>**
At least 11 behaviour change techniques, including the 3 universal ones	**zhou2024<empty citation>**: −9.68 vs −2.88 mmHg
Multifactorial, not single-factor intervention	**zhou2024<empty citation>**: −6.50 vs −1.54 mmHg
Explicit and declared theoretical foundation	**zhou2024<empty citation>**: only 30% of trials declare one
Priority output = therapeutic decision aid for the provider	**mihevc2025<empty citation>**: effect obtained without behavioural change
Privacy policy, data localisation and protection made explicit	**alessa2019<empty citation>**: 70% of applications share with third parties
Training treated as a component of the device, not a precondition	**mihevc2025<empty citation>**: 10.9% abandonment through inability to use
Assessment of effect maintenance beyond 12 months	**zhou2024<empty citation>** and **mihevc2025<empty citation>**: documented erosion

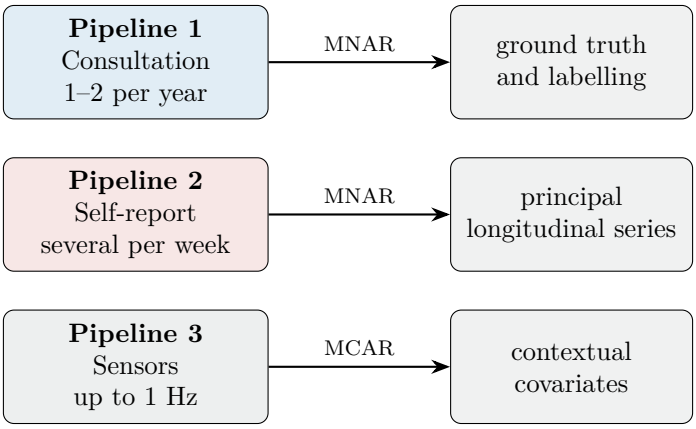


Figure 3.4: The three acquisition pipelines, their sampling rates and their roles. Two of the three carry missingness that depends on the patient's state (MNAR); only the instrumented pipeline fails at random (MCAR), and it is the one that cannot measure the parameters of interest.

3.2.1a Pipeline 1: clinical consultation data

Content. Hospital history, laboratory results, vital signs recorded at the visit, background variables and clinical context. The framing document specifies the record structure: systolic and diastolic blood pressure, blood glucose, heart rate and measurement timestamp, together with background variables—age, sex, body mass index—and clinical context—history, treatments.

Properties. This pipeline is low-frequency, high-reliability, and irregularly sampled by clinical rather than temporal logic. A measurement exists because a consultation occurred, and a consultation occurred because the patient attended—which Section 3.1.2a established happens for perhaps one to two contacts per year against the seven to eight the protocol requires. The sampling process is therefore not independent of the patient's state: a patient who deteriorates may consult more often, or may disappear entirely. In Rubin's terminology this is data missing not at random, and it has direct consequences for any model trained on the series: the missingness pattern is itself informative, and a model that ignores it will be fitted on a sample systematically unrepresentative of the states it is meant to detect.

Reference status. Within the system, this pipeline supplies the ground truth. The framing document states that risk level is derived from recognised clinical thresholds—ESH 2023 for blood pressure, ADA standards for blood glucose—and validated by the partner physician.[3]

3.2.1b Pipeline 2: mobile self-report

Content. Manual patient entry of home measurements.

Properties. This pipeline is medium-frequency, variable-reliability, and subject to behavioural artefacts. Section 2.3.1 established its failure modes precisely: in the commercial applications reviewed by **alessa2019<empty citation>**, 73 of 84 required manual entry of a value the device had just displayed. Three consequences follow. Transcription error—digit transposition, systolic–diastolic inversion—is undetectable downstream. The timestamp records the moment of entry, not the moment of measurement, so a patient entering three days of readings on a Sunday evening produces a series whose temporal structure is fictitious. And selective reporting is possible: a patient may omit a reading they find discouraging, which biases the series in precisely the direction that matters clinically.

The last of these deserves emphasis, because it is not a noise problem but a bias problem. Noise can be averaged away—this is the whole content of $\mathrm{Var}(\hat{\mu}_i) = \sigma_\varepsilon^2/n$

[3] **[To verify]** Add formal bibliographic entries for the 2023 European Society of Hypertension guideline and the American Diabetes Association Standards of Care edition actually used, and state the edition year. Both are cited as label sources and must be referenced as such.

established in Section 3.1.1b—whereas selective omission cannot, and it systematically removes the extreme values a detection model exists to find.

3.2.1c Pipeline 3: connected biomedical sensors

Content. Instrumented acquisition transmitting without human intervention. In the architecture described by **efendi2025<empty citation>**, a MAX30100 optical sensor for heart rate and oxygen saturation and a MAX30205 digital temperature sensor, both read by an ESP32 microcontroller.

Properties. This pipeline is high-frequency, instrument-limited, and free of transcription and reporting bias. **efendi2025<empty citation>** collect one record per second over ten minutes, yielding 600 observations per participant. Reliability is no longer a function of patient behaviour but of sensor accuracy, signal conditioning and transmission integrity, and its failures—a flat battery, a lost connection—are plausibly independent of the patient's clinical state.

Its limitation is different in kind: it measures what the sensor can measure, which, as the next subsection shows, does not include blood pressure or blood glucose at an acceptable level of validation.

3.2.1d The properties that matter for modelling

Table 3.7: Statistical properties of the three acquisition pipelines.

Property	**P1: consultation**	**P2: self-report**	**P3: sensors**
Sampling frequency	1–2 per year	several per week	up to 1 Hz
Sampling regularity	irregular, event-driven	irregular, behaviour-driven	regular
Missingness mechanism	MNAR: attendance depends on state	MNAR: omission possible	MCAR: technical failure
Dominant error	measurement conditions	transcription, timestamp, selection	instrument accuracy, movement artefact
Parameters available	BP, glucose, HR, laboratory	BP, glucose, HR	HR, SpO_2, temperature
Blood pressure available	yes, reference quality	yes, if validated cuff	**no**
Blood glucose available	yes, laboratory	yes, if glucometer	**no**
Role in the system	ground truth and labelling	principal longitudinal series	contextual covariates

The final row of Table 3.7 is the conclusion of this subsection and it should govern the architecture. The two parameters that define the clinical problem—blood pressure and blood glucose—are not obtainable from the optical sensor pipeline. That pipeline supplies heart rate, oxygen saturation and temperature, which are covariates: useful for

characterising the context of a reading, for detecting movement, for enriching the feature space, but not substitutes for the measurements on which the clinical decision rests.

The record structure of the framing document confirms this allocation: the fields listed are systolic pressure, diastolic pressure, blood glucose, heart rate and timestamp. Of these, only heart rate can come from an optical sensor.

3.2.2 Physical principles and hardware constraints

3.2.2a Photoplethysmography

Photoplethysmography is a non-invasive optical method that measures variations in blood volume in peripheral tissue: a light-emitting diode illuminates the skin and a photodetector captures the reflected or transmitted fraction, whose pulsatile variation follows the heartbeat. Transmissive sensors are placed where tissue is thin, typically the fingertip, and reflective sensors on the wrist, arm, chest or forehead.

efendi2025<empty citation> describe the same mechanism for the MAX30100: the sensor works "by shining both lights on the finger and detecting the amount of reflected light with a photo-detector"†, and they name this method photoplethysmography explicitly.

The underlying physics is Beer–Lambert attenuation. For light of incident intensity I_0 traversing an absorbing medium,

$$I = I_0\, e^{-\varepsilon c d}, \tag{3.4}$$

where ε is the molar extinction coefficient of the absorber at the wavelength used, c its concentration and d the optical path length. The measured signal decomposes into a large quasi-static component, conventionally written DC, arising from tissue, bone, venous blood and the non-pulsatile arterial fraction, and a small pulsatile component, written AC, arising from the arterial volume change at each systole (Figure 3.5). Heart rate is recovered from the periodicity of the AC component alone; it does not require absolute calibration, which is why heart rate from photoplethysmography is comparatively robust.

3.2.2b Dual wavelength and oxygen saturation

Oxygen saturation requires a second wavelength. **efendi2025<empty citation>** state that the MAX30100 light-emitting diodes "have wavelengths of 660 nm and 880 nm, respectively"†.

The principle rests on a differential absorption property. Oxyhaemoglobin and deoxyhaemoglobin have different extinction spectra: at 660 nm, in the red, deoxyhaemoglobin absorbs more strongly; in the near infrared, around 880 to 940 nm, oxyhaemoglobin absorbs more strongly. Measuring the pulsatile and static components at both wavelengths yields a ratio of ratios,

$$R = \frac{AC_{660}/DC_{660}}{AC_{880}/DC_{880}}, \tag{3.5}$$

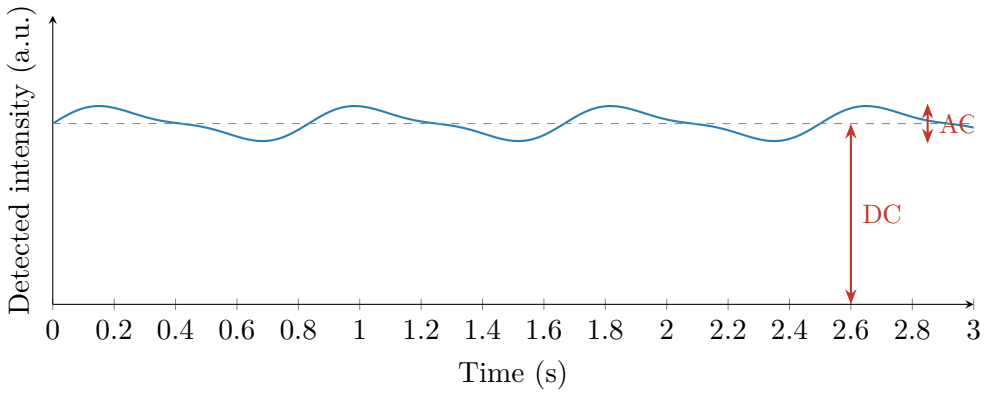


Figure 3.5: Schematic decomposition of a photoplethysmographic signal into its quasi-static (DC) and pulsatile (AC) components. Synthetic illustration, not recorded data. Heart rate follows from the period of the AC component and needs no absolute calibration; oxygen saturation needs the AC/DC ratio at two wavelengths and an empirical curve.

from which saturation is derived by an empirical calibration curve, conventionally of the form $SpO_2 \approx a - bR$, with coefficients established on human volunteers under controlled desaturation.

Two observations follow, and both are design constraints.

First, the calibration is empirical, not physical. The coefficients a and b are fitted, not derived. A sensor used outside the population and conditions of its calibration has no guaranteed accuracy, and skin pigmentation is one documented source of divergence in pulse oximetry. For a system intended for deployment in Cameroon this is not an academic point, and it must be acknowledged even though none of the sources in this corpus addresses it. It is also the same argument, in a different register, as the transferability warning that **segun2025<empty citation>** apply to externally trained algorithms (Section 2.2.4): a calibration is a model fitted on a population, and it travels no better than any other.

Second, the description given by efendi2025<empty citation> is physically imprecise. They write that "the saturated oxygen in arterial blood has the property of absorbing infrared radiation. The higher the hemoglobin level, the more IR light is absorbed."† This conflates two distinct quantities: the concentration of haemoglobin and the saturation of haemoglobin with oxygen. What the ratio-of-ratios method exploits is the relative difference in absorption between the oxygenated and deoxygenated forms at two wavelengths, not the total haemoglobin concentration—which is in fact largely cancelled by the ratio. The statement as written would imply that an anaemic patient reads as desaturated, which is not the case. We note this because the sentence may otherwise be reproduced uncritically.

3.2.3 A documented reference implementation

3.2.3a The three-layer architecture

efendi2025<empty citation> structure their system into the three layers conventional in IoT construction (Figure 3.6).

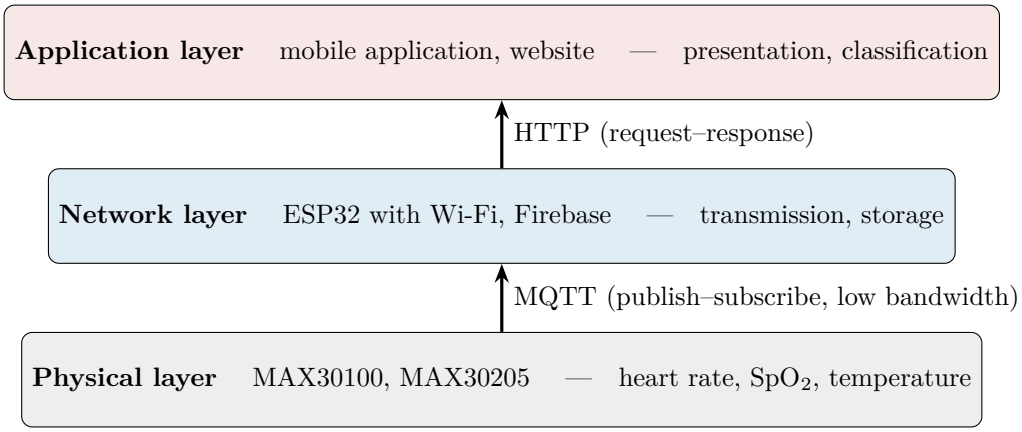


Figure 3.6: The three-layer architecture of **efendi2025<empty citation>**, with the two transport protocols used in sequence.

Transmission uses two protocols in sequence. The sensor-to-cloud path uses MQTT, a lightweight messaging protocol that the authors select because it "enables resource-constrained devices to send sensor data to a gateway which acts as a translator, then processes and stores the data with Firebase functionalities such as Cloud Functions or Real-time Database"†. The cloud-to-client path uses HTTP: "the data which has been stored to Firebase is accessed by mobile applications and websites using HTTP protocol"†.

The choice of MQTT upstream is appropriate and worth retaining. It is a publish–subscribe protocol designed for low bandwidth and unreliable links, and it is markedly more economical than HTTP for frequent small payloads—which is the exact profile of sensor telemetry over an intermittent mobile connection.

Firebase is selected for four properties the authors enumerate: real-time database services synchronising data instantaneously across clients and platforms; built-in authentication and authorisation ensuring secure access control; transmission in JSON; and persistent WebSocket connections, so that "whenever current sensor data arrives, Firebase sends it to linked devices"†.

The fourth property is the one that matters most for the target context: offline operation. The authors state that data are "stored locally and real-time events continue to occur even when the device is offline"†, and that "the real-time data will be synchronised between local data change remotely when the device is offline and auto merging when the device is going to be online"†. Given that the framing document identifies intermittent connectivity in the Cameroonian context as a source of irregular and missing measurements,

a storage layer with native offline-first behaviour and automatic merge on reconnection addresses a stated constraint directly.

3.2.3b Device validation

The authors validate the instrumentation against reference instruments before using it. The protocol is as follows.

Participants. Six individuals aged 50 to 65. Two of them—aged 63 and 65—have a chronic disease and are classified as not stable.

Procedure. Data collected every second for ten minutes in a seated resting position, yielding 600 records per participant. Readings from the monitoring device are compared against measurement tools, namely a digital oximeter and a digital thermometer. An interview established each participant's health history, and written consent was obtained.

Metric. Mean absolute percentage error, with a predefined acceptance threshold of 3%:

$$\mathrm{MAPE} = \frac{100}{n} \sum_{k=1}^{n} \left| \frac{A_k - P_k}{A_k} \right|. \tag{3.6}$$

Table 3.8: Validation results reported by **efendi2025<empty citation>**.

Parameter	Device	Reference tools	Gap	Percentage error
Heart rate (BPM)	84.37	85.79	1.42	1.68%
Oxygen saturation (%)	96.85	96.30	0.55	0.57%
Body temperature (°C)	36.53	36.69	0.16	0.44%
Overall MAPE				0.90%
Accuracy (100% − MAPE)				99.10%

The experimental protocol also defines four real-world scenarios—sleeping, walking, sitting, cardio exercising—though the validation results reported are from the seated condition.

3.2.3c Critical appraisal

Five reservations must accompany any citation of these figures.

First, the denominator of the error is the device, not the reference. The reported percentage errors can be reproduced only by dividing the gap by the device reading. For heart rate, $1.42/84.37 = 1.68\%$, which is the published value, whereas $1.42/85.79 = 1.66\%$. The quantity written A_k in the formula—conventionally the actual or reference value—is therefore the device output, and the comparator instrument occupies the position of the prediction. The study thus measures how far the reference deviates from the device. For the other two parameters the two conventions round to the same figure, so heart rate is the only row that discriminates; but the inversion is systematic and

it reverses the logic of a validation. The overall figure is the unweighted mean of the three, $(1.68 + 0.57 + 0.44)/3 = 0.90\%$, computed on the parameter means rather than on paired observations, which removes dispersion before the statistic is formed.[4]

Second, agreement with an unvalidated reference is not accuracy. The comparator instruments are a digital oximeter and a digital thermometer, neither identified by make or model, and neither reported as independently validated. Section 2.4.1 established, on WHO authority, that only about one device in five on the market carries evidence of accuracy validation. What the study demonstrates is concordance between two consumer devices, which is a necessary but not a sufficient condition for accuracy.

Third, MAPE is not an appropriate agreement statistic for this purpose. It averages the absolute relative difference and therefore conceals the structure of the disagreement: a device reading systematically 1.4 BPM high and a device oscillating randomly by ±1.4 BPM produce the same MAPE but have entirely different clinical implications. The standard instrument for method comparison is Bland–Altman analysis, which reports bias and limits of agreement separately; it is not used here. The authors' own figures reveal what the summary statistic hides: they note "a fairly wide gap between the two devices, especially at time 300 to 360"† for heart rate, and "a significant gap in the 340 and 380 time range"† for oxygen saturation. These transient divergences disappear entirely in a 0.90% average.

Fourth, the validation conditions are the most favourable possible. Six participants, ten minutes, seated at rest. Optical and oscillometric measurements are known to degrade during physical activity because of movement artefacts. The four scenarios including walking and cardio exercise are described in the methods but no comparative error figures are reported for them.

Fifth, and decisively for the present work, neither blood pressure nor blood glucose is measured. The validated parameters are heart rate, oxygen saturation and temperature. This architecture, however well executed, does not acquire the two variables on which the clinical problem of this thesis rests.

3.2.3d The machine learning component, and a structural flaw worth studying

efendi2025<empty citation> add a supervised classifier predicting whether the user's condition is stable or not stable.

Dataset. 4 500 records, combining participant data with data from a public Kaggle repository. Three features—heart rate, oxygen saturation, body temperature—and a binary label. The classes are balanced exactly: 2 250 records each.

Preprocessing. The authors apply a transformation in which each feature is recoded

[4] [**To verify**] Confirm from the Methods of **efendi2025<empty citation>** whether MAPE was computed per observation and then averaged, or on the aggregate means as the published figures suggest. The two give different values and only the first is a meaningful agreement statistic.

as 0, 1 or 2 to represent low, normal and high conditions, with reference to their Tables 9 to 11, and the label is recoded as 1 for stable and 0 for not stable.

Split and training. 70% training (3 150), 10% validation (450), 20% testing (900). Five models compared: AdaBoost, XGBoost, decision tree, random forest and a Bayesian classifier. Configuration: 300 epochs, learning rate 0.025, sigmoid activation, Adam optimiser, on an Intel Core i7-4770 at 3.40 GHz with 8 GB of memory under Python in a Jupyter notebook.

Results. All five models exceed 0.9 accuracy. XGBoost is selected with accuracy 0.973 and F_1 0.970; AdaBoost follows at 0.967 accuracy and 0.946 F_1; the decision tree, random forest and Bayesian classifiers reach F_1 scores of 0.933, 0.929 and 0.928. Loss values are below 0.1 throughout, with XGBoost lowest at 0.025 and the Bayesian classifier highest at 0.062.

The flaw. The label *stable* is defined as the condition in which all three parameters fall within the normal ranges of Tables 9 to 11. The feature transformation recodes each parameter as low, normal or high using those same ranges. After transformation, therefore, the label is a deterministic function of the three transformed features:

$$y = 1 \iff (x_1 = 1) \wedge (x_2 = 1) \wedge (x_3 = 1). \tag{3.7}$$

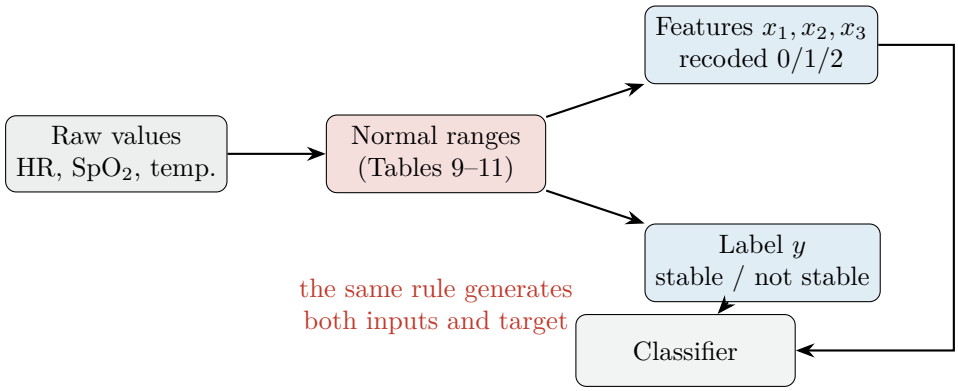


Figure 3.7: The leakage mechanism in **efendi2025<empty citation>**. The normal ranges that define the label also define the feature encoding, so the classifier recovers a rule already present in its inputs.

This is label leakage (Figure 3.7). The model is not learning a clinical relationship; it is recovering the rule that generated the labels, from features into which that rule has already been encoded. A decision tree of depth three should achieve 100% accuracy on this task. That XGBoost reaches 97.3% rather than 100% indicates inconsistency somewhere in the labelling or the transformation, not predictive difficulty.

Two further features of the setup confirm the diagnosis. The perfect class balance of 2 250 against 2 250 does not arise in clinical data, where, as the framing document notes for the present work, critical states are rare; it arises from construction. And the comparison

of five classifiers is uninformative when all five are solving a lookup problem: the 0.045 spread in F_1 between best and worst measures nothing of consequence.

This matters for the present thesis well beyond the appraisal of one paper. It is the precise error that must not be repeated. The framing document states that the risk label—stable, moderate, high, critical—is derived from recognised clinical thresholds, ESH 2023 for blood pressure and ADA standards for blood glucose, and validated by the partner physician. If the model's features then include the current values of blood pressure and blood glucose, the same circularity arises: the model will reproduce the threshold rule and report a high score that demonstrates nothing.

Two of the framing document's own commitments already point the way out, and should be stated explicitly as the defence against this failure.

The first is the anticipatory dual-task formulation: forecasting physiological degradation at 24 to 48 hours rather than classifying the current state. A model that predicts a future threshold crossing from a past window cannot leak its label, because the label does not exist in the input window. This is not merely a more ambitious objective; it is the formulation that makes the problem genuine. It is also, in different words, the distinction drawn in Section 2.2.2 between the act and the series: a classifier of the current reading is an instrument of the act, a forecaster over a window is an instrument of the series.

The second is the feature engineering the framing document describes—rolling aggregates, trends, variability, deviations from thresholds. Of these, deviations from thresholds computed at the same instant as the label is precisely the leaking feature, while the same quantity computed on a lagged window is legitimate. The distinction must be made explicit in the methodology and enforced in code.

3.2.4 Low-cost alternatives and a contradiction to resolve

3.2.4a What the framing document proposes

The framing document states the acquisition design as a deliberately spare collection device—manual entry and phone-camera photoplethysmography—adapted to infrastructure constraints, which makes the approach deployable where expensive medical equipment is unavailable. Its technological limitations section adds that this low-cost acquisition device is less precise than dedicated medical equipment and introduces measurement noise.

3.2.4b What the literature says about camera-based measurement

The sources of the review do not converge on a single answer, and the disagreement must be stated precisely.

who2023hypertension<empty citation> states that cuffless devices, including popular connected wristbands, are not currently recommended for hypertension manage-

ment, and that as of 2023 no alternative to the validated upper-arm cuff is accurate enough for diagnosis and management (Section 2.4.1).

alessa2019<empty citation> found three applications out of 186 claiming to turn the smartphone into a blood pressure measuring device, none of which provided any evidence of reliability or validity, and they warn that such smartphone-based cuffless devices may harm patients' health and safety.

alhaddad2022<empty citation> show that optical sensing, PPG included, can detect glucose trends but does not reach clinical-grade accuracy for absolute glucose measurement.

liu2025ppg<empty citation>, by contrast, report a strong correlation between blood pressure estimated from smartphone-camera PPG and reference devices on 127 participants, with interpretable features and Bland–Altman agreement analysis. This is genuine validation evidence, but from a single site, on a modest sample, with accuracy still below that of cuff devices and with generalisation across skin tones, devices and lighting not established.

3.2.4c Resolving the contradiction

The contradiction is real but narrower than it first appears, and it dissolves once the parameter is specified.

Photoplethysmography measures pulsatile blood volume. From that signal, heart rate is recovered from periodicity alone, requiring no absolute calibration—this is the robust case (Section 3.2.2a). Oxygen saturation requires two wavelengths and an empirical calibration curve; a smartphone camera with white-light illumination does not provide the controlled dual-wavelength illumination the method assumes, and any value derived from it is of unknown validity. Blood pressure is not measured by photoplethysmography at all: it is inferred, typically from pulse wave characteristics, through a model requiring individual calibration—which the WHO does not yet consider acceptable for clinical decisions, and which **liu2025ppg<empty citation>** have only begun to validate, on a single-site sample.

The design is therefore defensible if and only if it is stated with the precision of Table 3.9.

Read this way, the low-cost acquisition contribution survives intact. The device collects what low-cost optics can legitimately collect—heart rate, a contextual covariate—while blood pressure and blood glucose enter through manual entry from validated instruments, which is the pipeline the framing document's own record structure implies and which carries no inferential claim.

What must not appear in this thesis is any claim that clinical decisions rest on blood pressure obtained by camera. The evidence of **liu2025ppg<empty citation>** justifies evaluating camera PPG as an exploratory channel, recorded alongside validated cuff

Table 3.9: Admissible source for each parameter of the proposed system.

Parameter	Acceptable source	Not acceptable
Systolic and diastolic pressure	validated automated upper-arm cuff; manual entry or Bluetooth	cuffless wristband; camera PPG as a decision input (exploratory channel only, Section 2.4.2)
Blood glucose	glucometer; manual entry or Bluetooth	any optical method
Heart rate	camera PPG, cuff-derived, or optical sensor	—
Oxygen saturation	validated pulse oximeter	camera PPG
Temperature	digital thermometer or MAX30205	—

readings; it does not yet justify substituting it for them, which would contradict the WHO and **alessa2019<empty citation>**.

3.2.5 The device landscape, and what it omits

liu2025review<empty citation> survey the categories of IoT mobile sensing devices in chronic disease management. Their taxonomy is useful for situating the proposed system, subject to the reservation on their method stated at the outset (Table 3.10).

Table 3.10: Categories of IoT mobile sensing devices reported by **liu2025review<empty citation>**.

Category	Description and examples reported
Portable medical devices	battery-powered, handheld, compact: portable scanners collecting ECG, temperature, oxygen saturation, electromyography, respiratory rate; portable magnetic resonance scanners
Wearable devices	smartwatches and health tracking bands recording heart rate, blood pressure and blood glucose in real time, with analysis and health recommendations
Intelligent clinical devices	mobile nursing management, infusion monitoring, smart wards, bedside intelligent interaction
Remote health devices	wireless monitoring platforms for remote dynamic blood pressure monitoring and remote wireless health check management
Implantable devices	pacemakers, implantable cardioverter-defibrillators
Ingestible devices	capsule endoscopes, remotely controlled drug delivery

Two entries bear directly on this work.

They report that Vybornova and colleagues developed an IoT-based non-invasive cuffless blood pressure monitor measuring optical photoplethysmographic pulse wave signals on the wrist, calculating systolic and diastolic values using pulse wave analysis. This is cited without any accuracy figures and without reference to validation status. Given the WHO position, it should be cited only alongside that position, and never as evidence that cuffless measurement is ready for clinical use.

They also report the smart home system of **chatrati2022<empty citation>**, presented in Section 2.5, which classifies single home readings of blood pressure and glucose.

They also report, in their statement of the problem, that traditional chronic disease monitoring predominantly relies on manual processes, suffers from the phenomenon of information silos, and lacks real-time monitoring and feedback and effective prediction and early warning mechanisms—the diagnosis reported in Section 2.4.

What the landscape omits. Across both this survey and the architecture of **efendi2025<empty citation>**, no device category supplies clinically validated blood pressure or blood glucose without a cuff and without a finger-stick respectively. The entire instrumented pipeline, however elaborate, measures around the problem rather than into it. This is not a gap in the literature; it is a physical constraint, and the architecture of the proposed system must accommodate it rather than attempt to engineer past it.

3.2.6 Specification and the decision this section forces

3.2.6a Acquisition specification

Table 3.11: Acquisition requirements established in Section 3.2 and their basis.

Requirement	Basis
Three pipelines modelled separately, with distinct missingness mechanisms	Section 3.2.1; MNAR in pipelines 1 and 2
Blood pressure from a validated automated upper-arm cuff only	**who2023hypertension<empty citation>**; **alessa2019<empty citation>**
Blood glucose from a glucometer only	Section 3.2.5; no optical alternative exists
Camera PPG restricted to heart rate and exploratory blood pressure; never a decision input	**who2023hypertension<empty citation>**; **alessa2019<empty citation>**; **liu2025ppg<empty citation>**
Timestamp of measurement stored distinctly from timestamp of entry	Section 3.2.1; fictitious temporal structure otherwise
MQTT upstream, HTTP downstream	**efendi2025<empty citation>**
Offline-first storage with automatic merge on reconnection	**efendi2025<empty citation>**; intermittent connectivity stated in the research framing document
Link-layer encryption (WPA/WPA2) and authenticated cloud access	**efendi2025<empty citation>**; regulatory barrier in **dodoo2022<empty citation>**
Method comparison by Bland–Altman, not MAPE alone	Section 3.2.3c; MAPE conceals bias structure
Error computed with the reference instrument in the denominator	Section 3.2.3c; inverted in **efendi2025<empty citation>**
Sensor validation under movement, not seated rest only	Section 3.2.3c
Lagged features only; no threshold-derived feature at label time	Section 3.2.3d; label leakage in **efendi2025<empty citation>**
Reference instrument for validation must itself be validated	Section 2.4.1; one device in five carries validation evidence

3.2.6b The acquisition decision

The initial plan of this thesis described an instrumented pipeline built on the MAX30100, MAX30205 and ESP32. The framing document describes a deliberately spare device using manual entry and camera photoplethysmography. These are not variants of one design; they are two different systems with different costs, different failure modes and different claims. Three positions are coherent, and this thesis must adopt one explicitly.

Position A, spare acquisition. No microcontroller hardware. Blood pressure and blood glucose by manual entry from validated instruments; heart rate optionally by camera photoplethysmography. The IoMT hardware pipeline is then presented as state of the art examined and deliberately not adopted, with the reasons given: cost, maintenance, the sustainability failure documented in Section 2.2.4, and the fact that the sensors do not measure the parameters of interest.

Position B, instrumented acquisition. An ESP32-based device is built, measuring heart rate, oxygen saturation and temperature as covariates, alongside manual entry of blood pressure and blood glucose. This adds a hardware development workload and a device-validation obligation—Bland–Altman against a validated reference, under movement—and it contradicts the framing document as currently written, which would require amendment.

Position C, hybrid and staged. The system is specified to accept the sensor pipeline, with the data model and transport layer provided for, but the evaluated prototype uses spare acquisition. The hardware is then documented as the extension path.

The position adopted is A. It is the one consistent with the deployability argument that runs through Chapter 2: the instrumented pipeline acquires none of the parameters on which the clinical decision rests (Table 3.7), it carries a validation obligation that this work is not equipped to discharge, and the sustainability failures documented in Section 2.2.4 fall hardest on designs that require maintained devices in the field. The hardware examined in Sections 3.2.2 and 3.2.3 is therefore reported as state of the art and deliberately not adopted, and the transport and storage choices retained from **efendi2025<empty citation>**—MQTT upstream, offline-first storage with automatic merge—are retained independently of the sensors, because they address connectivity rather than instrumentation.

This decision determines the treatment of the following section: the gateway and transport layers are specified rather than implemented, and Section 3.3 examines them on that basis.

3.3 Cloud Infrastructure, Real-Time Transmission and Storage

Section 3.2 established what is acquired and by what means. This section addresses what happens to the data between the point of measurement and the point of clinical

decision: how it is layered, transported, stored and pushed. Beyond the IoT sources of the review (**efendi2025**; **liu2025review**), it relies on two reference architectures for chronic disease telemonitoring: **mucchi2021<empty citation>**, the most complete architectural specification available, with a preliminary usability evaluation; and **yazdani2018<empty citation>**, a conceptual design without implementation, cited for its modular decomposition and its scalability argument. The governing constraint, carried from Chapter 2, is that the system must triage.

3.3.1 The multi-tier model

3.3.1a The four-layer reference architecture

mucchi2021<empty citation> define a reference communication network architecture in four layers. We reproduce their definitions precisely, because the terminology of this chapter derives from them (Figure 3.8).

BODY layer. A set of body sensors and devices gathering the patient's parameters. The authors distinguish in-body devices—tiny sensors that are implanted or swallowed—from on-body devices worn by the patient for bio-parameter measurement, and they name the examples explicitly: blood pressure, pulse oximeter, ECG sensors.

ENVIRONMENT layer. Indoor and outdoor environmental sensors for information gathering—temperature, air quality, humidity—and user positioning, by satellite positioning or video monitoring, allowing location-based and personalised services based on the user's lifestyle.

EDGE layer. Responsible for short-term analysis of the acquired data and for user–system interactions, typically through a smartphone application. In the authors' words, it "allows the quick processing and identification of anomalous patient's conditions and the consequent warning notification"†. Structurally it comprises a gateway, a wide-area network device containing a reception and transmission unit, and a hub containing a data collection unit, a data processing unit, a local storage unit and a transmission unit.

CLOUD layer. In charge of the eHealth service and of long-term data analysis through the adoption of advanced algorithms for data cross-correlation. Its units are a cloud repository unit, a control unit, a cloud data processing unit, eHealth services, data analysis and alert management.

The division of labour between the last two layers is the single most useful idea in this section, and it is developed in Section 3.3.4.

3.3.1b The seven-layer platform of Liu and Wang

Two cross-cutting systems span all seven: a standard specification support system and an information security assurance system. Between the data analysis and application service

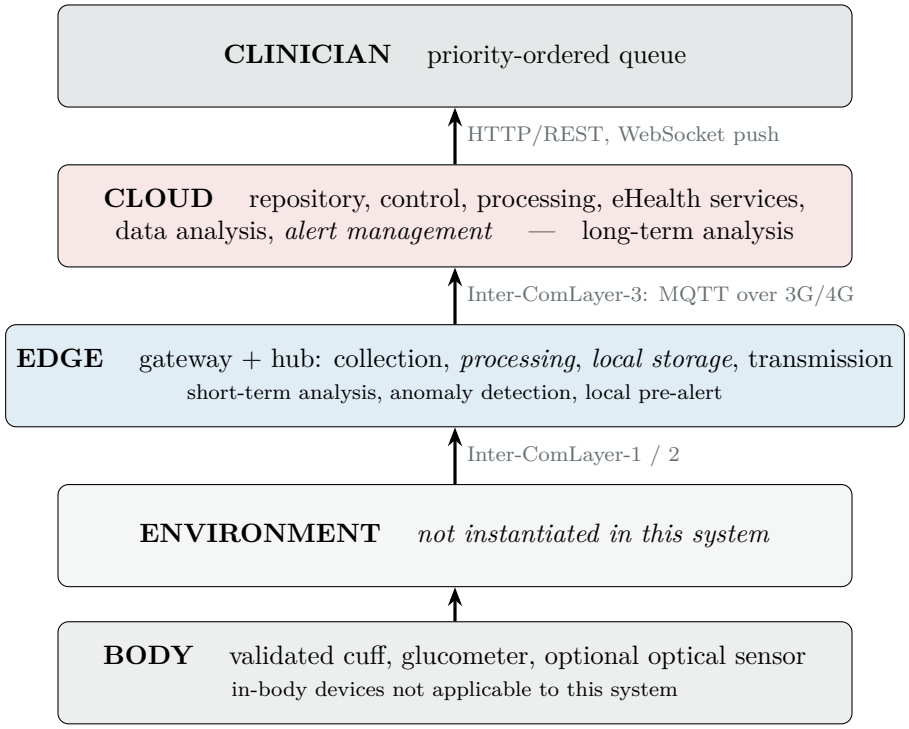


Figure 3.8: The four-layer reference architecture of **mucchi2021<empty citation>**, instantiated for the present system. The environment layer is retained in the model and not instantiated, since the system has no environmental sensors. Italicised units are the ones that carry the triage function.

Table 3.12: Seven-layer platform reported by **liu2025review<empty citation>**.

Layer	**Contents reported**
Access	connectivity to IoT devices, mobile applications and other data sources; blood pressure monitor, blood glucose meter, smartwatches
Data transfer	wireless communication: Bluetooth, Wi-Fi, 4G, 5G
Data storage	web (Java, Tomcat, SpringCloud); big data (Hadoop, Spark); database (MySQL, Redis cache); DevOps continuous integration and delivery; Linux server side
Data processing	data cleansing, data conversion, feature engineering, data extraction, data integration
Data analysis	support vector machine, decision tree, k-means, gradient boosting, AdaBoost, neural networks
Application service	master data services (remote monitoring, intelligent follow-up, online consultation); business applications (medication reminders, user feedback, research support)
Decision support	multi-source data fusion, feature learning, personalised medicine, interactive analysis, integration of interdisciplinary knowledge

layers, the authors place a multi-point triggering and dynamic response mechanism.

The explicit presence of a blood pressure monitor and a blood glucose meter at the access layer confirms the device allocation established in Table 3.9: these parameters enter the system as instrument readings, not as inferences from optical sensors.

3.3.1c The three functional layers of Yazdani et al.

Their architecture has a user base layer, where physicians, the healthcare team, patients and system administrators connect to the server through a client interface installable on a tablet or phone or presented as a web page; a function base layer containing the architectural modules; and a database layer storing physiological data alongside the physician's prescriptions, diagnoses and the recommendations of the decision support system.

The function base comprises five modules, and this decomposition is the paper's contribution (Table 3.13).

Table 3.13: Modules of the function base layer (**yazdani2018**).

Module	**Function**
Membership management	registration of patient, physician and healthcare team; definition of the checkup period, the intervals at which patient data must be collected and sent; rule selection, choosing the decision support system and rules appropriate to the disease and symptoms
Communication transmission	ensuring data aggregation tools are attached and data are sent
Supervisor	determining which decision support system corresponds to the data received, by parsing metadata attached to the patient record
CDSS setting	defining the decision support systems and their rules; adding and updating them
Notification	communicating analysis results to medical staff and informing patients of prescriptions

The data flow is also specified: "if no data is sent within the time limits set by the physician, an SMS reminder will be sent to the patient. The SMS reminder is sent repeatedly as long as the patient enters a new data."† The server verifies and stores the data, determines the related decision support system, analyses the patient data, and classifies the result as normal or alert. The physician can view patient status on a priority basis, this prioritisation being based on that status.

Two elements here are directly relevant and should be retained despite the paper's age. The checkup period as a configurable parameter operationalises the WHO follow-up requirement established in Section 3.1.2a—monthly until target is reached, three-monthly thereafter—as a system setting rather than a fixed constant. And the priority-ordered physician view is the user-facing form of the triage function.

Table 3.14: Mapping of the four architectures onto the reference layers of **mucchi2021<empty citation>**.

mucchi2021<empty citation>	**efendi2025<empty citation>**	**liu2025review<empty citation>**	**yazdani2018<empty citation>**
BODY	physical layer	access layer	—
ENVIRONMENT	—	access layer	—
EDGE	*absent*	data transfer layer (transport only)	*absent*
CLOUD	network layer	storage + processing + analysis	database layer
eHealth services (within cloud)	application layer	application service + decision support	user base + function base
Alert management	classifier in application layer	multi-point triggering	notification module

3.3.1d Comparative mapping

3.3.1e What the comparison establishes

The four architectures agree on sensing, transport, storage and presentation. They disagree on whether an edge layer exists, and that disagreement is the substantive finding of this subsection.

mucchi2021<empty citation> give the edge layer a data processing unit and a local storage unit and assign it short-term analysis and anomaly warning. **liu2025review<empty citation>** reduce their equivalent layer to transport only. **efendi2025<empty citation>** and **yazdani2018<empty citation>** have no edge layer: the ESP32 transmits without processing, and the client interface of Yazdani et al. is a presentation surface.

The consequence is architectural, not cosmetic. In an edgeless design, every reading travels to the cloud and every decision is taken there. In an edge design, the local tier can filter, aggregate and raise a first-level warning without a round trip. Given that Section 2.2.4 established approximately 550 million people in sub-Saharan Africa lacking reliable electricity and internet, and that **mucchi2021<empty citation>** themselves parameterise for "very limited connectivity"†, the edge design is not an optimisation but a precondition.

3.3.2 Communication protocols

3.3.2a The communication layer taxonomy

mucchi2021<empty citation> decompose the links into three intra-layer and three inter-layer communications (Table 3.15).

Only the last of these is named with specific technologies, and only the last is relevant to a system whose body layer reduces to a Bluetooth cuff and a glucometer. Intra-Com-1 and Intra-Com-2 concern implanted and ingestible devices and do not apply here; this thesis states so rather than reproducing a taxonomy whose upper half is inert.

Table 3.15: Communication links defined by **mucchi2021<empty citation>**.

Link	Between	Technology
Intra-Com-1	within the body: nano-devices, implants	molecular communications
Intra-Com-2	in-body ↔ on-body; sink nodes collect from in-body devices and deliver off-body	not specified
Intra-Com-3	optional: hub ↔ gateway, where the gateway is not integrated into the data collector	not specified
Inter-ComLayer-1	BODY ↔ EDGE: on-body sensors to hub or gateway, e.g. a smartphone	not specified
Inter-ComLayer-2	ENVIRONMENT ↔ EDGE	not specified
Inter-ComLayer-3	EDGE ↔ CLOUD	3G/4G/5G, Wi-Fi, SigFox, low-power wide-area network

3.3.2b MQTT upstream

efendi2025<empty citation> select MQTT for the sensor-to-cloud path, stating that it "enables resource-constrained devices to send sensor data to a gateway which acts as a translator, then processes and stores the data with Firebase functionalities such as Cloud Functions or Real-time Database"†, and describing it as "a messaging protocol which is commonly used in IoT devices"†.

The rationale is sound. MQTT is a publish–subscribe protocol: a client publishes to a topic on a broker, and subscribers receive without the publisher knowing who they are. Its header overhead is minimal relative to HTTP, it maintains a single long-lived connection rather than opening one per message, and it defines quality-of-service levels permitting at-least-once or exactly-once delivery over an unreliable link.

A reservation must be recorded. No source in this corpus quantifies the advantage. **efendi2025<empty citation>** assert that MQTT suits resource-constrained devices; they do not measure bandwidth, latency, power consumption or delivery reliability against HTTP, and they report no comparison. The choice is therefore justified by convention and by protocol design, not by evidence presented here. If this thesis claims an efficiency benefit, it must either cite a benchmark outside this corpus or measure it.

3.3.2c HTTP and REST downstream

efendi2025<empty citation> route the cloud-to-client path over HTTP: "the data which has been stored to Firebase is accessed by mobile applications and websites using HTTP protocol"†. Firebase, they note, "can support various protocols including HTTP and MQTT"†, and "data transmitted by IoT devices will be stored as JSON which is then accessed by users in real-time"†.

The asymmetry is deliberate and should be preserved. Upstream traffic is high-frequency, small-payload, from a constrained device over an unreliable link—the MQTT profile. Downstream traffic is request-driven, from a capable device, and must interoperate

with web clients—the HTTP profile.

3.3.2d WebSockets and real-time push

The real-time property does not come from HTTP but from a persistent connection beneath it. **efendi2025<empty citation>** state that Firebase "provides Realtime Database and designed for real-time updates. Whenever current sensor data arrives, Firebase sends it to linked devices via persistent WebSocket connections. This enables clients to examine the health vitals virtually immediately on mobile application and website."†

The distinction between polling and push matters for this system's economics as much as its responsiveness. A polling client issues requests whether or not data have changed; a pushed client consumes resources in proportion to events. For a cohort transmitting a handful of readings per week, polling would dominate the traffic entirely.

3.3.2e The XML choice in Yazdani et al., and why it is not adopted

yazdani2018<empty citation> use XML for patient data exchange, and give four reasons: that the label elements of XML content are independent of communication protocols, making XML documents suitable for internet and public data transfer networks; that XML is advantageous for editor applications, middleware and application tools, improving efficiency and yielding financial savings in information systems; that users can send and request patient data, view status and receive prescriptions and notifications anywhere with mobile or Wi-Fi coverage; and that metadata added to the XML file allows the supervisor module to determine which decision support system should analyse the patient data, this being performed by an XML parser.

The fourth reason is architecturally interesting—self-describing payloads enabling routing without a central registry—but the format choice is dated. The three later sources all use JSON, which is more compact, natively supported by the storage layer and by every mobile runtime, and does not require a parser stage. The idea to retain from **yazdani2018<empty citation>** is metadata-driven routing, not XML.

3.3.2f Protocol selection

3.3.3 The cloud backend

3.3.3a The NoSQL JSON schema, and a modelling consequence nobody states

Data are stored as a JSON tree. **efendi2025<empty citation>** reproduce the live structure, which in their system is as flat as it could be: a single node holding `heart-rate`, `oxygen-saturation` and `temperature` with their current values.

This is a document store, not a relational database. Three consequences follow for the present work, and none of the four sources addresses them.

Table 3.16: Protocol selected for each link, and the constraint it answers.

Link	Protocol	Constraint it answers
Device → gateway	Bluetooth Low Energy	validated cuff and glucometer transmit without manual entry (Section 3.2)
Gateway → cloud	MQTT over 3G/4G	constrained device, unreliable link, small frequent payloads
Cloud → client	HTTP/REST	interoperability, request-driven access
Cloud → client, real time	WebSocket	event-proportional traffic, immediate alert delivery
Payload format	JSON	native to storage and runtime; no parser stage

A JSON tree is not natively a time series. The structure above holds the current value and nothing else. To hold a series, the key design must carry time—typically a push key or timestamp-keyed child node per reading—and that design decision determines whether a range query over a window is cheap or expensive. Since the framing document specifies feature engineering by rolling aggregates, trends and variability, the data model must be designed around the window query, not around the current reading. The structure of **efendi2025<empty citation>** would require a full subtree read to compute a seven-day mean—which is precisely the quantity the measurement protocol of Section 3.1.4 defines as the reference value.

Denormalisation shifts the burden to write time. Document stores conventionally duplicate data across the shapes each client needs, so that reads are single-node. This is efficient for display and poor for ad hoc analysis, and it means the schema must anticipate the queries. A model-training export is an ad hoc analysis.

Schemalessness is not an advantage here. The record structure is fixed by the framing document—systolic, diastolic, blood glucose, heart rate, timestamp, plus background and context. A validated schema enforced at write time would catch the unit errors and carried-forward values that **liu2025review<empty citation>** identify as the characteristic defects of electronic medical records (Section 2.4). Firebase supports this through security rules; the opportunity should be taken rather than relying on client-side validation.

3.3.3b Offline-first synchronisation

This is the property that matters most for the target context, and the source states it directly: data are "stored locally and real-time events continue to occur even when the device is offline"†, and "the real-time data will be synchronised between local data change remotely when the device is offline and auto merging when the device is going to be online"†.

Set against the framing document's statement that intermittent connectivity in the Cameroonian context leads to irregular and missing measurements, and against the 550

million figure from Section 2.2.4, a storage layer whose default behaviour is local write with deferred merge removes one cause of missingness entirely: readings taken while offline are not lost, they are delayed.

One caution. Deferred synchronisation means the arrival timestamp differs from the measurement timestamp, sometimes by days. Section 3.2.1 identified this as a defect of manual entry; it reappears here as a property of the transport. The schema must therefore carry both times explicitly, and every temporal feature must be computed on the measurement time. A model trained on arrival order would learn the connectivity pattern rather than the physiology—and since connectivity correlates with geography and income, that model would encode exactly the inequalities the system exists to reduce.

3.3.3c Alerting and notification

The four sources place the alert function differently, and the differences are instructive.

mucchi2021<empty citation> place alert management in the cloud layer, while assigning identification of anomalous patient conditions and the consequent warning notification to the edge layer—that is, a fast local warning and a considered cloud-side decision. Their application implements this in the professional profile, which performs "pre-processing of the acquired data for a pre-alert generation in case of anomalous values and transmission to the cloud repository"† and "generation and management of alerts/warnings"†.

yazdani2018<empty citation> place it in a dedicated notification module with two distinct functions, patient notification and physician notification, and add the SMS reminder for missing data.

liu2025review<empty citation> place it in the multi-point triggering and dynamic response mechanism between analysis and application.

efendi2025<empty citation> do not implement alerting at all: their application layer classifies into stable or not stable and displays the result.

The convergent design is a two-level alert: a local pre-alert raised at the edge on the current reading, and a considered alert raised in the cloud on the series. This is structurally identical to the two-stage clinical protocol of **mihevc2025<empty citation>** documented in Section 3.1.4c—an abnormal reading triggers a verification profile, and only the profile triggers a teleconsultation. The architecture and the clinical protocol agree, which is a strong argument for adopting both.

3.3.3d What none of the sources reports

Operating cost. No source states what any of these architectures costs to run per patient per year. Given that Section 2.2.4 documented Babyl ceasing operations in August 2023 for financial reasons after reaching approximately 2.5 million registered users, and that

dodoo2022<empty citation> identified sustainability as the characteristic failure of telemedicine projects in the region, this silence is a gap this thesis names. The billing unit of a real-time database is the persistent connection and the bytes synchronised, and a design that holds an open WebSocket per client consumes resources continuously rather than per reading. This must be quantified for the proposed system rather than assumed negligible.

Vendor dependence. Firebase is a proprietary managed service. None of the sources discusses what happens if pricing, terms or availability change, or whether data can be exported and the system rehosted. For a system intended to outlive a thesis, this is a design risk, and it is the technical counterpart of the sustainability barrier.

Data residency. The database endpoint reported by **efendi2025<empty citation>** is hosted in the `asia-southeast1` region. Where patient data physically reside is a regulatory question, and **dodoo2022<empty citation>** identified the legal and regulatory dimension as one of six barriers in sub-Saharan Africa. This thesis must state the chosen region and justify it.

3.3.4 The edge layer and the triage requirement

This subsection is where the architectural literature answers the requirement that Chapter 2 established.

3.3.4a The requirement, restated

Section 2.2.3 concluded that a telemonitoring system deployed at scale in this region must triage, and gave three quantitative reasons. **bediang2021<empty citation>** found that remote expertise took 10.3 days on average against a 72-hour protocol and was not significantly faster than the control pathway, the bottleneck being specialist availability rather than transmission. **kingue2013<empty citation>** quantified the cost of systematic review at approximately six calls per provider per day, five minutes each, across 1 187 calls. **segun2025<empty citation>** reported that up to 18% of sub-Saharan African countries had no registered cardiologist as of 2020.

A system that forwards every reading to a clinician reproduces this scarcity. A system that sorts relieves it.

3.3.4b Where the function belongs

The division of **mucchi2021<empty citation>** assigns the edge layer short-term analysis, quick processing, identification of anomalous conditions and consequent warning notification, and the cloud layer long-term data analysis through advanced algorithms for

data cross-correlation. This is precisely a triage split:

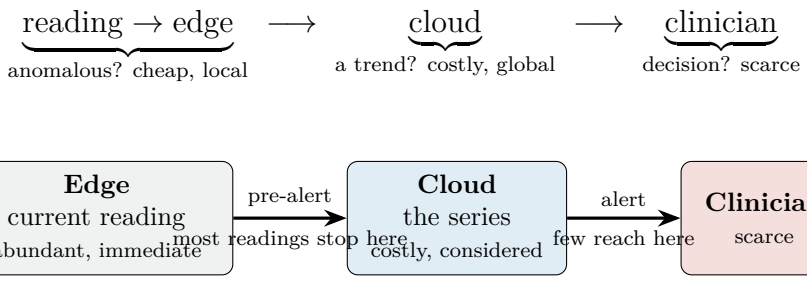


Figure 3.9: Triage as three filters in series, each stage costlier and scarcer than the last. The design objective is to maximise what each stage resolves, so that the final stage is reached only when necessary.

yazdani2018<empty citation> supply the complementary element at the clinician end: "the physician can have access to patient data at any time through the patient status and is able to view patients' status on a priority basis. This prioritisation is done based on the patient status."† Triage is not only filtering but ordering: the clinician sees a queue sorted by urgency, not a chronological stream.

3.3.4c Where the learned model sits

liu2025review<empty citation> place their data analysis layer—support vector machine, decision tree, k-means, gradient boosting, AdaBoost, neural networks—between data processing and application service, with a multi-point triggering and dynamic response mechanism connecting it upward.

This is the correct placement for the model developed in this thesis, and it differs from that of **efendi2025<empty citation>**, who place their classifier in the application layer. The difference is not pedantic: a model in the application layer classifies for display, while a model in the analysis layer classifies for routing. The anticipatory formulation of the framing document—forecasting degradation at 24 to 48 hours—is a routing function, since its output determines who is contacted and when.

The rule selection and CDSS setting modules of **yazdani2018<empty citation>** suggest a further refinement worth considering. Their architecture allows different decision rules to be attached to different diseases, configured by medical staff through an administrator interface, so that the platform supports several decision support systems rather than being rebuilt for each. For a system covering both hypertension and diabetes, with thresholds anchored in the ESH and ADA references respectively (Section 3.2.1), this separation of the rule base from the platform is a sound design and a defensible extensibility claim.

3.3.5 Security, privacy and the gap between recommendation and practice

3.3.5a What is recommended

mucchi2021<empty citation> set out the principle clearly: "health data can flow from patient to cloud and from cloud to others (health operators) by using different channels (often wireless), which not always can be assumed secure. A trusted and secure domain for the health data should be assured, so that there is no segment which has a lower security than others."† They add that beyond classical cryptography, "modern security techniques should be added, such as physical-layer security and crypto-key extraction from health signals"†, that access policy should be addressed "by making the patient aware of where the data is and who is requesting access to it"†, and that blockchain "can also be used to trace the access log to data and for non-repudiation of actions ordered or made on the patient's data"†.

liu2025review<empty citation> devote a section to the same question, surveying differential privacy frameworks using convolutional autoencoders and generative adversarial networks, k-anonymity through the Mondrian algorithm, RFID-based secure authentication for remote patient monitoring, lightweight encryption with access tokens, permissioned blockchain using elliptic curve digital signatures, and private set intersection. They observe that regulatory compliance is an imperative for healthcare providers, necessitating a prioritisation of data governance security.

efendi2025<empty citation> report that the ESP32 supports IEEE 802.11 security including WAPI, WFA and WPA/WPA2, and that Firebase provides built-in authentication and authorisation.

3.3.5b What was implemented

mucchi2021<empty citation> state it themselves, and the admission deserves to be quoted: "this paper does not focus on this topic, for the sake of space, but anyway this paragraph wants to summarise the main characteristics of a good policy of security and privacy when a tele-medicine architecture is on. Concerning our experimental activity, we only relied on the application layer security protocols, and on the fact that the data where stored in a private server without public access."†

The gap between the recommendation—physical-layer security, key extraction from biosignals, blockchain audit trails—and the practice—application-layer transport encryption and a server that is not publicly addressable—is the characteristic pattern of this literature. It is not a criticism of the authors, who are candid about it. It is a warning that security sections in this field frequently describe what ought to be done rather than what was done, and that this thesis states its own implementation plainly rather than

enumerating techniques it does not use.

3.3.5c The minimum defensible position

Table 3.17: Minimum security and privacy requirements, and their basis.

Requirement	Basis
Transport encryption end to end, with no segment weaker than the others	**mucchi2021<empty citation>**
Authenticated access, role-separated (patient, nurse, physician)	**efendi2025<empty citation>**; user roles in **yazdani2018<empty citation>**
Direct identifiers removed before any modelling	the research framing document
Declared data residency region	endpoint of **efendi2025<empty citation>**; regulatory barrier in **dodoo2022<empty citation>**
Access traceability: who read what, when	**mucchi2021<empty citation>**
Documented consent, with purpose limited to research	the research framing document
Written statement of what was implemented, distinct from what is recommended	Section 3.3.5b

mucchi2021<empty citation> also record the applicable legal instrument in their jurisdiction, Italian Legislative Decree 196/2003. The Cameroonian equivalent must be identified and cited; none of the sources in this corpus provides it.[5]

3.3.6 What these architectures have actually been validated on

This subsection is necessary because the four sources are frequently cited as if they were equivalent, and they are not.

Table 3.18: Implementation and validation status of the four architectural sources.

Source	Implementation	Validation	Real patients
yazdani2018<empty citation>	conceptual UML design only	none	no
liu2025review<empty citation>	no system; taxonomy diagram	none	no
efendi2025<empty citation>	full hardware and cloud	6 participants, 10 min, seated	no: non-patient volunteers, two with a chronic disease
mucchi2021<empty citation>	Android application, cloud repository	10 nurses, 20 days, simulated patients	no

[5] **[To verify]** Identify and cite the Cameroonian legal instrument governing personal and health data processing, together with any national health research ethics requirement applicable to the cohort. This cannot be drawn from the present corpus and must come from national sources.

3.3.6a The validation of Mucchi et al.

The protocol is documented carefully. Ten family and community nurses of the Florence city district were selected, with inclusion criteria of at least six months' length of service and possession of an Android smartphone; no requirement was placed on information technology skills, the authors noting that one aim was to develop an application as user-friendly as possible. A training session presented the protocol and functionalities, the application was installed on the nurses' personal phones, and simulated vital parameters were entered. Over the following twenty days each nurse was asked to simulate at least one patient check-up per day, entering parameters and checking the historical record including alert or warning messages. A questionnaire followed, assessed on the mean opinion score scale.

No real patients were involved. The authors state that the second phase, "whose Ethical Committee approval process is still ongoing, will involve the introduction of the App on an experimental basis in clinical practice"[†].

3.3.6b The result the authors do not discuss

Table 3.19: Usability questionnaire results (**mucchi2021**), Table 1. The scale runs from insufficient to excellent; on the mean opinion score convention, "sufficient" ranks below "fair".

Investigated aspect	**Insuff.**	**Suff.**	**Fair**	**Good**	**Excel.**
Nurse ICT skills	—	—	67%	33%	—
App functionalities w.r.t. operating conditions	—	—	33%	50%	17%
App usability in operative contexts	—	—	33%	50%	17%
Approval of the service and the app	—	—	67%	33%	—
System efficiency for managing chronic diseases	16%	33%	17%	17%	17%

The final row is the only aspect on which any respondent returned an insufficient rating, and the published percentages place half of the respondents at sufficient or below. It is also the only aspect spread across all five categories. The results discussion of **mucchi2021<empty citation>** states that the main tested functionalities of the application can effectively support the nursing staff, which is supported by the second and third rows, and does not comment on the fifth.

A caution on the denominator is required before any of these figures is quoted. Ten nurses were recruited, but every percentage in the table is an exact multiple of one sixth: 16.7, 33.3, 50.0, 66.7. The five categories of the final row correspond to one, two, one, one and one respondent out of six. The questionnaire therefore appears to have been answered by six respondents, not ten, and the figures should be reported as counts out of six—three of six rating efficiency at sufficient or below—rather than as the percentages 16% and 33%,

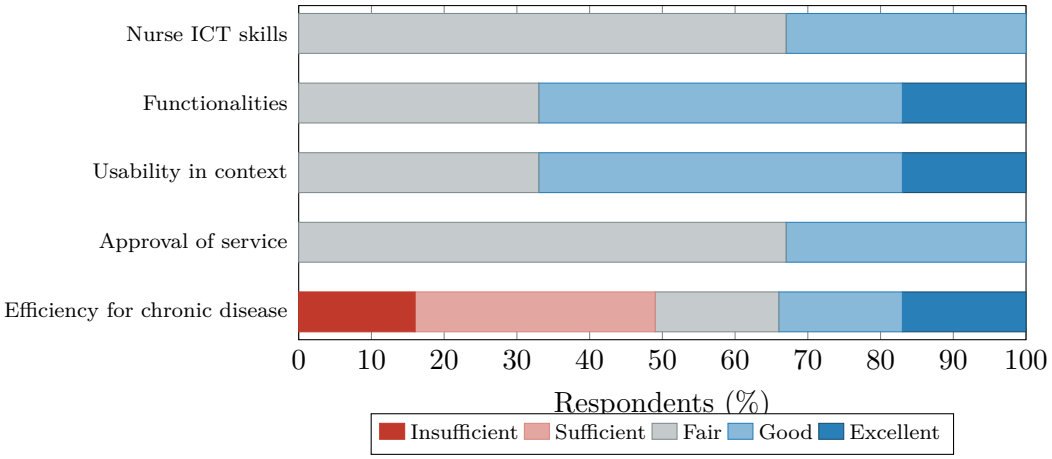


Figure 3.10: The usability results of Table 3.19. Only the bottom row—efficiency for the clinical purpose—receives any rating below "fair", and it is the only aspect spread across all five categories.

whose sum of 49% is a rounding artefact of three sixths.[6]

This matters because the fifth row is the aspect closest to the clinical question. Usability was rated well; efficiency for the purpose was not. This thesis cites this architecture for its structure, which is sound and well specified, and does not cite it as evidence that such a system is effective—which its own usability data do not support and which it never claims.

The first row is also worth retaining: two-thirds of respondents rated their own information technology skills only fair and none rated them excellent, in a high-income European setting with no skill requirement for inclusion. Read with the 10.9% equipment-related dropout in **mihevc2025<empty citation>** (Section 2.3.3) and with the individual barrier of **dodoo2022<empty citation>**, this converges on a consistent finding: digital competence of front-line staff is a binding constraint, not a transitional one.

3.3.6c The feedback the nurses gave

mucchi2021<empty citation> report the comments, and three bear directly on the present design: "the alerts notification should be improved especially in case of urgent situations; the integration with the protocols adopted in different real operating conditions should be considered"†; "interfacing with the current management information systems in use in the healthcare sector may be useful"†; and "the possibility of interfacing with other healthcare professionals and the need for an immediate feedback from the doctor in the event of an alert should be considered"†.

The first and third are the same request in different words: the alert must reach a decision-maker, and reach them fast. This is the professional-oversight finding of

[6] **[To verify]** Confirm the number of questionnaire respondents in **mucchi2021<empty citation>**. Ten nurses were enrolled, but every value in their Table 1 is a multiple of 1/6, which implies six respondents. If six is correct, report counts rather than percentages throughout.

Section 2.3.2 arriving from a fourth direction—through the stated preference of the nurses using the system.

3.3.7 Specification

3.3.8 What this section establishes, and what it does not

Established. The multi-tier model is stable across four independent sources and admits a clean mapping (Table 3.14). The protocol choices are coherent and defensible. The cloud backend offers the offline-first behaviour the deployment context requires. And the edge–cloud division of labour in the reference architecture is structurally the triage function that Chapter 2 demonstrated to be necessary—which means the architecture required by the clinical analysis and the architecture proposed by the engineering literature coincide. That convergence is the main result of this section.

Not established. No architecture in this corpus has been validated on real patients in routine care. Two have no implementation at all. The one architectural source with a usability evaluation returned half of its respondents rating system efficiency for chronic disease management at sufficient or below, a result its authors do not discuss. No source quantifies the protocol efficiency it claims, nor the operating cost, nor the behaviour of any of these designs under the intermittent connectivity they nonetheless parameterise for. The architecture is well specified and poorly evidenced, and this thesis says so.

3.4 Machine Learning Algorithms and Predictive AI for Risk Assessment

Sections 3.2 and 3.3 established what is acquired and how it travels. This section addresses what is computed on it, building on the machine learning sources of the review (Section 2.5). The governing requirement, refined in Section 3.3.4, is that the model exists to triage: to decide which patients reach a clinician's attention and in what order. That purpose determines which algorithms are candidates, which metrics matter, and which failure modes are unacceptable.

3.4.1 The algorithm families and their inductive biases

Before comparing reported performances it is necessary to state what each family assumes, because the comparison in Section 3.4.2 is otherwise uninterpretable.

Table 3.20: Architectural requirements established in Section 3.3 and their basis.

Requirement	Basis
Four-layer architecture with an explicit edge tier	**mucchi2021<empty citation>**; edge absent in **efendi2025<empty citation>** and **yazdani2018<empty citation>**
Edge performs short-term analysis and local pre-alert	**mucchi2021<empty citation>**
Cloud performs long-term analysis and considered alert	**mucchi2021<empty citation>**
Two-level alerting mirroring the clinical two-stage protocol	**mucchi2021<empty citation>**; **mihevc2025<empty citation>**, Section 3.1.4c
Learned model in the analysis layer, not the application layer	**liu2025review<empty citation>**; routing, not display
Clinician view ordered by priority, not chronology	**yazdani2018<empty citation>**
Configurable checkup period per patient	**yazdani2018<empty citation>**; WHO follow-up intervals, Section 3.1.2a
Rule base separable from platform, to carry both ESH and ADA thresholds	CDSS setting module of **yazdani2018<empty citation>**
MQTT upstream, HTTP/REST downstream, WebSocket for push	**efendi2025<empty citation>**
JSON payloads; XML not adopted	**efendi2025<empty citation>**; contra **yazdani2018<empty citation>**
Schema keyed for window queries, not current value only	Section 3.3.3a; rolling-aggregate features
Measurement time and arrival time stored separately	Section 3.3.3b; deferred sync decouples them
Server-side schema validation at write time	Section 3.3.3a; EMR defects in **liu2025review<empty citation>**
Offline-first local storage with automatic merge	**efendi2025<empty citation>**; connectivity limitation in the research framing document
Local storage unit at the edge	hub composition in **mucchi2021<empty citation>**
End-to-end encryption with no weaker segment	**mucchi2021<empty citation>**
Role-separated authentication: patient, nurse, physician	**efendi2025<empty citation>**; **yazdani2018<empty citation>**
Declared data residency and applicable national instrument	endpoint of **efendi2025<empty citation>**; **dodoo2022<empty citation>**
Access traceability	**mucchi2021<empty citation>**
Implemented security stated separately from recommended security	Section 3.3.5b
Operating cost per patient quantified	Section 3.3.3d; Babyl collapse, Section 2.2.4
Export path and vendor independence assessed	Section 3.3.3d
Front-line digital competence treated as a binding constraint	Section 3.3.6b; **mihevc2025<empty citation>**

3.4.1a Decision trees

A decision tree recursively partitions the feature space by axis-aligned splits, choosing at each node the split that most reduces an impurity criterion. For a binary outcome with proportion p of positives in a node, the Gini impurity is

$$G = 2p(1-p), \tag{3.8}$$

and a split is selected to minimise the weighted impurity of the children. Trees are interpretable and require no feature scaling, and they handle mixed categorical and continuous inputs natively. Their weakness is variance: a small change in the data can produce a different tree.

3.4.1b Bagging: random forest

A random forest fits many trees on bootstrap resamples and, at each split, considers only a random subset of features; predictions are the majority vote. The variance of an average of B identically distributed variables with pairwise correlation ρ and variance σ^2 is

$$\rho\sigma^2 + \frac{1-\rho}{B}\,\sigma^2, \tag{3.9}$$

so the random feature subsetting, which lowers ρ, is what makes the ensemble work rather than the averaging alone: the second term vanishes as B grows, the first does not. **mendoza2025<empty citation>** describe the method as "ensemble learning methods which construct decision trees during training and output the mode of the classes"†, noting its "robustness against overfitting"† and "ability to handle large datasets with high dimensionality"†. They configured 500 trees, three variables considered per split, and node size one.

3.4.1c Boosting: AdaBoost and XGBoost

Boosting fits trees sequentially, each correcting the residual errors of its predecessors. XGBoost minimises a regularised objective

$$\mathcal{L} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|w\|^2, \tag{3.10}$$

where T is the number of leaves and w the leaf weights, so that complexity is penalised explicitly. **mendoza2025<empty citation>** configured maximum depth six, learning rate 0.3, 100 boosting rounds and a fixed random seed of 123 for reproducibility.

Boosting typically outperforms bagging on tabular data of moderate size, but it is more sensitive to noise and to class imbalance, because misclassified examples receive increasing

weight—and in an imbalanced problem the misclassified examples are disproportionately the rare class, or disproportionately the noise.

3.4.1d Support vector machines

A support vector machine seeks the separating hyperplane maximising the margin, solving

$$\min_{w,b} \ \tfrac{1}{2}\|w\|^2 + C\sum_i \xi_i \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i, \ \ \xi_i \geq 0, \tag{3.11}$$

with non-linear boundaries obtained by kernel substitution. **liu2025review<empty citation>** report support vector machines widely used in this field, citing an accuracy of 92.19% for dementia prediction and 92.19% for respiratory pathology classification among their examples.[7] Support vector machines scale poorly with sample size and require feature scaling, and they do not produce calibrated probabilities without an additional step.

3.4.1e Recurrent networks: LSTM

A long short-term memory cell maintains a state vector updated through gates that control what is retained, discarded and emitted:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f), \qquad i_t = \sigma(W_i[h_{t-1}, x_t] + b_i), \qquad o_t = \sigma(W_o[h_{t-1}, x_t] + b_o), \tag{3.12}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c[h_{t-1}, x_t] + b_c), \qquad h_t = o_t \odot \tanh(c_t). \tag{3.13}$$

The gating permits gradient flow across long sequences, which is why such networks are the conventional baseline for clinical time series.

liu2025review<empty citation> report one directly relevant application: Reddy and Delen used a recurrent LSTM network to predict readmission of lupus patients, with the deep method outperforming traditional classification and achieving an area under the curve of 0.70. This figure deserves emphasis, because it is the only realistic recurrent-network result in the corpus and it sits far below the 95–100% accuracies that dominate the surrounding literature.

3.4.1f Comparison of inductive biases

The decisive line of Table 3.21 is the last. Only the recurrent network treats the input as a sequence. The other four receive a feature vector and have no notion of order; any temporal information they use must be engineered into that vector as lags, rolling means,

[7] **[To verify]** The same figure, 92.19%, is reported for two unrelated applications. One of the two is probably a transcription error in **liu2025review<empty citation>**. Check the two primary references before reproducing either value.

Table 3.21: Inductive biases of the five candidate families.

Family	Assumes	Handles	Fails on	Interpretability
Decision tree	axis-aligned splits suffice	mixed types, missing values	high variance	direct
Random forest	ensembling reduces variance	high dimension, noise	smooth boundaries	importance measures
XGBoost	residuals are learnable sequentially	tabular structure	noise, imbalance	SHAP
SVM	margin separability after kernel mapping	small n, high p	large n, calibration	low
LSTM	temporal dependence carries signal	sequences, irregular lags	small n, needs volume	low

slopes or variability. This is precisely the comparison the framing document poses in its third contribution and first hypothesis.

3.4.2 What the literature reports, and the contradiction

3.4.2a Efendi et al.: XGBoost best

Table 3.22: Classifier comparison reported by **efendi2025<empty citation>**, Table 16.

Model	Precision	Recall	Accuracy	F_1
XGBoost	0.973	0.968	0.973	0.970
AdaBoost	0.956	0.937	0.967	0.946
Decision tree	0.941	0.925	0.931	0.933
Random forest	0.936	0.922	0.946	0.929
Bayesian classifier	0.925	0.931	0.931	0.928

All five exceed 0.92 on every metric. XGBoost is selected and embedded in the deployed system.

3.4.2b Mendoza-Cano et al.: XGBoost worst

Table 3.23: Model comparison reported by **mendoza2025<empty citation>**, Table 3.

Model	AUC (95% CI)	Prec.	Sens.	Spec.	Acc.	F_1
Random forest	0.75 (0.72–0.77)	0.94	0.51	0.98	0.84	0.659
XGBoost	0.54 (0.49–0.60)	0.42	0.23	0.85	0.65	0.294
Logistic regression	0.61 (0.59–0.64)	0.70	0.98	0.03	0.70	0.818

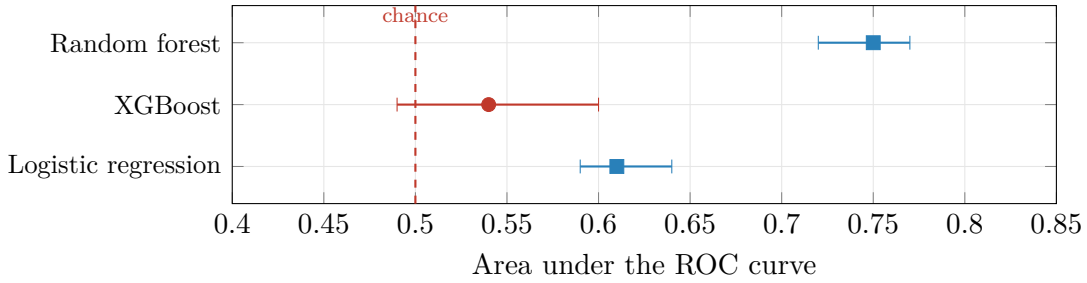


Figure 3.11: Discrimination reported by **mendoza2025<empty citation>** with 95% confidence intervals. The XGBoost interval spans the chance line: on this dataset the model is not distinguishable from a coin flip.

The confidence interval for the XGBoost area under the curve spans 0.49 to 0.60 and therefore includes 0.50 (Figure 3.11). The authors state it plainly: "XGBoost documented the lowest predictive performance"†.

3.4.2c The ambient literature

liu2025review<empty citation> report a long series of accuracies from the chronic-disease literature: gradient boosting at 99% for chronic kidney disease risk; an artificial neural network for kidney disease detection nearing 100%; a deep recurrent network with supervised multi-labelling achieving 99.76% in an intrusion detection system for the internet of medical things; a deep-learning-assisted AdaBoost variant at 97.2% accuracy for breast cancer; a convolutional network at 84.5% against k-nearest neighbours at 83% for disease prediction; AdaBoost and random forest exhibiting accuracy over 95%.

The distribution of these figures is itself informative. A literature in which reported accuracies cluster between 95% and 100% is not a literature reporting hard problems. **liu2025review<empty citation>** themselves note that the limitations of machine learning methods include their black-box nature and dependence on a large amount of labelled data, and that the efficacy of a model "is contingent upon the dataset's representation rather than its volume"†.

3.4.2d The contradiction stated

Both are peer-reviewed, both published in 2025, both use the same algorithm on a health-monitoring classification task, and they disagree by 0.676 of F_1 (Figure 3.12). A thesis that cited only the first would conclude that XGBoost is the obvious choice. A thesis that cited only the second would conclude it is unusable. Neither conclusion is correct, and the reason is the subject of the next subsection.

Table 3.24: The same algorithm on two health-monitoring classification tasks, both published in 2025.

	efendi2025<empty citation>	**mendoza2025<empty citation>**
XGBoost accuracy	0.973	0.65
XGBoost F_1	0.970	0.294
XGBoost AUC	not reported	0.54 (0.49–0.60)
Best model	XGBoost	random forest
Outcome predicted	"stable" vs "not stable"	uncontrolled hypertension
Sample	4 500 records, balanced 50/50	1 308 adults, 30.3% positive
Features	3 vital signs, discretised	demographic and administrative

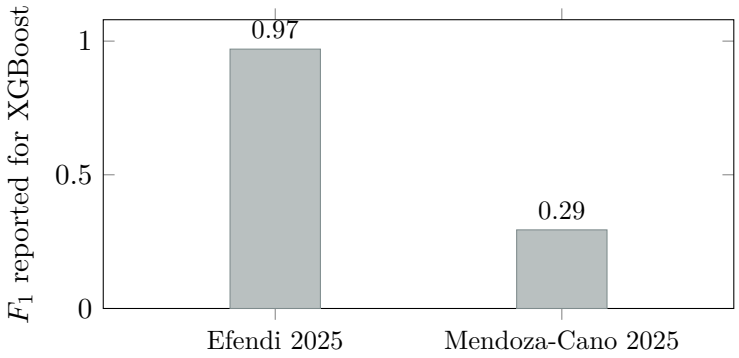


Figure 3.12: The same algorithm, the same year, two peer-reviewed health-monitoring tasks, and a difference of 0.676 in F_1. The explanation lies in how each task was constructed, not in the algorithm.

3.4.3 Resolving the contradiction: the task, not the algorithm

3.4.3a Efendi et al. measure label recovery, not prediction

Section 3.2.3d established the mechanism. The class label *stable* is defined as the condition in which heart rate, oxygen saturation and temperature all fall within the normal ranges of the paper's reference tables. The feature transformation then recodes each parameter as 0, 1 or 2—low, normal, high—with reference to those same ranges. After transformation, the label is a deterministic function of the three transformed features, $y = 1 \iff (x_1 = 1) \wedge (x_2 = 1) \wedge (x_3 = 1)$.

Any classifier capable of representing a three-way conjunction achieves perfect accuracy on this task. The reported spread of 0.042 in F_1 across five algorithms measures nothing about their relative merit; it measures implementation noise on a problem that is solved by construction.

Three further observations confirm that the result is not a measurement of predictive capability.

The deployed model contradicts the paper's own thresholds. In the real-world experiment, a single 50-year-old participant was monitored in four conditions. Sleeping, the system recorded 70.1 BPM and predicted *stable*. One reference table gives the normal range at age 50 as 85–119 BPM, the other 85–145 BPM. A value of 70.1 BPM lies below the lower bound of both, and the model nonetheless returned *stable*. During cardio exercise it recorded 121 BPM and again returned *stable*, which the second table permits but the first does not. The classifier therefore does not implement the labelling rule it was trained on, or the ranges stated in the paper are not those used.

The training configuration is not applicable to the models trained. The authors report 300 epochs, a learning rate of 0.025, a sigmoid activation function and the Adam optimiser, and plot training loss over 300 epochs for all five models including the decision tree, the random forest and the Bayesian classifier. None of these three trains by gradient descent; none has epochs, a sigmoid activation or an Adam optimiser. The paper also describes XGBoost as a deep learning model in its contribution table; XGBoost is an ensemble of gradient-boosted decision trees and is not a deep learning method.

The reported metrics are not mutually consistent. On a test set of 900 records balanced 50/50, precision and recall determine accuracy. For random forest, a recall of 0.922 and a precision of 0.936 imply an accuracy of 0.930, not the 0.946 reported; for AdaBoost, 0.937 and 0.956 imply 0.947, not 0.967. The remaining three rows are consistent within rounding.[8]

These are not pedantic objections. They indicate that the comparison was not conducted

[8] **[To verify]** Recompute the confusion matrix of each model in **efendi2025<empty citation>**, Table 16, against the stated 50/50 test balance. Two of the five rows do not close. Either the test split is not balanced as stated, or the metrics were computed on different subsets.

under conditions in which it could be informative, and they mean that Table 3.22 cannot be cited as evidence that XGBoost outperforms random forest.

3.4.3b Mendoza-Cano et al. measure genuine prediction from weak features

The opposite situation obtains. The outcome—uncontrolled blood pressure, defined as systolic $\geq$ 130 mmHg or diastolic $\geq$ 80 mmHg on the average of the second and third of three readings—is measured independently of the predictors. The predictors, selected by LASSO at $\lambda = 0.0056528$ with ten-fold cross-validation, are sex, age, socioeconomic status, years since diagnosis, diabetes history, source of medical care and adherence to a nutritional plan. Physical activity engagement was eliminated with a zero coefficient.

None of these is a physiological measurement. They are demographic and administrative variables. The question being asked is whether one can predict who has uncontrolled blood pressure without measuring anything about their physiology—and the answer, on 1 308 participants, is: weakly. Random forest reaches an area under the curve of 0.75; logistic regression 0.61; XGBoost 0.54.

This is not a failure of the algorithms. It is a correct measurement of how much signal demographic variables carry about blood pressure control, and the answer is some, but not much. Note also that this outcome is defined at the 130/80 mmHg threshold rather than the 140/90 mmHg convention retained throughout this thesis (Section 2.1), so its prevalence of 30.3% is not directly comparable with the control figures of Table 2.2.

3.4.3c The principle, and why it governs this thesis

$$\text{reported performance} = f(\text{task difficulty}, \text{label integrity}, \text{feature informativeness}, \text{algorithm}). \tag{3.14}$$

The algorithm is the last and smallest term. A reported accuracy figure is uninterpretable without the other three, and comparing accuracies across studies with different tasks is meaningless.

For the present work this yields a direct and uncomfortable implication. The framing document specifies that the risk label—stable, moderate, high, critical—is derived from recognised clinical thresholds, validated by the partner physician. If the model's features then include the current blood pressure and blood glucose values, the task becomes that of **efendi2025<empty citation>**: recovering a threshold rule from the variables that define it. The resulting accuracy will be high and will demonstrate nothing.

The anticipatory formulation of the framing document—forecasting physiological degradation at 24 to 48 hours—is the escape, and it must be presented as such rather than as an additional ambition. Formally, the leaking task is

$$\hat{y}_t = g(x_t) \quad \text{where} \quad y_t = h(x_t), \tag{3.15}$$

and the legitimate task is

$$\hat{y}_{t+\Delta} = g(x_{t-k}, \ldots, x_t), \qquad \Delta \in [24\,\mathrm{h}, 48\,\mathrm{h}], \tag{3.16}$$

in which the label lies outside the input window by construction and cannot be recovered from it (Figure 3.13). The expected accuracy of the second task is far lower than the first, and that lower figure is the honest one. This thesis anticipates the difference explicitly, because a jury that sees 0.97 on a threshold-classification task will ask the question posed above.

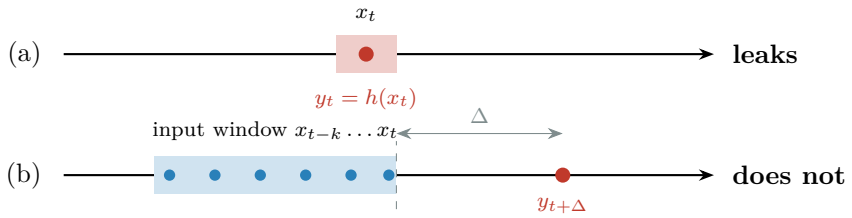


Figure 3.13: (a) Classifying the current state from the current reading, when the label is a rule applied to that same reading. (b) Forecasting a state outside the input window. In (b) the label cannot be recovered from the input, so the measured performance is a measurement.

3.4.4 Metrics: what to report, and three errors to avoid

3.4.4a Definitions

With TP, FP, TN and FN the entries of the confusion matrix,

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall (sensitivity)} = \frac{TP}{TP + FN}, \qquad \text{Specificity} = \frac{TN}{TN + FP}, \tag{3.17}$$

$$F_1 = \frac{2\,\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \qquad \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{3.18}$$

The area under the ROC curve is the probability that a randomly chosen positive is ranked above a randomly chosen negative; 0.5 is chance.

3.4.4b The accuracy trap

With prevalence π, accuracy decomposes as

$$\text{Accuracy} = \pi\,\text{Sensitivity} + (1 - \pi)\,\text{Specificity}. \tag{3.19}$$

When π is small, specificity dominates and a model that never predicts the positive class attains accuracy $1 - \pi$. The framing document states that in the target data critical states

are rare. Accuracy must therefore not be the headline metric of this thesis.

mendoza2025<empty citation> illustrate the trap concretely. Their logistic regression achieves an accuracy of 0.70, which is approximately the prevalence of controlled hypertension in their sample ($912/1308 = 0.697$), while its specificity is 0.03. It is a model that has learned to answer the same way almost every time.

3.4.4c First error: the positive class is not stated, and is not consistent

Applying the decomposition above to Table 3.23, with uncontrolled prevalence $\pi = 396/1308 = 0.303$:

$$\text{random forest:} \quad 0.303 \times 0.51 + 0.697 \times 0.98 = 0.838 \approx 0.84 \checkmark \tag{3.20}$$

$$\text{XGBoost:} \quad 0.303 \times 0.23 + 0.697 \times 0.85 = 0.662 \approx 0.65 \checkmark \tag{3.21}$$

$$\text{logistic regression:} \quad 0.303 \times 0.98 + 0.697 \times 0.03 = 0.318 \neq 0.70. \tag{3.22}$$

The F_1 values confirm the first two: $2(0.94)(0.51)/1.45 = 0.661$ against a reported 0.659, and $2(0.42)(0.23)/0.65 = 0.297$ against a reported 0.294. For the logistic regression row, taking *controlled* hypertension as the positive class instead reproduces every published value: accuracy $0.697 \times 0.98 + 0.303 \times 0.03 = 0.692 \approx 0.70$, precision equal to the controlled prevalence 0.697 against a reported 0.70, and $F_1 = 2(0.70)(0.98)/1.68 = 0.817$ against a reported 0.818.

The three rows of the same table are therefore not computed with the same positive class. Two treat uncontrolled hypertension as positive; the third treats controlled hypertension as positive. The paper never states which convention is used. The consequence is that the logistic regression row is not comparable with the other two, and its apparently excellent F_1 of 0.818 describes performance on the clinically uninteresting class.

3.4.4d Second error: sensitivity reported as specificity

The Results section of **mendoza2025<empty citation>** states correctly that "Random Forest achieved the highest precision (0.94, 95% CI 0.91–0.97), minimizing false positives, whereas Logistic Regression demonstrated high sensitivity (0.98, 95% CI 0.98–0.99)"†.

The Discussion then states, of random forest, that "its precision (0.94, 95% CI 0.91–0.97) was notably high, ensuring reliable identification of at-risk individuals, while its sensitivity (0.98, 95% CI 0.97–0.99) confirmed its effectiveness in detecting cases of uncontrolled HTN"†.

Table 3.23 gives random forest a sensitivity of 0.51 (0.46–0.55) and a specificity of 0.98 (0.97–0.99). The Discussion has substituted the second for the first.

The correction is material. Random forest detects 51% of uncontrolled cases. It misses almost half of them. The claim that its sensitivity confirmed its effectiveness in detecting

cases of uncontrolled hypertension is the reverse of what the paper's own table shows. This thesis cites the table and not that sentence.

For a triage system this is the central result of the section. The best model in the only genuine prediction study available misses half the patients it exists to find.

3.4.4e Third error: loss curves for models that do not minimise a loss by iteration

efendi2025<empty citation> report a training loss over 300 epochs for the decision tree, random forest and Bayesian classifier alongside AdaBoost and XGBoost. The first three are not fitted by iterative gradient minimisation and have no epoch-indexed loss. The quantity plotted is therefore not defined for those models, and the comparison of loss values across the five—0.025 for XGBoost, 0.062 for the Bayesian classifier—is not reproduced here.

3.4.4f Confidence intervals

mendoza2025<empty citation> report an interval for every metric; **efendi2025<empty citation>** report none. The difference matters. An area under the curve of 0.54 would be reported as a number slightly above chance; the interval 0.49–0.60 establishes that it is not distinguishable from chance (Figure 3.11). With the sample the framing document specifies, intervals in the present work will be wide, and reporting them is what makes a modest result honest rather than a weak one disguised.

3.4.5 Interpretability

mendoza2025<empty citation> apply two complementary approaches, and both are directly transferable.

Random forest variable importance, by percentage increase in mean squared error on permutation and by increase in node purity. The two rankings differ, which is itself instructive: by permutation the order is years since diagnosis (11.9), age (9.4), source of medical care (4.6), then diabetes history and sex; by node purity it is age (189.8), source of medical care (42.1), years since diagnosis (32.9), then diabetes history and sex.

SHAP values for XGBoost, giving the marginal contribution of each predictor: age 0.115, source of medical care 0.065, years since diagnosis 0.055, sex 0.035, with diabetes history lowest.

Two points for the present work. First, variable importance and SHAP need not agree, and reporting only one invites the suspicion that it was chosen. Second—and this requires care—SHAP explains a model, not a phenomenon. Here it explains an XGBoost model whose discrimination is not distinguishable from chance. Explaining the decisions of a model that does not discriminate is an exercise of limited value, and the paper does not

note this. If the present work reports SHAP, it must do so for a model whose discrimination has first been shown to be adequate.

liu2025review<empty citation> observe more generally that the black-box nature of machine learning is among its principal limitations, and that regulatory and clinical acceptance depends on interpretability. Section 2.2.3 reached the same conclusion from the clinical side: a triage system is adopted only if the clinician can understand why a patient was flagged and, more importantly, trust the ones it did not flag.

3.4.6 Class imbalance and the rare critical state

The framing document anticipates this correctly: it identifies strong class imbalance, critical states being rare, as a data limitation, and proposes as methodological responses a metric centred on recall of the critical class, and oversampling.

The results of **mendoza2025<empty citation>** demonstrate why this is not optional. At a prevalence of 30.3%—which is not extreme—the sensitivity of random forest falls to 0.51 and that of XGBoost to 0.23. At the prevalence the framing document anticipates for critical states, considerably lower than 30%, the untreated behaviour of these algorithms will be to predict the majority class almost always.

The design response has three components, and the first is the most important.

Choose the operating point from the cost of error, not from the default threshold. A classifier outputs a score; the 0.5 threshold is a convention with no clinical meaning. For triage, a false negative is a patient in deterioration who is not seen; a false positive is a clinician's time spent on a patient who turns out to be stable. Section 2.2.3 established that clinician time is the scarce resource, so false positives are not free—but Section 2.1.2 established that the dominant failure of the existing system is non-detection. The threshold should be set to a target sensitivity on the critical class and the resulting alert volume reported, rather than the reverse.

Weight or resample during training, as the framing document proposes, while noting that resampling changes the calibration of the predicted probabilities and that any probability reported to a clinician must be recalibrated afterwards.

Report per-class metrics, never aggregates alone. A macro-averaged F_1 conceals exactly the failure that matters. The random forest row of Table 3.23—precision 0.94, sensitivity 0.51—shows what a single aggregate would have hidden.

3.4.7 From classification to forecasting

The hypotheses of the framing document can now be assessed against the state of the art.

3.4.7a The three hypotheses against the evidence

H1: a deep temporal model obtains better recall on high and critical risk states than robust classical models, on locally collected telemonitoring data. The only comparable datum in the corpus is the recurrent network of Reddy and Delen for lupus readmission, reported by **liu2025review<empty citation>** as outperforming traditional classification with an area under the curve of 0.70. This is a modest figure, and it is the realistic order of magnitude for a genuine clinical prediction task on a sequence. It supports the plausibility of H1 while setting expectations well below the 95–100% band that dominates the ambient literature.

H2: short-horizon forecasting of physiological parameters allows anticipation of degradation events with a clinically useful lead time. No source in this corpus attempts this. The architectures of Section 3.3 provide for it—**mucchi2021<empty citation>** assign the cloud layer long-term data analysis through advanced algorithms for data cross-correlation—but none implements it. H2 is therefore untested in the available literature, which is the strongest claim to novelty this thesis can make, and also its principal risk.

H3: given the limited size and irregularity of the local data, the gain of the deep model over classical models may remain modest. This is the hypothesis best supported by the state of the art. **liu2025review<empty citation>** note that deep learning methods present high computational resource demands, increased model complexity and the need for data privacy protection, and that efficacy is contingent upon the representation of the dataset rather than its volume. With the sample size the framing document targets, a model with the parameter count of a recurrent network is at substantial risk of overfitting, and a negative result would be a genuine finding rather than a failure.

3.4.7b The horizon must match the sampling interval

One constraint on H2 follows directly from the acquisition decision of Section 3.2.6b and is not stated in the framing document, although it governs whether the hypothesis is testable at all.

Under the spare acquisition adopted, blood pressure and blood glucose arrive through pipeline 2 at the routine frequency established in Section 3.1.4: of the order of two readings per week. The expected number of readings falling inside a forecast horizon of length Δ is then

$$\mathbb{E}[N_\Delta] = \lambda\Delta, \qquad \lambda = \frac{2}{168\,\mathrm{h}}, \tag{3.23}$$

which gives $\mathbb{E}[N_{24\mathrm{h}}] = 0.29$ and $\mathbb{E}[N_{48\mathrm{h}}] = 0.57$. Fewer than one reading falls inside the horizon on average, so for most prediction points there is no observation to predict (Figure 3.14). Pipeline 1 is worse still: at one to two consultations per year it contributes nothing at this horizon. A 24 to 48 hour forecast is therefore not estimable from routine-

frequency data.

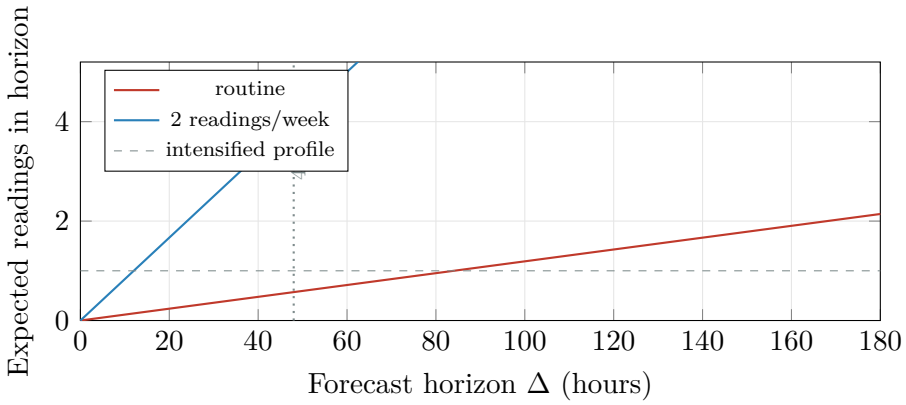


Figure 3.14: Expected number of observations inside a forecast horizon, by sampling regime. At routine frequency the 48-hour horizon contains 0.57 readings on average; the intensified verification profile of Section 3.1.4c contains four.

Three resolutions are available and one must be adopted. The horizon may be lengthened to match the sampling interval, predicting the next scheduled reading or the seven-day mean that Section 3.1.4 defines as the reference value—which preserves the anticipatory character while making the target estimable. The routine frequency may be raised to daily, which costs adherence against the trade-off set out in Section 3.1.4. Or the forecast may be defined only within the intensified verification regime, where two readings per day give $\mathbb{E}[N_{48h}] = 4$—which is coherent with the two-stage architecture of Section 3.3.3c, since that is precisely the regime in which a clinical decision is pending.

The third is the most defensible, because it aligns the forecasting task with the moment at which its output would change a decision. This thesis adopts it, and H2 is restated accordingly: short-horizon forecasting applies to patients in the verification regime, and the routine regime is served by trend detection over the longer interval rather than by a 48-hour forecast.[9]

3.4.7c A methodological commitment to add

The framing document specifies group k-fold cross-validation by patient, which is correct and prevents the most common leakage in clinical modelling. For a forecasting task it is not sufficient: the split must also be temporal. Training on later data and testing on earlier data leaks the future, even across patients, if any global preprocessing—normalisation, imputation, feature scaling—is fitted on the full dataset. The split must therefore be by patient and forward in time, and every preprocessing step must be fitted on the training fold alone.

[9] **[To verify]** Confirm the routine sampling frequency actually achieved in the collected cohort before fixing the horizon. If observed adherence yields fewer than two readings per week, even the seven-day target becomes marginal and the design must fall back on consultation-anchored prediction.

3.4.8 Specification for the experimental phase

Table 3.25: Requirements for the experimental phase established in Section 3.4.

Requirement	**Basis**
Target is $\hat{y}_{t+\Delta}$, not $\hat{y}_t$	label leakage in **efendi2025<empty citation>**; Section 3.4.3c
Horizon matched to the sampling regime; forecast defined in the verification regime	Section 3.4.7b
No feature computed from the same instant as the label	Section 3.4.3c
Preprocessing fitted on the training fold only	Section 3.4.7
Split by patient and forward in time	framing document, extended
Baselines in order: majority class, threshold rule, logistic regression, random forest, XGBoost, LSTM	**mendoza2025<empty citation>** include logistic regression as reference
The clinical threshold rule is a mandatory baseline	a model that does not beat the ESH and ADA thresholds has no justification
Headline metric: recall on the critical class	framing document; Section 3.4.6
Accuracy never reported alone	Section 3.4.4b
Positive class stated explicitly for every metric	Section 3.4.4c
Per-class metrics reported, not only aggregates	Section 3.4.4d
Confidence intervals on every metric	**mendoza2025<empty citation>**; absent in **efendi2025<empty citation>**
AUC reported with interval, chance level marked	XGBoost interval 0.49–0.60 includes 0.50
Operating threshold chosen from error cost, then alert volume reported	Section 3.4.6
Probabilities recalibrated after any resampling	Section 3.4.6
Interpretability reported only for a model with adequate discrimination	Section 3.4.5
Both permutation importance and SHAP, with disagreement discussed	**mendoza2025<empty citation>**
Fixed random seed, stated	**mendoza2025<empty citation>**: seed 123
Code, feature definitions and labelling rule published	**mendoza2025<empty citation>** archived data and code
Training configuration appropriate to each model family	Section 3.4.3

The penultimate line deserves a note. **mendoza2025<empty citation>** deposited their dataset and their code in public repositories. The framing document states that the data are real, anonymised and supplied by a partner physician, with direct identifiers removed before any processing, collected with the agreement of the physician and the institution for research purposes only. Full data publication will not be possible. Publishing the code, the feature definitions and the labelling rule is possible, and is committed to here, since it is what allows the result to be checked without exposing the patients.

3.4.9 What this section establishes

Established. The algorithm families and their inductive biases are well characterised, and only one of the candidates treats the input as a sequence. Reported performance in this literature is governed by task construction far more than by algorithm choice, and the two principal sources demonstrate both poles of that observation: a task solved by construction yields 0.97, and a genuine prediction task from weak features yields 0.54 to 0.75. The metric apparatus required for an honest report is available and is largely absent from the sources.

Not established. No source predicts future physiological state. No source works on a clinical time series of blood pressure and blood glucose. No source operates in a low-resource setting. The best sensitivity achieved on genuine prediction of uncontrolled hypertension anywhere in this corpus is 0.51, and the model that achieved it is a random forest—not the gradient-boosted ensemble that the surrounding literature treats as the default choice.

3.5 Clinical Decision Support Systems and Alert Generation

Sections 3.2 to 3.4 established what is acquired, how it travels, and what is computed on it. This section addresses the last link: how a computed score becomes a clinical action. It relies on the decision flow of **yazdani2018<empty citation>**, the only source whose explicit subject is the coupling of telemonitoring with clinical decision support, on the rule-engine techniques described by **liu2025review<empty citation>**, on the behaviour change evidence of **zhou2024<empty citation>**, and on the only clinically validated alert protocol, that of **mihevc2025<empty citation>**.

3.5.1 What a decision support telemonitoring system is

3.5.1a Definition

yazdani2018<empty citation> give the definition we adopt: "Decision Support Telemedicine Systems (DSTS) is a combination of clinical decision support systems and telemedicine. A hybrid system that transmits information from the patient's home to the health monitoring center and allows them to support their monitoring data is an example of DSTS."†

They situate its necessity in a problem Chapter 2 established independently: "producing a large amount of patient data that requires careful evaluations can lead to unacceptable increase in workloads of physicians. Therefore, a combined decision support tools and telemedicine can be used as a possible solution to overcome human resource limitation and analyze generated data."†

This is the same argument reached in Section 2.2.3 from the clinical side—the six calls per provider per day of **kingue2013<empty citation>**, the 18% of countries without a cardiologist reported by **segun2025<empty citation>**—arriving here from the informatics side. Decision support is not an enhancement of telemonitoring; it is the condition under which telemonitoring does not consume more clinical time than it saves.

3.5.1b The scalability problem

The stated contribution of **yazdani2018<empty citation>** is a diagnosis worth retaining. They observe that existing decision-support telemonitoring architectures are "largely confined to a specific disease and it means these architectures are not scalable"†, so that "if that architecture is to be used to monitor the health status of other diseases, it should be re-established with the new knowledge base (the new decision support system). This issue will result in the creation of multiple island systems for monitoring the status of patients."†

Their response is to separate the platform from the knowledge base, so that several decision support systems can be loaded onto a common infrastructure, each with its own rule set, selected per patient according to disease and symptoms.

For a thesis whose object covers two pathologies with two distinct guideline sources (Section 3.2.1), this separation is not an abstraction. A system in which the hypertension thresholds are hard-coded alongside the glycaemic thresholds is one that must be rewritten to add a third condition. One in which the rule base is a configurable artefact is extensible by configuration.

3.5.1c Three families of decision support

Table 3.26: Families of clinical decision support and their trade-offs.

Family	Basis	Strength	Weakness
Rule-based	explicit thresholds from clinical guidelines	transparent, auditable, no training data, defensible to a clinician	no personalisation, no trend detection, coverage gaps
Model-based	learned from data	personalisation, trend detection, adaptive	opaque, needs data, may not transfer
Hybrid	rules as a floor, model as a refinement	guideline conformity plus personalisation	two components to validate

The sources distribute across these. **chatrati2022<empty citation>** are purely rule-based for hypertension and purely model-based for diabetes, with no coordination between the two. **efendi2025<empty citation>** are purely model-based, on a model that reproduces a rule (Section 3.2.3d). **mihevc2025<empty citation>** are purely rule-based, and are the only ones clinically validated. **liu2025review<empty citation>**

advocate the hybrid: "by integrating a sophisticated rule engine with machine learning algorithms, the chronic disease management platform is capable of dynamically monitoring and conducting real-time analysis of patients' health status"†.

The hybrid is the position this thesis adopts, and Section 3.4.8 already established why in a different form: the clinical threshold rule is a mandatory baseline, because a model that does not outperform the guideline thresholds has no justification for existing. Making the rule a component rather than merely a baseline means the system degrades to guideline conformity when the model is uncertain, rather than degrading to nothing.

3.5.2 The reference decision flow

3.5.2a The activity diagram of Yazdani et al.

Their specification covers three actors—patient as client, supervisor as server, physician as client—and it is the most detailed account of a telemonitoring decision cycle in this corpus. Traced step by step:

Acquisition. "Patient data is collected by sensors or manually by the patient interface. The data are proportionate with the intervals that the physician has determined, and are sent to the server for storage and processing."†

Non-transmission handling. "If no data is sent within the time limits set by the physician, a SMS reminder will be sent to the patient. The SMS reminder is sent repeatedly as long as the patient enters a new data."†

Verification and routing. "The server, after verifying the data and store them in the database, automatically analyzes the data to determine the related CDSS."†

Classification. "After determining the CDSS, patient data will be analyzed. The result of this analysis is to determine the patient's condition in one of the two Normal or Alert status."†

Branch on normal. The patient is notified, the record is updated, no clinician is involved.

Branch on alert. A treatment proposal is generated and the physician notified: "the CDSS in the Alert mode displays the proposed prescription to the physician. After reviewing the patient's data, the physician can approve the proposed CDSS prescription or prescribe his own recommendation."†

Physician sub-decision. Three outcomes are modelled: therapy unchanged; new therapy required, in which case the physician prescribes; or more information needed, in which case "the patient will be informed and coordinated to have face to face meeting with the physician through the patient interface"†.

Closure. "All events in this telehealth monitoring architecture, including physician and CDSS prescriptions, are stored as patient record in the database."†

3.5.2b The three decision points

Table 3.27: Decision points of the reference flow (**yazdani2018**).

Point	Decided by	Output
Did data arrive on schedule?	system	reminder, repeated until compliance
Is the patient normal or alert?	CDSS	notify patient, or escalate with proposal
Is the proposal accepted?	physician	approve, override, or request consultation

The third is the one that matters most and is the most frequently omitted in engineering designs. The system proposes; the physician disposes. No prescription reaches the patient without passing a clinician. **yazdani2018<empty citation>** do not argue for this; they simply build it in. The argument is supplied by Section 3.4: the best model of **mendoza2025<empty citation>** detects 51% of uncontrolled cases and the second best is indistinguishable from chance. A system that acted autonomously on such a model would be acting on a coin flip half the time.

The first decision point is also significant and connects to Chapter 2. Non-transmission is itself an event that triggers an action. Section 2.2.2 established loss to follow-up at 50% or more in low-resource settings, and Section 2.2.3 measured 33.3% three-month attendance in Cameroonian control centres. A system that only reacts to data received is blind to exactly the patients most at risk. The repeated SMS of **yazdani2018<empty citation>**, and the HEARTS indicator for patients with no visit in the last three and twelve months (Section 3.1.3a), are the same mechanism: silence must be alarmed.

3.5.2c Comparison with the only validated protocol

Table 3.28: The reference flow against the only clinically validated protocol.

	yazdani2018<empty citation>	**mihevc2025<empty citation>**
Classification	two states: normal, alert	three states: green, yellow, red
On normal	notify patient	no change
On intermediate	—	verification profile: 7-day BP profile or 1-day 6-point glucose profile
On critical	notify physician with proposal	immediate teleconsultation
Who decides	physician approves or overrides	general practitioner evaluates, may modify therapy
Validated	no	yes: RCT, −6.9 mmHg, −0.5% HbA1c

The yellow tier is the critical addition. In a two-state system, every reading that is not normal escalates to a clinician. In the three-state system, an abnormal reading escalates to more data, and only the aggregate of that data escalates to a clinician. This is the architectural two-stage split of Section 3.3.4 expressed as a clinical protocol, and it is the single most transferable design element in the corpus—not least because it is the only one attached to a demonstrated clinical effect.

3.5.2d The convergent flow

Combining the two, with the model placed where Section 3.4 established it belongs, gives the flow of Figure 3.15.

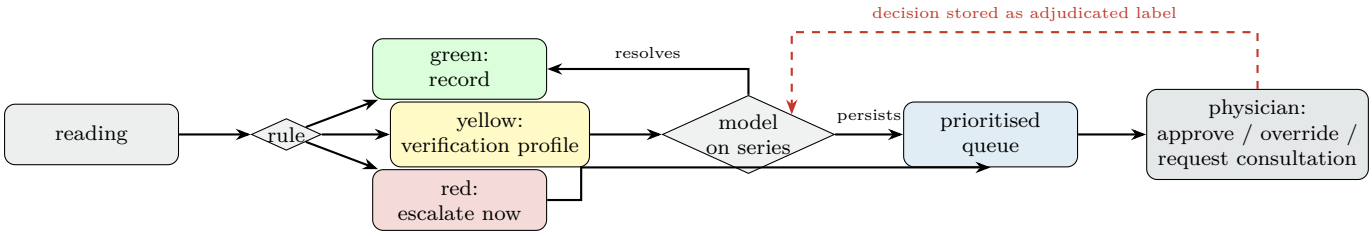


Figure 3.15: The convergent decision flow: rule-based triage at the edge, model-based assessment of the verification profile, prioritised escalation, and clinician disposition. The dashed return path is the closed loop of Section 3.5.5d.

A parallel branch runs alongside: non-transmission within the configured interval raises a reminder, repeated, and escalates as non-adherence if unanswered.

3.5.3 The rule engine coupled to the model

liu2025review<empty citation> provide the only technical specification of this coupling in the corpus. Four elements are directly usable.

3.5.3a Characterise the series before choosing the algorithm

"The process of feature analysis necessitates the identification of variation patterns in time-series data derived from historical monitoring indicators, such as periodicity, drift, and stationarity, to select the most suitable algorithms. For example, autocorrelation functions can be used to identify periodic changes, while median smoothing and trend component detection can capture data drift. Stationarity changes are assessed using unit root tests to determine if the time series is stationary."†

This prescribes a diagnostic step that precedes modelling. Blood pressure has a known circadian structure—the nocturnal dipping discussed by **gafane2024<empty citation>**—so periodicity is expected; therapeutic drift is the clinical quantity of interest (Section 2.2.2); and stationarity determines whether differencing is required before any trend feature is meaningful. This diagnostic belongs in the experimental chapter before any model is fitted, and it is the kind of step that distinguishes a considered analysis from an undirected sweep of classifiers.

A caution attaches. These tools assume regularly sampled series. The framing document anticipates irregular and missing measurements from intermittent connectivity, and Section 3.2.1 established that pipeline 2 has missingness that is not at random. Autocorrelation and unit-root tests on irregularly spaced data require either resampling

onto a regular grid, which fabricates observations, or methods designed for unevenly spaced series. The choice must be stated, because resampling a series of two true readings per week onto a daily grid manufactures five interpolated points per week, and any trend feature computed on that grid is five-sevenths fiction.

3.5.3b Anomaly detection as a distinct problem

"When selecting an algorithm, it is essential to choose the appropriate anomaly detection algorithm based on the data's distribution characteristics. Commonly utilized algorithms in this field include box plots, absolute median differences, and extreme value theory, among others."†

These are unsupervised and distribution-based, and they complement rather than compete with the supervised classifier of Section 3.4. The median absolute deviation in particular is directly useful for the per-patient normalisation the framing document proposes, since it is robust to the outliers a telemonitoring series will contain:

$$\mathrm{MAD}_i = \mathrm{median}\Big(|x_{i,t} - \mathrm{median}(x_i)|\Big), \qquad z_{i,t} = \frac{x_{i,t} - \mathrm{median}(x_i)}{1.4826\,\mathrm{MAD}_i}. \tag{3.24}$$

This yields the personalised threshold identified in Section 2.7 as a gap of the fixed-threshold protocol of **mihevc2025<empty citation>**: a deviation is assessed relative to the individual patient's own distribution rather than against a population constant. It is also the empirical counterpart of the quantity $\sigma_{\varepsilon,i}$ introduced in Section 3.1.1b.

liu2025review<empty citation> add that "machine learning algorithms, such as random forests or autoencoders, are employed to promptly capture anomalies in data and trigger early warning systems"†.

3.5.3c The offline and online split

"Two crucial aspects of system design include model training and real-time detection. Real-time detection typically employs stream processing technologies such as Flink for online anomaly detection, while offline training involves reading training data from a data warehouse, training the model, and saving it for use in real-time detection."†

This is the standard separation between a batch training path and a low-latency serving path, and it maps directly onto the architecture of Section 3.3: offline training in the cloud layer, where **mucchi2021<empty citation>** place long-term data analysis, and online detection at the edge, where they place quick processing and identification of anomalous conditions.

The specific technology named is not appropriate here. A stream processing framework is dimensioned for throughput that a cohort transmitting a few readings per week will never approach. The separation is the transferable element; the tooling is not. For this

work, offline training on a periodic schedule and online evaluation of a saved model at the point of reception is the same architecture at the appropriate scale.

3.5.3d The adaptive trigger

"An adaptive triggering strategy, based on multi-source data, is proposed. This strategy involves real-time monitoring of patients' health statuses by integrating data from various channels such as electronic health records, wearable device monitoring data, lifestyle habits, and biomarkers. The comprehensive health profile established for each patient is then transmitted to a central processing system, thereby creating an intelligent multi-point triggering disease prediction and assessment system."†

The term *multi-point* carries two meanings in their framework, and the distinction is useful: multiple data sources triggering jointly, and multiple monitoring points—hospital, community, family—contributing. The first is precisely the three-pipeline integration specified in Section 3.2.1.

liu2025review<empty citation> also report a directly relevant precedent: Zhang and colleagues developed an ontology-based framework amalgamating type 2 diabetes patient data with medical domain knowledge and assessment criteria, and a clinical decision support system that "autonomously selects and modifies standard assessment protocols to align with individual patient conditions"†. This is the rule-selection module of **yazdani2018<empty citation>** implemented, and it is the closest published instance of the configurable-knowledge-base design.

3.5.4 Alert prioritisation, and the arithmetic of alert volume

This subsection contains the most important quantitative result of the chapter, and no source in this corpus states it.

3.5.4a Prioritisation as specified

yazdani2018<empty citation>: "the physician can have access to patient data at any time through the patient status and is able to view patients' status on a priority basis. This prioritisation is done based on the patient status (Normal and Alert)."† They report a precedent in which a system identifies patients with gestational diabetes and prioritises them on the basis of their metabolic status, sending automatic recommendations from a rule-based database but routing insulin dosing decisions to the physician first.

The principle is sound but the resolution is coarse. A binary sort produces two buckets, and when the alert bucket contains more patients than the clinician can see in a session, the sort has not helped: it has merely relabelled the problem. Prioritisation requires an ordering, not a partition, and an ordering requires a continuous score—which is what

the model of Section 3.4 produces and what the rule of **mihevc2025<empty citation>** does not.

3.5.4b The arithmetic

Consider a cohort of N patients each transmitting m readings per week, with π the proportion of readings that genuinely warrant clinical attention, and a classifier of sensitivity Se and specificity Sp. Alerts per week number

$$A = Nm\Big[\pi\,\mathrm{Se} + (1-\pi)(1-\mathrm{Sp})\Big], \tag{3.25}$$

of which the proportion that are genuine is the positive predictive value

$$\mathrm{PPV} = \frac{\pi\,\mathrm{Se}}{\pi\,\mathrm{Se} + (1-\pi)(1-\mathrm{Sp})}. \tag{3.26}$$

The second term of the denominator is the one that governs, and it does not depend on sensitivity at all. False alerts are generated by the volume of normal readings multiplied by the false-positive rate, and in a monitoring system the volume of normal readings is overwhelming.

Substituting the measured values of **mendoza2025<empty citation>** at their prevalence $\pi = 0.303$ reproduces their reported precisions, which confirms the model: for random forest, $0.303(0.51)/[0.303(0.51) + 0.697(0.02)] = 0.92$ against a reported 0.94; for XGBoost, $0.303(0.23)/[0.303(0.23) + 0.697(0.15)] = 0.40$ against a reported 0.42.

Now apply it to a telemonitoring setting, where critical events are rare. Taking $\pi = 0.02$—two per cent of readings genuinely critical, which is optimistic given the framing document's statement that critical states are rare—gives Table 3.29.

Table 3.29: Positive predictive value and false-alert burden at $\pi = 0.02$.

Se	Sp	PPV at $\pi = 0.02$	False alerts per 1 000 readings
0.51	0.98	0.34	19.6
0.90	0.98	0.48	19.6
0.90	0.99	0.65	9.8
0.90	0.995	0.79	4.9
0.90	0.999	0.95	1.0

At two per cent prevalence, reaching 80% precision requires 99.5% specificity. Solving for the requirement directly,

$$\mathrm{Sp} \;\geq\; 1 - \frac{\pi\,\mathrm{Se}}{1-\pi}\cdot\frac{1-\mathrm{PPV}}{\mathrm{PPV}}. \tag{3.27}$$

This is the quantitative form of the triage requirement, and it is considerably more demanding than anything the literature reports. The best model of **mendoza2025<empty citation>**

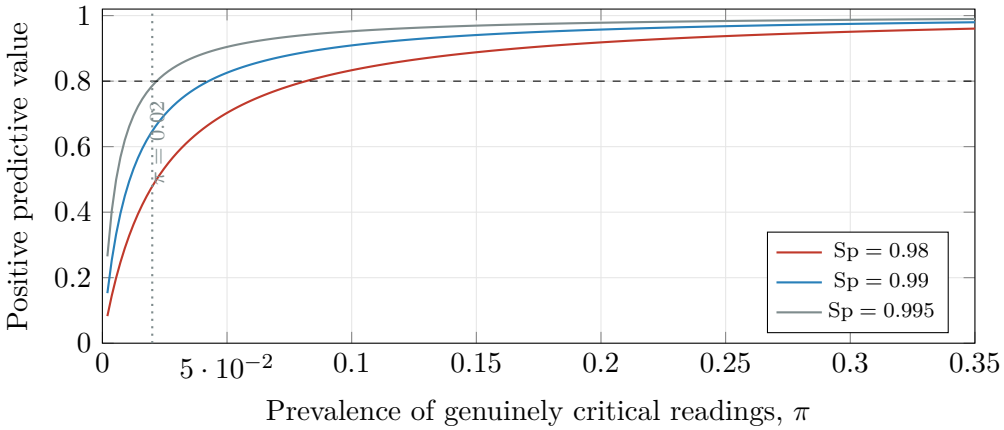


Figure 3.16: Positive predictive value against prevalence at Se = 0.90, for three specificities. The dashed line marks an 80% precision target. At $\pi = 0.02$ it is met only beyond Sp = 0.995.

reaches a specificity of 0.98 at a prevalence fifteen times higher.

3.5.4c Why sampling frequency decides the problem

The term Nm is the multiplier, and it varies across the architectures of this chapter by four orders of magnitude (Table 3.30 and Figure 3.17).

Table 3.30: False-alert burden by acquisition design, at Sp = 0.98 and $\pi = 0.02$.

Design	Readings per patient per week	False alerts per patient per week
mihevc2025<empty citation>: 2 BP + 1 glucose	3	0.06
Manual entry, daily	7	0.14
efendi2025<empty citation>: 1 Hz continuous	604 800	11 854

For a cohort of thirty patients on the protocol of **mihevc2025<empty citation>**, the system generates roughly two false alerts per week—entirely manageable. The same specificity applied to continuous sensor streaming generates over eleven thousand false alerts per patient per week, which is not a tuning problem but a categorical impossibility.

This resolves a tension that has run through the chapter. Section 3.2 found that the optical sensor pipeline cannot measure the parameters of clinical interest. The present subsection establishes a second and independent reason to restrict its role: per-reading alerting at sensor frequency is arithmetically unusable regardless of model quality. High-frequency data can be aggregated into features—a daily mean, a variability measure, a trend slope—and alerted on at the aggregate level, but it cannot be alerted on reading by reading. The data processing unit at the edge, in the architecture of **mucchi2021<empty citation>**, is exactly where that aggregation belongs.

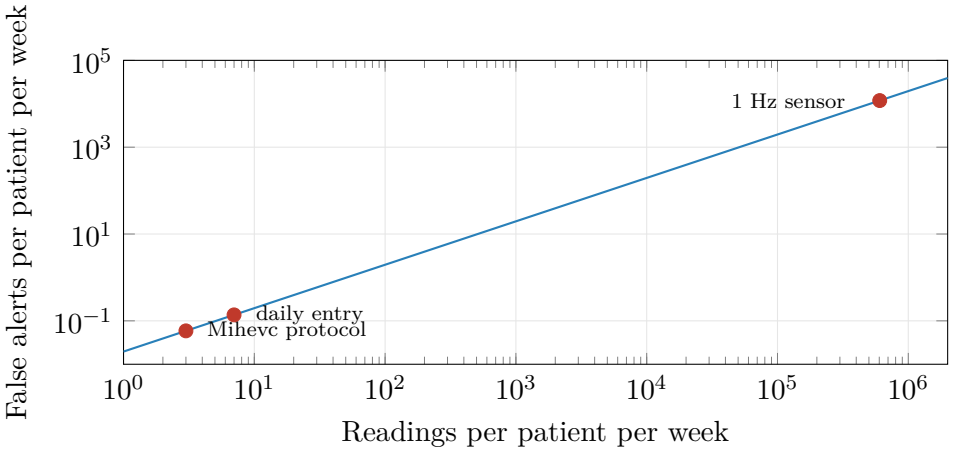


Figure 3.17: False-alert burden scales linearly with sampling frequency at fixed specificity. The clinical protocols and the sensor pipeline differ by more than five orders of magnitude on this axis.

3.5.4d What the sources say about alert burden, and what they do not

No source in this corpus uses the term *alert fatigue* or measures it. This thesis does not cite these sources for a phenomenon they do not address. What they do establish, which is adjacent and usable:

dodoo2022<empty citation>, reported in Section 2.2.4, found that health professionals in the region view telemedicine as an additional task entailing increased workload. An alerting system whose precision is low converts this perception into fact.

mucchi2021<empty citation>, reported in Section 3.3.6b, record nurses requesting that alert notification be improved especially in urgent situations, and that immediate feedback from the doctor in the event of an alert be considered. They were asking for alerts that are fewer and more consequential, not more numerous.

bediang2021<empty citation>, reported in Section 2.2.3, supply an indirect measurement that deserves notice. In the intervention centres, 23.7% of local providers' referrals were judged unnecessary by the remote expert, against 11.5% in control centres ($p = 0.01$). The system that improved access to expertise also doubled the proportion of inappropriate escalations recorded. The authors do not comment on this. It is the closest thing in this corpus to a measured instance of over-escalation accompanying a decision support system, and it is cited here as such—with two reservations. The study is not randomised and the groups differ at baseline. More importantly, the ascertainment mechanism is not symmetric: in the intervention arm a cardiologist reviewed every case and could judge a referral unnecessary, whereas no such reviewer existed in the control arm, so the difference may measure how closely referrals were examined rather than how often they were unwarranted.[10]

[10] [**To verify**] Establish from **bediang2021<empty citation>** how referrals in the control centres were adjudicated as unnecessary, given that those centres had no remote expert. If the two arms were

who2023hypertension<empty citation> place an upper bound on tolerable workload from the other direction: the readiness-to-scale criterion requires a missed-visit rate below 25% (Section 3.1.3a), which presupposes that follow-up is actionable at the scale of the cohort.

3.5.4e The design response

Four measures follow, and they are testable rather than aspirational.

Alert on aggregates, not on readings. Every alert is raised on a statistic over a window—a seven-day mean, as in the verification profile of **mihevc2025<empty citation>**—never on a single value, except at the red tier where a single extreme value justifies immediate escalation.

Use the yellow tier as the volume regulator. An intermediate reading costs the patient one verification profile and costs the clinician nothing. This is the mechanism by which the arithmetic above is made tractable. Averaging n readings divides the variance of the estimate by n (Section 3.1.1b), so the separation between the normal and abnormal distributions, measured in standard deviations, grows as $\sqrt{n}$. The whole receiver operating characteristic curve moves outward, and a higher specificity becomes attainable at the same sensitivity. With the fourteen readings of a seven-day profile, that separation is 3.7 times larger than for a single reading.

Report alert volume alongside every performance figure. This thesis states, for its chosen operating point, how many alerts per clinician per week the system would generate for a cohort of a given size. A sensitivity figure without an alert volume is not an evaluation of a triage system, and this requirement extends the specification of Section 3.4.8.

Let the clinician set the threshold. The operating point is a clinical resource decision, not a technical one. Exposing it as a configurable parameter—the CDSS-setting module of **yazdani2018<empty citation>** is the structural place for it—allows a service with one physician and a service with five to run the same system at different sensitivities.

3.5.5 Feedback loops

3.5.5a To the physician: a proposal requiring approval

The mechanism of **yazdani2018<empty citation>**—the system displays a proposed prescription, the physician approves or overrides, the approved prescription reaches the patient, every event is recorded—is the correct pattern, and three independent findings support it.

assessed by different procedures, the 23.7% against 11.5% comparison measures ascertainment and must not be presented as over-escalation.

It addresses therapeutic inertia, which Section 2.2.2 identified on WHO authority as the mechanism behind the weakest link of the care cascade, 56% of treated patients not at target. The WHO states that a simple standardised protocol, by clarifying the treatment steps for patient and clinician, contributes to reducing it. A system that presents the next protocol step at the moment the data show the target is not met is that protocol made operative.

It is where the clinical value was measured. bediang2021<empty citation> found the remote expert modified the local provider's diagnosis in 46.2% of cases against 17.9% in controls. The value of the system lay in altering a clinical decision, not in transmitting data.

It matches the mechanism of the only randomised trial on the target population. mihevc2025<empty citation> attributed their effect to improved communication, medication adherence, early detection of exacerbations and therapy adjustments based on accurate telemonitoring data, and explicitly not to behaviour change, which did not occur (Section 2.3.3).

The precedent reported by **yazdani2018<empty citation>** encodes the safety boundary precisely: automatic recommendations are sent to patients from a rule-based database, but recommendations about the amount of insulin to inject are sent first to the physician. Lifestyle advice may be automated; dosing may not.

3.5.5b To the patient: what the evidence supports

zhou2024<empty citation> provide the only quantitative evidence here, from twenty randomised trials and 4 168 participants. Table 3.31 maps the techniques most frequently employed in effective interventions onto their implementation in the proposed system.

Table 3.31: Behaviour change techniques of **zhou2024<empty citation>** and their implementation here.

Technique	Frequency	Implementation
Self-monitoring of outcomes of behaviour	17/20	the measurement protocol itself
Feedback on outcomes of behaviour	15/20	returning the reading with its classification
Instruction on how to perform the behaviour	15/20	measurement guidance in the application
Pharmacological support	15/20	adherence tracking
Biofeedback	14/20	trend display
Prompts and cues	13/20	measurement reminder
Credible source	13/20	attribution to the clinical team
Action planning	11/20	scheduling of the next measurement

Their quantitative findings, carried from Table 2.7: interventions with at least eleven

techniques achieved −9.68 mmHg against −2.88 mmHg for those with fewer; multifactorial interventions achieved −6.50 mmHg against a non-significant −1.54 mmHg; and only 30% of trials declared a theoretical foundation. **alessa2019<empty citation>** established that reminders appear in only 21% of 186 commercial applications and feedback in 28% (Section 2.3.1), so the techniques with the strongest evidence are among the least implemented in the market.

3.5.5c The asymmetry that must be stated

The pooled effect of **zhou2024<empty citation>** on systolic pressure is −5.78 mmHg, at low GRADE certainty, with 75% of included trials at high risk of bias. The clinician-coupled system of **mihevc2025<empty citation>** achieved −6.9 mmHg and found no behaviour change at all. **mao2020<empty citation>** found mobile health alone at −2.16 mmHg, not significant, against −6.17 mmHg when combined with professional management (Section 2.3.2).

The patient-facing loop is worth building and is not where the effect comes from. The design conclusion, stated once and carried into the implementation: behaviour change techniques are implemented because they are cheap—reminders, display, scheduling—and because **zhou2024<empty citation>** show their quantity correlates with effect size. The clinical decision loop is implemented because that is where the measured effect originates.

3.5.5d The closed loop

liu2025review<empty citation> describe the full circuit: "establishing effective feedback channels allows these responses to be used to further optimize algorithms and enhance services. A closed-loop feedback mechanism enables the system to update and optimize treatment plans in real-time based on the patient's actual response and health changes."†

This has an implication for the model that is easily overlooked. Every physician decision—approve, override, request consultation—is a label. An override on a red alert is a false positive confirmed by a clinician; an approval is a true positive. **yazdani2018<empty citation>** store all of these as patient record. A system that captures them systematically accumulates, over months of operation, a validation set that is clinically adjudicated rather than threshold-derived—which is the escape from the circularity diagnosed in Section 3.4.3.

This is designed in from the start, because it costs nothing at build time and cannot be reconstructed retrospectively. It is also the most credible route to the kind of contextually valid training data that **segun2025<empty citation>** identified as absent from sub-Saharan Africa (Section 2.2.4).

Table 3.32: The boundary of automation implied collectively by the sources.

May be automated	**Must pass a clinician**
Measurement reminders	any dose change
Return of the reading and its classification	any medication change
Trend display	any instruction to stop a treatment
Generic lifestyle guidance	any urgency-graded instruction
Scheduling the next measurement	any diagnosis
Non-transmission follow-up	any referral

3.5.5e Safety constraints on automated advice

Two reasons, both established. The best model of **mendoza2025<empty citation>** misses 49% of uncontrolled cases, so autonomous action would act on an unreliable signal. And the WHO establishes that the effect of lifestyle modification is approximately 1.3 mmHg systolic (Section 2.2.2), so automated lifestyle advice is both the safest thing to automate and the least consequential.

One caution on personalisation. **liu2025review<empty citation>** report a self-care recommendation system for diabetes integrating clinical guidelines with social, economic, cultural and geographical conditions such as food intake and physical exercise. This is the right principle for a Cameroonian deployment—dietary advice derived from European guidelines is unusable where the foods it names are unavailable—but it is a principle, not a validated method, and adapting guideline content to local context requires clinical authorship, not an algorithm.

3.5.6 Interoperability and the knowledge base

liu2025review<empty citation> note that adherence to healthcare industry regulations is essential for platform compliance, naming the Health Insurance Portability and Accountability Act, the Health Information Technology for Economic and Clinical Health Act and the 21st Century Cures Act, and that the adoption of standardised medical terminology and coding systems such as ICD and SNOMED CT is crucial for supporting accurate medical records and analysis.

The regulatory instruments named are United States law and do not apply here. The terminology standards do. Coding diagnoses in ICD and clinical findings in SNOMED CT costs little at design time and is the difference between a record that can later be reconciled with a hospital information system and one that cannot. The nurses of **mucchi2021<empty citation>** raised precisely this, asking that interfacing with the management information systems in use in the healthcare sector be considered.

Section 3.3.5 noted that the Cameroonian legal instrument governing health data processing must be identified from national sources, none of the documents in this corpus

providing it. That requirement stands and now extends to the knowledge base: the version and date of every clinical rule must be recorded with it, since the framing document anchors the labels in guideline editions that will be superseded. A rule base without version metadata cannot be audited after a guideline revision.

3.5.7 Specification

Table 3.33: Decision-support requirements established in Section 3.5.

Requirement	Basis
Hybrid design: rule engine as floor, model as refinement	**liu2025review<empty citation>**; mandatory baseline, Section 3.4.8
Knowledge base separable from platform, one rule set per pathology	**yazdani2018<empty citation>**; ontology precedent in **liu2025review<empty citation>**
Three-tier classification, not binary	**mihevc2025<empty citation>**; only validated protocol
Yellow tier triggers data collection, not clinician escalation	**mihevc2025<empty citation>**; volume regulator, Section 3.5.4e
Alerts raised on window aggregates, not single readings	Section 3.5.4c; variance reduction raises specificity
Red tier may escalate on a single extreme value	**mihevc2025<empty citation>**; immediate treatment at $\geq$ 160/100, Table 3.1
Non-transmission is an alarmed event	**yazdani2018<empty citation>**; HEARTS retention indicator
Escalation produces a prioritised ordering, not a partition	**yazdani2018<empty citation>**; binary sort insufficient at volume
Model outputs a continuous score to support the ordering	Section 3.5.4
System proposes, physician disposes; no autonomous prescribing	**yazdani2018<empty citation>**; Se = 0.51 in **mendoza2025<empty citation>**
Physician may approve, override, or request consultation	**yazdani2018<empty citation>**
Every physician decision stored as an adjudicated label	Section 3.5.5d
Alert volume reported with every performance figure	Section 3.5.4b; stated in no source
Operating threshold exposed as a clinical configuration parameter	Section 3.5.4e
Time-series diagnostics before model selection	Section 3.5.3a
Irregular sampling handled explicitly, not by naive resampling	Section 3.5.3a
Per-patient robust normalisation by median and MAD	**liu2025review<empty citation>**; personalised threshold gap, Section 2.7
Offline training path separated from online detection path	**liu2025review<empty citation>**; edge–cloud split in **mucchi2021<empty citation>**
At least eleven behaviour change techniques, including the eight most frequent	**zhou2024<empty citation>**: -9.68 vs -2.88 mmHg
Automated advice limited to the permitted column of Table 3.32	**mendoza2025<empty citation>**; WHO lifestyle effect 1.3 mmHg
ICD and SNOMED CT coding	**liu2025review<empty citation>**
Every rule versioned and dated	Section 3.5.6
All events recorded as patient record	**yazdani2018<empty citation>**

3.5.8 What this section establishes

Established. The decision flow of a telemonitoring decision support system is specified completely in one source and validated clinically in another, and the two agree apart from the tier count—where the validated one has three. The rule engine coupled to a learned model is a described technique with named components. The physician-in-the-loop constraint is both designed and independently justified by the measured sensitivity of the best available model. The patient-facing feedback loop has quantified effect sizes and a quantified dose-response in the number of techniques employed.

Established here and not in the sources. The alert volume arithmetic of Section 3.5.4, and its two consequences: that at 2% prevalence an 80% precision target requires 99.5% specificity, and that per-reading alerting at sensor sampling frequency is arithmetically unusable regardless of model quality. These follow from the definitions and from the measured values of **mendoza2025<empty citation>**; they are stated in no source in this corpus.

Not established. No source measures alert fatigue, alert volume, or the time cost of an alert to a clinician. No source evaluates a decision support system of this kind in sub-Saharan Africa. The architecture of **yazdani2018<empty citation>**, which is the most complete specification available, has never been implemented. The closed feedback loop of Section 3.5.5d is described by **liu2025review<empty citation>** but demonstrated by no one. The decision support layer is the least evidenced layer of the entire system, which is notable given that Section 2.3.2 established it is the layer where the clinical effect originates.

Chapter 4

Data Collection, Preprocessing and Predictive Risk Modelling

4.1 Multi-Source Data Collection Strategy and Dataset Construction

Purpose and status of the sources

This section specifies the dataset the present work will construct. It differs in kind from Chapters 2 and 3: it is not a review of what others have done but a statement of what will be done, and it is therefore governed by the constraints those chapters established rather than by new evidence.

Two sources are used. The research framing document is the research framing document submitted to the supervisory committee; it states the intended contributions, hypotheses, limitations, dataset constitution and ethical framework, and it is the authority for everything this section commits to. **efendi2025<empty citation>** supplies a worked example of a vital-signs record schema and of the consent and participant-characterisation apparatus around it; its limitations, established in Sections 3.2.3c and 3.2.3d, mean it is used as a template for structure and not as a model for method.

Table 4.1 lists the constraints carried from the preceding chapters that govern what follows.

4.1.1 The three collection streams

4.1.1a Stream A: clinical consultation records

Content. The framing document specifies real, anonymised clinical data supplied by a partner physician in a hospital setting, structured as a time series per patient. From Section 3.2.1, this stream carries systolic and diastolic pressure, blood glucose, heart rate,

Table 4.1: Constraints on the dataset established in Chapters 2 and 3.

Constraint	Established in
Blood pressure only from a validated automated upper-arm cuff	Sections 2.4.1 and 3.2.4c
Blood glucose only from a glucometer; no optical substitute	Section 3.2.5
Camera photoplethysmography restricted to heart rate	Table 3.9
Measurement timestamp distinct from entry and arrival timestamp	Sections 3.2.1 and 3.3.3b
The three pipelines have distinct missingness mechanisms	Section 3.2.1
Reference value is the seven-day mean of second readings	Section 3.1.4
The label must lie outside the input window	Section 3.4.3c
Split by patient and forward in time	Section 3.4.7
Headline metric is recall on the critical class; report alert volume	Sections 3.4.8 and 3.5.4b
Physician decisions captured as adjudicated labels	Section 3.5.5d
Data residency, consent and national legal instrument stated	Sections 3.3.5 and 3.5.6

laboratory results where available, and the background and context variables: age, sex, body mass index, history, treatments.

Role. This stream is the reference. Measurements are taken by trained staff on hospital equipment under the standardised conditions the WHO specifies and which Section 3.1.1a set out. It supplies the ground truth against which home measurements are assessed, and it supplies the clinical context—diagnosis, treatment regimen, comorbidity—that no home device can provide.

Statistical character. Low frequency, high reliability, missing not at random. A record exists because a consultation occurred; a consultation occurred because the patient attended; and Section 3.1.2a established that attendance in this setting falls far below the seven to eight annual contacts the protocol requires, with 33.3% three-month attendance measured in Cameroonian control centres by **bediang2021<empty citation>**. A patient who deteriorates may consult more, or may disappear entirely. The sampling process is not independent of the outcome, and any model trained on this stream alone is partly modelling attendance behaviour.

4.1.1b Stream B: home self-measurement

Content. Systolic pressure, diastolic pressure, blood glucose and heart rate, with measurement timestamp, acquired by the patient at home on a validated cuff and a glucometer.

Role. This is the principal longitudinal series and the reason the system exists. It is the stream that converts the episodic record of Stream A into something on which a trend can be computed.

Protocol. Following **mihevc2025<empty citation>**, the only clinically validated protocol available (Section 3.1.4): blood pressure twice weekly, two readings morning and evening with a one-minute interval enforced by the device and the second retained; blood glucose weekly, fasting and ninety minutes after a meal; verification profiles on amber classification, being a seven-day blood pressure profile or a one-day six-point glycaemic

profile.

Statistical character. Medium frequency, variable reliability, missing not at random in a second and distinct way. Section 3.2.1 established three specific failure modes: transcription error undetectable downstream; entry timestamp substituted for measurement timestamp; and selective omission, in which a patient omits a reading they find discouraging, which removes precisely the extreme values the model exists to detect. The first two are eliminated by Bluetooth transmission; the third is eliminated by no technical means and must be handled analytically.

4.1.1c Stream C: instrumented sensors

Content. Heart rate, oxygen saturation, body temperature, and, if present, activity and posture.

Role. Covariates only. Section 3.2.5 established that no optical sensor in the reviewed literature delivers clinically validated blood pressure or blood glucose, and Section 3.5.4c established a second and independent objection: at one-hertz sampling, a specificity of 0.98 produces more than eleven thousand false alerts per patient per week. Stream C may characterise the context of a reading—whether the patient was moving, whether the reading was nocturnal—and may enrich the feature space. It may not carry a clinical decision.

Statistical character. High frequency, instrument-limited, missing completely at random through technical failure rather than behaviour.

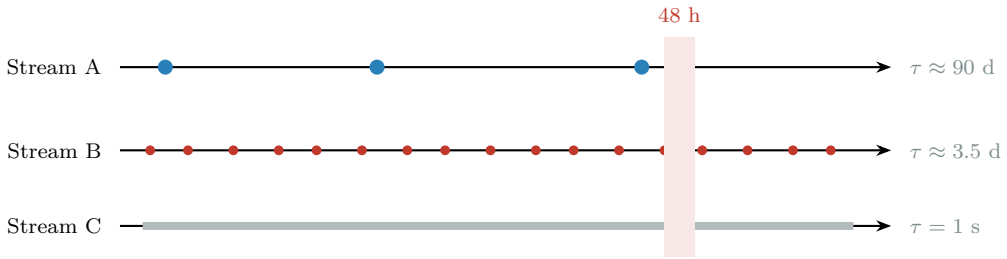


Figure 4.1: The three streams on a common time axis over one quarter, with a 48-hour horizon marked. Stream A contributes no observation inside the horizon; Stream B contributes at most one; Stream C contributes 172 800 but cannot measure the parameters of interest.

4.1.1d What is actually available

The framing document states that the data are real anonymised clinical data from a partner physician in a hospital. This describes Stream A. It does not describe Stream B or Stream C, and the readings differ in everything that follows (Table 4.2).

Scenario D is not a variant of Scenario A and must be distinguished from it. An inpatient observation chart records vital signs once or more per day for the duration

Table 4.2: Collection scenarios and their consequences. The thesis must declare which obtains.

	A: retrospective outpatient	**B: prospective**	**C: mixed**	**D: retrospective inpatient**
Source	existing consultation records	newly recruited cohort, home devices	records plus prospective sub-cohort	existing in-hospital observation charts
Streams	A only	A + B (+ C)	A + B on a subset	A only, at ward density
Interval τ	$\approx$ 90 days	3–4 days	both	$\approx$ 1 day
Ethics burden	secondary use	prospective recruitment, individual consent	both	secondary use
Device cost	none	one cuff and one glucometer per patient	partial	none
Feasible horizon	next consultation	$\approx$ 7 days	depends on subset	24–48 h, marginal
24–48 h horizon	impossible	marginal	marginal	marginal
Contribution C4 has an object	no	yes	yes	no

of a stay, so a cohort of inpatients yields a series two orders of magnitude denser than the same number of outpatients, over a much shorter span. The target of the framing document—at least thirty patients and at least three hundred measurements, which is about ten measurements per patient—is consistent with a stay of about ten days at daily observation, and is not consistent with ninety-day outpatient intervals unless the records span several years per patient. If the available data are inpatient series, the horizon arithmetic of Section 4.1.3 changes materially and favourably, and the thesis should say so rather than describing the data generically as hospital records.

The point to register in all four cases is that the low-cost acquisition contribution presupposes Scenario B or C. A retrospective record review acquires nothing, and that contribution would have no object. Conversely, Scenario B requires recruitment, individual consent, device provision and a collection period of several months, none of which the framing document budgets.

4.1.2 The record schema

4.1.2a Design principles

Four principles, each traceable to a constraint established earlier.

One row per measurement event, not one row per patient visit. Section 3.3.3a established that a document store keyed on the current value cannot answer a window query, and that the rolling aggregates the framing document specifies require exactly that.

The atomic unit is therefore the measurement.

Provenance travels with every value—which stream, which device, who performed it. Section 3.2.1 established that the three streams have different error structures; a model cannot account for that if the schema discards the distinction.

Two timestamps, always. Measurement time and record time. Section 3.3.3b established that offline-first synchronisation decouples them by up to days, and that a model trained on arrival order learns the connectivity pattern rather than the physiology.

No derived field stored alongside its inputs in the same row. Section 3.4.3 established label leakage as the dominant failure mode in this literature. Features are computed at training time from lagged windows; they are not persisted next to the label.

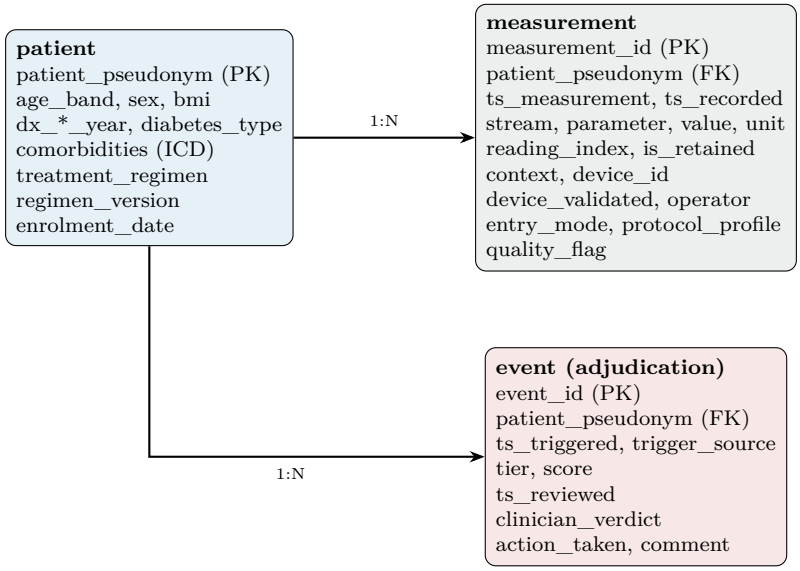


Figure 4.2: The three-table data model. The event table is the only source of labels not derived from the thresholds they are meant to predict (Section 3.5.5d).

4.1.2b Measurement record

Two fields deserve comment. `device_validated` enforces at the schema level the restriction that Sections 2.4.1 and 3.2.4c derived from the WHO: a blood pressure value from an unvalidated device must not be insertable. `quality_flag` encodes the three artefacts that **liu2025review<empty citation>** identify in electronic medical records—duplicates, unit errors, carried-forward values—as a first-class field rather than something discovered at cleaning time. Section 3.1.3 identified the carried-forward value as the most dangerous of the three for a trend model, since it simulates a stability that did not occur.

4.1.2c Patient record

`dx_hypertension_year` is included because **mendoza2025<empty citation>** found years since diagnosis to be the single strongest predictor of uncontrolled hypertension

Table 4.3: Fields of the measurement record.

Field	Type	Notes
measurement_id	UUID	primary key
patient_pseudonym	string	pseudonymous; not derived from identity
ts_measurement	ISO 8601 with offset	when the reading was taken
ts_recorded	ISO 8601 with offset	when it entered the system
stream	enum	consultation | home | sensor
parameter	enum	sbp | dbp | glucose | hr | spo2 | temp
value	float	—
unit	enum	see Table 4.5
reading_index	int	1 or 2 within a measurement pair
is_retained	bool	true for the second reading, per protocol
context	enum	fasting | postprandial_90 | morning | evening | unspecified
device_id	string	links to the device register
device_validated	bool	must be true for sbp, dbp, glucose
operator	enum	patient | nurse | physician
entry_mode	enum	bluetooth | manual | transcribed
protocol_profile	enum	routine | bp_7day | glucose_6point
quality_flag	enum	ok | out_of_range | duplicate | carried_forward | unit_suspect

Table 4.4: Fields of the patient record.

Field	Type	Notes
patient_pseudonym	string	—
age_band	enum	five-year bands, not exact age (Section 4.1.5c)
sex	enum	—
bmi	float	as the framing document specifies
dx_hypertension_year	int	year only
dx_diabetes_year	int	year only
diabetes_type	enum	t1 | t2 | unspecified
comorbidities	array	ICD-coded (Section 3.5.6)
treatment_regimen	array	drug class, dose, start date
regimen_version	int	increments on every change
enrolment_date	date	offset per patient (Section 4.1.5c)
site	string	excluded from the analysis file

by permutation importance, at 11.9% increase in mean squared error against 9.4 for age (Section 3.4.5). It is cheap to record and it is the best-evidenced feature available.

`regimen_version` exists because Section 2.2.2 established therapeutic inertia as the mechanism behind the weakest link of the care cascade. Without a versioned regimen it is impossible to distinguish a patient whose pressure is drifting under an unchanged treatment from one whose treatment was intensified last week—and that distinction is the clinical content of the entire system.

4.1.2d Units and conventions

Table 4.5: Units, plausible ranges and conventions.

Parameter	**Unit**	**Plausible range**	**Note**
SBP, DBP	mmHg	60–260 / 30–160	disjunctive rules, no coverage gaps
Blood glucose	mmol/L stored	1.0–35.0	Cameroonian practice uses g/L; convert at the boundary
Heart rate	bpm	30–220	—
SpO_2	%	50–100	covariate only
Temperature	°C	30.0–45.0	covariate only

Blood glucose is the field most likely to produce silent unit errors, since three conventions are in use:

$$1 \text{ mmol/L} = 18.0182 \text{ mg/dL} = 0.180 \text{ g/L}. \tag{4.1}$$

The thresholds of **mihevc2025<empty citation>** in local units: fasting target below 7 mmol/L, about 1.26 g/L; postprandial target below 10 mmol/L, about 1.80 g/L; hypoglycaemia alert below 4 mmol/L, about 0.72 g/L; diagnostic threshold 11.1 mmol/L, about 2.00 g/L. Values are stored in one unit, converted at the boundary, and `quality_flag` is set to `unit_suspect` on any value whose magnitude is consistent with the wrong convention—a stored value of 1.26 where mmol/L is expected is almost certainly g/L.

4.1.2e Event record: the adjudication table

Section 3.5.5d established that every clinician decision is a clinically adjudicated label, that it is the only route in this corpus to training data not derived from the thresholds it is meant to predict, and that it cannot be reconstructed retrospectively.

`clinician_verdict` is the label that breaks the circularity of Section 3.4.3. The difference `ts_reviewed−ts_triggered` measures the response latency that **bediang2021<empty citation>** found to be the real bottleneck: 10.3 days against a 72-hour protocol, and not significantly faster than no system at all (Section 2.2.3).

Table 4.6: Fields of the event record.

Field	Type	Notes
`event_id`	UUID	—
`patient_pseudonym`	string	—
`ts_triggered`	timestamp	—
`trigger_source`	enum	rule | model | non_transmission | manual
`tier`	enum	green | amber | red
`score`	float	model output, where applicable
`ts_reviewed`	timestamp	null until reviewed
`clinician_verdict`	enum	confirmed | not_clinically_significant | uncertain
`action_taken`	enum	none | advice | dose_change | drug_change | consultation | referral
`comment`	text	free text

One scope note. This table is populated only by a system in operation. Under Scenario A or D of Table 4.2 it is empty, and the secondary endpoint of Section 4.1.4b is unavailable. The table is nonetheless specified here, because the system is designed to be deployed and the cost of adding it later is the loss of every event that occurred before it existed.

4.1.3 Temporal design: the compatibility condition

4.1.3a The condition

A forecast horizon Δ is meaningful only if the sampling interval τ satisfies $\tau \ll \Delta$. If $\tau \geq \Delta$, the forecast target falls between observations and the model is asked to predict a value against which it will never be scored. A reasonable floor is two or three observations within the horizon.

Section 3.4.7b established this in its general form and Figure 3.14 plots it. Table 4.7 applies it to the scenarios of Table 4.2.

Table 4.7: Observations falling inside the forecast horizon, by sampling regime.

Regime	τ	In 24 h	In 48 h	Compatible
Consultation records, quarterly	$\approx$ 90 d	0	0	no
Inpatient chart, daily	1 d	1	2	marginal
Home routine protocol	$\approx$ 3.5 d	0	0 or 1	no
Home protocol, daily	1 d	1	2	marginal
Verification profile, twice daily	0.5 d	2	4	yes
Sensor stream, 1 Hz	1 s	86 400	172 800	yes, but see Section 3.5.4c

Under the routine home protocol this thesis has adopted, a 24-hour horizon contains zero observations and a 48-hour horizon at most one. The contribution as originally stated is not realisable on routine-frequency data.

4.1.3b The resolution, restated

Section 3.4.7b resolved this and the resolution is restated here because it governs the dataset. The forecast horizon is not single-valued: it is matched to the regime the patient is in.

In the **routine regime**, the target is the classification of the next scheduled measurement, or whether the seven-day mean at the next window crosses threshold. This gives a horizon of three to seven days, which is clinically meaningful—the WHO confirmation visit is one to four weeks (Table 3.1) and the verification profile of **mihevc2025<empty citation>** is seven days—and is compatible with the data.

In the **verification regime**, where the protocol prescribes two readings per day, a 24 to 48 hour horizon contains two to four observations and is estimable. This is also the regime in which a clinical decision is pending, so it is the regime in which a short-horizon forecast would change an action rather than merely be recorded.

The anticipatory character of the contribution is preserved by both; what is abandoned is the claim that a single 24 to 48 hour horizon applies uniformly. Should the available data prove to be inpatient series at daily observation (Scenario D), the 24-hour horizon becomes marginally estimable in the routine regime as well, and the restriction relaxes accordingly.

4.1.3c Sample size against model capacity

The framing document targets at least 30 patients and at least 300 measurements, which is approximately ten observations per patient.

For feature engineering with a seven-day rolling window at two observations per week, the window contains two points and the first usable sample arrives after four observations, leaving roughly six usable samples per patient, or 180 in total.

For a recurrent model the position is worse. An LSTM layer has

$$P = 4\Big[u(f+u)+u\Big] \tag{4.2}$$

parameters, where u is the unit count and f the input dimension. For a deliberately small configuration—a single layer, $u = 16$, $f = 10$—this gives $4(16 \times 26 + 16) = 1\,728$ parameters, plus a small output head. Against 180 training sequences that is approximately ten parameters per sample, inverted from the conventional rule of thumb by two orders of magnitude. A two-layer network at $u = 32$ reaches about 13 800 parameters, or seventy-five per sample.

Hypothesis H3—that the gain of the deep model may remain modest—is therefore not a prudent hedge. On the sample size as planned, a negative result for the recurrent model is close to arithmetically guaranteed, and the comparison will measure sample size rather

than architecture. That can still be reported honestly, and Section 3.4.9 establishes that an honest negative result is worth more than an inflated positive one. But it must be framed as such in advance, not discovered at the defence.

A second risk compounds this. With group k-fold at five folds and 30 patients, each fold holds six patients. If critical events are rare—the framing document states they are—and clustered within a few patients, some folds will contain zero positive cases and the fold-level recall will be undefined. The split must therefore be stratified by patient-level event presence, not merely grouped.

4.1.3d Four responses to the sample-size constraint

Increase the sampling frequency for a sub-cohort. A subset measuring twice daily for a defined period yields the density the shorter horizon requires. This costs patient burden and must be weighed against the 10.9% equipment-related dropout that **mihevc2025<empty citation>** observed at a far lighter protocol (Section 2.3.3).

Use a public dataset for the comparative study and the local dataset for external validation. liu2025review<empty citation> catalogue public chronic disease datasets, several containing both blood pressure and glucose. The architecture comparison would be conducted where sample size permits, and the local dataset would answer the question **segun2025<empty citation>** raise—whether an externally trained model transfers to an African population (Section 2.2.4). This converts a sample-size weakness into a contribution, and it addresses a gap that Section 2.7 established is unfilled: no published African validation study exists for the tools in circulation.

Reduce the scope of the architecture comparison. Compare regularised logistic regression, random forest and XGBoost, drop the recurrent model, and state why. This is defensible—**mendoza2025<empty citation>** compared exactly these three—but it abandons a stated contribution.

Report the recurrent model as a capacity-limited negative result, with the parameter-per-sample ratio stated. This is the least satisfying option and the most honest one if none of the others is available.

4.1.4 Labelling

4.1.4a Threshold sources and the convention conflict

The framing document anchors the risk label in the ESH and ADA guideline editions, validated by the partner physician. Section 3.5.6 adds the requirement that every rule carry its version and date, since both will be superseded.

One convention conflict must be resolved explicitly. The WHO surveillance definition is 140/90 mmHg with a disjunctive rule (Section 2.1); the ESH classification uses 130/80 as a

threshold within its grading, which is also the definition **mendoza2025<empty citation>** adopt; and the home target of **mihevc2025<empty citation>** is 135/85 (Section 3.1.4). Three different numbers, for three different purposes. Table 4.8 fixes the allocation, and no section of this thesis departs from it.

Table 4.8: Allocation of blood pressure thresholds by function.

Function	Threshold	Source
Risk label for modelling	130/80 mmHg	ESH classification, as specified in the research framing document
Alerting tiers	135/85 mmHg home target	**mihevc2025<empty citation>**, Table 3.5
Epidemiological comparison	140/90 mmHg	**who2023hypertension<empty citation>**, Chapter 2

4.1.4b The circularity constraint, restated as a rule

If $y_t = h(x_t)$ and the features include x_t, the task is label recovery and not prediction. Section 3.4.3 demonstrated this on **efendi2025<empty citation>**, where three features were discretised using the very ranges that define the class and an accuracy of 0.973 measured nothing. The operational rules that follow:

1. The label is computed on $x_{t+\Delta}$ and the features on $x_{t-k}, \ldots, x_t$, with no overlap.
2. No feature expressing deviation from a threshold may be computed at the label instant.
3. The clinical threshold rule applied to the last observed value is a mandatory baseline. The model must beat it, and the margin by which it does is the actual contribution.
4. Where the adjudication table of Section 4.1.2e has accumulated verdicts, performance is reported against clinician-adjudicated labels as a secondary endpoint. These are not derived from thresholds and are therefore immune to the circularity entirely.

4.1.4c Adjudication and inter-rater agreement

The framing document commits to validation of the labelling by the partner physician. Two additions are needed for that commitment to be checkable.

A **written labelling protocol**, stating the rule, the inputs it sees and the handling of ambiguous cases, so that the labelling is reproducible by someone other than its author.

A **second reader on a sample**. **zhou2024<empty citation>** report inter-rater agreement for their technique coding; **alessa2019<empty citation>** report a Cohen's kappa of 0.85; **dodoo2022<empty citation>** report 86% agreement between two experts. A thesis whose central construct is a risk label, assigned by one person, with no agreement statistic, is weaker than any of its own sources on exactly this point. A second

clinician reading 10 to 20% of the records, with kappa reported, closes that gap at low cost.

4.1.5 Ethical framework

4.1.5a What the framing document commits to

Three commitments are recorded: the data are real and anonymised; direct identifiers are removed before any processing; collection proceeds with the agreement of the partner physician and the institution, for research purposes only. These are necessary and they are not sufficient.

4.1.5b What is missing

Table 4.9: Elements of the ethical framework not yet documented.

Required	Status	Note
Ethics committee reference number	absent	**mihevc2025<empty citation>** cite a trial registration; **bediang2021<empty citation>** cite a national ethics reference
Cameroonian legal instrument governing health data	absent	**mucchi2021<empty citation>** cite their national decree; flagged in Sections 3.3.5 and 3.5.6
Written institutional authorisation	referenced, not evidenced	must be appended
Individual consent, if prospective	absent	**efendi2025<empty citation>** obtained written consent; mandatory under Scenario B or C
Retention period and destruction date	absent	—
Data residency region	absent	the endpoint of **efendi2025<empty citation>** was in south-east Asia
Re-identification risk assessment	absent	see Section 4.1.5c
Secondary-use status, if retrospective	absent	see below

The last line matters. **mendoza2025<empty citation>** could dispense with approval because their source survey is a public, already de-identified dataset. Hospital records are neither, and secondary use of them requires whatever national and institutional process applies—which must be identified rather than assumed.[1]

4.1.5c Re-identification risk in a small cohort

Removing direct identifiers is not anonymisation. The residual risk comes from quasi-identifiers: attributes that are not identifying alone but are jointly unique. In a cohort of

[1] [**To verify**] Obtain and cite the ethics committee reference, the institutional authorisation, and the Cameroonian legal instrument governing processing of personal health data. None of the documents in this corpus supplies the last of these, and it has been outstanding since Section 3.3.5.

thirty patients at a single facility, the combination of exact age, sex, exact consultation dates and site is almost certainly unique for every individual. A clinician at that facility, or anyone with access to its appointment book, could re-identify the entire dataset. The commitment to removing direct identifiers does not address this.

Four measures, each cheap:

Age in five-year bands, not exact years. This costs the model very little—**mendoza2025<empty** used three age groups and found age among the strongest predictors nonetheless.

Date shifting. A random offset δ_i, drawn once per patient and uniformly over ± 180 days, is applied to every timestamp for that patient. Intervals within a patient are preserved exactly, which is all a time-series model needs, while absolute dates are destroyed. The offset table is held separately from the analysis file.

Exclusion of the site identifier from the analysis file. With one or two sites it carries no information and considerable risk.

Suppression or banding of rare values. A single patient with an exceptional diagnosis year or comorbidity is identifiable by that fact alone.

One consequence of date shifting must be recorded, because it conflicts with a requirement established earlier. Per-patient offsets destroy alignment *between* patients, so no calendar-level event—a medicine stock-out, a strike, a seasonal effect—can be identified in the shifted data. Section 2.2.2 showed that retention depends on medicine supply, as in the Indian example in which stock-outs discouraged return visits; medicine availability over the observation period must therefore be documented as a control variable. That requirement is met by holding a separate calendar-event file, unshifted and containing no patient data, and by expressing each event in the analysis file as an elapsed-time flag relative to the patient's own enrolment.

A final consideration: this thesis will be published, and any figure plotting an individual patient's trajectory against real dates reproduces the risk the measures above remove. Figures plot shifted dates or elapsed time from enrolment.

4.1.5d The analysis file

The output of this section is one file, and its contents are stated exactly. The analysis file contains: the patient pseudonym; age band; sex; body mass index; years since each diagnosis; diabetes type; ICD-coded comorbidities; the treatment regimen with its version; and, per measurement, the shifted measurement timestamp, the stream, the parameter, the value and unit, the reading index and retention flag, the context, the operator, the entry mode, the protocol profile and the quality flag.

It contains no name, no record number, no telephone number, no address, no exact date of birth, no exact calendar date, no site identifier and no device serial number. The pseudonym is randomly generated and the mapping to identity is held by the partner physician and is not transferred.

4.1.6 Quality controls at collection

Controls applied at write time are worth more than cleaning applied afterwards, because the information needed to correct an error is available at the point of entry and lost by the time analysis begins. Section 3.3.3a established that server-side schema validation is available and should be used.

Table 4.10: Quality controls applied at write time.

Control	Rule	Flag
Plausibility	value within the range of Table 4.5	`out_of_range`
Internal coherence	SBP > DBP, and SBP − DBP ≥ 10 mmHg	`out_of_range`
Duplicate	same patient, parameter and timestamp to the minute	`duplicate`
Carried forward	three or more consecutive identical values of a continuous parameter	`carried_forward`
Unit suspicion	magnitude consistent with another convention	`unit_suspect`
Device validation	`device_validated` false on sbp, dbp, glucose	reject at write
Temporal coherence	`ts_measurement` > `ts_recorded`	`out_of_range`
Protocol conformity	reading pair present, interval 45–180 s	informational

Two notes. Flagged records are retained, not deleted—a carried-forward sequence is itself informative about data-collection behaviour, and deleting it hides a pattern. And the last control measures whether the protocol was actually followed; **mihevc2025<empty citation>** enforced the one-minute interval in the device, and the degree to which that enforcement held is a reportable quantity.

4.1.7 Specification

4.1.8 Decisions and their consequences

Three decisions govern everything downstream, and two of them remain open at the time of writing.

The collection scenario determines the streams available, the ethics pathway, the device budget, the collection period, and whether the low-cost acquisition contribution has an object at all. Table 4.2 sets out the four cases. This thesis declares its scenario in Section 4.2, where the preprocessing pipeline is specified, because imputation strategy, normalisation scope and window construction all depend on which streams exist and at what interval.

The forecast horizon is settled. Section 3.4.7b established the incompatibility between a uniform 24 to 48 hour horizon and routine-frequency sampling, and Section 4.1.3 adopts the regime-dependent resolution: next scheduled measurement or next seven-day window in the routine regime, 24 to 48 hours in the verification regime. The framing document is amended accordingly, and the anticipatory contribution is preserved in both.

Table 4.11: Dataset requirements established in Section 4.1.

Requirement	**Basis**
One row per measurement event	Section 3.3.3a window queries
Both `ts_measurement` and `ts_recorded` stored	Sections 3.2.1 and 3.3.3b
`stream` recorded on every row	Section 3.2.1 distinct error structures
`device_validated` enforced at write for BP and glucose	Section 2.4.1
`quality_flag` as a first-class field	EMR artefacts in **liu2025review<empty citation>**
Glucose stored in one unit, converted at the boundary	Table 4.5
`regimen_version` incremented on every therapy change	therapeutic inertia, Section 2.2.2
`dx_hypertension_year` recorded	**mendoza2025<empty citation>**: strongest predictor
Adjudication table in place from first deployment	Section 3.5.5d; cannot be reconstructed
Horizon matched to the regime, not single-valued	Sections 3.4.7b and 4.1.3
Label on $x_{t+\Delta}$, features on $x_{t-k..t}$, no overlap	Section 3.4.3c
Clinical threshold rule as mandatory baseline	Section 3.4.8
Threshold allocation of Table 4.8 observed throughout	Section 4.1.4
Guideline version and date recorded with every rule	Section 3.5.6
Written labelling protocol	Section 4.1.4c
Second reader on 10–20% of records, kappa reported	$\kappa = 0.85$ in **alessa2019<empty citation>**; 86% in **dodoo2022<empty citation>**
Stratified group k-fold, forward in time	Sections 3.4.7 and 4.1.3c
Parameter-per-sample ratio stated for every model	Section 4.1.3c
Age in five-year bands	Section 4.1.5c
Per-patient date shift, intervals preserved	Section 4.1.5c
Separate unshifted calendar-event file for supply disruptions	Sections 2.2.2 and 4.1.5c
Site identifier excluded from the analysis file	Section 4.1.5c
Ethics reference, legal instrument, retention and residency stated	Table 4.9
Figures plot shifted dates or elapsed time	Section 4.1.5c

The sample-size strategy remains open. At thirty patients and three hundred measurements, the architecture comparison will measure sample size rather than architecture, for the reasons computed in Section 4.1.3c. The strongest available response is to run the comparison on a public dataset and use the local cohort for external validation, which converts the limitation into a contribution and addresses the absence of African validation studies documented in Section 2.7. Whichever response is adopted, the parameter-per-sample ratio is reported alongside every model.

4.2 Ingestion, Multi-Channel Temporal Aggregation and Missing Data Handling

Purpose, sources and a terminological correction

Section 4.1 specified what is collected and in what form. This section specifies what is done to it between the database and the model. It is the stage at which most of the failures diagnosed in Chapter 3 are either prevented or built in permanently: label leakage, fabricated temporal structure, and the systematic bias that imputation introduces when data are missing for clinical reasons.

Two sources are used. The research framing document commits this work to masking, imputation, per-patient normalisation and robustness to missingness as responses to the irregular and missing measurements arising from intermittent connectivity, and to rolling aggregates, trends, variability and threshold deviations as the feature construction. **liu2025review<empty citation>** describe the data-processing layer of a chronic-disease platform and name specific techniques for cleaning, outlier handling and time-series characterisation; the reservation on their method stands, and the techniques they name are used because each is independently verifiable, not on their authority.

A terminological correction is required before proceeding. There is no spatial dimension in this dataset. It is a single-site, single-cohort study: no patient coordinates, no geographic stratification and no multi-site structure is specified. The operation performed is temporal aggregation across several channels—blood pressure, blood glucose, heart rate—each sampled on its own schedule. This work therefore uses *multi-channel temporal aggregation* throughout and does not describe the operation as spatiotemporal.

4.2.1 The pipeline and its ordering constraint

liu2025review<empty citation> describe the data-processing layer as comprising data cleansing, data conversion, feature engineering, data extraction and data integration, and state that it "utilizes various preprocessing techniques, including data cleaning, format

conversion, and outlier handling, to integrate and conduct preliminary analysis on the collected raw data"†.

The list is correct but unordered, and the order is the part that determines validity. Section 3.4.7 established that every preprocessing step which learns a parameter from data must be fitted on the training fold alone. The pipeline therefore partitions into two phases separated by the split (Figure 4.3).

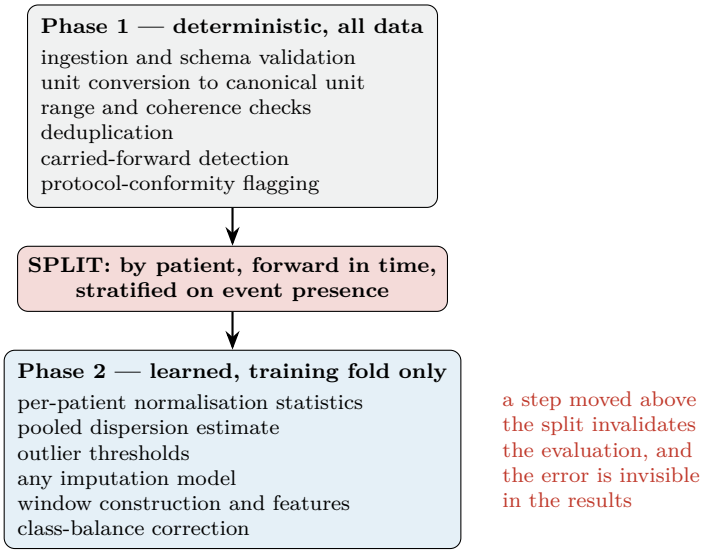


Figure 4.3: The two-phase preprocessing pipeline. Phase 1 operations learn nothing from the distribution and cannot leak; Phase 2 operations estimate a parameter and must therefore be fitted on the training fold alone.

A step placed on the wrong side of the split invalidates the evaluation, and the error is invisible in the results: it manifests only as an optimistic estimate that fails on deployment. Stating this partition explicitly, and implementing it as a pipeline object rather than as a sequence of notebook cells, is the cheapest guarantee of validity available to this work.

4.2.2 Ingestion and deterministic validation

Section 4.1.6 specified the controls; this subsection specifies how they behave.

The governing principle is to flag and not to delete. **liu2025review<empty citation>** report a finding that one cleaning tool was "the only method capable of cleaning adult data and identifying pervasive carry-over and duplicate data in EHRs"†, the errors concerned being "unit errors, incorrect figures, duplicates, and carry-over values"†. Three of those four are the artefacts specified in Table 4.10, and the fourth—incorrect figures—is the transcription error that Bluetooth acquisition eliminates (Section 3.2.1).

The consequence of flagging rather than deleting is that the flags themselves become data. A patient whose series contains three carried-forward sequences is telling the system something about how their measurements are being produced, and a model that never sees that fact cannot account for it.

Table 4.12: Ingestion operations, their phase and their behaviour on violation.

Operation	Learns from data?	Phase	On violation
Schema validation	no	1	reject at write
Unit conversion	no	1	convert; flag if magnitude suspect
Range check	no	1	flag `out_of_range`; retain
SBP > DBP and SBP − DBP ≥ 10 mmHg	no	1	flag; retain
Exact duplicate	no	1	retain one, flag the rest
Three or more identical consecutive values	no	1	flag `carried_forward`; retain
Reading-pair interval 45–180 s	no	1	informational flag
Statistical outlier detection	yes	2	flag for review (Section 4.2.3)

4.2.3 Outliers: artefact or event?

This subsection addresses a danger that none of the sources in this corpus raises, and that is specific to risk-detection systems.

4.2.3a The problem

liu2025review<empty citation> report three approaches—box plots, absolute median differences and extreme value theory—and a use of robust linear regression for outlier detection "allowing for the automatic cleaning of anomalies in longitudinal electronic health records"†.

The phrase *automatic cleaning of anomalies* is exactly the danger. In this system the anomaly is the event of interest. A systolic pressure of 185 mmHg is a statistical outlier in a patient whose median is 138. It is also a red-tier alert under the protocol of Table 3.5. An outlier-removal step applied before labelling deletes the positive class, and the deletion is silent: the model trains on a cleaned series in which hypertensive crises do not occur, achieves excellent apparent performance, and detects nothing in deployment.

The danger is amplified by the class imbalance the framing document anticipates. If critical states are rare, they are also by construction statistically extreme, and any automatic outlier rule targets them preferentially (Section 3.4.6).

4.2.3b The discrimination rule

A systolic reading of 400 mmHg is impossible and is an artefact. A reading of 195 mmHg is possible, and Table 3.1 records that the WHO prescribes same-day treatment initiation above 160/100—so it is both plausible and clinically urgent. The dividing line is physiological plausibility, not statistical rarity.

Table 4.13: Three categories of extreme value and their handling.

Category	Criterion	Action
Physiologically impossible	outside the range of Table 4.5	flag `out_of_range`; exclude from features and from labels
Physiologically possible, extreme	within range, beyond the statistical threshold	retain; this may be the event
Internally contradicted	extreme value contradicted by a reading within hours, with no plausible trajectory	flag for clinician review; never auto-delete

4.2.3c The statistical threshold, and a constraint on it

Where a statistical threshold is used, for flagging and not for deletion, the median absolute deviation is preferred over the standard deviation because it is not itself inflated by the outliers it is meant to detect:

$$\mathrm{MAD}_i = \mathrm{median}_t \left| x_{i,t} - \mathrm{median}_t(x_{i,t}) \right|, \qquad z_{i,t} = \frac{x_{i,t} - \mathrm{median}_t(x_{i,t})}{1.4826\,\mathrm{MAD}_i}. \tag{4.3}$$

The constant 1.4826 makes the denominator consistent with the standard deviation under normality, so z is interpretable on the usual scale.

The constraint is sample size. With approximately ten observations per patient (Section 4.1.3c), a per-patient median and MAD are estimated from ten points and are extremely unstable—a single extreme value among ten shifts the MAD substantially, which is the opposite of the robustness intended.

The response is shrinkage toward the pooled estimate:

$$\widehat{\mathrm{MAD}}_i = \lambda_i\,\mathrm{MAD}_i + (1-\lambda_i)\,\mathrm{MAD}_{\mathrm{pooled}}, \qquad \lambda_i = \frac{n_i}{n_i + n_0}, \tag{4.4}$$

where n_i is the observation count for patient i and n_0 a tuning constant representing the sample size at which the individual estimate is weighted equally with the pooled one. With $n_0 = 20$, a patient with ten observations draws two-thirds of their dispersion estimate from the cohort and one-third from themselves. As the series lengthens the estimate becomes progressively personal—which is precisely the personalisation that Section 3.1.6 identified as the gap in the fixed-threshold protocol of **mihevc2025<empty citation>**, implemented in a way that degrades gracefully when data are scarce. The quantity $\mathrm{MAD}_{\mathrm{pooled}}$ is computed on the training fold only.

Table 4.14: Strategies for handling irregular sampling.

Strategy	Mechanism	Cost
Resample to a regular grid	interpolate onto fixed intervals	fabricates observations
Preserve irregularity, encode the interval	keep raw timestamps; supply Δt as a feature	model must handle variable spacing
Time-aware model	architecture consumes Δt directly	more complex, more parameters

4.2.4 Irregular sampling

4.2.4a Three strategies, and why the obvious one is wrong

Section 3.5.3a established the objection to the first: at the routine protocol frequency, resampling onto a daily grid manufactures five interpolated points for every two real ones, and any trend feature computed on that grid is five-sevenths fiction. The fabricated points are also smoother than reality, since interpolation is a low-pass operation—so variability features computed on a resampled grid are systematically understated, and variability is one of the four feature families the framing document specifies. Resampling would thus bias downward precisely the quantity $\sigma_{\varepsilon,i}$ that Section 3.1.1b identified as an independent risk factor.

The second strategy is the default for this work, with the third available for the recurrent model if the sample size permits, which Section 4.1.3c suggests it will not.

4.2.4b What must be encoded

Preserving irregularity means the interval structure must enter the feature vector explicitly. Per channel c and per sample at time t:

Table 4.15: Interval-structure features, computed per channel.

Quantity	Definition
$\Delta t_{\text{last}}^{(c)}$	time since the most recent observation of channel c
$n_W^{(c)}$	observation count of channel c within the window
$\text{span}_W^{(c)}$	$t_{\max}^{(c)} - t_{\min}^{(c)}$ actually covered within the window
$\overline{\Delta t}_W^{(c)}$	mean inter-observation interval within the window
$\text{CV}(\Delta t)_W^{(c)}$	coefficient of variation of intervals: regularity of adherence

The per-channel subscript is essential. Under the protocol of Section 4.1.1, blood pressure arrives twice weekly and blood glucose weekly, so the channels are never synchronous. A representation that assumes a shared time index silently aligns measurements taken days apart.

4.2.4c Time-series characterisation before modelling

liu2025review<empty citation> prescribe a diagnostic step identifying periodicity, drift and stationarity. This belongs in the experimental chapter, before any model is fitted, with one qualification repeated from Section 3.5.3a: autocorrelation and unit-root tests assume regular sampling. Applying them here requires either resampling, which the preceding subsection rejects, or methods for unevenly spaced series such as the Lomb–Scargle periodogram. This thesis states which was used and does not report an autocorrelation function computed on a resampled series as though it described the raw one.

Table 4.16: What the pre-modelling diagnostic should establish.

Question	If yes	Implication
Is there circadian periodicity in blood pressure?	morning and evening readings differ systematically	`context` must be a feature; do not pool morning and evening
Is there drift under stable therapy?	the median moves across windows	trend features are informative; this is the forecast target
Is the series stationary?	no	differencing required before trend features are interpretable
Does variability change with level?	yes	consider a variance-stabilising transform

4.2.5 Window construction

4.2.5a The sample

A training sample is a pair: an input window and a label computed outside it,

$$\mathcal{S}_{i,t} = \Big(\underbrace{\{x_{i,s} : t - L \le s \le t\}}_{\text{input window of length } L},\ \underbrace{g\big(x_{i,t'}\big)}_{\text{label}} \Big), \qquad t' \in [\, t + \varepsilon,\ t + \Delta \,], \tag{4.5}$$

where $\varepsilon > 0$ is an embargo between the end of the window and the earliest admissible label observation, so that no observation contributes to both. Because sampling is irregular, the label is not located at a fixed instant $t + \Delta$ but is the first qualifying observation inside the label search interval; if no observation falls in that interval the sample has no label and is discarded.

The embargo is the operational form of the rule derived in Section 3.4.3c: the label must lie outside the input window by construction, so that the model cannot recover it (Figure 4.4).

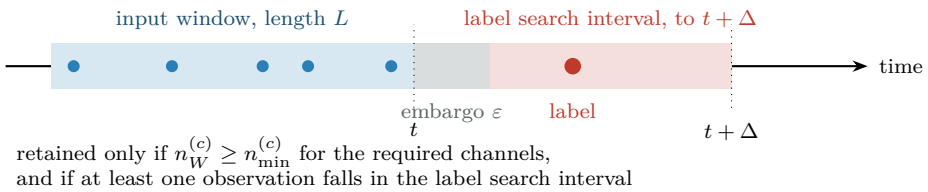


Figure 4.4: Construction of one training sample. Irregular spacing means the label is the first qualifying observation inside the search interval, not a value at a fixed instant.

Table 4.17: Window definitions compared.

	Count-based: last k observations	**Time-based: last L days**
Information content	constant	variable
Time span	variable	constant
Can be empty	no	yes
Clinical meaning	weak: "the last five readings" may span a year	strong: "the seven-day mean" is the protocol unit

4.2.5b Time-based, not count-based

Time-based windows are correct here, because every clinical reference in this work is expressed in time: the seven-day profile of **mihevc2025<empty citation>**, the one-to-four-week confirmation visit of Table 3.1, the monthly follow-up until target of Section 3.1.2a. A count-based window of five readings could span four days in an adherent patient and seven months in a lapsed one, and the resulting feature would mean different things for different patients.

The cost is that a time-based window may contain too few observations to compute a feature. The response is a minimum-observation condition, retained if and only if $n_W^{(c)} \geq n_{\min}^{(c)}$ for the required channels, with $n_{\min} \geq 2$ for any level feature and $n_{\min} \geq 3$ for any slope. Samples failing the condition are discarded, and the number discarded is reported: it is a direct measure of adherence and of how much the dataset shrinks under the protocol.

4.2.5c Window length, and what each length can support

The choice of L is constrained by the minimum-observation condition just stated. At the routine frequency of two retained blood pressure values per week, Table 4.18 gives what each candidate length can actually support.

This corrects a natural assumption. Seven days is the obvious candidate because it is the verification profile of **mihevc2025<empty citation>** and the averaging period that Section 3.1.4 specified for the reference value—but at routine frequency a seven-day window holds two observations and cannot support the slope feature on which the entire anticipatory contribution rests. The working choice is therefore $L = 14$ days for the

Table 4.18: Window length against feature feasibility, at two retained readings per week.

L (days)	Expected n_W	Features supported under $n_{\min}$
7	2	level only; no slope ($n_{\min} = 3$ not met), no variability
14	4	level and slope; variability marginal
28	8	level, slope and variability

routine regime, with $L = 7$ retained only inside the verification regime, where the protocol prescribes two readings per day and a seven-day window holds fourteen. Results are reported for more than one L, as a cheap sensitivity analysis that pre-empts the question.[2]

4.2.5d Overlapping windows and the independent sample count

With approximately ten observations per patient, the temptation to use a small stride and generate overlapping windows is strong, because it multiplies the apparent sample count. Overlapping windows are near-duplicates: two windows sharing three of four observations are not independent samples. Three consequences follow.

Overlapping windows must never straddle the train–test boundary; a window in the test set sharing observations with a training window leaks. The effective sample size for any confidence interval is the number of independent windows, not the number generated. And where overlap is used to augment training data, it is used in training only, with validation and test built from non-overlapping windows.

The arithmetic deserves to be stated, because it sharpens the capacity estimate of Section 4.1.3c. A patient contributing ten observations at two per week spans about thirty-five days. The number of *non-overlapping* windows of length L is approximately span$/L$, which gives two to three per patient at $L = 14$, or sixty to ninety across thirty patients. The figure of 180 usable samples quoted in Section 4.1.3c is the overlapping count, obtained by sliding the window forward one observation at a time; the independent count is between a third and a half of it. Every confidence interval in the experimental chapter is computed on the independent count, and both figures are reported.

Table 4.19: Missingness mechanisms and the validity of imputation.

Mechanism	Definition	Imputation valid?
MCAR	missingness independent of all values	yes
MAR	missingness depends on observed values	yes, under the right model
MNAR	missingness depends on the unobserved value itself	no

4.2.6 Missingness: mechanism, and why imputation is the wrong default

4.2.6a The three mechanisms

4.2.6b Which mechanism governs each stream

Sections 3.2.1 and 4.1.1 established the answers, and they are unfavourable.

Stream A is MNAR. A record exists because the patient attended. Section 3.1.2a established attendance far below the required frequency, and **bediang2021<empty citation>** measured 33.3% three-month attendance in Cameroonian controls. Attendance depends on the patient's state: deterioration may drive attendance up, or may drive the patient out of the system entirely. The probability that a value is observed depends on the value.

Stream B is MNAR by a different route. Section 3.2.1 identified selective omission: a patient may decline to record a reading they find discouraging. The omitted readings are disproportionately the abnormal ones.

Stream C is approximately MCAR. Battery exhaustion, connectivity loss and device failure are, to a first approximation, independent of the physiological value.

4.2.6c The direction of the bias

Suppose observation occurs with probability depending on the value, and that extreme values are less likely to be recorded. Then

$$\mathbb{E}\big[x \mid \text{observed}\big] \neq \mathbb{E}[x], \tag{4.6}$$

and specifically, if omission is concentrated on high values, $\mathbb{E}[x \mid \text{observed}] < \mathbb{E}[x]$.

Mean imputation fills every gap with $\bar{x}_{\text{observed}}$, which is biased toward normality. Forward-fill carries the last observed value, which—since the patient was willing to record it—is also disproportionately a normal one. Either way, the model learns that absence predicts stability. The truth is the reverse: absence predicts deterioration.

[2] [**To verify**] The routine frequency is read here as two retained blood pressure values per week, consistent with the rate used in Section 3.4.7b. The source is ambiguous between two measurement sessions and two sessions each yielding a morning and an evening value, which would double the count and make $L = 7$ viable. This is the same ambiguity flagged in Section 3.1.4 and it must be resolved before L is fixed.

This is not a quantitative inaccuracy that larger samples would fix. It is a sign error in the learned relationship, and it is unrecoverable by any amount of data under the same mechanism.

It is also, independently, a clinical error, and here the two analyses converge. Section 2.2.2 established loss to follow-up at 50% or more as the central mechanism of care-cascade failure. Section 3.5.2 established that non-transmission must itself be an alarmed event. A preprocessing step that imputes silence as stability contradicts the system's own alerting logic.

4.2.6d Why the reported pipelines do not transfer

liu2025review<empty citation> describe a pipeline that "addressed missing data by imputing features, standardized the range of dataset values through scaling, selected features to eliminate redundancy with minimal contribution to the prediction outcome, and augmented features to correct class imbalance"†. That work uses the Pima Indians Diabetes Database—a cross-sectional dataset in which missingness is a recording artefact, not a behavioural signal. Imputation is defensible there and is not defensible here. It is also the same dataset on which **chatrati2022<empty citation>** trained, and whose transferability to an African population is not discussed (Section 2.5).

They also report a differentially private synthesis using autoencoders and generative adversarial networks, "capable of capturing temporal information and feature correlations inherent in the original patient data"†. Synthetic generation is a plausible response to small sample size, but it inherits whatever bias the source data carry: a generator trained on MNAR-missing data reproduces the missingness mechanism rather than correcting it. It is not a solution to this problem and is not presented as one.

4.2.7 Masking: representing absence as information

The framing document commits to masking, and the correct implementation inverts the problem: absence is not a hole to be filled but a channel to be read.

4.2.7a The representation

For each channel c and each sample, the feature vector carries, alongside whatever values are present:

The fourth deserves emphasis. The protocol specifies a known number of measurements per week, so the expected count in a window of length L is known, and the ratio of observed to expected is a direct, interpretable measure of protocol adherence available at every time point without additional instrumentation. Given that Section 2.2.3 established retention as the one robust finding across both Cameroonian studies, a feature that measures it

Table 4.20: Missingness features, per channel.

Feature	Meaning
$m^{(c)} \in \{0, 1\}$	was channel c observed in this window at all
$n_W^{(c)}$	how many times
$\Delta t_{\text{last}}^{(c)}$	how long since the last observation
$r^{(c)} = n_W^{(c)} / n_{\text{expected}}^{(c)}$	adherence ratio against the protocol

continuously is likely to be among the most informative in the set—and it exists only because missingness was treated as signal.

4.2.7b Model-specific handling

Tree ensembles handle this natively, and that is an argument in their favour. XGBoost learns a default branch direction at each split for instances with a missing value, so missingness is used as a splitting signal without any imputation at all. Combined with the explicit mask features above, a gradient-boosted model receives the missingness information twice: implicitly through the default direction and explicitly through the mask channel.

This is a substantive point for the hypotheses of Section 3.4.7. Section 4.1.3c established that the sample size makes a negative recurrent-model result near-certain; this subsection adds a second and independent reason to expect tree ensembles to do well here, namely that the dominant irregularity of the data is one that trees absorb and that recurrent networks must be specially modified to accommodate.

Recurrent models require explicit construction. A sequence model consumes a fixed-width vector at each step and has no native representation of "not measured". Three elements are needed: the mask as an additional input channel; a decay on the carried value, so that the influence of the last observed reading diminishes as the gap widens rather than persisting unchanged; and loss masking, so that no gradient is computed on positions imputed for shape.

A decay formulation implementing the second, proposed here as a design choice and not taken from any source in this corpus:

$$\tilde{x}_t^{(c)} = m_t^{(c)} x_t^{(c)} + \left(1 - m_t^{(c)}\right)\left[\gamma_t^{(c)}\, x_{t'}^{(c)} + \left(1 - \gamma_t^{(c)}\right)\mu_i^{(c)}\right], \qquad \gamma_t^{(c)} = \exp\left(-\kappa_c\, \Delta t_{\text{last}}^{(c)}\right), \quad (4.7)$$

where $x_{t'}^{(c)}$ is the last observed value of the channel, $\mu_i^{(c)}$ the patient's running mean and $\kappa_c > 0$ a learned or fixed decay rate. For a short gap $\gamma \to 1$ and the last value carries; for a long gap $\gamma \to 0$ and the representation falls back to the patient's own baseline rather than asserting that the last reading still holds.

The decay rate should differ by channel, because the channels have different physiological

persistence: a blood pressure reading three days old is more informative about today than a fasting glucose reading three days old. Setting κ_c per channel, and reporting the fitted values, is itself a result.

One caution on this construction. The fallback target $\mu_i^{(c)}$ is the patient's own baseline, so a long gap returns the representation to that patient's normal—which is, in a weaker form, the same assertion of stability that Section 4.2.6c rejects. The mask channel and the gap feature are what prevent the model from reading that fallback as evidence; they must be present, and the decay must never be used without them.

4.2.7c The one case where imputation is acceptable

Sensor-stream gaps, which are approximately MCAR, and only for covariate channels carrying no clinical decision. Even there the mask is retained and the imputation is fitted on training data only.

Blood pressure and blood glucose are never imputed. They carry the decision; absence is reported as absence.

4.2.8 Per-patient normalisation without leakage

The framing document commits to per-patient normalisation. It is the right instinct—Section 3.1.1b established that the clinically meaningful quantities are each patient's own habitual level μ_i and variability $\sigma_{\varepsilon,i}$, not a population constant—but the naive implementation leaks.

4.2.8a The leak

Computing a patient's mean and dispersion over their whole series, then normalising and splitting, gives every training sample access to statistics computed partly from test-period observations. The leak is small in magnitude and systematic in direction, and it inflates the result.

4.2.8b Expanding-window normalisation

The correct construction is causal: at time t, normalise using only observations strictly before t,

$$\hat{\mu}_{i,t}^{(c)} = \operatorname{median}\Big\{x_{i,s}^{(c)} : s < t\Big\}, \qquad \hat{\sigma}_{i,t}^{(c)} = 1.4826 \cdot \operatorname{MAD}\Big\{x_{i,s}^{(c)} : s < t\Big\}, \tag{4.8}$$

$$z_{i,t}^{(c)} = \frac{x_{i,t}^{(c)} - \hat{\mu}_{i,t}^{(c)}}{\max\Big(\hat{\sigma}_{i,t}^{(c)},\ \sigma_{\text{floor}}\Big)}. \tag{4.9}$$

This has four properties that make it the right choice independently of the leakage argument. It is leak-free by construction, since no future observation enters any statistic. It

works for patients never seen in training, which the naive approach does not: under group k-fold a test-fold patient has no training statistics, and expanding-window normalisation simply builds them from that patient's own early observations. It matches deployment, since in production the system has only the past. And it degrades gracefully when combined with the shrinkage of Section 4.2.3c: early in a patient's series $\hat{\sigma}$ is dominated by the pooled estimate, and as observations accumulate it becomes progressively personal.

Two practical requirements follow. A burn-in period is needed—the first n_{burn} observations of each patient cannot be normalised against their own history and must either be dropped or normalised against the pooled statistics; the choice is stated and the number dropped reported. And σ_{floor} must be set to prevent division by a near-zero dispersion, which arises exactly in the carried-forward sequences that Section 4.2.2 flags.

4.2.9 The feature set

The framing document specifies four families—rolling aggregates, trends, variability, threshold deviations—to which this section adds a fifth, missingness, and a sixth, static context.

Table 4.21: The feature set, per channel c and window W of length L.

Family	Feature	Definition
Level	last value	$x_t^{(c)}$
	window median	$\text{median}\{x_s^{(c)} : s \in W\}$
	reference mean	mean of retained second readings over seven days (Section 3.1.4)
Trend	slope	OLS coefficient of x on t over W, in units per day
	net change	$\left(x_t^{(c)} - x_{t-L}^{(c)}\right)/L$
Variability	robust dispersion	$1.4826 \cdot \text{MAD}$ over W
	mean absolute successive difference	$\frac{1}{n-1}\sum_{k=2}^{n}\left\lvert x_k^{(c)} - x_{k-1}^{(c)}\right\rvert$
Deviation	personal z	$z_{i,t}^{(c)}$ per Section 4.2.8b, expanding window
	guideline margin	$x_t^{(c)} - \theta_c$, lagged only
Missingness	mask, count, gap, adherence ratio	Table 4.20
Context	time of day, fasting state, operator, entry mode	from the record schema
Static	age band, sex, BMI, years since diagnosis, regimen version, drug count	Table 4.4

Three notes.

The **guideline-margin** feature is the one most likely to leak, since θ_c is the same quantity used to construct the label (Table 4.8). It is admissible only when computed on a value strictly inside the input window, never at the label instant. If its inclusion proves

difficult to police, the safest course is to omit it: the personal z carries most of the same information without the exposure.

Regimen version and drug count are among the most informative static features available and are frequently omitted. Section 2.2.2 established therapeutic inertia as the mechanism behind the weakest link of the cascade; a feature encoding whether therapy has recently changed distinguishes a drifting patient under an unchanged regimen from one whose treatment was just intensified.

Years since diagnosis is the best-evidenced feature in the entire set. **mendoza2025<empty citati** found it the strongest predictor of uncontrolled hypertension by permutation importance, and it costs one integer to record.

4.2.10 What must be reported

A preprocessing chapter that does not quantify what it did is not checkable.

Table 4.22: Quantities reported for the preprocessing stage.

Quantity	**Why**
Patients enrolled, patients retained, reasons for loss	10.9% equipment-related dropout in **mihevc2025<empty citation>**
Observations per patient: median, interquartile range, minimum, maximum	the sample-size claim of Section 4.1.3c
Inter-observation interval per channel: distribution	justifies the window length
Observed-to-expected ratio per channel	protocol adherence
Missingness per channel, overall and per patient	Section 4.2.6
Flag counts by type	Section 4.2.2
Windows generated, windows discarded for $n_{\min}$, with reason	Section 4.2.5
Independent against overlapping window counts	Section 4.2.5d
Burn-in observations dropped	Section 4.2.8b
Class distribution before and after any balancing	Section 3.4.6

The observed-to-expected ratio is the row to foreground. It is simultaneously a data-quality statistic, a clinical finding about adherence in this cohort, and a model feature. It is the kind of number a jury will ask for and that cannot be reconstructed after the fact.

4.2.11 Specification

4.2.12 Dependency on the collection scenario

This section is written parametrically in L, Δ, ε and $n_{\min}$, and those parameters are fixed by the scenario that Table 4.2 sets out. Table 4.24 states the consequence of each.

The scenario is declared at the head of the experimental chapter, once the data are in hand, and the parameters of Table 4.23 are fixed accordingly at that point. Nothing in the present section changes with the scenario except the numerical values of those four

Table 4.23: Preprocessing requirements established in Section 4.2.

Requirement	**Basis**
Pipeline partitioned into deterministic and learned phases	Sections 3.4.7 and 4.2.1
Pipeline implemented as an object, not as notebook cell order	Section 4.2.1
Flags retained, never silently deleted	Section 4.2.2
Impossible values excluded; extreme values retained	Table 4.13
Statistical outlier detection flags for review, never auto-deletes	Section 4.2.3
MAD shrunk toward a pooled estimate fitted on training only	Section 4.2.3c
No resampling onto a regular grid	Sections 3.5.3a and 4.2.4
Inter-observation intervals encoded per channel	Table 4.15
Periodicity tested with methods valid for uneven sampling	Section 4.2.4
Time-based windows with a minimum-observation condition	Section 4.2.5
$L = 14$ days in the routine regime, $L = 7$ in the verification regime	Table 4.18
Embargo between window end and label search interval	Sections 3.4.3c and 4.2.5
Overlapping windows confined to training; independent count reported	Section 4.2.5d
Blood pressure and blood glucose never imputed	Section 4.2.7
Missingness encoded as mask, count, gap and adherence ratio	Table 4.20
Per-channel decay for recurrent models, used only with mask and gap; κ_c reported	Section 4.2.7b
Imputation permitted only for MCAR covariate channels, mask retained	Section 4.2.7
Expanding-window per-patient normalisation	Section 4.2.8b
Burn-in stated and count reported	Section 4.2.8b
Dispersion floor to guard carried-forward sequences	Section 4.2.8b
Guideline-margin feature lagged only, or omitted	Section 4.2.9
Full reporting table of Table 4.22 included	Section 4.2.10

Table 4.24: Preprocessing parameters by collection scenario.

Scenario	L	Δ	**Consequence**
A: retrospective outpatient	not applicable	next consultation	a 14-day window is empty for almost every sample; the design reduces to visit-to-visit features and the recurrent model is not applicable
B: prospective home	14 days	next measurement or next 7-day window	workable; the discard rate under $n_{\min}$ becomes a headline statistic
C: mixed	two definitions	per stream	the streams require different window definitions and must be joined at the feature level, not the observation level; this is the only configuration in which the three-pipeline design is actually exercised
D: retrospective inpatient	3–7 days	24–48 h	daily observation makes short windows viable; the series is short, so few windows per patient

parameters; the ordering constraint, the treatment of missingness, the normalisation and the reporting requirements hold in all four cases.

4.3 Feature Engineering for Physiological Parameters

Purpose, sources and a notational convention

Section 4.2 specified how raw measurements become windows. This section specifies what is computed from each window and how those quantities are assembled into the vector the model consumes. It is the last stage at which a design error remains invisible: a badly constructed feature set produces a model that trains, scores and reports, and that is wrong in ways no metric reveals.

Two sources are used. The research framing document commits this work to rolling aggregates, trends, variability and threshold deviations, and anchors the labelling in the ESH and ADA guidelines. **liu2025review<empty citation>** describe feature engineering within the data-processing layer and survey feature-selection methods used in the chronic-disease literature; the standing reservation on their method applies, and the methods they name are used because each is independently verifiable.

Two findings from earlier chapters govern everything here. Section 3.4.3c established that a feature computed at the label instant converts prediction into label recovery. Section 4.2.5d established that the number of *independent* windows available is between sixty and ninety, and that figure is the binding constraint of Section 4.3.6.

A notational convention is fixed here and used for the remainder of the thesis. The symbol λ denotes the penalty parameter of a regularisation path, as in the LASSO procedure of **mendoza2025<empty citation>**. The current value of a channel is written x_{last}, and the shrinkage weight introduced in Section 4.2.3c is written w_i.

4.3.1 The information ceiling of a window

This subsection derives a constraint that determines the window length, and it is the reason the rest of the section is shaped as it is.

A window of channel c contains n numbers. Every descriptor computed from it—mean, standard deviation, slope, last value, median, dispersion, successive difference—is a function of those same n numbers. No set of descriptors can carry more than n independent quantities. Computing seven descriptors from a two-point window produces seven columns, not seven pieces of information.

4.3.1a The degeneracy at two observations

The degeneracy is exact and can be shown. For $n = 2$, with observations x_1, x_2 at times t_1, t_2, the mean is $\bar{x} = (x_1 + x_2)/2$ and $x_{\text{last}} = x_2$. The sample standard deviation is

$$s = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2}{n - 1}} = \sqrt{\frac{(x_2 - x_1)^2}{2}} = \frac{|x_2 - x_1|}{\sqrt{2}}, \tag{4.10}$$

while the mean absolute successive difference is $\text{MASD} = |x_2 - x_1|$, so that

$$\boxed{s = \frac{\text{MASD}}{\sqrt{2}}} \tag{4.11}$$

exactly. The two are proportional with correlation one, by construction and not by coincidence. The ordinary least squares slope is

$$\beta = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\pm\,\text{MASD}}{\Delta t}, \tag{4.12}$$

which adds only the sign and a rescaling by the interval. And the median of two points is their mean.

With two observations, the four descriptors the framing document specifies—level, dispersion, slope, last value—are not four quantities. They are two: a level and a signed difference. Supplying all four provides two pieces of information in four columns, which inflates the dimension without inflating the content, and Section 4.3.6 shows that dimension is the scarce resource here (Figure 4.5).

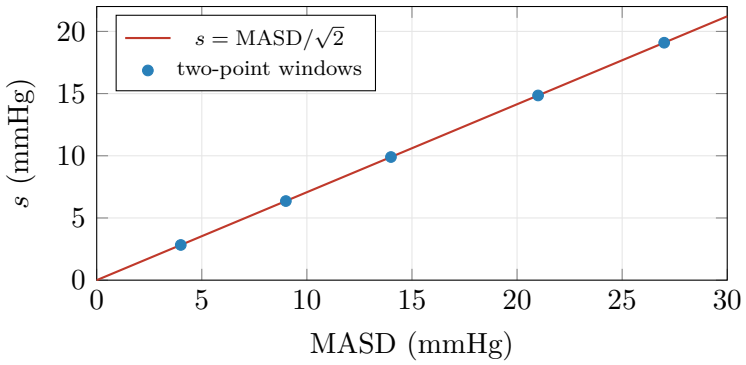


Figure 4.5: At $n = 2$ every two-point window lies exactly on the line $s = \text{MASD}/\sqrt{2}$. The two descriptors are one quantity in two columns.

4.3.1b The consequence for window length

The rule that follows resolves a parameter that would otherwise be set arbitrarily: L must satisfy $n_W^{(c)}(L) \geq k_c$, where k_c is the number of descriptors intended for channel c.

Under the routine protocol of two blood pressure values and one glucose value per week, Table 4.25 gives the ceiling.

Table 4.25: Observations per window and the ceiling on independent descriptors.

L	n_W, **blood pressure**	n_W, **glucose**	**Independent descriptors available, blood pressure**
7 days	2	1	2
14 days	4	2	4
28 days	8	4	up to 8

To support the four descriptors the framing document specifies, the window must be at least fourteen days—and glucose reaches four observations only at twenty-eight. This confirms from an independent direction the choice of $L = 14$ made in Table 4.18, which was derived there from the minimum-observation condition rather than from the information ceiling. The seven-day window inherited from the verification profile of **mihevc2025<empty citation>** is the right unit for *alerting*, where it averages readings to reduce variance, and is too short for *feature construction*, where it supports two descriptors. These are different operations and use different windows; stating that explicitly is better than silently using one length for both.

4.3.2 Statistical descriptors

4.3.2a Definitions

For channel c, patient i, window W of length L ending at t and containing n observations $\{(t_k, x_k)\}$:

Level.

$$\mu^{(c)} = \frac{1}{n}\sum_{k=1}^{n} x_k, \qquad \tilde{x}^{(c)} = \operatorname{median}_k x_k, \qquad x^{(c)}_{\text{last}} = x_n. \tag{4.13}$$

Trend. The ordinary least squares slope on time, which is correct for irregular spacing and requires no resampling:

$$\beta^{(c)} = \frac{\sum_k (t_k - \bar{t})(x_k - \bar{x})}{\sum_k (t_k - \bar{t})^2}, \tag{4.14}$$

in units of the channel per day. This estimator is undefined when all t_k coincide and unstable when they nearly do, which occurs when a patient records several readings in one sitting and none in the rest of the window. Its sampling variance is proportional to $\sigma^2 / \sum_k (t_k - \bar{t})^2$, so it explodes precisely when the window is temporally clustered. A guard is therefore required: β is computed only when $\operatorname{span}^{(c)}_W \geq L/2$, and the mask is set otherwise.

Variability.

$$s^{(c)} = \sqrt{\frac{1}{n-1}\sum_{k}(x_k - \bar{x})^2}, \qquad \mathrm{MAD}^{(c)} = \mathrm{median}_k\left|x_k - \tilde{x}\right|, \qquad \mathrm{MASD}^{(c)} = \frac{1}{n-1}\sum_{k=2}^{n}\left|x_k - x_{k-1}\right|. \tag{4.15}$$

The third measures variation between consecutive readings rather than around the mean, and it is the only one of the three that uses the ordering. For $n > 2$ it is not a rescaling of s: a series that oscillates has high MASD and moderate s, while one that drifts monotonically has low MASD relative to s. At $n = 2$ the distinction vanishes, which is a further reason to prefer $L \geq 14$ days.

Personal deviation. The expanding-window standardised deviation $z_{i,t}^{(c)}$ of Section 4.2.8b.

4.3.2b Small-sample behaviour

Each estimator degrades differently as n falls, and the differences matter at $n = 2$ to 4.

Table 4.26: Behaviour of each descriptor at small n.

Descriptor	**Minimum** n	**Behaviour at small** n
μ, x_{last}	1	stable
$\tilde{x}$	1	equals μ at $n = 2$; no robustness gained
s	2	very high variance, $\mathrm{Var}(s^2) \propto 1/(n-1)$
MAD	3	degenerate at $n = 2$, where it equals half the range
β	2	variance $\propto \sigma^2/\sum_k (t_k - \bar{t})^2$; explodes on a temporally clustered window
MASD	2	equals $\sqrt{2}\,s$ at $n = 2$

The robust descriptors—median and MAD—are chosen for their resistance to outliers, but that resistance requires observations to be resistant with. At $n = 2$ the median is the mean and the MAD is half the range, so nothing is gained and a column is spent. Robust statistics belong in this design at the level of the patient's history, where n is larger (Section 4.2.8b), and not at the level of a short window.

4.3.2c Why learned representations are not used

The account that **liu2025review<empty citation>** give of physiological feature extraction is largely about deep unsupervised representation learning. They report a convolutional model applied "using an unsupervised feature learning method to learn meaningful feature representations from unlabeled physiological signals"†, with the aim of eliminating "the limitations of current feature extraction and feature selection in physiological signal anomaly detection"†, the representations then feeding an anomaly detector.

The architectural goal is the same as this work's. The method does not transfer. Learned representations require sample sizes orders of magnitude beyond those available here, and convolutional feature learning is designed for densely sampled signals—electrocardiograms, photoplethysmographic waveforms—not for two blood pressure readings per week. That direction is recorded here as where the field is going, and hand-constructed descriptors are adopted as the correct choice at this sample size. The reasoned rejection is itself part of the methodological contribution, since it is a decision taken rather than an option overlooked.

4.3.3 Clinical threshold features

4.3.3a The family

For channel c with guideline threshold θ_c, computed on window observations only:

Table 4.27: Threshold-derived features.

Feature	Definition	Interpretation
Exceedance indicator	$\mathbb{1}[x_{\text{last}} > \theta_c]$	binary, current
Signed margin	$x_{\text{last}} - \theta_c$	how far, in clinical units
Normalised margin	$(x_{\text{last}} - \theta_c)/\theta_c$	comparable across channels
Time-above fraction	$\frac{1}{n}\sum_k \mathbb{1}[x_k > \theta_c]$	proportion of the window out of range
Exceedance burden	$\sum_{k=2}^{n} \max\big(0,\ \bar{x}_{k-1,k} - \theta_c\big)(t_k - t_{k-1})$	magnitude × duration, in mmHg·days
Grade band	ordinal index of the guideline band containing x_{last}	preserves the graded structure

The fifth deserves comment. Exceedance burden integrates how far above threshold the patient was and for how long, approximating the integral of $\max(0, x - \theta)$ by evaluating the excess at the midpoint of each inter-observation interval. It is the natural quantity when the clinical concern is cumulative exposure rather than instantaneous level—and Section 2.1 established exactly that framing, since the WHO burden figures are expressed in disability-adjusted life years, which are themselves a magnitude integrated over time. Two qualifications. The midpoint approximation differs from the true integral on any segment that crosses θ, and the error is small only when the inter-observation interval is short, which at routine frequency it is not. And no source in this corpus defines this quantity: it is presented as an original construction to be evaluated, not as an established index.

The sixth matters because guidelines are graded, not binary. Collapsing a graded classification to a single exceedance discards the distinction between grade 1 and grade 3, which is precisely the distinction that determines urgency—and **kingue2013<empty citation>**, reported in Section 2.2.3, found the effect of their intervention confined to stage III

hypertension and absent for stages I and II. A feature set that cannot represent the grade cannot represent where the effect lives.

4.3.3b The leakage boundary, precisely

Section 3.4.3c established the rule. Applied here it requires a distinction that is easily blurred.

Admissible: the exceedance of the threshold computed on observations inside the input window. If the label is a threshold crossing at $t + \Delta$ and the feature is the exceedance at t, the two are correlated through the autocorrelation of the physiology, which is a real relationship and not a leak.

Inadmissible: the exceedance computed at or after the label instant, or on any observation inside the embargo of Section 4.2.5.

The distinction is enforced by construction: features are computed from the window, labels from a disjoint interval, and no code path reads the label interval during feature computation. This is enforced by the pipeline object of Section 4.2.1 and not by discipline.

4.3.3c The threshold table as configuration

Section 3.5.6 required every rule to carry its version and date. The same applies here: thresholds are a configuration artefact, not constants embedded in feature code. The table carries, per row, the channel, the context (home or office), the band label, the lower and upper bounds, the unit, the source, the source version and the date from which it is effective.

Three reasons. The guidelines will be superseded and the features must be recomputable under the new table. Three conventions are in play—the WHO surveillance threshold, the ESH grading and the home target of **mihevc2025<empty citation>**—and Table 4.8 fixed which is used for which function, so the table makes that allocation machine-readable. And Section 4.3.11 records that two of the three sources are not yet in hand, so the table will be populated later and the code must not assume otherwise.

Table 4.28 records what the corpus does support, usable as verified placeholders until the guidelines are obtained.

Table 4.28: Threshold values verifiable from the present corpus.

Source	Channel	Values
who2023hypertension<empty citation>	office BP	surveillance ≥ 140/90; same-day treatment ≥ 160/100; confirm 140–159/90–99 at one to four weeks
mihevc2025<empty citation>	home BP	green 110–134/60–84; amber 135–174/85–104 and <110/<60; red ≥ 175/≥ 105
mihevc2025<empty citation>	glucose	green 4–7 pre-meal, 7–10 post-meal; amber 7.1–10 and 10.1–14; red <4, >10 pre-meal, >14 post-meal (mmol/L)

4.3.4 Cross-channel and derived features

The multi-parametric fusion the framing document proposes is partly this: quantities computed across channels rather than within one.

Table 4.29: Cross-channel features and their justification.

Feature	Definition	Justification
Pulse pressure	SBP − DBP	marker of arterial stiffness; Section 2.1.3 established stiffness as the physiological link between diabetes and hypertension
Mean arterial pressure	$\text{DBP} + \frac{1}{3}(\text{SBP} - \text{DBP})$	the perfusion-relevant summary
Glycaemic excursion	post-prandial minus fasting, same day	distinguishes an elevated baseline from poor post-prandial control
Rate–pressure product	HR × SBP	combines the two channels the system reliably acquires

A technical justification is required for including exact functions of existing features. A decision tree partitions by axis-aligned splits on single variables. To represent the condition $\text{SBP} - \text{DBP} > 60$ it must approximate a diagonal boundary with a staircase of splits, which costs depth and sample efficiency—and Table 3.21 recorded that axis-aligned splits are precisely the inductive bias of the tree families. Supplying pulse pressure explicitly converts that staircase into a single split. For a linear model the feature adds nothing, being a linear combination of existing columns; for a recurrent model it may ease optimisation without adding information. The right answer therefore depends on the model family, and this thesis states so rather than presenting the derived features as universally beneficial.

One asynchrony caution. The glycaemic excursion requires a fasting and a post-prandial reading on the same day. Under the routine protocol both occur weekly, so the feature is computable once per week at best and its mask will be set frequently. Pulse pressure requires systolic and diastolic values, which always arrive together from the same cuff reading, so it is available whenever blood pressure is. Cross-channel features inherit the worst availability of their inputs, and their masks are computed accordingly.

4.3.5 Assembling the vector

The feature vector for sample (i, t) concatenates six blocks,

$$\mathbf{v}_{i,t} = \Big[\underbrace{\mathbf{d}^{(c)}}_{\text{descriptors}} \,\Big\|\, \underbrace{\boldsymbol{\theta}^{(c)}}_{\text{thresholds}} \,\Big\|\, \underbrace{\mathbf{x}^{\text{cross}}}_{\text{cross-channel}} \,\Big\|\, \underbrace{\mathbf{m}^{(c)}}_{\text{missingness}} \,\Big\|\, \underbrace{\mathbf{k}}_{\text{context}} \,\Big\|\, \underbrace{\mathbf{s}_i}_{\text{static}} \Big], \tag{4.16}$$

with every block carrying its own availability mask per Table 4.20. Two assembly rules.

Channel-wise, not timestamp-wise. Each channel is windowed on its own schedule and its descriptors computed independently (Section 4.2.4b). Concatenation happens at

the feature level, after each channel has been reduced to a fixed-width summary, and never at the observation level, which would require aligning measurements taken days apart.

Every feature carries its mask. A missing descriptor is represented as a sentinel plus an explicit indicator, never as zero. Zero is a valid blood pressure slope.

4.3.6 The dimensionality budget

This subsection contains the operative constraint of the section, and no source in this corpus states it.

4.3.6a The count

A feature set constructed without restraint reaches the count of Table 4.30.

Table 4.30: Feature count of an unrestrained set.

Block	**Count**
4 clinical channels $\times$ 7 descriptors	28
4 channels $\times$ 4 threshold features	16
Cross-channel	4
4 channels $\times$ 4 missingness features	16
Window-dependent subtotal	64
Context	4
Static	9
One window length	77
Three window lengths (7, 14, 28 days): $64 \times 3 + 13$	205

Only the window-dependent blocks multiply with the number of window lengths; context and static features do not, which is why three lengths give 205 rather than 3×77.

4.3.6b Why $p \approx n$ is fatal, and what n actually is

When the feature count approaches the sample count, any sufficiently flexible model interpolates the training data exactly. Training error goes to zero, cross-validation variance becomes enormous, and the selected model is determined by which noise directions happened to align with the labels in a particular fold. The reported performance then measures the random seed.

The working constraint for tabular problems with modest signal is $n/p \geq 10$, preferably $n/p \geq 20$.

The value of n must be the *independent* window count, not the generated one. Section 4.2.5d established that sliding a window forward one observation at a time yields about 180 windows from the planned cohort, but that non-overlapping windows number only sixty to ninety. Overlapping windows inflate the row count without inflating the information, so

they cannot relax a budget whose purpose is to control overfitting. The budget is therefore computed at $n = 60$ to 90, and Table 4.31 gives the admissible dimension.

Table 4.31: Admissible feature count under the two sample-size conventions.

Convention	n	p **at** $n/p = 10$	p **at** $n/p = 20$
Generated windows (overlapping)	180	18	9
Independent windows, upper	90	9	4
Independent windows, lower	60	6	3

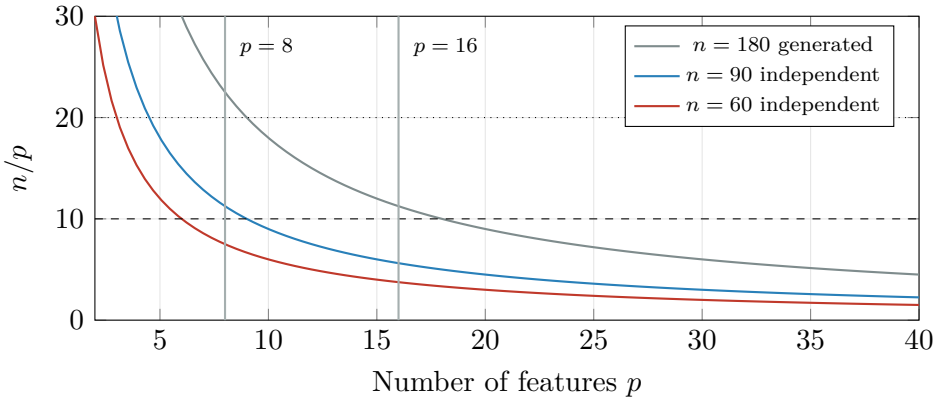


Figure 4.6: The dimensionality budget. Dashed and dotted lines mark $n/p = 10$ and 20. A sixteen-feature set clears the threshold only under the generated-window count; against independent windows it does not.

The unrestrained set of 205 features exceeds even the most permissive bound by a factor of more than eleven. The comparison with the corpus is instructive: **mendoza2025<empty citation>** had $n = 1\,308$ and retained about eleven predictors after LASSO, giving $n/p \approx 119$—and their best model still reached only an area under the curve of 0.75 and a sensitivity of 0.51 (Table 3.23). A study operating near $n/p \approx 5$ cannot expect to do better.

4.3.6c Two candidate sets

Table 4.32 gives two sets, with $L = 14$ days so that four descriptors are supported.

The primary set of eight features is the one reported as the main result. It clears $n/p = 10$ at the upper independent count and falls marginally short at the lower, which is stated rather than concealed. The extended set of sixteen is reported as a secondary analysis, with its n/p ratio quoted alongside every figure, so that a reader can discount it appropriately.

Four structural economies make the primary budget work, and each is a decision rather than an omission.

One window length, not three. A three-length sweep triples the window-dependent dimension for information that is largely redundant, since a seven-day and a fourteen-day

Table 4.32: Primary and extended feature sets.

Block	**Primary set**	p	**Extended set**	p
SBP	x_{last}, β, signed margin	3	x_{last}, β, s, signed margin	4
Glucose	x_{last}, signed margin	2	x_{last}, β, s, signed margin	4
DBP	—	0	x_{last}, β	2
Cross-channel	—	0	pulse pressure	1
Missingness	adherence ratio	1	adherence ratio, maximum gap	2
Static	years since diagnosis, regimen version	2	age band, years since diagnosis, regimen version	3
Total		**8**		**16**
n/p **at** $n = 90$		**11.3**		**5.6**
n/p **at** $n = 60$		**7.5**		**3.8**

mean on the same series correlate strongly. Sensitivity to L is reported by re-running the pipeline, not by concatenating lengths into one vector.

Asymmetric treatment of channels. Systolic pressure and glucose carry the clinical decision and receive the most columns; diastolic pressure enters only the extended set; oxygen saturation and temperature are excluded entirely, consistent with Table 3.7 establishing them as context rather than decision variables.

Dispersion dropped from the primary set. At $L = 14$ and four observations, s is estimated from four points and Table 4.26 shows its variance is then very high. It is retained in the extended set so that its contribution can be measured, not assumed.

Missingness aggregated. Per-channel masks are retained in the data but enter the model as one or two summary features rather than sixteen.

4.3.6d Reporting

This thesis reports n, p and n/p for the final feature set alongside every performance figure, and states which sample-size convention n refers to. A model result without a dimensionality statement is not interpretable at this sample size, and stating it pre-empts the first methodological question a jury will ask.

4.3.7 Feature selection without leakage

4.3.7a What the corpus reports

liu2025review<empty citation> survey the methods in use: correlation feature selection reported as superior in accuracy and execution time for chronic disease prediction, best-first search among wrapper methods, a hybrid of enhanced k-means clustering with filter and wrapper stages, recursive feature elimination with gradient boosting and with

boosted decision trees, a genetic algorithm, and one coronary heart disease model retaining sixty-five variables.

mendoza2025<empty citation> provide the best-documented procedure: LASSO regression with $\alpha = 1$ and a binomial family, the optimal penalty chosen by ten-fold cross-validation at $\lambda = 0.0056528$, and variables with zero coefficients excluded. Of twelve candidates, engagement in physical activity was eliminated.

4.3.7b The leak these methods invite

Feature selection is a learned operation and must sit inside the cross-validation loop. Selecting on the full dataset and then cross-validating the selected set gives every fold's test data a say in which features exist, and the resulting estimate is optimistic. The bias grows as p/n grows, which is exactly the regime of Section 4.3.6.

The correct construction is nested cross-validation: the outer loop estimates performance; the inner loop, on the outer training fold only, performs selection and tuning; the selected set is applied to the outer test fold without modification. The selected set may differ between outer folds, and reporting how often each feature is selected across folds is more informative than reporting one set—it measures stability, which at this n/p is the quantity in doubt.

A note on **mendoza2025<empty citation>**. Their methods describe LASSO with ten-fold cross-validation, and separately an 80/20 train–test split for the tree models. The paper does not state whether selection preceded or followed the split. If it preceded, the selected set has seen the test partition. At $n = 1\,308$ with twelve candidates the resulting bias is small; the point is the reporting gap and not a demonstrated error, and this thesis states its own ordering explicitly rather than leaving the same ambiguity.

4.3.7c The procedure adopted

The feature set is fixed a priori on clinical and structural grounds, as the primary set of Table 4.32. Performance on that fixed set is the primary result. Nested selection is run as a secondary analysis, to test whether a data-driven set improves on it, and the selection frequency of each feature across outer folds is reported.

Fixing the set in advance is the stronger design at this sample size. It removes a source of variance, it makes the result reproducible, and it means the primary claim does not depend on a selection procedure whose stability cannot be established on sixty to ninety independent samples.

4.3.8 Scaling and model-specific preparation

Scaling is fitted on the training fold only (Section 4.2.1). For the recurrent model, standardisation to zero mean and unit variance per feature is conventional; for the

Table 4.33: Preparation required by model family.

Model	Scaling required	Monotone transform tolerated	Missing handled natively
Logistic regression	yes	no, it affects coefficients	no
Random forest	no	yes, splits are rank-based	partially
XGBoost	no	yes	yes, learned default direction
LSTM	yes	no	no, requires Section 4.2.7b

penalised linear baseline it is required for the regularisation path to be meaningful, since the LASSO penalises coefficients on the scale of their features.

One asymmetry is recorded as a finding rather than a detail. XGBoost requires no scaling, tolerates monotone transforms and handles missing values natively through a learned default split direction. The recurrent model requires all three to be solved by hand. Given that Section 4.2.6 established missingness as the dominant irregularity of this dataset, and Section 4.1.3c established that the sample size disfavours the recurrent model, this is a third independent reason to expect the tree ensemble to prevail in the architecture comparison. The three reasons—sample size, native missingness handling, and the absence of preparation requirements—are stated together here rather than discovered in the results.

4.3.9 The baselines this feature set must beat

Section 3.4.8 required the clinical threshold rule as a mandatory baseline. Feature engineering introduces a second, and it is the harder one.

Persistence. Predict that the classification at $t + \Delta$ equals the classification at t. No features, no model, no training.

Physiological series are strongly autocorrelated over the horizons considered here, so persistence is a genuinely strong baseline—and because the label is a threshold function of the same channel that supplies x_{last}, the elaborate feature set is competing against simply carrying the current value forward. Any trend, variability or threshold feature earns its place only by the margin it adds over that.

The contribution of this chapter is the gap between rungs 2 and 5. If that gap is small, the honest conclusion is that at this sampling density and sample size engineered features add little over the guideline rule—which is a result, is publishable, and is considerably more useful to a clinician than an inflated accuracy figure. Section 3.4.9 established that the literature is full of the latter.

Table 4.34: The baseline ladder, in order.

Rung	Description
0	majority class
1	persistence: the classification at t carried to $t + \Delta$
2	clinical rule: guideline thresholds applied to x_{last}
3	penalised logistic regression on level features only
4	penalised logistic regression on the primary set
5	random forest and XGBoost on the primary set
6	LSTM on the raw masked sequence

4.3.10 Specification

Table 4.35: Feature-engineering requirements established in Section 4.3.

Requirement	Basis
$L \geq 14$ days for feature construction; 7 days retained for alerting	Section 4.3.1
Descriptors per channel not exceeding observations per window	Section 4.3.1
s and MASD never both included when $n = 2$	$s = \text{MASD}/\sqrt{2}$ exactly
Median and MAD at patient level, not window level	Table 4.26
OLS slope guarded by $\text{span}_W \geq L/2$	Section 4.3.2
Threshold features on window observations only; embargo enforced in code	Section 4.3.3b
Threshold table as versioned configuration, not constants	Section 4.3.3c
Grade band retained as ordinal, not collapsed to binary	effect confined to stage III in **kingue2013<empty citation>**
Pulse pressure supplied explicitly for tree models	Section 4.3.4
Cross-channel features inherit the worst input mask	Section 4.3.4
Budget computed on independent windows, not generated ones	Sections 4.2.5d and 4.3.6b
Primary set $p = 8$; extended set $p = 16$ reported separately	Table 4.32
One window length per run; sensitivity by re-running	Section 4.3.6c
n, p, n/p and the convention for n reported with every result	Section 4.3.6
Feature set fixed a priori as the primary analysis	Section 4.3.7
Any selection nested inside cross-validation	Section 4.3.7b
Selection frequency per feature reported across folds	Section 4.3.7b
Scaling fitted on the training fold only	Section 4.2.1
Persistence baseline reported	Section 4.3.9
Contribution stated as the margin over the clinical rule	Table 4.34

4.3.11 Limits of this specification

Three limitations are recorded here rather than left to be discovered.

The guideline sources are not yet in hand. The framing document names the ESH and ADA guidelines as the labelling sources, and neither is in the corpus assembled for this thesis. The threshold table of Section 4.3.3c cannot be populated from them, and the values of Table 4.28 are operational thresholds from **mihevc2025<empty citation>**, adapted for an older population, standing as placeholders.[3]

[3] **[To verify]** Obtain the ESH guideline edition and the ADA Standards of Care edition named in the

No descriptor in this set is validated against a clinical outcome by any source in this corpus. The mean, slope and variability features are standard constructions, but no document assembled for this thesis demonstrates that any of them predicts anything in hypertension or diabetes. The only feature in the entire set with published predictive evidence is years since diagnosis, from **mendoza2025<empty citation>**. Blood pressure variability as a risk marker has a literature in cardiology; it is not represented here, and until it is, the variability features are presented as candidates proposed on physiological reasoning and not as established markers.[4]

The exceedance burden is an original construction. It is defined in Table 4.27 by analogy with the magnitude-integrated-over-time logic of the disability-adjusted life year, and it is evaluated here for the first time. It is reported as a candidate feature whose contribution is measured, not as an index with prior standing.

4.4 Learning Problem Formulation, Model Training and Risk Score Derivation

Purpose, sources and working assumptions

Sections 4.1 to 4.3 produced a feature matrix. This section specifies what is predicted from it, how the models are fitted and compared, and how a model output becomes the score that drives the three-tier alert of Section 3.5.2. It closes the chain from measurement to clinical action.

Three sources are used. The research framing document states the contributions and hypotheses and commits to class weighting, oversampling, a recall-centred metric and group k-fold by patient. **mendoza2025<empty citation>** supply the only fully documented gradient-boosting implementation in the corpus with reported discrimination and confidence intervals. **efendi2025<empty citation>** supply a second implementation whose configuration and results were shown in Section 3.4.3 to be unusable, and which is retained here as a worked example of what to avoid.

This section is written under the assumptions set out in Section 4.4.9, which also states what would have to be revised if any of them is overturned. Where an assumption bears directly on a quantity, it is marked at that point.

research framing document, populate the threshold table from them with version and date, and recompute every threshold-derived feature. Until then no grade-band feature can be constructed, since the bands are not known.

[4] [**To verify**] Obtain at least one primary source on blood pressure variability as an independent cardiovascular risk marker, or state in the discussion that the variability features rest on physiological reasoning alone.

4.4.1 Formalising the learning problem

4.4.1a Three candidate formulations

For patient i with window ending at t and feature vector $\mathbf{v}_{i,t}$ constructed per Section 4.3:

Formulation 1, forecast the parameter. Regression on the continuous value,

$$\hat{x}^{(c)}_{i,t+\Delta} = f\big(\mathbf{v}_{i,t}\big), \qquad \mathcal{L} = \big(\hat{x} - x\big)^2, \tag{4.17}$$

with the risk class obtained by applying the clinical rule to the forecast. The target is continuous, so every sample is informative and the class-imbalance problem of Section 3.4.6 does not arise at training time. Against this, squared error weights a 5 mmHg miss at 120 the same as at 175, where it crosses a threshold.

Formulation 2, classify the future state. Direct classification onto the four risk levels. The target is the clinical decision itself, but the imbalance is severe and all information about how far a patient lies from a boundary is discarded.

Formulation 3, forecast the margin. Regression on the signed distance to a reference threshold,

$$\hat{m}^{(c)}_{i,t+\Delta} = \hat{x}^{(c)}_{i,t+\Delta} - \theta^{\text{ref}}_c, \tag{4.18}$$

after which the tier is obtained by applying the full band table of Section 4.3.3c to the reconstructed value $\hat{x} = \hat{m} + \theta^{\text{ref}}_c$. This is Formulation 1 recentred: it inherits the sample efficiency of the continuous target while placing the principal clinical boundary at zero. The reference threshold is a presentational choice—it makes the score interpretable—and the multi-tier decision still comes from the band table, not from the single margin.

4.4.1b Disambiguating the dual-task formulation

The phrase used in the framing document admits two readings, and this thesis states which it means, since the ambiguity is visible to a reader and costs nothing to remove.

Reading A, two heads on one model. A shared representation with a regression head forecasting the parameter and a classification head predicting the risk class, trained jointly under

$$\mathcal{L} = \alpha\, \mathcal{L}_{\text{reg}} + (1 - \alpha)\, \mathcal{L}_{\text{cls}}. \tag{4.19}$$

The regression head supplies a dense training signal that regularises the classification head, which is valuable precisely when positives are rare: the regression loss is informative on every sample, while the classification loss is informative mainly on the few positives.

Reading B, two pathologies. Separate targets for blood pressure and for blood glucose, predicted jointly from a shared feature vector. This is justified by Section 2.1.3, which established the comorbidity at 58.1% of diabetic patients, and by the WHO target coupling the two (Section 2.1.3).

The two are not exclusive, and this thesis adopts both: Reading B as the outer structure, one target per pathology, with Reading A inside each where the model family permits a shared representation.

4.4.1c The horizon

Sections 3.4.7b and 4.1.3 established that a uniform 24 to 48 hour horizon contains zero or one observations under the routine protocol, and adopted a regime-dependent resolution. In the routine regime, which is the regime of the training data,

$$\Delta \longrightarrow \text{the next scheduled measurement occasion,} \tag{4.20}$$

so that the task is: given the fourteen-day window ending at t, predict the classification of the next measurement. At the routine frequency this is a horizon of three to four days. It is anticipatory, it is clinically meaningful—the confirmation interval of Table 3.1 is one to four weeks and the verification profile of **mihevc2025<empty citation>** is seven days—and it is estimable. The 24 to 48 hour horizon is retained only for patients inside the verification regime, where two readings per day make it estimable; that case is reported separately and on far fewer samples.

This is presented as a deliberate adaptation to the sampling density available, with the arithmetic of Section 4.1.3 as its justification. That is a stronger position than retaining an unrealisable horizon.

4.4.1d The formulation adopted

$$\boxed{\hat{m}^{(c)}_{i,\,t+\Delta} = f\Big(\mathbf{v}_{i,t}\Big), \qquad c \in \{\text{SBP},\ \text{glucose}\}, \qquad \Delta = \text{next occasion}} \tag{4.21}$$

with the tier derived by applying the versioned band table of Section 4.3.3c to the reconstructed forecast, and a classification head trained jointly where the model family permits it.

Three reasons. It maximises the information extracted from a small sample, since every window contributes a continuous residual rather than a label that is almost always negative. It places the principal clinical boundary at zero, so the score is directly interpretable in clinical units. And it decouples the model from the thresholds, so a guideline revision changes the decision layer without retraining—which is the separation that Section 3.5.1 required of the knowledge base.

4.4.2 The label

4.4.2a Ordinal, not nominal

The framing document specifies four levels, and they are ordered:

$$\text{stable} \prec \text{moderate} \prec \text{high} \prec \text{critical}. \tag{4.22}$$

The consequence is in the loss. Under multi-class cross-entropy, predicting *critical* when the truth is *stable* costs exactly what predicting *moderate* costs. Clinically it does not. Either an ordinal formulation is used—a cumulative-link model, or a set of nested binary classifiers $\mathbb{1}[y \succeq k]$—or Formulation 3 is adopted, in which the ordering is carried by the continuous forecast and no choice of ordinal loss is required. This is a second argument for the margin formulation.

4.4.2b Two label sources, reported separately

Section 4.1.4b established two label definitions, and both are used.

Table 4.36: The two label sources and their status.

Label	Source	Status
Rule-derived	band table applied to $x_{t+\Delta}$	primary; available for every sample
Clinician-adjudicated	`clinician_verdict` of Table 4.6	secondary; available only for triggered events

Performance against the second is the more credible result, because it is immune to the circularity of Section 3.4.3: the label was not computed from the same thresholds the features see. It will be available on far fewer samples, and it is reported as a secondary endpoint with its own sample size rather than merged with the primary. Under a retrospective collection scenario it is unavailable entirely (Section 4.1.2e).

4.4.2c Prevalence

Class prevalence determines everything downstream and is measured, not assumed. It is reported per class, overall and per patient, before any balancing. The 30.3% of **mendoza2025<empty citation>** is the only comparable figure in the corpus and is for a cross-sectional outcome; the prevalence of a transition to the critical tier over a three-day horizon will be far lower, and the regime to plan for is the $\pi = 0.02$ of Table 3.29.

4.4.3 Training protocol

4.4.3a The cross-validation structure

The constraints established earlier compose into one structure: split by patient, forward in time, stratified on patient-level event presence, with preprocessing, feature selection and hyperparameter tuning all fitted inside the training fold (Sections 3.4.7, 4.1.3c, 4.2.1 and 4.3.7b).

The resulting design is nested cross-validation with $K_{\text{out}} = 5$ outer folds estimating performance and $K_{\text{in}} = 4$ inner folds, on the outer training set only, performing selection and tuning. The outer test fold is touched once, after the inner loop has chosen everything (Figure 4.7).

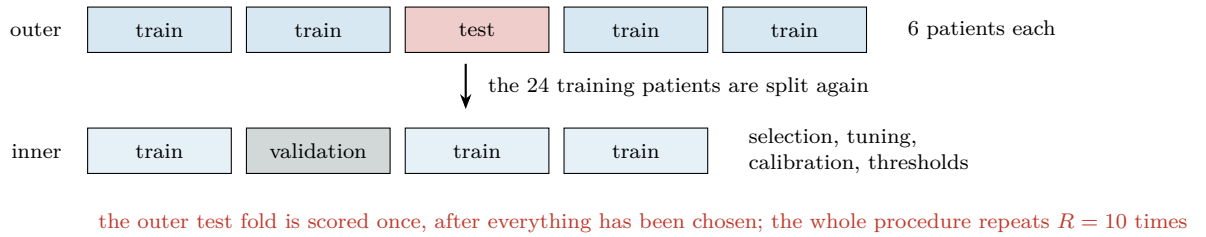


Figure 4.7: Nested cross-validation. Every learned quantity—preprocessing statistics, selected features, hyperparameters, the calibrator and the decision thresholds—is estimated inside the inner loop.

With thirty patients each outer fold holds six. Stratification on event presence is mandatory, because an unstratified fold may contain no positives at all and the fold-level recall is then undefined (Section 4.1.3c).

4.4.3b Repeated cross-validation, because variance is the problem

At the independent sample size established in Section 4.3.6b, with six patients per fold, the fold-to-fold variance of any metric is large, and a single five-fold estimate is one draw from a wide distribution. Reporting that draw as a point estimate overstates what is known.

The entire nested procedure is therefore repeated $R = 10$ times with different fold assignments, and the distribution is reported: median and interquartile range across repetitions, not a single number. This is nearly free at this scale and it converts an unfounded point estimate into an honest interval. It also supplies the confidence intervals that Section 3.4.8 required and that **efendi2025<empty citation>** omitted entirely.

4.4.3c Gradient-boosting hyperparameters

mendoza2025<empty citation> report their configuration precisely, and it is the right template—but their values are tuned for $n = 1\,308$ and must not be copied to a sample two orders of magnitude smaller.

Table 4.37: Gradient-boosting configuration, against the reference implementation.

Parameter	mendoza2025<empty citation>	Adopted	Reason
`max_depth`	6	2–3	depth 6 permits 64 leaves; at the sample size available that is one to three samples per leaf
`eta`	0.3	0.03–0.05	small steps with more rounds generalise better at small n
rounds	100	early stopping on inner validation	fixed round counts overfit or underfit unpredictably
`min_child_weight`	not stated	5–10	forces leaves to be supported by several samples
`subsample`	not stated	0.7–0.8	variance reduction
`colsample_bytree`	not stated	0.7–0.8	decorrelates trees; matters at the p/n of Section 4.3.6b
λ (L2)	not stated	1–10	explicit regularisation, the term $\Omega(f)$ of Section 3.4.1
objective	logistic	logistic or squared error	per the formulation adopted
seed	123, fixed	fixed and stated	reproducibility

The depth reduction is the important one. A depth-six tree on sixty to ninety independent samples is a memorisation device, and the discipline imposed in Section 4.3.6 on feature count is undone if the model is then permitted that much capacity.

The practice of fixing and reporting the seed is copied exactly from **mendoza2025<empty citation>** **efendi2025<empty citation>** report none.

4.4.3d Recurrent-model configuration

Section 4.1.3c computed that a single layer of sixteen units on ten input channels has about 1 800 parameters. Against the independent window count of sixty to ninety, that is twenty to thirty parameters per sample, not the ten per sample obtained from the generated count.

Table 4.38: Recurrent-model configuration.

Parameter	Adopted	Reason
Layers	1	two layers triple the parameter count
Units	8–16	the smallest that can represent the dynamics
Dropout	0.3–0.5	heavy, given the ratio
Recurrent dropout	0.2–0.3	—
Sequence length	4–8 observations	all a fourteen-day window contains
Masking	mandatory	no native missing handling (Section 4.2.7b)
Early stopping	on inner-validation recall	not on loss
Optimiser	Adam, learning rate 10^{-3}	conventional
Seed	fixed and reported	—

This thesis states in advance that this configuration is expected to underperform. Four

independent reasons have now accumulated: the parameter-to-sample ratio just computed; the absence of native missing-value handling where missingness is the dominant irregularity (Section 4.2.7b); the need for manual scaling and decay construction (Table 4.33); and the window length of fourteen days containing at most four observations, which is a very short sequence for an architecture whose advantage is long-range dependence.

Pre-stating this is not pessimism. A negative finding predicted in advance from sample-size arithmetic is evidence about the data regime; the same finding reported as a surprise is evidence about nothing.

4.4.3e Why the second implementation transfers nothing

The configuration of **efendi2025<empty citation>**—three hundred epochs, a learning rate of 0.025, a sigmoid activation and the Adam optimiser—was shown in Section 3.4.3 to be inapplicable to three of the five models they fitted, which do not train by gradient descent. Their reported performance was shown in the same place to measure label recovery on a task whose features were discretised by the ranges that define the label.

That paper is cited in this thesis for its system architecture and its device validation, never for its model configuration or its results. Its one transferable element is the deployment pattern of a trained model applied to incoming readings—which Section 3.3.4 relocated from the application layer to the analysis layer.

4.4.3f Compute is not the constraint

efendi2025<empty citation> report their hardware as though it bounded the work. At this sample size and dimension it does not. One gradient-boosting fit is a matter of milliseconds, and the full nested design—five outer folds, four inner, ten repetitions, a grid of fifty configurations, of the order of 10^4 fits—completes in minutes on an ordinary laptop.

Compute is therefore not presented as a constraint in this work. The constraint is sample size, and reporting hardware specifications in its place invites the question of why the sample is not larger.

4.4.4 Class imbalance at training time

4.4.4a Three mechanisms

The framing document commits to the first two. The third is absent from it and is in several respects the best: it leaves the probability model intact and moves only the operating point, which Section 3.5.4e already established is a clinical configuration parameter rather than a modelling choice.

Table 4.39: Mechanisms for handling imbalance and their effect on calibration.

Mechanism	Implementation	Effect on calibration
Class weighting	a positive-class weight w in the loss	distorts
Oversampling the minority	duplicate or synthesise positives	distorts
Threshold adjustment	leave training untouched; move the decision point	none

4.4.4b Inside the fold

Any resampling is fitted on the training fold only. Oversampling before the split places duplicates of the same patient's window in both training and test, and the resulting estimate is meaningless. This is among the most common errors in small-sample clinical machine learning and it is invisible in the output.

Under group k-fold the error is partly prevented by the patient-level split, but only partly: synthetic samples generated from a training patient remain in training, while duplicated windows from an overlapping stride (Section 4.2.5d) can still straddle the boundary if the stride and the split are applied in the wrong order.

4.4.4c The calibration cost

Weighting and oversampling both change the prior the model sees, so the output is a probability under a prevalence that does not exist. A score of 0.6 from a model trained with a positive-class weight of twenty does not mean a sixty per cent chance of the event.

If the score is only ever compared against a threshold this does not matter, since any monotone distortion preserves the ranking. If the score is displayed to a clinician as a risk it matters entirely, because a number presented as a probability that is not one is a safety problem and not a technical imprecision. Section 3.5 established that the clinician decides; a clinician deciding on a miscalibrated probability is deciding on a misleading number. Section 4.4.5 resolves this.

4.4.5 Deriving the risk score

4.4.5a From margin to probability

A gradient-boosted model with a logistic objective outputs a margin $\eta = \sum_k f_k(\mathbf{v})$, mapped to a score by the logistic function $s(\mathbf{v}) = \sigma(\eta)$. This is a score in $[0, 1]$. It is not a calibrated probability, for three reasons: the boosting objective optimises ranking and loss reduction rather than calibration; regularisation shrinks margins toward zero, compressing scores toward one half; and any class weighting shifts the implied prior.

Under the margin formulation adopted in Section 4.4.1, the model outputs a forecast

in clinical units, which is interpretable but is not a risk. Conversion requires a second step mapping the forecast and its uncertainty to the probability of crossing:

$$P\Big(x_{t+\Delta} > \theta\Big) = \Phi\left(\frac{\hat{m}_{t+\Delta}}{\hat{\sigma}_{\text{resid}}}\right), \tag{4.23}$$

where $\hat{\sigma}_{\text{resid}}$ is the residual standard deviation estimated on the inner validation fold and the residuals are assumed approximately normal, which is checked. This is a strength of the margin formulation: the uncertainty is estimated explicitly rather than implied by an uncalibrated classifier output.

4.4.5b Prior correction after weighting

Where class weighting is used the distortion has a closed form and is undone. With a positive-class weight w, the fitted model estimates odds inflated by w, so that $\text{odds}'(\mathbf{v}) = w\,\text{odds}(\mathbf{v})$. Inverting, with s' the raw output and s the corrected probability:

$$\boxed{s(\mathbf{v}) = \frac{s'(\mathbf{v})}{s'(\mathbf{v}) + w\Big(1 - s'(\mathbf{v})\Big)}} \tag{4.24}$$

Two worked checks. With $w = 20$ and a raw output of $s' = 0.5$, the corrected probability is $0.5/(0.5 + 20 \times 0.5) = 1/21 \approx 0.048$: a raw score suggesting an even chance corresponds to a true probability of about five per cent, so a clinician shown 0.5 would be shown a number about ten and a half times too large. The second check is sharper. The conventional setting is $w = (1 - \pi)/\pi$, which at the planning prevalence $\pi = 0.02$ gives $w = 49$; then $s' = 0.5$ maps to $0.5/(0.5 + 49 \times 0.5) = 1/50 = 0.02$, exactly the base rate—which is what a raw score of one half ought to mean, and is nowhere near what it appears to mean (Figure 4.8).

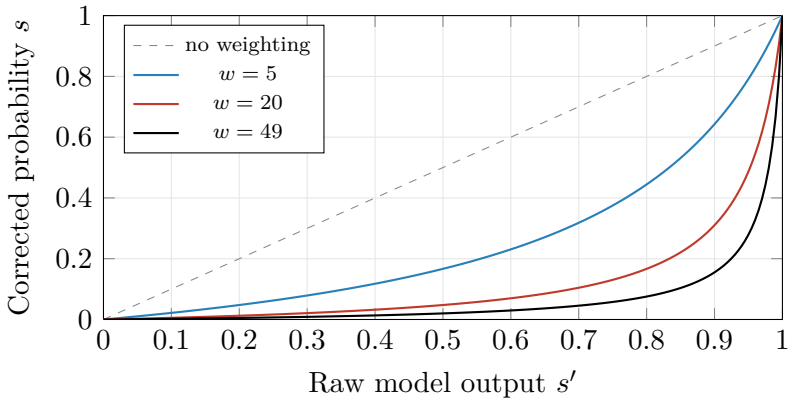


Figure 4.8: The prior correction. At $w = 49$, the setting conventional at a two per cent prevalence, a raw output of 0.5 corresponds to a true probability of 0.02.

The same correction applies to oversampling by a factor r, with w replaced by r.

4.4.5c Calibration

Prior correction fixes the location; it does not fix the shape. A second step is required, fitted on held-out data.

Table 4.40: Calibration methods and their suitability.

Method	Form	Suited to
Platt scaling	$\hat{p} = \sigma(a\eta + b)$, two parameters	small samples; assumes a sigmoidal distortion
Isotonic regression	monotone step function	larger samples; flexible but overfits at small n
Beta calibration	three parameters on $[0, 1]$	intermediate flexibility

At the sample size available, Platt scaling is the correct choice. Isotonic regression fits a free monotone function and will overfit a calibration set of a few dozen samples.

The calibrator is fitted on the inner validation fold and applied to the outer test fold. Fitting it on the test fold is a leak; fitting it on the training fold yields a calibrator fitted to data the model has already memorised, which is useless.

4.4.5d Assessing calibration

Discrimination and calibration are distinct, and a model may have a good area under the curve with poor calibration, since that measure depends only on ranking.

The Brier score is the mean squared error of the probability, $\mathrm{BS} = \frac{1}{n}\sum_i (\hat{p}_i - y_i)^2$, and it decomposes as

$$\mathrm{BS} = \underbrace{\mathrm{REL}}_{\text{miscalibration, lower better}} - \underbrace{\mathrm{RES}}_{\text{discrimination, higher better}} + \underbrace{\mathrm{UNC}}_{\pi(1-\pi),\ \text{fixed}} . \tag{4.25}$$

Reporting the decomposition rather than the raw score separates the two failure modes, which matters here because they have different remedies: miscalibration is fixed by Platt scaling, poor resolution is not fixable at all.

The calibration curve bins the predictions and plots observed frequency against mean predicted probability, with the diagonal as reference. At this sample size quantile bins of at least twenty samples are used, giving at most four or five bins against the independent count, and a confidence interval is plotted on each—a calibration curve without intervals at this sample size is uninterpretable.

A caution on the uncertainty term. Since $\mathrm{UNC} = \pi(1 - \pi)$ depends only on prevalence, at $\pi = 0.02$ it equals 0.0196, so a model predicting the base rate for everyone achieves a Brier score of 0.0196. A low Brier score at low prevalence is not evidence of anything, and it is always reported alongside the base-rate value.

4.4.6 Threshold selection and the three tiers

4.4.6a Two cut-points

The three-tier protocol of Table 3.4 requires two cut-points on the calibrated score,

$$\text{tier}(\mathbf{v}) = \begin{cases} \text{green} & s(\mathbf{v}) < \tau_{\text{amber}}, \\ \text{amber} & \tau_{\text{amber}} \le s(\mathbf{v}) < \tau_{\text{red}}, \\ \text{red} & s(\mathbf{v}) \ge \tau_{\text{red}}, \end{cases} \tag{4.26}$$

with distinct selection criteria, because the two tiers have different costs. Amber costs the patient a verification profile and costs the clinician nothing, so it is set for high sensitivity. Red costs clinician time, so it is set for acceptable precision.

4.4.6b Selection by target sensitivity

The default of one half is a convention with no clinical content. Section 3.4.6 established that the threshold is set from the cost of error. Operationally: a target sensitivity on the critical class is fixed in advance, for instance $\text{Se}^{\star} = 0.90$ for amber; on the inner validation fold the smallest τ achieving it is found; that τ is applied unchanged to the outer test fold; and the sensitivity actually obtained, the resulting precision and the alert volume are reported.

The third step is the one usually violated. Choosing the threshold on the test fold inflates every metric and is a second form of the leak that the nested design of Section 4.4.3a exists to prevent.

4.4.6c Alert volume at the chosen operating point

Section 3.5.4b established this as a mandatory reporting requirement. At the selected τ_{red} the weekly alert count is

$$A_{\text{week}} = Nm\Big[\pi\,\text{Se} + (1-\pi)(1-\text{Sp})\Big] \tag{4.27}$$

for a stated cohort size N and measurement rate m. For a pilot cohort of thirty patients at three readings per week, with $\pi = 0.02$, $\text{Se} = 0.90$ and $\text{Sp} = 0.98$:

$$A_{\text{week}} = 30 \times 3 \times \Big[0.02(0.90) + 0.98(0.02)\Big] = 90 \times 0.0376 \approx 3.4 \text{ alerts per week}, \tag{4.28}$$

of which approximately 1.6 are genuine. A sensitivity figure without this calculation is not an evaluation of a triage system, and it appears next to every reported operating point.

4.4.6d The rule engine as a floor

Section 3.5.1 established the hybrid design. At the decision layer this means the model may raise a tier but may not lower one set by the clinical rule:

$$\text{tier}_{\text{final}} = \max\Big(\text{tier}_{\text{rule}}(x_{\text{last}}),\ \text{tier}_{\text{model}}\big(s(\mathbf{v})\big)\Big) \tag{4.29}$$

under the ordering green $\prec$ amber $\prec$ red.

Three justifications, all established earlier. A patient at 185/110 mmHg is red under the same-day-treatment criterion of Table 3.1 regardless of any model. The best model of **mendoza2025<empty citation>** misses 49% of positives, so no model in this corpus is reliable enough to override a guideline. And the composition makes the system's worst case equal to guideline conformity: if the model fails entirely, the system degrades to the standard of care rather than to nothing. That property is what makes the system defensible to a clinician, and it is stated explicitly rather than left implicit.

4.4.7 Reporting and pre-registration

4.4.7a What is reported

Table 4.41: Reporting requirements for the modelling stage.

Item	Why
n, p, n/p, the convention for n, class prevalence per class	Section 4.3.6
The full baseline ladder, rungs 0 to 6	Table 4.34
The persistence baseline explicitly	Section 4.3.9; the hardest to beat
Per-class sensitivity, precision and F_1, never aggregates alone	Section 3.4.4d
Positive class stated for every metric	Section 3.4.4c
AUC with confidence interval, chance level marked	Section 3.4.4
Distribution across the R repetitions, not a point estimate	Section 4.4.3b
Brier score with its base-rate value and decomposition	Section 4.4.5d
Calibration curve with intervals	Section 4.4.5d
τ_{amber}, τ_{red} and how they were chosen	Section 4.4.6b
Alert volume at the operating point	Section 3.5.4b
Seeds, library versions, the full hyperparameter grid	**mendoza2025<empty citation>**
Secondary endpoint against clinician-adjudicated labels	Section 4.4.2b

4.4.7b Pre-registration

The failures diagnosed throughout Chapter 3—selection on the full dataset, thresholds chosen on test data, metrics reported for whichever class flatters, post-hoc sensitivity analyses presented as primary—share one remedy: declaring the analysis before seeing the test data.

Four items are fixed in advance and stated here, in the methodology, rather than in the results:

1. **The primary endpoint:** recall on the critical class at the pre-specified τ_{amber}.
2. **The primary comparison:** gradient boosting on the primary feature set against the clinical rule baseline. One comparison, not a matrix.
3. **The threshold rule:** target sensitivity 0.90, selected on inner validation.
4. **The feature set:** the eight-feature primary set fixed a priori per Table 4.32.

Everything else is secondary and labelled as such. This costs nothing, it is verifiable by a reader comparing this chapter with the results, and it is the strongest available answer to the question of whether the result was found or constructed.

4.4.8 Specification

Table 4.42: Modelling requirements established in Section 4.4.

Requirement	Basis
Target is the forecast margin to a reference threshold, not the class	Section 4.4.1
Tier derived from the band table, not from a single margin	Section 4.3.3c
Horizon is the next measurement occasion in the routine regime	Section 4.4.1c
The dual-task formulation disambiguated explicitly	Section 4.4.1b
Ordinal structure preserved in the loss or carried by the margin	Section 4.4.2
Nested cross-validation, 5 outer by 4 inner, grouped, stratified, forward in time	Section 4.4.3a
Repeated $R = 10$ times; the distribution reported	Section 4.4.3b
Tree depth 2–3, not 6	Table 4.37
Early stopping on inner validation, not fixed rounds	Table 4.37
Recurrent model at most one layer and sixteen units; underperformance pre-stated	Section 4.4.3d
Seeds fixed and reported	**mendoza2025<empty citation>**
Resampling strictly inside the training fold	Section 4.4.4
Prior correction applied after any weighting	Section 4.4.5b
Platt scaling, fitted on inner validation	Section 4.4.5c
Brier score reported with base rate and decomposition	Section 4.4.5d
Two thresholds, selected on inner validation by target sensitivity	Section 4.4.6b
Alert volume reported at every operating point	Section 3.5.4b
$\text{tier}_{\text{final}} = \max(\text{rule}, \text{model})$	Section 4.4.6d
The four pre-registration items stated in the methodology	Section 4.4.7b
efendi2025<empty citation> cited for architecture only	Section 3.4.3

4.4.9 Assumptions on which this section rests

Sections 4.1 to 4.3 were written parametrically in quantities that the collection scenario fixes. This section cannot be, since a training protocol must name its sample size. Table 4.43 states the values assumed, and what would have to be revised if each is overturned.

Two items are not assumptions but missing documents, carried forward from Section 4.3.11: the guideline editions that define the labels. The band table of Section 4.3.3c cannot be populated without them, and the labelling on which this entire chapter rests is

Table 4.43: Working assumptions and their consequences.

Quantity	**Assumed**	**What changes if overturned**
Collection scenario	B: prospective cohort on the protocol of Section 4.1.1	under a retrospective scenario the adjudicated secondary endpoint of Section 4.4.2b is unavailable; under the inpatient variant the horizon shortens and the sequence lengthens
Horizon Δ	next measurement occasion, three to four days	a denser protocol restores the 24–48 hour horizon of the original contribution
Feature window L	14 days	Table 4.25 fixes the minimum; a denser protocol permits 7 days
Independent samples n	60 to 90 (Section 4.2.5d)	every n/p figure, the depth limit and the bin count for calibration
Feature count p	8 primary, 16 extended	Table 4.32
Planning prevalence π	0.02	the weight w, the alert volume, and the Brier base rate

defined by them.[5]

[5] [**To verify**] Until the guideline editions are obtained, every threshold-derived quantity in Section 4.4—the reference threshold θ_c^{ref}, the band boundaries, the rule baseline and the rule floor—is computed from the operational values of Table 4.28 and must be recomputed once the guidelines are in hand.

Chapter 5

Overall System Architecture and Software Solution Design

5.1 Multi-Tier System Architecture

Purpose, sources and a correction on applicable law

Chapter 4 specified the data and the model. This chapter specifies the system that carries them. Section 5.1 fixes the layer decomposition, the data flow between layers, the latency budget, the failure modes, and the security and legal framework.

Four sources are used, all examined in Chapter 3. **mucchi2021<empty citation>** supply the four-layer reference architecture, derived from the ETSI SmartBAN high-level model, and the security principle adopted in Section 5.1.5. **efendi2025<empty citation>** supply the only implemented stack in the corpus with a working cloud backend; the reservations of Section 3.4.3 bear on their modelling, not on their architecture, which is why they are cited here. **liu2025review<empty citation>** survey security techniques and interoperability standards, under the standing reservation of Section 3.3.3d. **mihevc2025<empty citation>** supply the only clinically validated system in the corpus and the only statement of a regulatory framework with which the authors actually complied.

The correction. mihevc2025<empty citation> state that their data "were encrypted and complied with the General Data Protection Regulation"†. Their trial was conducted in Slovenia, within the European Union, where that regulation is binding. A system deployed in Cameroon, processing the data of Cameroonian patients, operated by a Cameroonian institution, falls outside its territorial scope. Asserting compliance with an instrument that imposes no obligation is an error of category, and worse, it displaces the work of identifying the instrument that does apply. Section 5.1.6 treats this properly rather than reproducing the claim.

5.1.1 The Four Layers

5.1.1a The model and its provenance

The decomposition is that of **mucchi2021<empty citation>**, with one modification. Their architecture comprises BODY, ENVIRONMENT, EDGE and CLOUD layers. This system has no environmental sensors, so that layer is omitted; and the application tier, which **mucchi2021<empty citation>** fold into the cloud layer as "E-Health Services"†, is promoted to a layer of its own, following **efendi2025<empty citation>** and **liu2025review<empty citation>**.

Table 5.1: Layer correspondence across the four reference architectures.

#	mucchi2021<empty citation>	efendi2025<empty citation>	liu2025review<empty citation>	This system
1	BODY	Physical	Access	Physical
—	ENVIRONMENT	—	Access	omitted
2	EDGE	(absent)	Data transfer	Edge
3	CLOUD	Network	Storage, processing, analysis	Cloud
4	(within CLOUD)	Application	Application service, decision support	Application

Section 3.3.1 established that the layer on which the four sources disagree is the Edge, and that its presence or absence is the architectural decision that matters. This system has one, for three reasons developed below: it is where triage happens (Section 3.3.4); it is where transmission priority is decided (Section 5.1.2c); and it is what allows the system to function during the connectivity interruptions that Section 2.2.4 quantified.

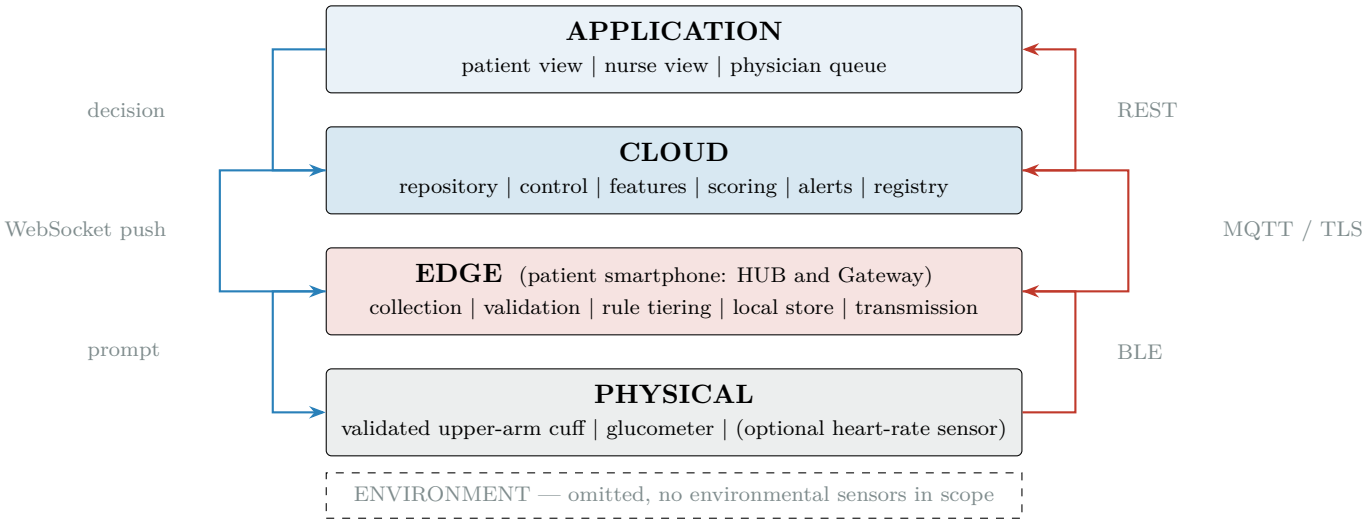


Figure 5.1: Layer decomposition and data flow. Red denotes the upstream measurement path, blue the downstream decision path. The Edge is the layer on which the four reference architectures of Table 5.1 disagree.

5.1.1b Physical layer

Table 5.2: Physical-layer specification.

Component	Specification	Basis
Blood pressure monitor	automated upper-arm cuff, independently validated, Bluetooth Low Energy	Section 3.2.5; **who2023hypertension<empt**
Glucometer	strip-based, Bluetooth Low Energy	Section 3.2.5
Measurement enforcement	two readings, interval $\geq$ 45 s enforced by the device, second retained	Section 2.3.3
Optional covariate sensor	heart rate only; never blood pressure, never SpO_2	Section 2.4.2
Device register	make, model, serial, validation reference, calibration date	Section 4.1.2

Excluded by design: cuffless wrist devices, smartphone-camera blood pressure, and any inferential blood-pressure method. Section 3.2.5 established the WHO position, that no alternative to the validated upper-arm cuff existed at the time of the 2023 report, and Section 2.4.2 showed that the evidence on camera-based measurement does not yet justify its use as a decision input. This is Position A of Section 3.2.6b expressed as a hardware specification.

5.1.1c Edge layer

mucchi2021<empty citation> decompose the Edge into a Gateway containing a reception and transmission unit, and a HUB containing data collection, data processing, local storage and transmission units. In this system all of these run on the patient's smartphone, which is simultaneously HUB and Gateway; their *Intra-Com-3* link is therefore absent, being the link that exists only when HUB and Gateway are separate devices.

Table 5.3: Edge-layer specification.

Unit	Function	Basis
Data collection	BLE ingestion; capture of `ts_measurement` at source	Section 4.1.2
Data processing	deterministic validation; rule-engine tier evaluation; aggregation of any high-frequency covariate stream	Sections 3.5.3, 3.5.4 and 4.2.2
Local storage	offline-first queue, encrypted at rest, surviving application restart	Section 3.3.3b
Transmission	MQTT upstream; tier-dependent priority; SMS fallback on red	Sections 3.3.2 and 5.1.2c
Patient interface	measurement prompts, reading display, trend view, therapy view	Section 3.5.5

Three functions sit at the Edge that most designs place in the Cloud, and each has a reason.

The **rule-engine tier evaluation** runs here because the tier determines transmission urgency (Section 5.1.2c). A rule evaluated only in the cloud cannot tell the phone that a reading is urgent, which is circular: the phone would have to transmit in order to learn whether to transmit quickly.

Aggregation of high-frequency streams runs here because Section 3.5.4b established that per-reading alerting on a 1 Hz stream generates of the order of ten thousand false alerts per patient per week. The Edge reduces such a stream to a daily summary before anything is transmitted or scored.

Local storage is what makes the system usable under intermittent connectivity, which Section 2.2.4 established as a structural feature of the deployment context rather than an occasional fault.

5.1.1d Cloud layer

Table 5.4: Cloud-layer specification.

Unit	Function	Basis
Repository	longitudinal record keyed for window queries; both `ts_measurement` and `ts_recorded` stored	Sections 3.3.3a and 4.1.2
Control	identity, roles, access policy, audit log	Section 3.3.5
Data processing	feature construction; quality flags; expanding-window normalisation	Sections 4.2 and 4.3
Analysis	model scoring; prior correction; Platt calibration; tier mapping	Sections 4.4.5 and 4.4.6
Alert management	queue construction, priority ordering, escalation, non-transmission detection	Sections 3.5.2 and 3.5.4
Offline training	periodic retraining from the warehouse; model registry with versions	Section 3.5.5
Services	REST for clients; WebSocket for push	Section 3.3.2

The separation of online scoring from offline training is taken from **liu2025review<empty citation>** Section 3.5.5 established that the separation transfers while their tooling does not, being dimensioned for a throughput this cohort will never approach.

The **model registry** appears in no source and is required by Sections 3.5.6 and 4.3.3b: every score must be attributable to a model version and a threshold-table version, or no result can be reproduced once either changes. Since Section 4.4.9 records that the guideline editions defining the threshold table are not yet in hand, the registry is also the mechanism by which scores computed under provisional thresholds remain identifiable after the definitive table is installed.

5.1.1e Application layer

Two profiles follow the SELF and PRO distinction of **mucchi2021<empty citation>**, with a third added for the task-sharing arrangement that Section 3.1.3a established as the regional norm.

Table 5.5: Application-layer profiles and capabilities.

Profile	Capabilities
Patient	enter or confirm readings; view own history and trend; receive prompts; view prescribed therapy; receive the outcome of a clinician decision
Nurse	the above for assigned patients; perform and attribute measurements; view flagged readings; *cannot* modify therapy
Physician	priority-ordered queue; patient record with trend; model score with its explanation; proposed protocol step; approve, override or request consultation; every action written to the adjudication table

The priority-ordered queue is from **yazdani2018<empty citation>**, and Section 3.5.4 established that an ordering rather than a partition is what the alert arithmetic requires. The adjudication write-back is from Section 3.5.5e and is the mechanism that accumulates the clinician-adjudicated labels of Section 4.4.2b.

5.1.1f Traceability

Table 5.6: Provenance of each design element.

Design element	Origin
Four-layer decomposition	**mucchi2021<empty citation>**, ETSI SmartBAN
Edge with local storage and processing	**mucchi2021<empty citation>**
MQTT upstream, HTTP downstream, WebSocket push	**efendi2025<empty citation>**
Offline-first with deferred merge	**efendi2025<empty citation>**
Three-tier alerting with verification profile	**mihevc2025<empty citation>**
Priority-ordered clinician queue	**yazdani2018<empty citation>**
Configurable checkup period and rule selection	**yazdani2018<empty citation>**
Offline training, online scoring split	**liu2025review<empty citation>**
ICD and SNOMED CT coding	**liu2025review<empty citation>**
Validated-cuff restriction	**who2023hypertension<empty citation>**; Section 3.2.5
Tier-dependent transmission priority	this work, Section 5.1.2c
Alert-volume-driven aggregation at the Edge	this work, Section 3.5.4b
Rule-engine floor $\max(\text{rule}, \text{model})$	this work, Section 4.4.6d
Model and threshold-table version registry	this work, Section 5.1.1d
Latency decomposition and the dominance of t_5	this work, Section 5.1.3
Failure-mode and degradation table	this work, Section 5.1.4

5.1.2 Inter-Layer Data Flow

5.1.2a The nominal path

$$\underbrace{\text{cuff, glucometer}}_{\text{Physical}} \xrightarrow{\text{BLE}} \underbrace{\text{phone}}_{\text{Edge}} \xrightarrow{\text{MQTT/TLS}} \underbrace{\text{broker} \rightarrow \text{repository} \rightarrow \text{features} \rightarrow \text{score} \rightarrow \text{tier}}_{\text{Cloud}} \rightarrow \underbrace{\text{queue} \rightarrow \text{c}}_{\text{Applica}} \tag{5.1}$$

5.1.2b Where each decision sits

Table 5.7: Decision placement by layer and latency class.

Decision	Layer	Latency class	Basis
Is the reading well-formed?	Edge	milliseconds	Section 4.2.2
Which tier under the clinical rule?	Edge	milliseconds	Section 5.1.2c
How urgently to transmit?	Edge	milliseconds	this work
Which tier under the model?	Cloud	seconds	Section 4.4.6
Final tier = max(rule, model)	Cloud	seconds	Section 4.4.6d
Where in the queue?	Cloud	seconds	Section 3.5.4
Confirm, override or escalate?	Application, human	hours to days	Section 3.5.2

The last row dominates the total, by a margin that Section 5.1.3 quantifies.

5.1.2c Tiered transmission priority

This is the design decision that justifies running the rule engine at the Edge, and it appears in no source in the corpus.

Section 3.3.3b established that offline-first storage with deferred merge removes one cause of missing data, since readings taken offline are delayed rather than lost. It also noted the cost: arrival time decouples from measurement time, sometimes by days, which is why Section 4.1.2 stores both.

For routine readings that trade is favourable. For a red-tier reading it is not. A hypertensive crisis queued on a handset until the next period of connectivity is a clinical failure that the architecture itself caused, and the fact that the reading eventually arrives is no mitigation.

The resolution is to make transmission priority a function of the tier, which requires the tier to be known at the Edge.

The SMS fallback is available because **dodoo2022<empty citation>** report 816 million mobile connections in sub-Saharan Africa against a projection of 1.05 billion, and because **yazdani2018<empty citation>** already use SMS in their reminder loop. SMS reaches where data does not, and a red alert is short enough to fit one message.

Table 5.8: Transmission behaviour by tier, evaluated at the Edge.

Tier	Transmission behaviour
Green	queue locally; batch synchronisation on next connectivity; no retry urgency
Amber	attempt immediate send; exponential backoff; synchronisation within hours
Red	immediate send; aggressive retry; SMS fallback after n failures; local on-device warning to the patient regardless of connectivity

Two cautions attach to that justification and are stated rather than elided. A connection count is not a count of people, since multiple subscriptions per user are common in the region, so the figure overstates population coverage; and a national connection total establishes nothing about signal availability at a particular patient's residence, which is the quantity that actually governs whether the fallback works.[1]

The local on-device warning matters independently of transmission. A patient whose reading is critical should be told so by the handset, immediately, even when nothing can be sent. That instruction is generic and therefore safe under Section 3.5.5e: advising a patient to seek care is not a dose change, and the prohibition established there is on autonomous therapeutic action, not on autonomous information.

5.1.2d The reverse path

$$\text{clinician decision} \rightarrow \text{adjudication table} \rightarrow \text{Cloud} \xrightarrow{\text{push}} \text{phone} \rightarrow \text{patient} \tag{5.2}$$

Two requirements. Any therapy change passes a clinician, and nothing on this path is generated autonomously (Section 3.5.5e). And every decision is written to the adjudication table of Section 4.1.2e *before* it is delivered, so that the label is captured even when delivery fails — which is the common case precisely for the patients whose connectivity is worst, and therefore the case in which losing the label would bias the adjudicated sample.

5.1.3 The Latency Budget

No source in this corpus decomposes end-to-end latency, and **bediang2021<empty citation>** supply the one field measurement that makes the decomposition worth performing.

5.1.3a Decomposition

The bound on t_3 follows from Section 4.4.3: at the independent sample size and the primary feature count fixed in Table 4.43, a single forward pass through a gradient-boosted ensemble of shallow trees is a sub-millisecond operation, and the second is consumed by ingestion and feature construction, not by the model.

[1] [**To verify**] The **dodoo2022<empty citation>** projection is stated to a horizon now past. Confirm whether the realised figure is available, and replace the projection with it; a projection to a date already elapsed invites the obvious question.

Table 5.9: Latency segments.

Segment	Symbol	Expected	Governed by
Measurement to phone	t_1	seconds	BLE
Phone to cloud, online	t_2	seconds	network
Phone to cloud, offline	t_2'	hours to days	connectivity; Section 5.1.2c
Ingestion, features, scoring	t_3	< 1 s	Section 4.4.9; Section 4.4.3
Queue to clinician view	t_4	seconds	push
Clinician review	t_5	hours to days	staffing
Decision to patient	t_6	seconds to hours	reverse path

5.1.3b Where the time actually goes

bediang2021<empty citation> measured both ends of this pathway in a Cameroonian district-hospital setting.

Table 5.10: Measured delays, from **bediang2021<empty citation>**.

Measured quantity	Intervention	Control	p
Time to perform the examination	1.3 ± 4.1 days	7.0 ± 14.5 days	< 0.01
Time to receive expert opinion	10.3 ± 11.6 days	11.2 ± 27.9 days	0.82

The protocol under which these were collected required expert opinion within 72 hours. The measured mean in the intervention arm was 10.3 days, which exceeds that target by a factor of approximately 3.4.

The technology accelerated acquisition by a factor of about five, from 7.0 days to 1.3. It did not accelerate the clinical decision at all: the expert-opinion delay was not distinguishable from the control pathway, in which patients travelled to a referral centre.

One caution on these statistics. In three of the four cells the standard deviation exceeds the mean, by a factor of three in the intervention examination time and by a factor of two and a half in the control expert-opinion time. A non-negative quantity with that dispersion cannot be approximately normal; the distributions are heavily right-skewed, the means are therefore poor central estimates, and a p-value computed on means and standard deviations under a normality assumption is not reliable.[2]

This is the quantified form of the argument of Section 3.5.4: t_1 through t_4 are engineering problems, solved and small, while t_5 is a resource problem and dominates the total by orders of magnitude (Figure 5.2).

[2] [**To verify**] Confirm whether **bediang2021<empty citation>** report medians, interquartile ranges or a non-parametric test alongside these means, and whether the fifty-day maximum attributed to them in Section 5.1.4 appears in the paper or is inferred. The qualitative conclusion — acquisition accelerated, decision not — survives either way, since a factor of five and a null result are both robust to

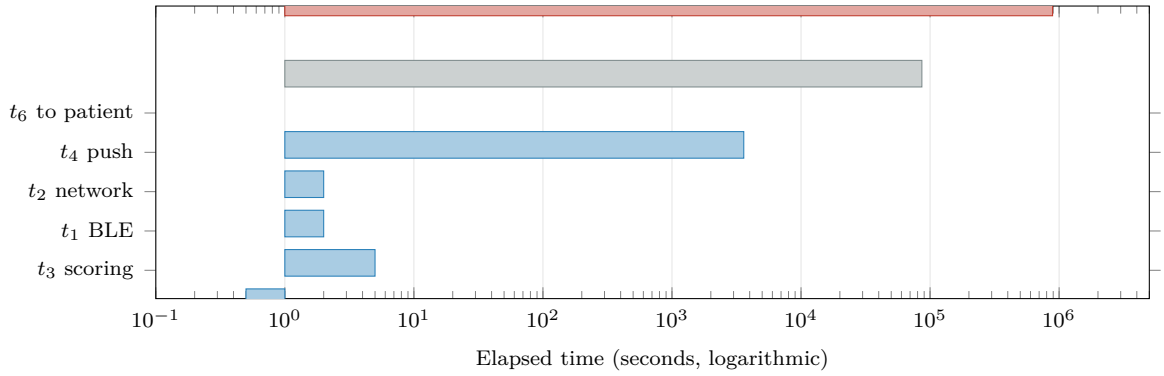


Figure 5.2: Latency segments on a logarithmic scale. The engineering segments t_1 to t_4 occupy under ten seconds in total; t_5, taken at the 10.3-day mean of **bediang2021<empty citation>**, exceeds them by about five orders of magnitude.

5.1.3c What follows for the architecture

Optimising t_1 through t_4 has almost no effect on the outcome. A design change reducing scoring latency from 800 ms to 200 ms saves 0.6 s against a total of 10.3 days, which is 8.9×10^5 s: a saving of about seven parts in ten million, or 7×10^{-5} per cent of the total. No plausible engineering improvement to those segments is clinically detectable.

Reducing t_5 is the only material improvement available, and the architecture can act on it in exactly two ways.

Reduce queue length, by triage. This is the arithmetic of Section 3.5.4, delivered by the tiering of Section 4.4.6: fewer items in the queue means less waiting per item, for a fixed clinician capacity.

Reduce per-item review time, by what is presented. A clinician who must open a record, scroll a table and mentally reconstruct a trajectory spends minutes per item. One shown the fourteen-day trajectory, the threshold line, the calibrated score with its two or three contributing features, and the proposed protocol step, spends seconds. The dashboard design of the next section is therefore a latency intervention and is justified on those grounds, not as a presentational refinement. This is the one place in the system where interface design and clinical outcome are connected by an arithmetic argument rather than by assertion.

t_2' is bounded by the tiering of Section 5.1.2c, which is the reason that mechanism exists: it caps the offline delay for the readings where delay is harmful, while leaving it unbounded where it is not.

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the distributional shape; the point estimates are what should not be quoted precisely.

5.1.4 Failure Modes and Degradation

No source in this corpus analyses what happens when a layer fails. Given that Section 2.2.4 documented a regional telemedicine service ceasing operations after reaching 2.5 million users, and that **dodoo2022<empty citation>** identify sustainability as the characteristic failure mode of the region's projects, graceful degradation is not an optional refinement.

Table 5.11: Failure modes, detection and degraded behaviour.

Failure	Detected by	Degradation	Residual risk
Device battery exhausted	absence of readings	non-transmission alarm	patient unmonitored until noticed
Device lost or broken	as above	as above	as above
Phone unavailable	as above	none at the Edge	no local alerting at all
Connectivity lost	Edge	local queue; red escalates by SMS	amber delayed
Cloud unavailable	Edge retry	local queue; Edge rule still tiers; red by SMS	no model score, no queue
Model service fails	Cloud	rule floor: tier = $\text{tier}_{\text{rule}}$	loss of personalisation only
Calibration drifts	monitoring	thresholds revert to rule	over- or under-alerting
Clinician unavailable	queue age	escalation after a defined interval	unbounded t_5
Guideline superseded	manual	versioned table swapped; registry preserves provenance	historical scores not comparable

Three rows carry the design argument.

Model failure degrades to guideline conformity, not to nothing. This is the composition $\text{tier}_{\text{final}} = \max(\text{tier}_{\text{rule}}, \text{tier}_{\text{model}})$ of Section 4.4.6d, and it is the property that makes the system defensible to a clinician: its worst case equals the current standard of care. Given that the best model of **mendoza2025<empty citation>** attains a sensitivity of 0.51, a design whose failure mode is to behave like the guideline is the correct one, and the composition is what converts that observation into a structural guarantee rather than a hope.

Phone unavailability is a single point of failure. All five Edge functions of Table 5.3 run on one device. No source addresses this, and the honest statement is that a patient without a working handset is outside the system. The only available mitigation is the non-transmission alarm, which detects the condition without remedying it — and which depends on the Cloud noticing an absence, this being the one detection that cannot by construction be performed at the Edge. A device that has failed cannot report its own failure.

Clinician unavailability requires an escalation rule, which no source specifies.

The HEARTS indicators of Section 3.1.3a include patients with no visit in three and in twelve months; the analogous system quantity is queue age. A red item unreviewed after a defined interval must escalate to a named alternate. Without such a rule t_5 is unbounded, and the extreme tail of the **bediang2021<empty citation>** distribution becomes the expected behaviour rather than an outlier — which is precisely what the skew noted in Section 5.1.3 implies, since in a right-skewed delay distribution the long waits are not rare accidents but a stable fraction of the caseload.

5.1.5 Security

5.1.5a The organising principle

mucchi2021<empty citation> state it, and it is the correct organising idea:

> "Health data can flow from patient to cloud and from cloud to others by using different channels, which not always can be assumed secure. A trusted and secure domain for the health data should be assured, so that there is no segment which has a lower security than others."†

Security is determined by the weakest segment. The design task is therefore to enumerate the segments and specify each, rather than to accumulate techniques.

5.1.5b The seven segments

Table 5.12: Security segments and their protection.

#	Segment	Protection	Basis
1	Device to phone	BLE with LE Secure Connections; pairing bound to the patient account; legacy pairing refused	—
2	Phone at rest	operating-system full-disk encryption required; application-sandboxed store; no identifiers in logs or crash reports	Section 4.1.5c
3	Phone to cloud	TLS 1.2 or later; certificate pinning; MQTT over TLS	**efendi2025<empty citation>**
4	Cloud at rest	encryption at rest; server-side access rules enforcing role separation	**efendi2025**; **liu2025review<empty citatio**
5	Cloud to clinician	TLS; authenticated session; short expiry; re-authentication before a therapy change	Section 5.1.1e
6	Clinician device	outside system control	—
7	Export and backup	the analysis file of Section 4.1.5c	Section 4.1.5c

5.1.5c The two weakest segments, which no source addresses

Segment 6. The clinician reviews on a personal handset or a shared hospital workstation. The pilot of **mucchi2021<empty citation>** installed their application on the nurses' own smartphones. The system cannot enforce device encryption, screen locking or the

absence of malicious software there, and in practice this is the weakest link in almost every deployment of this kind.

What the architecture can do: enforce short session expiry; require re-authentication before any therapeutic action; avoid caching patient data on the client; and log access, so that a compromise is detectable even where it is not preventable. What it cannot do is secure the device, and this thesis states that rather than implying otherwise.

Segment 7. The research export is the one copy of the data that leaves every protection of segments 1 to 5, being a file on a laptop. Section 4.1.5c specified its contents and the measures that reduce re-identification risk. Those measures are the security control for this segment and are presented as such, not solely as an ethical matter.

5.1.5d What the survey offers, and what is proportionate

The survey of **liu2025review<empty citation>** is extensive: differentially private synthesis using convolutional autoencoders and generative adversarial networks; k-anonymity through the Mondrian algorithm; RFID-based authentication for remote patient monitoring providing "mutual authentication, user untraceability, prevention of replay attacks, forward and backward secrecy, and data integrity"†; lightweight encryption with access tokens; permissioned blockchain with elliptic-curve digital signatures; blockchain combined with a distributed file system; private set intersection with k-anonymity; and steganographic concealment of clinical text within medical images.

Almost none of this is proportionate to a thirty-patient pilot, and adopting it would be security theatre: a catalogue of mechanisms whose presence in a dissertation signals diligence while protecting nothing that is actually at risk here. The proportionate set is transport encryption, encryption at rest, role-separated authentication, access logging, and the de-identification measures of Section 4.1.5c.

Two items from the list nonetheless deserve mention, for different reasons. **Access logging for non-repudiation**, which both **mucchi2021<empty citation>** and one of the surveyed works reach for blockchain to achieve, is obtainable from an append-only audit table at negligible cost, and Section 3.3.5 already required it. And **k-anonymity** is directly relevant, because Section 4.1.5c established that thirty patients with exact ages and dates are uniquely identifiable; the banding and date-shifting specified there are the operational form of that idea, and naming the correspondence is more useful than citing the algorithm.

5.1.5e Implemented versus recommended

Section 3.3.5b identified the characteristic pattern of this literature: a security section cataloguing techniques the authors did not use. **mucchi2021<empty citation>** are candid about it, writing that "we only relied on the application layer security protocols,

and on the fact that the data where stored in a private server without public access"† — having just recommended physical-layer security, cryptographic key extraction from biosignals, and blockchain audit trails.

This thesis therefore presents two separate statements: what the system implements, and what a production deployment would require beyond it. Conflating them is the failure the corpus demonstrates, and the partition of Section 5.1.8 is where that separation is made concrete.

One empirical note is worth including. **efendi2025<empty citation>** surveyed their users across four dimensions, and security scored lowest, at 4.06 out of 5, or 81.2%, against 4.56 for effectiveness. Users notice. A security section that is candid about its limits is also the one most likely to be believed.[3]

5.1.6 Legal Framework

5.1.6a The General Data Protection Regulation does not apply

mihevc2025<empty citation> complied with the GDPR because they processed the data of European Union residents within the Union. This system does neither. To write that it is "GDPR compliant"† would assert compliance with an instrument imposing no obligation here, and — the more serious cost — would stand in place of identifying the instrument that does.

The position adopted is instead the following. The design adopts the data-protection principles of the GDPR — lawfulness, purpose limitation, data minimisation, storage limitation, integrity and confidentiality — as a voluntary design standard, while recording that the applicable legal framework is Cameroonian. That framing is accurate, it preserves the substantive protections, and it is not open to the objection that the claim invites.

5.1.6b The applicable instrument

No document in the corpus states the Cameroonian instrument governing the processing of health data, and this thesis does not assert one: an incorrect statutory reference is a more serious defect than an acknowledged gap, because it is a positive misstatement of law rather than an absence.

What the corpus does establish around the gap is informative. **mucchi2021<empty citation>** name their national instrument explicitly, which is the practice to follow. **liu2025review<empty citatio** name HIPAA, the HITECH Act and the 21st Century Cures Act, all United States law and none applicable here — which is the same error as the GDPR claim, committed by a

[3] **[To verify]** A mean of 4.06 is not attainable from ten respondents on a single five-point item, since any such mean is a multiple of 0.1. It is attainable as the mean of a five-item subscale over ten respondents, which yields multiples of 0.02. Confirm the instrument's item count before quoting the figure, and report n and the item count alongside it.

survey rather than a trial. And **dodoo2022<empty citation>** identify the legal and regulatory dimension as one of six barriers to telemedicine in sub-Saharan Africa, covering "policies, legislation, and standards governing confidentiality and privacy"†. The gap is therefore a documented feature of the regional context and not an oversight peculiar to this work, which is the correct way to present it.

Four instruments must be identified and cited before deployment: the Cameroonian personal-data and health-data instrument; the institutional policy of the partner facility; the national or regional ethics committee reference for this study; and any African Union instrument on data protection that Cameroon has ratified.[4]

5.1.6c Data residency is a legal choice

The published endpoint of **efendi2025<empty citation>** was hosted in a south-east Asian region. Where data physically reside determines which jurisdiction's law governs their storage, so selecting a region is a legal decision taken by means of a configuration flag — which is exactly the kind of decision that is made by default and never revisited.

Three consequences. Selecting a European region may bring European law into scope unnecessarily, for a system that otherwise has no European nexus, and would make true the claim that Section 5.1.6a declines to make — by accident rather than by design. Selecting a region distant from Cameroon adds latency to t_2 and t_4, which Section 5.1.3 showed to be negligible in the total, so latency is not a legitimate ground for the choice in either direction. And the chosen region must be stated in the thesis, because an unstated region is an unstated jurisdiction.

5.1.6d Consent

Under the working assumption of Scenario B recorded in Table 4.43, individual consent is required and is not optional. **efendi2025<empty citation>** record that "each participant assign the letter of consent that willing to be involved in the experiment"†, following an interview establishing health history; **mihevc2025<empty citation>** obtained consent under a registered protocol.

The consent document must cover eight points: what is collected; who sees it; where it is stored and under which jurisdiction; the retention period and destruction date; that participation does not replace clinical care; that withdrawal is possible; what happens to already-collected data on withdrawal; and that the pseudonym mapping is held by the partner physician and is not transferred (Section 4.1.5c).

[4] **[To verify]** These four references are outstanding and are the author's to supply. The third has a documented pathway: **bediang2021<empty citation>** cite ethical clearance CE00398/N°CRERSHC/2016, so a Cameroonian study with an identifiable Cameroonian ethics reference exists within this corpus, and the procedure is therefore known. Until the first is supplied, Section 5.1.6a is the operative statement of the legal position and Section 5.1.6c cannot be resolved.

The third and fourth of these depend on Sections 5.1.6b and 5.1.6c respectively, so the consent document cannot be finalised before those are resolved. This dependency is stated because it determines the order of the remaining work: the legal instrument is not a formality to be appended at the end but a precondition for the consent form, which is itself a precondition for collection.

5.1.7 Specification

Table 5.13: Architectural requirements established in Section 5.1.

Requirement	Basis
Four layers: Physical, Edge, Cloud, Application	**mucchi2021<empty citation>**
Environment layer omitted, and the omission stated	Section 5.1.1
Physical layer restricted to validated upper-arm cuff and glucometer	**who2023hypertension<empty citation>**; Section 3.2.6b
Edge performs validation, rule tiering, aggregation, local storage, transmission	Section 5.1.1c
Transmission priority determined by tier, at the Edge	Section 5.1.2c
SMS fallback on red tier, with its coverage caveat stated	Section 5.1.2c
Local on-device warning on red, independent of connectivity	Sections 3.5.5e and 5.1.2c
Cloud: repository, control, processing, analysis, alert management, offline training	Section 5.1.1d
Model and threshold-table version registry	Sections 3.5.6 and 4.3.3b
Three application profiles; physician sees a priority-ordered queue	**yazdani2018<empty citation>**
Clinician decisions written to the adjudication table before delivery	Section 3.5.5e
Latency budget reported; t_5 identified as dominant	**bediang2021<empty citation>**; Section 5.1.3
Point estimates from **bediang2021<empty citation>** quoted with their skew caveat	Section 5.1.3
Dashboard design justified as a t_5 intervention	Section 5.1.3
Failure table published; model failure degrades to the rule	Section 5.1.4
Queue-age escalation rule for unreviewed red items	Section 5.1.4
Handset single point of failure stated explicitly	Section 5.1.4
Seven security segments enumerated, each specified	Table 5.12
Clinician device acknowledged as outside system control	Section 5.1.5
Append-only access log	Section 3.3.5
Implemented security stated separately from recommended	Sections 3.3.5b and 5.1.8
GDPR not claimed; its principles adopted voluntarily, and said so	Section 5.1.6a
Cameroonian instrument, institutional policy and ethics reference cited	Section 5.1.6b
Data residency region stated	Section 5.1.6c
Consent covering the eight points, with its dependencies stated	Section 5.1.6d

5.1.8 What Is Built and What Is Specified

Section 3.3.6 established that of the four reference architectures in this corpus, two have no implementation at all, one was validated on six healthy volunteers over ten minutes, and the fourth was validated by a small panel on simulated patients with no real patients involved. None has been evaluated in routine care.

This thesis therefore states plainly which parts of the architecture above are built, which are specified, and which are deferred. That distinction is what separates a design chapter from a list of aspirations, and the corpus demonstrates that it is frequently not drawn.

Table 5.14: Implementation status of the architecture.

Status	Components
Built and exercised	Physical acquisition; Edge ingestion, validation, local storage, rule tiering, transmission; Cloud repository, feature pipeline, model scoring, tier mapping; Application patient and physician views
Specified, not built	SMS fallback; queue-age escalation; offline retraining loop; model registry; ICD and SNOMED CT coding
Deferred, with reasons	Environment layer; nurse profile; multi-site deployment; production security beyond the proportionate set of Section 5.1.5

Nothing in the second or third column weakens the work. An unbuilt component that is specified, justified and scoped is a contribution; one that is silently absent is a gap. The difference is visible to a reader only if the partition is published, which is why it appears here rather than being left to inference.

Two document dependencies now block Chapter 5 as they blocked Chapter 4: the guideline editions required by Table 4.43 for the threshold table, and the Cameroonian legal instrument required by Section 5.1.6b. Neither can be substituted by reasoning from the corpus, and both are recorded as outstanding rather than provisionally filled.

5.2 IoT Hardware Subsystem and Communication Protocols

Purpose, sources and scope

Section 5.1 placed four layers. This section specifies the optional instrumented branch of the Physical layer and the protocol that carries its output to the Edge.

Two sources are used. **efendi2025<empty citation>** supply the component inventory, pin lists and electrical characteristics, and the MQTT/HTTP protocol split; the reservations established in Sections 3.2.3c and 3.4.3 mean that their components are taken while their configuration and results are not. The research framing document state contribution C4, a deliberately spare acquisition device using manual entry and smartphone-camera photoplethysmography, and identify measurement noise and intermittent connectivity as the technological limitations of the approach.

A scoping statement opens this section, because without it the subsystem appears to contradict Chapter 3.

> The instrumented subsystem specified here acquires heart rate, oxygen saturation and body temperature. It does *not* acquire blood pressure or capillary glycaemia, which Section 3.2.5 established may come only from independently validated upper-arm cuffs and glucometers. The subsystem is therefore a covariate channel, optional to the system's clinical function; its purpose is to enrich the feature space and to demonstrate contribution C4. The clinical decision does not depend on it.

Stating this at the head of the section forecloses the reading in which the microcontroller appears to be the system's primary acquisition device, which would contradict Section 3.2 and is the first point a reader would challenge.

5.2.1 What the Subsystem Measures, and Why That Is a Limited Claim

Table 5.15: Parameter provenance and clinical status.

Parameter	**Acquired by**	**Enters the clinical decision?**
Systolic, diastolic pressure	validated BLE cuff	yes, primary
Capillary glycaemia	BLE glucometer	yes, primary
Heart rate	cuff pulse reading; or camera PPG; or MAX30100	no, covariate
Oxygen saturation	MAX30100	no, covariate
Body temperature	MAX30205	no, covariate

Three consequences follow and are stated rather than left implicit.

The subsystem is removable. If it is not built, or fails in deployment, the system loses feature columns and loses nothing clinically. The failure table of Table 5.11 already reflects this.

Its validation burden is correspondingly lighter. Section 3.2.3b required Bland–Altman comparison against a validated reference, under movement, for any device carrying a clinical decision. A covariate channel requires that the measurement be reproducible and plausible, not that it be clinically accurate — a weaker and attainable standard.

It is where contribution C4 lives. The low-cost acquisition claim of the research framing document cannot rest on the cuff and the glucometer, which are commercial validated instruments costing what they cost. It rests on this subsystem and on the camera alternative of Section 5.2.6.

A fourth consequence is less comfortable and is stated here because a reader will reach it unaided. An automated oscillometric cuff reports pulse rate alongside blood pressure,

so heart rate is already available at every clinical measurement occasion without any additional instrument. Neither the camera nor the MAX30100 therefore adds a parameter the system lacks; what they add is temporal density, heart rate at moments when no cuff reading is taken. That is a real but modest gain, and C4 is defensible on that basis and on the device-cost argument of Section 5.2.7, not on the claim that these channels are otherwise unobtainable.

5.2.2 Component Specification

Taken from **efendi2025<empty citation>**, who give electrical characteristics and pin lists but no wiring diagram, no GPIO assignment and no register configuration.

Table 5.16: Microcontroller characteristics, after **efendi2025<empty citation>**.

Characteristic	**Value**
Processor	Tensilica Xtensa
ROM	448 kB
SRAM	520 kB
Sleep-mode memory	RTC fast SRAM 8 kB; RTC slow SRAM 8 kB
Pin count	36, comprising EN, GND, VIN, 3.3 V and GPIO
Connectivity	integrated Wi-Fi and Bluetooth
Wireless security	IEEE 802.11, supporting WAPI, WFA, WPA and WPA2

The RTC sleep memory is the characteristic **efendi2025<empty citation>** report and do not use, and Section 5.2.4 shows it to be the one that determines whether the device is deployable at all.

Table 5.17: Pulse-oximetry sensor characteristics, after **efendi2025<empty citation>**.

Characteristic	**Value**
Function	heart rate and oxygen saturation
Principle	reflective dual-wavelength photoplethysmography
Wavelengths	660 nm (red), 880 nm (infrared)
Supply	3.3–5 V optimal
Interface	I^2C
Pins (7)	VIN, SCL, SDA, INT, IRD, RD, GND
INT	interrupt; pulse detection
IRD	infrared LED indicator

One electrical constraint is derivable from Tables 5.17 and 5.18 and is not stated by **efendi2025<empty citation>**. The temperature sensor's supply range tops out at 3.3 V; the pulse-oximetry sensor's optimal range begins at 3.3 V. The two intervals intersect at exactly one point. The bus and both sensors must therefore run at 3.3 V, which is also the microcontroller's logic level — convenient, but it places one sensor at the bottom of its stated range and the other at the top of its. A single regulated 3.3 V

Table 5.18: Temperature sensor characteristics, after **efendi2025<empty citation>**.

Characteristic	**Value**
Function	body temperature
Range	up to 50 °C
Accuracy	0.1 °C
Supply	2.7–3.3 V
Operating current	600 µA optimal
Converter	high-resolution sigma-delta ADC
Interface	I^2C-compatible two-wire
Pins (8)	VCC, GND, SDA, SCL, OS, A0, A1, A2
A0–A2	address selection
Alarm functions	over-temperature alarm, interrupt, shutdown

rail is the only configuration satisfying both, and supplying the oximeter from 5 V while the temperature sensor runs at 3.3 V would impose a level-translation requirement on the shared bus that neither device provides.

5.2.3 Interconnection

5.2.3a The bus and the address constraint

Both sensors are I^2C and both expose SDA and SCL. They therefore share one two-wire bus, which requires distinct slave addresses.

The pin lists resolve this, and the resolution is visible in an asymmetry **efendi2025<empty citation>** report without remarking on it: the temperature sensor has A0, A1 and A2 address-selection pins, and the oximeter has none. The oximeter's address is fixed in silicon; the temperature sensor's is configurable.

The address-selection pins therefore exist to avoid exactly this collision, and must be strapped deliberately rather than left floating. A floating address pin on an I^2C device produces an indeterminate address and intermittent bus failure — which is the class of fault that manifests as missing readings and would be indistinguishable, in the collected data, from a patient who did not measure. That indistinguishability matters beyond the circuit: Section 4.2.6 classified missingness by mechanism, and a hardware fault masquerading as patient non-adherence corrupts that classification at source.

5.2.3b Pin assignment

efendi2025<empty citation> give pin names and no GPIO assignment. The following uses the conventional default I^2C mapping for this microcontroller. It is a design choice, not a specification taken from the source, and is presented as such.

Two notes on the interrupt lines. Using INT rather than polling the oximeter is what makes the duty-cycled operation of Section 5.2.4 possible: the processor sleeps until the

Table 5.19: Proposed GPIO assignment. This table is design, not source material.

Microcontroller	MAX30100	MAX30205	Function
3.3 V	VIN	VCC	supply
GND	GND	GND	reference
GPIO21	SDA	SDA	I^2C data
GPIO22	SCL	SCL	I^2C clock
GPIO19 (input)	INT	—	pulse, data-ready interrupt
GPIO18 (input)	—	OS	over-temperature alarm
—	—	A0, A1, A2	strapped to GND or 3.3 V; must not float

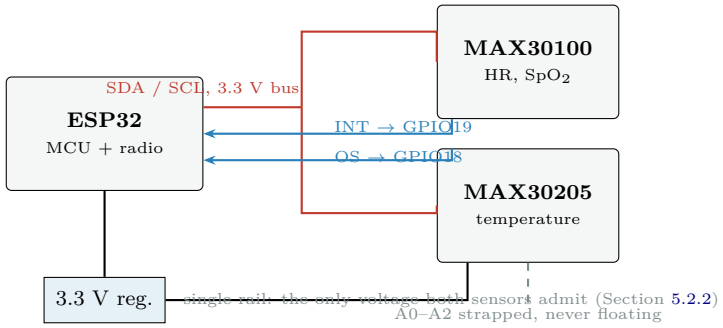


Figure 5.3: Interconnection. One shared I^2C bus on a single 3.3 V rail, with both sensor interrupt lines wired so that the processor need not poll.

sensor signals data ready, instead of waking on a timer and polling. Using OS as a hardware over-temperature alarm yields a threshold crossing that does not require the processor to be awake at all, which is the hardware analogue of the red-tier escalation of Section 5.1.2c.

5.2.3c Datasheet dependencies

The source does not supply the following, which must be obtained from manufacturer documentation.

These are stated as datasheet dependencies rather than filled with invented values. A fabricated resistor value in a thesis is worse than an acknowledged gap, because it will be tested by anyone who builds the circuit.[5]

5.2.4 Power and Duty Cycle

This subsection contains the derivation that determines whether the subsystem is deployable, and it follows from a limitation **efendi2025<empty citation>** state but do not act upon.

[5] **[To verify]** The datasheets for all three components are publicly available from the manufacturer and are not in the present corpus. Obtain them and replace the illustrative currents of Table 5.21 with measured or specified values before the power budget is quoted as a result rather than as an order-of-magnitude argument.

Table 5.20: Values required from datasheets, not available in the corpus.

Missing	Why it matters
I^2C base addresses of both devices	address collision; A0–A2 strapping
Pull-up resistor values for SDA and SCL	bus rise time; reliability at the chosen bus speed
Oximeter register configuration	LED current, pulse width, sample rate, averaging
Oximeter supply and LED currents	the power budget of Section 5.2.4
Microcontroller active and deep-sleep currents	the power budget of Section 5.2.4
Whether the breakout board carries its own regulator and pull-ups	may conflict with the single-rail decision of Section 5.2.2

5.2.4a The limitation they identify

They write that "one of the main limitations is the dependence on battery power, which can limit the duration of continuous monitoring"†. Their own protocol explains why: data collected every second for ten minutes, producing 600 records per participant, with the radio and both sensors continuously active.

5.2.4b The arithmetic

With duty cycle d, active current I_{act}, sleep current I_{slp} and battery capacity C:

$$d = \frac{t_{\text{act}}}{T}, \qquad I_{\text{avg}} = d\, I_{\text{act}} + (1 - d)\, I_{\text{slp}}, \qquad \text{life} = \frac{C}{I_{\text{avg}}}. \tag{5.3}$$

Under the protocol adopted in Section 4.1.1, two measurement occasions per week, with a 60-second active window covering sensor settling, acquisition and transmission:

$$T = 3.5 \text{ days} = 302\,400 \text{ s}, \qquad d = \frac{60}{302\,400} = 1.98 \times 10^{-4}, \tag{5.4}$$

a duty cycle of 0.020%.

Table 5.21: Power regimes. The currents are illustrative, of the right order for this class of device, and are to be replaced by datasheet values.

Regime	I_{avg}	Life on 2 000 mAh
Continuous, as **efendi2025<empty citation>** operate	120 mA	17 h
Duty-cycled at $d = 1.98 \times 10^{-4}$	0.0338 mA	$\approx$ 6.7 years

With $I_{\text{act}} = 120$ mA and $I_{\text{slp}} = 10$ µA, the duty-cycled average is

$$I_{\text{avg}} = \underbrace{1.98 \times 10^{-4} \times 120}_{0.0238\ \text{mA, active}} + \underbrace{0.0100}_{\text{mA, sleep}} = 0.0338\ \text{mA}. \tag{5.5}$$

The improvement is a factor of $120/0.0338 = 3.5 \times 10^3$: between three and four orders of magnitude, obtained not by better hardware but by matching the sampling regime to the clinical protocol. It is worth noting the ceiling: even with a sleep current of exactly zero the factor would be $1/d = 5.0 \times 10^3$, so no amount of sleep-current optimisation reaches four orders at this duty cycle.

5.2.4c Which term dominates, and the design rule that follows

The decomposition above shows that the active term contributes 70% of the average current and the sleep term 30%. The average is therefore governed principally by the active window, not by the sleep floor — which inverts the intuition that a duty cycle of two parts in ten thousand must make the active phase irrelevant.

The crossover is derivable. The two terms are equal when $d\,I_{\text{act}} = I_{\text{slp}}$, that is at an active window of

$$t^{\star} = \frac{I_{\text{slp}}\,T}{I_{\text{act}}} = \frac{0.010 \times 302\,400}{120} \approx 25\ \text{s}. \tag{5.6}$$

The design rule follows directly and is more useful than the lifetime figure. Above a 25-second active window, shortening the window is the effective lever and reducing sleep current buys little; below it, the sleep floor dominates and further shortening buys nothing (Figure 5.4). Since 60 seconds is above the crossover, the first optimisation to attempt is a shorter acquisition — faster sensor settling, a pre-established network association, a smaller payload — and not a lower-leakage sleep mode.

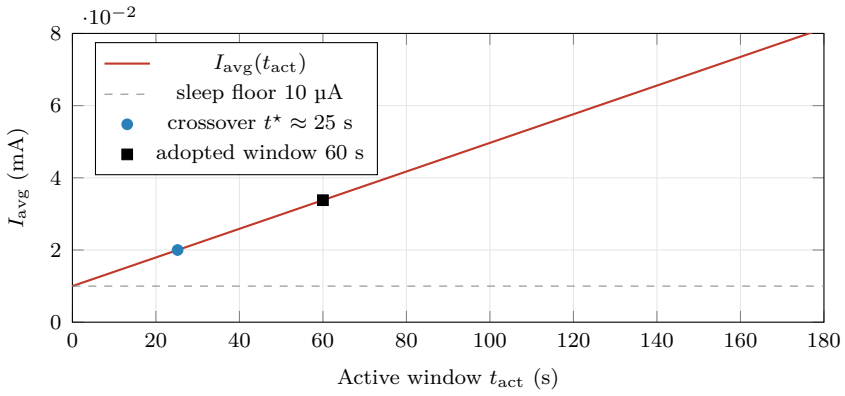


Figure 5.4: Average current against active window. The adopted 60-second window lies above the crossover, so the active term dominates and shortening the window — not lowering sleep current — is the effective optimisation.

5.2.4d The same finding as the alert-volume argument

Section 3.5.4b established that one-hertz sampling makes alerting arithmetically unusable: at a specificity of 0.98, continuous streaming generates of the order of twelve thousand false alerts per patient per week, against approximately 0.06 under a three-readings-per-week protocol.

The battery limitation and the alert-volume impossibility are the same error viewed from two directions. Sampling continuously at a frequency the clinical protocol does not require exhausts the battery and floods the alerting layer. Duty-cycling to the protocol resolves both at once, and Section 5.2.5e shows it resolves a third problem as well. This convergence is stated explicitly because it is the kind of argument that demonstrates the architecture was reasoned rather than assembled.

5.2.4e Implementation

The RTC slow SRAM, 8 kB retained through deep sleep, holds the state needed across cycles: last transmission timestamp, retry counter, queued readings not yet acknowledged, and the tier of the last evaluation. **efendi2025<empty citation>** report that this memory exists; using it is what makes deep sleep practical, since without retained state every wake is a cold start.

The wake schedule should be event-driven where possible and timed where not: wake on the cuff's BLE advertisement, so that the covariate channel samples when the clinical measurement happens rather than on an independent timer firing at unrelated moments. This also guarantees temporal alignment between the covariate and the parameter it contextualises, which Section 4.2.4 identified as a problem when channels are sampled on independent schedules.

5.2.5 MQTT Client Design

5.2.5a What the source specifies

efendi2025<empty citation> establish the split: MQTT upstream from device to gateway, HTTP downstream to clients, with the gateway translating into cloud functions or the realtime database. Their justification is that MQTT "enables resource-constrained devices to send sensor data to a gateway"[†]. Section 3.3.2 accepted the choice and recorded that no source in this corpus quantifies the advantage: no bandwidth, latency, power or reliability comparison against HTTP is reported anywhere in the corpus.

The elements below are design specification, not source material, and are presented as such.

5.2.5b Topic structure

```
telemon/v1/<site>/<patient_pseudonym>/<channel>/<measurement_id>
```

Four properties. The version prefix permits schema migration without breaking deployed devices. The pseudonym, rather than a name or record number, satisfies Section 4.1.5c at the transport layer, which matters because topic names appear in broker logs. The channel segment allows subscription filtering, so that the scoring service subscribes to clinical channels and ignores covariates. And the hierarchy supports wildcard subscription for monitoring.

5.2.5c Quality of service, and why exactly-once is the wrong choice for red

MQTT defines three delivery guarantees, and Section 5.1.2c established that transmission urgency should follow the alert tier. The natural mapping and the one adopted differ, and the difference is worth the paragraph.

Table 5.22: Quality of service by tier.

Tier	QoS	Guarantee	Rationale
Green, routine covariate	0	at most once	loss of one heart-rate sample is immaterial
Amber, clinical reading	1	at least once	must arrive; duplicates removed by `measurement_id`
Red, critical	1	at least once	see below; the real guarantee is the SMS fallback

Assigning QoS 2 to the red tier is the obvious choice and is the wrong one. Exactly-once delivery is obtained by a four-step handshake, two round trips more than at-least-once, and the links on which red-tier delay is actually harmful are precisely the degraded links where those extra round trips are slowest and most likely to fail. Meanwhile the property QoS 2 provides, suppression of duplicates, is already provided at the application layer: Section 4.1.2 makes `measurement_id` a UUID, the repository deduplicates on it, and Section 4.2.2 flags duplicates rather than silently discarding them. QoS 2 therefore purchases, at the cost of latency, a guarantee the system already holds.

The resolution is QoS 1 with application-layer idempotency for both amber and red, with urgency expressed through retry policy and the SMS fallback of Section 5.1.2c rather than through the delivery mode. This is a correction to the intuitive mapping and is stated as such.

5.2.5d Last Will and Testament, and the non-transmission alarm

MQTT permits a client to register, at connection time, a message the broker will publish on its behalf should the connection drop without a clean disconnect.

This is the transport-layer implementation of the requirement that silence be alarmed. Section 3.5.2 established, from the repeated SMS reminder of **yazdani2018<empty citation>** and the HEARTS retention indicator, that non-transmission must itself trigger an action; Table 5.11 placed battery exhaustion, device loss and handset unavailability in the failure table with the non-transmission alarm as the detection mechanism. The Last Will supplies that signal directly: the broker announces the disconnection, and the Cloud escalates according to how long the device has been absent relative to its configured checkup period, which is the configurable parameter of **yazdani2018<empty citation>**.

The distinction between a graceful disconnect, the device sleeping as designed, and an ungraceful one is exactly the distinction between normal duty cycling and failure, and the protocol supplies it at no cost. It is also the one detection that Section 5.1.4 identified as impossible at the Edge — a failed device cannot report its own failure — so locating it in the broker rather than in either endpoint is not merely convenient but necessary.

5.2.5e Payload and the data-cost derivation

The payload is JSON, per Section 3.3.3a, carrying the fields of Section 4.1.2. A record is of the order of 300 bytes. A measurement occasion stores three records: the retained blood-pressure reading, the glycaemia reading, and the covariate bundle — approximately 900 bytes per occasion.

Table 5.23: Transmitted volume by regime, at 300 bytes per record.

Regime	**Per patient, per month**	**30 patients, per month**
Continuous 1 Hz, as **efendi2025<empty citation>** operate	778 MB	23 GB
Duty-cycled, 2 occasions per week	7.7 kB	231 kB

The derivation, stated so that the two columns can be checked against each other. Continuous: $86\,400 \times 300$ B $= 25.9$ MB per patient per day, $\times 30 = 778$ MB per month, $\times 30$ patients $= 23.3$ GB. Duty-cycled: 2×900 B $= 1.8$ kB per week, $\times 4.29 = 7.7$ kB per patient per month, $\times 30 = 231$ kB. The ratio is 1.0×10^5: five orders of magnitude.

On Cameroonian mobile data tariffs this is the difference between a recurring cost borne by the patient and a cost that rounds to zero. This is the third convergence on the same design decision, and the one tied most directly to Chapter 2: Section 3.3.3d recorded that no source in the corpus states what any of these architectures costs to operate, and Section 2.2.4 documented a service ceasing operations, for financial reasons, after reaching approximately 2.5 million registered users. A system consuming 23 GB per month for thirty patients does not scale; one consuming 231 kB does. Duty-cycling is a sustainability decision as much as a power decision.

5.2.5f Security

Per segment 3 of Table 5.12: MQTT over TLS, client authentication by per-device credential, and broker-side access control restricting each device to its own topic subtree. **efendi2025<empty citation>** report that the microcontroller supports IEEE 802.11 with WPA and WPA2, which protects the wireless link but not the application-layer connection; the two are distinct and both are required. Reporting the former as though it addressed the latter is a smaller instance of the pattern identified in Section 3.3.5b.

5.2.6 The Camera-PPG Alternative

5.2.6a What it may and may not measure

Section 2.4.2 established the position, and it is not reopened here.

Table 5.24: Admissibility of camera photoplethysmography by parameter.

Parameter	Camera PPG	Basis
Heart rate	admissible	periodicity requires no absolute calibration (Section 3.2.2a)
Oxygen saturation	not admissible	requires controlled dual-wavelength illumination and empirical calibration (Section 3.2.2a)
Blood pressure	not admissible	**who2023hypertension<empty citation>**; **alessa2019<empty citation>**: "highly inaccurate and unfeasible"†

The camera alternative is therefore a substitute for one function of the instrumented subsystem, heart rate, and for neither clinical parameter. Contribution C4 is defensible on that basis and on no wider one.

5.2.6b Signal chain

Proposed design; no source in this corpus validates it.

1. **Acquisition.** Fingertip over the rear camera, LED flash as illumination, 30 fps, fixed exposure and white balance. Automatic exposure compensation is the dominant artefact source and must be disabled, since the control loop responds to the very intensity variation that constitutes the signal.
2. **Reduction.** Mean intensity of the red channel over a central region of interest, one value per frame, giving a 30 Hz series. Nyquist is 15 Hz, or 900 beats per minute, far above the physiological band.
3. **Detrend.** Remove the slow baseline, the DC term of Section 3.2.2a, by moving-average subtraction or first differencing.
4. **Bandpass.** 0.7–3.5 Hz, corresponding to 42–210 beats per minute, which covers the physiological range with margin at both ends.

5. **Rate estimation.** Dominant frequency by autocorrelation or spectral peak, cross-checked against inter-peak intervals in the time domain. Agreement between the two estimators is itself a quality signal.
6. **Aggregation.** Median over a 30-second acquisition, with the interquartile range retained as an uncertainty measure.

The passband is chosen on physiological grounds rather than from the reference ranges of **efendi2025<empty citation>**, because Section 3.2.3c established that those ranges contradict one another between two of their tables and cannot be used to set a filter boundary.

5.2.6c Quality gating

A camera reading that is not gated is worse than no reading, because it enters the series indistinguishable from a good one. Three gates, each cheap:

Table 5.25: Quality gates for camera acquisition.

Gate	Test	On failure
Contact	DC level within an expected band	reject; prompt re-acquisition
Motion	frame-to-frame variance above threshold	reject
Periodicity	spectral peak prominence below threshold	reject

A rejected acquisition sets the mask of Section 4.2.7 rather than producing a value, which is the behaviour Section 4.2.6 requires: absence is reported as absence. The rejection rate is itself reported, per Section 5.2.6d.

5.2.6d The validation study that comes free

Section 3.2.3b required Bland–Altman comparison against a validated reference. For heart rate that reference already exists within the system at no additional cost: an automated oscillometric cuff reports pulse rate alongside blood pressure, and under the protocol of Section 4.1.1 a cuff reading is taken twice weekly.

Pairing a camera acquisition with every cuff reading therefore yields, over thirty patients across a collection span of approximately five weeks, of the order of $30 \times 2 \times 5 = 300$ paired observations.

One methodological caution attaches to that count and must be respected in the analysis. Those 300 observations come from thirty patients, approximately ten per patient, so they are not independent: a systematic per-patient offset, from skin tone, perfusion or technique, inflates agreement if the repetitions are treated as independent samples. The limits of agreement must therefore be computed by the variant of the Bland–Altman method for repeated measurements, which partitions within-subject from between-subject

variance, and the effective sample size for the between-subject component is thirty, not 300.

Table 5.26: What the method-comparison sub-study reports.

Statistic	**Why**
Bland–Altman bias and limits of agreement, repeated-measures variant	separates systematic offset from random disagreement, without treating repetitions as independent
Proportion within ±5 and ±10 bpm	clinically interpretable
Rejection rate by gate	measures usability, not only accuracy
Between- and within-subject variance components	identifies whether disagreement is per-patient or per-reading
Not MAPE alone	Section 3.2.3c: averages away the structure of the disagreement, and its denominator is ambiguous

This converts the weakest element of contribution C4, an unvalidated acquisition method, into a self-contained method-comparison sub-study with a cost-free reference. It is the strongest available answer to the objection that camera photoplethysmography is unproven, and it must be planned into the protocol rather than attempted afterwards — because the pairing requires the two acquisitions to be simultaneous, which cannot be reconstructed retrospectively.

5.2.7 The Three Options, Compared

Section 3.2.6b adopted Position A and left the hardware question formally open. The analysis above closes it on cost and risk rather than on preference.

Option C is adopted. Its relationship to the earlier decision should be stated plainly, because it looks like a reversal and is not: the *deployed* configuration under C is identical to Position A, and C differs only in adding a bench-evaluated prototype that no patient receives. The decision of Section 3.2.6b is therefore unchanged for the cohort; what C adds is an artefact and a set of measurements, not a change to what patients use.

The reasoning, in the form it should take in the defence:

> The instrumented subsystem is specified and prototyped in order to demonstrate the hardware design and the duty-cycled MQTT transport, and is evaluated on the bench. The deployed cohort uses spare acquisition — validated Bluetooth cuff and glucometer, with camera photoplethysmography for heart rate — because the covariate channels the hardware would add do not enter the clinical decision, because the per-patient device cost would be incurred for no clinical gain, and because the camera alternative admits a cost-free method-comparison study against the cuff's own pulse reading. The hardware path

Table 5.27: Acquisition options compared.

	A: spare	**B: instrumented**	**C: staged hybrid**
BP and glucose	validated BLE devices	validated BLE devices	validated BLE devices
Heart rate	cuff pulse, camera PPG	MAX30100	cuff pulse, camera PPG
SpO_2, temperature	—	MAX30100, MAX30205	bench only
Hardware build	none	full, 30 units	prototype, not deployed
Device cost per patient	zero above the cuff	MCU, sensors, battery, enclosure	zero deployed
Validation burden	camera against cuff, free	Bland–Altman, both sensors, under movement	camera only
Power design required	no	yes, Section 5.2.4	bench only
C4 demonstrated by	camera PPG, manual entry	the hardware	both, staged
Clinical risk on failure	none, covariates only	none, covariates only	none

is documented as the extension route for a deployment in which continuous covariate monitoring is clinically indicated.

That position preserves contribution C4, preserves the hardware work as a contribution, and does not require thirty devices to be built and maintained by one researcher.

5.2.8 Specification

5.2.9 Limits of What This Section Establishes

Three elements of this section are design proposals rather than established results, and are labelled as such wherever they are used.

The circuit is incomplete. efendi2025<empty citation> give pin names and electrical characteristics; they give no wiring diagram, no GPIO assignment, no pull-up values, no decoupling, no power-supply design and no register configuration. Table 5.19 and Figure 5.3 are design proposals built from their pin lists together with the conventional default I^2C mapping, and the remaining values come from datasheets outside this corpus (Table 5.20).

Camera photoplethysmography is unvalidated within this corpus. No source provides accuracy, precision or failure rate for camera-based heart rate. The signal chain of Section 5.2.6b is a proposal; Section 5.2.6d is how the evidence is to be obtained; and until that study is run the claim is stated as untested rather than as supported.

The protocol choice is unquantified. Section 3.3.2 recorded that **efendi2025<empty citation>** assert the advantage of MQTT over HTTP and measure nothing, and that position is

Table 5.28: Requirements established in Section 5.2.

Requirement	**Basis**
Subsystem acquires covariates only; scoping statement opens the section	Sections 3.2.5 and 5.2
Redundancy of camera heart rate with cuff pulse acknowledged	Section 5.2.1
Single regulated 3.3 V rail for both sensors and the bus	Section 5.2.2, derived from the two supply ranges
Shared I^2C bus; A0–A2 strapped, never floating	Section 5.2.3
INT and OS used as interrupts, not polled	Section 5.2.3b
GPIO assignment presented as design, not as sourced	Section 5.2.3b
Datasheet dependencies listed; values not invented	Table 5.20
Duty-cycled operation matched to the clinical protocol	Section 5.2.4b
Active window identified as the dominant term; $t^\star \approx 25$ s	Section 5.2.4c
RTC slow SRAM used for cross-cycle state	**efendi2025<empty citation>**; Section 5.2.4
Wake on the cuff's BLE advertisement, not an independent timer	Section 5.2.4
Power budget recomputed with datasheet currents before quoting	Table 5.21
MQTT topic versioned; pseudonym only, never an identifier	Sections 4.1.5c and 5.2.5
QoS 0 for green, QoS 1 for amber and red; QoS 2 rejected with reasons	Section 5.2.5c
Idempotency on `measurement_id`	Section 4.1.2
Last Will registered; drives the non-transmission alarm	Sections 3.5.2 and 5.1.4
JSON payload; data volume computed, with the derivation shown	Section 5.2.5e
MQTT over TLS; per-device credential; topic-scoped broker ACL	Table 5.12
Camera PPG restricted to heart rate only	Section 2.4.2
Automatic exposure disabled during acquisition	Section 5.2.6b
Passband set physiologically, not from **efendi2025<empty citation>** ranges	Sections 3.2.3c and 5.2.6b
Three quality gates; rejection sets the mask, not a value	Sections 4.2.7 and 5.2.6
Camera PPG validated against cuff pulse, repeated-measures Bland–Altman	Section 5.2.6d
Rejection rate and variance components reported alongside agreement	Section 5.2.6d
Option C adopted; its identity with Position A when deployed stated	Sections 3.2.6b and 5.2.7

unchanged. No efficiency benefit is claimed in this work on the strength of the corpus.[6]

Two document dependencies remain outstanding from Chapter 4 and are unchanged: the guideline editions required by Table 4.43, and the Cameroonian legal instrument required by Section 5.1.6b.

5.3 Cloud Backend Infrastructure and Real-Time Management

Purpose and sources

Section 5.1.1d specified the Cloud layer's units; Section 5.2 specified what arrives at it. This section specifies the backend itself: how the record is stored, who may read and write it, what fires when data arrive, and how changes reach the clinician.

Two sources are used. **efendi2025<empty citation>** supply the only implemented backend in the corpus, including a published live schema, the protocol split and the offline synchronisation behaviour; the reservations of Section 3.2.3c stand, and their schema in particular is used as a counter-example rather than as a template. **yazdani2018<empty citation>** supply the functional decomposition — five modules over a three-layer structure — which Section 5.3.7 shows maps cleanly onto a serverless realisation despite having been designed for a client–server architecture.

5.3.1 The Property That Governs Everything

efendi2025<empty citation> describe their platform as "a cloud-hosted database provided by Google"† supporting "various protocols including HTTP and MQTT"†, in which "data transmitted by IoT devices will be stored as JSON which is then accessed by users in real-time"†, and enumerate four advantages: integrated features, zero maintenance, connection to the wider cloud platform, and code privacy.

One architectural property underlies all of it and determines the design: the realtime database is a single JSON tree, and a read of any node returns that node together with everything beneath it.

Three consequences follow, and the schema of Section 5.3.3 exists to satisfy them.

Depth and fan-out are performance decisions, not organisational ones. A node that accumulates children indefinitely becomes unreadable, because every read of its parent downloads the whole accumulation. In a longitudinal system this is not a corner case: history grows monotonically by design.

[6] [**To verify**] A measurement of the MQTT/HTTP comparison is within reach and would convert an inherited assertion into an original result: both protocols are available on the same hardware, the comparison is bytes on the wire, round trips and energy per transmitted record, and the measurement is a few hours of bench work. If it is run, it belongs in **??** and this paragraph should be rewritten to cite it.

Query capability is deliberately narrow. The tree supports ordering by key, by child value or by value, combined with range and limit operators — but one ordering per query and no compound conditions. Anything the system must query by, the key must encode.

Denormalisation is mandatory. Data are duplicated into whatever shapes the readers require, and consistency is maintained at write time rather than by joining at read time.

5.3.2 Why the Published Schema Cannot Be Used

efendi2025<empty citation> publish their live database, and it is worth reproducing because it states the problem precisely:

```
<endpoint>.asia-southeast1.firebasedatabase.app/
  |- heart-rate:  91
  |- oxygen-saturation:  97
  `- temperature:  37
```

Three keys at the root, each holding a scalar.

This structure stores the present and discards the past. Section 3.3.3a established the objection and it is not reopened: the research framing document specify rolling aggregates, trends, variability and threshold deviations, and Section 4.3 made the fourteen-day window the unit of feature construction. A structure holding one value per channel cannot answer a window query, because the window does not exist within it.

Two further defects follow from the same shape. There is no patient dimension: the tree is single-tenant, and admitting a second patient would require restructuring every path. And there is no distinction between measurement time and record time, which Sections 3.2.1 and 3.3.3b established as mandatory once offline synchronisation decouples them.

Their own system exhibits the consequence. They report that the heart-rate graph "will update every 5 min"†, and their application displays a current value with a short live trace. That is all the schema supports. It is adequate for a live monitor and inadequate for a longitudinal record, which is what this system is. The distinction is worth stating in these terms rather than as a criticism: their schema is correct for their stated purpose, and the purposes differ.

Table 5.29: Schema design rules and their justification.

Rule	Reason
A separate top-level tree per access pattern	a read returns the whole subtree (Section 5.3.1)
The key encodes what is queried by	one ordering per query, no compound conditions
Time series keyed by measurement timestamp, at millisecond resolution	range query by key; window construction (Section 4.3)
No unbounded fan-out under a node that is read whole	read cost grows with history
Duplicate freely; maintain consistency at write time	denormalisation is the model
Composite keys zero-padded to fixed width	keys sort lexicographically (Section 5.3.3c)
Nothing identifying anywhere in the tree	Section 4.1.5c

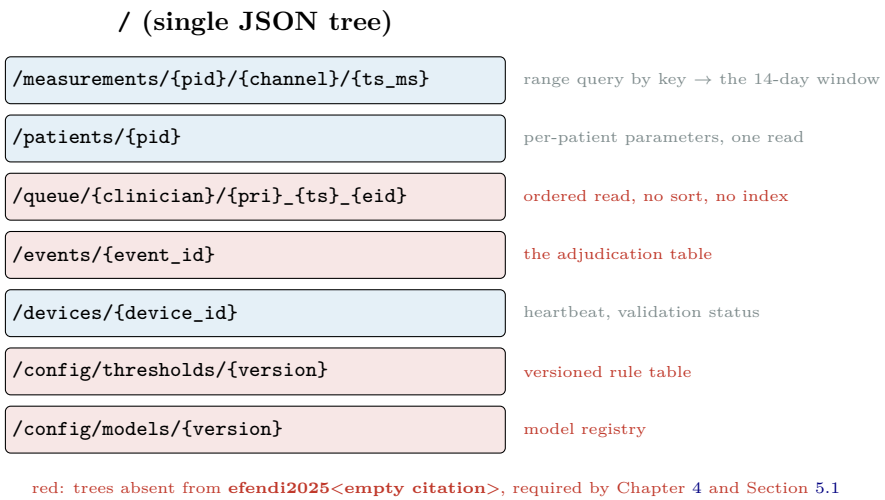


Figure 5.5: The seven top-level trees. Each exists to serve one access pattern, because a read returns an entire subtree.

Table 5.30: Tree contents.

Path	Fields
/measurements/{pid}/{channel}/{ts_ms}	value, unit, ts_recorded, reading_index, is_retained, context, device_id, operator, entry_mode, protocol_profile, quality_flag, measurement_id, supersedes
/patients/{pid}	age_band, sex, bmi, dx_htn_year, dx_dm_year, dm_type, comorbidities[], regimen[], regimen_version, enrolment_date_shifted, checkup_period_days, assigned_clinician, threshold_version, last_seen_ts
/queue/{clinician_id}/{pri}_{ts}_{eid}	pid, tier, score, trigger_source, summary
/events/{event_id}	pid, ts_triggered, trigger_source, tier, score, model_version, threshold_version, ts_reviewed, clinician_verdict, action_taken, comment
/devices/{device_id}	make, model, serial_hash, validated, calibration_date, assigned_pid, last_heartbeat_ts
/config/thresholds/{version}	channel, context, band_label, lower, upper, unit, source, source_version, effective_from
/config/models/{version}	algorithm, trained_at, feature_set_hash, calibration_params, tau_amber, tau_red, metrics

5.3.3 The Schema

5.3.3a Design rules

5.3.3b The seven trees

Four of these trees do not appear in **efendi2025<empty citation>** and are required by earlier chapters. `/events` is the adjudication table of Section 4.1.2e, the only route to the clinician-adjudicated labels of Section 4.4.2b. `/config/thresholds` is the versioned rule table of Sections 3.5.6 and 4.3.3b. `/config/models` is the registry of Section 5.1.1d, without which a score cannot be attributed to the model that produced it. And `/queue` is the priority ordering of **yazdani2018<empty citation>**, which Section 5.3.3c shows the key design delivers directly.

yazdani2018<empty citation> specify that the database holds "physiological data… along with the physician's prescriptions, diagnosis and recommendations of decision support system"†, and that "all events in this telehealth monitoring architecture, including physician and CDSS prescriptions, are stored as patient record"†. The `/events` tree is that requirement, with the addition that each entry carries the model and threshold versions that produced it — an addition Section 4.4.9 makes necessary, since scores computed under provisional thresholds must remain identifiable once the definitive table is installed.

5.3.3c The key design, and a correction to the composite key

The window query. With the measurement timestamp as the key, a fourteen-day window is a key range:

```
/measurements/{pid}/sbp .orderByKey() .startAt(t_minus_14d) .endAt(t)
```

This returns exactly the window of Section 4.3 and nothing else. Under the schema of Section 5.3.2 it is not expressible at all; under a schema keyed by an opaque push identifier it would require ordering by a child value, which is slower and requires an index.

The queue query. The intention is that the key be composed so that lexicographic ordering yields a list sorted by priority and then by age within priority, making the read

```
/queue/{clinician_id} .orderByKey() .limitToFirst(20)
```

This works, but only under a constraint the draft design omits, and omitting it produces a queue that is silently mis-ordered rather than one that fails visibly.

Keys sort as strings. An unpadded numeric priority therefore sorts in dictionary order, not numeric order, and a timestamp of varying digit width does the same within a priority band.

Table 5.31: Lexicographic ordering of composite keys, unpadded and padded.

Unpadded, as sorted	Padded, as sorted	Correct?
`1_1760_a`	`p1_0000001760_a`	red, older
`10_1759_d`	`p1_0000009817_b`	red, newer
`2_1759_b`	`p2_0000001759_c`	amber
`2_9817_c`	`p3_0000001759_d`	green

In the unpadded column a priority-10 item sorts above every priority-2 item, and within priority 2 the ordering by timestamp happens to be correct only because both values have the same digit count. The defect is invisible in testing with a small, homogeneous set of keys and appears in deployment as a queue whose head is not the most urgent item — which defeats the entire purpose of Section 3.5.4, where the ordering was established as the response to alert volume.

The corrected specification: priority is a fixed-width alphanumeric token with the urgent tier sorting first (`p1` red, `p2` amber, `p3` green), the timestamp is epoch milliseconds zero-padded to fixed width, and the event identifier breaks remaining ties. The non-numeric prefix is also required for a second reason: a key consisting only of digits may be treated as an array index rather than an object key, which changes the ordering semantics.

Under that correction, the priority ordering required by Section 3.5.4 is obtained for free from the key, with no sort, no index and no secondary query. This is the kind of result that falls out only if the access pattern is designed before the schema, which is why Table 5.29 places that rule first.

5.3.3d Why timestamp keys also solve the offline merge

efendi2025<empty citation> describe the offline behaviour: data are "stored locally and real-time events continue to occur even when the device is offline"†, and are "synchronized between local data change remotely when the device is offline and auto merging when the device is going to be online"†.

Automatic merge resolves conflicts leaf by leaf, last write winning. For mutable state that is a hazard. For an append-only series keyed by measurement timestamp it is harmless, because every key is written exactly once by exactly one writer, so no two writers ever contend for the same leaf.

That guarantee depends on the key being collision-free, which is why Table 5.29 specifies millisecond resolution. At second resolution two legitimate records can share a key — a nurse entering a reading twice in the same second, or a device batching — and the append-only rule of Section 5.3.4 would then *reject the second record as a duplicate*. The interaction between two otherwise sound rules would thus discard real data silently, and the resolution is resolution: a key fine enough that collision does not occur, with `measurement_id` available as the disambiguator if it ever does.

The timestamp key is therefore correct for three independent reasons — the range query, the queue ordering, and conflict-free offline merge — which is a reasonable indication that it is the right structure rather than merely a convenient one.

5.3.4 Authentication and Authorisation

5.3.4a Two distinct mechanisms

efendi2025<empty citation> report that the platform offers "built-in authentication and authorization mechanisms, ensuring secure access control for IoT devices and users"†, and that enabling "secure access directly to the database from client-side"† supports collaborative applications.

These are two mechanisms, and conflating them is a common error. Authentication establishes identity and issues a token. Authorisation is enforced by server-side rules evaluated on every read and write. Since clients address the database directly, the rules constitute the entire access-control layer: there is no application server between client and data in which to place policy. Table 5.12 placed this at segment 4.

5.3.4b Roles belong in the token, not in the database

The role — patient, nurse, physician, administrator — must be carried as a custom claim on the authentication token, not as a field in the user's database record.

Two reasons. A database field must itself be protected by a rule, and a rule that reads a role from the database in order to decide whether that role may be read is circular. And

a role stored as data is a role the client may attempt to write; a role minted server-side into a signed token is not.

One consequence must be recorded. Token claims are cached until the token refreshes, so a role revocation does not take effect immediately but at the next refresh. This bounds how quickly access can be withdrawn and is the concrete reason Section 5.1.5 required short session expiry: the expiry interval *is* the revocation latency, and choosing it long for convenience lengthens the window in which a withdrawn clinician retains access.

yazdani2018<empty citation> identify four user types — physician, healthcare team, patient, system administrator — which maps onto four claims. This system adds the constraint of Table 5.5: a nurse may measure and view but may not modify therapy.

5.3.4c Rule structure, and the cascade

Table 5.32: Access matrix. Rules grant; they cannot revoke (see below).

Path	Patient	Nurse	Physician	Device
`/measurements/{own}`	read; append-only write	read, write if assigned	read if assigned	write own channel
`/measurements/{other}`	none	if assigned	if assigned	none
`/patients/{own}`	read, no write	read, limited write	read, write regimen	none
`/queue/{clinician}`	none	none	own queue only	none
`/events`	none	read if assigned	read, write verdict	none
`/config/*`	none	read	read	read thresholds
`/devices`	own device	read assigned	read	own heartbeat

One property of the rule language governs how this table must be implemented and is a frequent source of error. Rules cascade downward and are monotone: a read granted at an ancestor path cannot be narrowed at a descendant. A permissive rule high in the tree therefore silently overrides every restriction beneath it, and Table 5.32 is implementable only because no row grants access at a path above another row that needs less. This must be verified when the rules are written, not assumed from the table, since the table reads as though each row were independent and the rule language does not treat them so.

Three constraints deserve emphasis.

Measurement writes are append-only. A rule denying writes where the key already exists makes the clinical record immutable, which is what a clinical record must be.

Immutability requires an amendment path. An immutable record with no way to correct an entry error is unusable — a transposed reading entered as 1200 instead of 120 would be permanent — so correction proceeds as it does in clinical practice: a new record superseding the old by a `supersedes` pointer, with both retained and the original flagged rather than erased. The feature pipeline of Section 4.2 reads only non-superseded

records; the audit trail retains both. This is why `supersedes` appears in Table 5.30 and did not appear in the draft schema.

Only a physician may write `clinician_verdict`, because Section 3.5.5e establishes that the system proposes and the physician disposes, and the adjudication table is the record of that division. And no client writes `/config`: threshold and model versions are deployment artefacts, not user data.

5.3.4d Write-time validation

Section 4.1.6 required a set of controls applied at write time. The rule language supports validation expressions alongside access conditions, and this is where that requirement becomes executable rather than aspirational.

Table 5.33: Write-time validation predicates.

Validation	Enforces
Required fields present	schema conformity
`value` within the plausible range of Section 4.1.2	range check
`unit` matches the canonical unit for the channel	unit-error prevention
Key parses as a timestamp, not in the future	temporal coherence
Key does not already exist	append-only
For `sbp`, `dbp`, `glucose`: referenced device has `validated = true`	the WHO restriction, enforced in the database

The last row is the one worth drawing attention to. The restriction to validated upper-arm cuffs, established in Section 3.2.5 and carried through four chapters, reduces to a single server-side predicate that makes a non-conforming write impossible. A constraint enforced by the database cannot be forgotten by a developer, which is a materially stronger guarantee than a constraint enforced by convention — and it is the only point in the system where a clinical guideline becomes a mechanical impossibility rather than a documented intention.

5.3.5 Cloud Functions: The Event-Driven Pipeline

efendi2025<empty citation> mention cloud functions as one of the facilities their gateway writes into, alongside the realtime database, but do not use them. This system does, and the decomposition follows the module structure of **yazdani2018<empty citation>**.

Table 5.34: Event-driven functions.

Trigger	Fires on	Action
`onMeasurementWrite`	create under `/measurements`	re-validate; set `quality_flag`; recompute the rule tier server-side; record any discrepancy with the Edge tier
`onScoringDue`	a measurement completing a window	build features (Section 4.3); score; prior-correct and calibrate (Section 4.4.5); map to tier; apply max(rule, model) (Section 4.4.6d)
`onTierEscalation`	tier amber or red	write `/events`; fan out to `/queue/{clinician}` with the composed key of Section 5.3.3c
`onRedAlert`	tier red	push notification; SMS fallback after n failed deliveries (Section 5.1.2c)
`onVerdictWrite`	`clinician_verdict` set	close the event; append to the adjudicated label store (Section 4.4.2b); notify the patient of the outcome
`onDeviceWillMessage`	MQTT Last Will (Section 5.2.5)	mark the device offline; start the non-transmission clock

5.3.5a Triggers

The first deserves a note. The rule tier is computed twice: at the Edge, to set transmission priority (Section 5.1.2c), and again in the Cloud. The second computation is not redundant, because it uses the authoritative threshold version, and a discrepancy between the two indicates a device running a stale configuration. Recording the discrepancy makes configuration drift detectable, which is otherwise a silent failure — and a silent failure of exactly the kind Section 5.1.4 identified as the dangerous class, since a device tiering by superseded thresholds produces plausible output that is wrong.

Under the fourteen-day window and the twice-weekly protocol, every measurement completes a rolling window, so `onScoringDue` fires on essentially every arrival. The separation from `onMeasurementWrite` is nonetheless kept, because the two have different failure consequences: a failed validation must block ingestion, whereas a failed scoring must not, the rule floor of Section 4.4.6d being sufficient on its own.

5.3.5b Two scheduled sweeps

Neither is event-driven, and both implement requirements from earlier chapters that no source implements.

Non-transmission sweep. For each patient, compare the time since the last measurement against `checkup_period_days` — the configurable parameter of **yazdani2018<empty citation>** which they define as the intervals "in which patient data should be collected and sent are determined by medical staff"†. On exceedance, raise an event with `trigger_source = non_transmission`. This is the requirement of Section 3.5.2 that silence be alarmed, and the repeated SMS reminder of **yazdani2018<empty citation>** is its patient-facing half.

Queue-age sweep. For each unreviewed event, compare its age against a tier-dependent limit and escalate to a named alternate on exceedance. Section 5.1.4 identified

clinician unavailability as the failure with unbounded consequence, and Section 5.1.3 established both the measured expert-opinion delay and the right-skew of its distribution. The skew is the argument for the sweep: in a right-skewed delay distribution the long waits are a stable fraction of the caseload rather than rare accidents, so without this sweep the tail is the system's expected behaviour for some patients every month.

5.3.5c Idempotency

Serverless triggers may fire more than once for a single event, and Section 5.2.5c assigns at-least-once delivery to clinical readings, which permits duplicates by construction. Every function must therefore be safe to re-run: keyed writes rather than appends, `measurement_id` as the deduplication key, and event creation conditional on absence.

Without this, a retried trigger produces a duplicate alert — which, given the arithmetic of Section 3.5.4b, is the one failure mode this system can least afford, since the entire alerting design rests on keeping the weekly count near three.

5.3.6 Synchronisation and Notification

5.3.6a Who should hold a persistent connection

efendi2025<empty citation> describe the mechanism: "Firebase provides Realtime Database and designed for real-time updates. Whenever current sensor data arrives, Firebase sends it to linked devices via persistent WebSocket connections. This enables clients to examine the health vitals virtually immediately."†

The capability is real and Section 3.3.2 accepted it. The design question is which clients should hold one, and the answer is not all of them.

Table 5.35: Connection policy by client.

Client	Persistent?	Reason
Clinician dashboard	yes	the queue must update live; this is a t_5 intervention (Section 5.1.3)
Patient application	no	connect, synchronise, disconnect
Device or gateway	no	MQTT carries the upstream; no subscription required

Under the protocol of Section 4.1.1 the patient application transmits twice weekly. Holding an always-open socket for a client with something to say for sixty seconds a week consumes a connection continuously to serve a duty cycle of 2×10^{-4}, which is the argument of Section 5.2.4 applied to the network rather than to the battery.

5.3.6b Push, and the tier mapping

Notification is the delivery mechanism, and it follows the tier exactly as transmission priority does.

Table 5.36: Notification by tier and recipient.

Tier	Patient	Clinician
Green	silent; visible in the application	not notified; appears in the queue
Amber	prompt for the verification profile	queued at normal priority
Red	immediate high-priority push; local on-device warning regardless of connectivity (Section 5.1.2c)	immediate push; SMS fallback

The Notification module of **yazdani2018<empty citation>** has exactly two functions, patient notification and physician notification, and Table 5.36 is that module parameterised by tier.

Two safety constraints from Section 3.5.5e apply to the content. A patient notification may say that a reading is high and that the patient should contact their health centre; it may not convey a dose change, a medication change, or any instruction to stop a treatment. And a clinician notification carries the proposal, never an executed action.

5.3.6c Concurrent connections are the dominant cost term

Section 3.3.3d recorded that no source in this corpus states operating cost, and Section 2.2.4 documented a service ceasing operations for financial reasons. The realtime database bills on three axes — stored volume, downloaded volume, and concurrent connections — and the third is the one a careless design inflates.

The saving must be computed on the right quantity. The billed figure is *concurrent* connections, a peak, not a time average. Thirty clinician-style persistent sockets is a peak of thirty held continuously. Thirty patient clients each connecting twice weekly for sixty seconds gives a time-average of $30 \times 1.98 \times 10^{-4} \approx 0.006$ connections, but the peak depends entirely on whether those connections coincide: randomly distributed, the peak is one or two, and the saving is between one and two orders of magnitude.

This yields a design requirement that is invisible unless the peak is computed rather than the average. If the application prompts every patient to measure at the same hour — which is the natural implementation of a measurement reminder, and the one **yazdani2018<empty citation>** describe — then all thirty synchronise together, the peak returns to thirty, and the entire saving is lost. Measurement prompts must therefore be jittered across a window, which couples the reminder schedule to the cost model. Matching system behaviour to the clinical protocol produces the saving; matching it carelessly gives it back.

This is the third place in this chapter where that matching produces a large reduction, after battery (Section 5.2.4) and transmitted volume (Section 5.2.5e).

5.3.7 The Five Modules, Realised

The architecture of **yazdani2018<empty citation>** is a client–server design in XML. Its functional decomposition survives translation to a serverless stack essentially unchanged, which is worth stating: the module boundaries were drawn on functional grounds and are therefore portable, while the technology was not.

Table 5.37: The modules of **yazdani2018<empty citation>** and their realisation here.

Module	Components specified	Realisation here
Membership management	registration; checkup period; rule selection	`/patients` with `checkup_period_days` and `threshold_version`
Communication transmission	deliver data; ensure device connection	MQTT ingest (Section 5.2.5); `/devices` heartbeat; Last Will
Supervisor	XML parser; decision-support detection	`onMeasurementWrite`, `onScoringDue`; routing by channel and assigned rule version
Decision-support setting	add system; update rules	`/config/thresholds/{version}`, administrator-write only
Notification	patient; physician	`onTierEscalation`, `onRedAlert`, `onVerdictWrite`

Two of their elements are more valuable than their age suggests.

Checkup period as a per-patient parameter. They define it as the interval at which data must be collected and sent, set by medical staff. Section 3.1.2a established the WHO follow-up requirement: monthly after initiation or change until target, three- to six-monthly once controlled. A per-patient configurable interval makes that requirement a setting rather than a constant, and it is the input to the non-transmission sweep of Section 5.3.5b.

Rule selection per patient. They attach a decision-support system and its rules to a patient according to disease and symptoms. For a system covering two pathologies with two guideline sources, this is what allows a hypertensive patient, a diabetic patient and a comorbid patient to be managed on one platform under different rule sets — the scalability argument of Section 3.5.3, implemented as a reference from `/patients` into `/config/thresholds`. Given that Section 2.1.3 put the comorbidity at 58.1% of diabetic patients, the comorbid case is not an edge case and the per-patient reference is not an optional refinement.

5.3.8 Cost, Portability and Residency

5.3.8a Cost structure

The billing axes are named above; the figures must come from current provider documentation and are not asserted here. What the design controls:

Table 5.38: Cost drivers and the design decisions that control them.

Axis	Driver	Controlled by
Stored volume	measurement count × record size	duty cycling (Section 5.2.4); 231 kB per month for 30 patients (Table 5.23)
Downloaded volume	window reads per dashboard session	range queries, never subtree reads (Section 5.3.3c)
Concurrent connections	clients holding sockets, at peak	Section 5.3.6a; clinicians only, with jittered patient prompts (Section 5.3.6c)

An estimated monthly cost per patient should be reported at the pilot scale and at a hypothetical thousand-patient scale. Section 3.3.3d recorded that no source in the corpus does this; doing it is a contribution in its own right, and the case documented in Section 2.2.4 establishes why it matters more here than the absence of such figures from the literature would suggest.[7]

5.3.8b Portability

efendi2025<empty citation> list zero maintenance among the platform's advantages. The corresponding cost is that security rules, function bindings and the tree structure are provider-specific and do not port.

The mitigation is a boundary. The domain logic — the rule engine, the feature pipeline of Section 4.3, the scoring and calibration of Section 4.4.5 — is kept in plain, provider-agnostic code, and the cloud functions do nothing but receive an event, call that code, and write the result. Under that arrangement the irreplaceable part of the system is portable and only the plumbing is not. This also has a consequence for **??**: the model can be evaluated offline, outside the platform entirely, which is how the nested cross-validation of Section 4.4.3a will in fact be run.

A scheduled export to a neutral format serves three purposes simultaneously: portability insurance, the research analysis file of Section 4.1.5c, and a backup. It is also segment 7 of Table 5.12, the segment no source addresses, so its protection is specified there rather than left to the backup's defaults.

[7] **[To verify]** Current pricing, quotas and free-tier limits are not in the corpus and change over time. Obtain them from provider documentation and cite the access date. The structure of Table 5.38 is stable; only the unit prices are not.

5.3.8c Residency is a one-time irreversible decision

The published endpoint of **efendi2025<empty citation>** was hosted in a south-east Asian region, which is visible in their own figure. Section 5.1.6c established that the storage region determines the governing jurisdiction, making it a legal decision taken by a configuration setting.

A database instance's region is fixed at creation and cannot subsequently be changed. Moving the data later requires creating a new instance and migrating, with a new endpoint and reconfiguration of every client. The decision must therefore be taken deliberately, before the first write, in conjunction with the legal determination required by Section 5.1.6b — which makes that outstanding document not merely a citation gap but a precondition for creating the database. Combined with the consent dependency identified in Section 5.1.6d, the applicable instrument now gates two separate steps that must precede any collection.

5.3.9 Specification

Table 5.39: Backend requirements established in Section 5.3.

Requirement	Basis
Seven top-level trees, one per access pattern	Section 5.3.3
Time series keyed by `ts_measurement` at millisecond resolution	Sections 3.3.3b and 4.1.2
Window retrieved by key range, never a subtree read	Section 5.3.3c
Queue key composed with fixed-width padded fields and a non-numeric priority prefix	Section 5.3.3c
`/events` adjudication tree present from first deployment	Sections 3.5.5e and 4.1.2e
`/config/thresholds` and `/config/models` versioned	Sections 3.5.6 and 5.1.1d
No identifying field anywhere in the tree	Section 4.1.5c
Roles as custom token claims, never database fields	Section 5.3.4b
Session expiry chosen as the revocation latency it implies	Sections 5.1.5 and 5.3.4b
Rule cascade verified: no grant above a path needing a narrower grant	Section 5.3.4c
Measurements append-only; existing key rejected	Section 5.3.4c
Corrections by superseding record, never by edit	Section 5.3.4c
Only physicians write `clinician_verdict`; no client writes `/config`	Sections 3.5.5e and 5.3.4c
Write-time validation rejects unvalidated devices for BP and glucose	Sections 3.2.5, 4.1.6 and 5.3.4d
Rule tier recomputed server-side; discrepancy recorded as drift	Section 5.3.5
$\max(\text{rule}, \text{model})$ applied in the scoring function	Section 4.4.6d
Non-transmission sweep against `checkup_period_days`	Section 3.5.2; **yazdani2018<empty citation>**
Queue-age escalation sweep, justified by the delay skew	Sections 5.1.3 and 5.1.4
All functions idempotent; `measurement_id` as deduplication key	Sections 5.2.5c and 5.3.5
Persistent connection for clinicians only	Section 5.3.6a
Patient measurement prompts jittered, to protect the peak	Section 5.3.6c
Push content constrained by the safety rule	Section 3.5.5e
Domain logic provider-agnostic; functions thin	Section 5.3.8
Scheduled export to a neutral format, protected as segment 7	Table 5.12 and Section 4.1.5c
Estimated cost per patient per month reported at two scales	Section 3.3.3d
Database region chosen deliberately before first write; irreversible	Sections 5.1.6c and 5.3.8c

5.3.10 Limits of What This Section Establishes

Three limits apply and are stated wherever the corresponding material is used.

Pricing and quotas are outside the corpus. Table 5.38 gives the structure of the cost model; the unit figures must come from current provider documentation with the access date cited.

The choice between the realtime database and a document store is not settled here. efendi2025<empty citation> used the realtime database and it suits this workload well, a timestamp-keyed range query being exactly what a single JSON tree does cheaply. A document store offers richer querying under different billing, and the comparison must be made against current documentation rather than from this corpus. The schema of Section 5.3.3 would require restructuring into collections and documents under that alternative; the access patterns it is designed around would not change, and those are the part worth preserving. This is the practical value of designing the access pattern first: it survives the storage decision.

Nothing here has been validated in routine care. Section 3.3.6 established that of the four architectures in the corpus, two have no implementation, one was validated on six healthy volunteers over ten minutes, and the fourth on a small panel with simulated patients. This backend design has no precedent in the corpus that has been tested in service, and that is stated rather than implied.

The three outstanding documents are unchanged — the two guideline editions required by Table 4.43 and the Cameroonian instrument required by Section 5.1.6b — with the instrument now determining the irreversible residency decision of Section 5.3.8c in addition to the consent document of Section 5.1.6d.

5.4 Application Interfaces Design

Purpose and sources

Sections 5.1 to 5.3 specified the architecture, the hardware and the backend. This section specifies the two interfaces through which the system is actually used, and it is the point at which the system either works in practice or is abandoned.

efendi2025<empty citation> supply the only implemented mobile interface in the corpus: registration, login and monitoring screens, a prediction display, a live chart, and a minimum device specification. **yazdani2018<empty citation>** supply the clinician side — the priority-ordered physician view, the role structure, and the configurable checkup period. The research framing document supply the resource-constrained deployment context that motivates local inference. **zhou2024<empty citation>** supply the only quantified evidence in the corpus about which interface features change outcomes.

One source is excluded. A link compilation consulted during the preparation of this work contains several incorrect author attributions and identifiers, and a document whose attributions cannot be relied upon cannot support a design claim. Where a paper it lists is used below, it is cited from the paper itself.[8]

5.4.1 What the Corpus Offers, and the Gap

The application of **efendi2025<empty citation>** has one user type. Registration collects username, password, full name, gender, address and date of birth with age derived, together with a condition dropdown offering optimal and less optimal. Login offers credentials, password recovery and a registration link. The monitoring screen shows heart rate, oxygen saturation, temperature, a prediction field reading *Stable* or *Not Stable*, a heart-rate graph updating every five minutes, and a logout control. Their flowchart runs registration and login, then sensing, microcontroller and radio, the cloud database, a REST interface, and the monitoring screen, concluding that the "user receives real-time information"†.

There is no clinician anywhere in it. The monitored person is the only user, and the prediction is displayed to that person directly.

yazdani2018<empty citation> have the clinician but no implementation. Their architecture is given in UML: three layers, five modules, an activity diagram, XML exchange and SMS notification, with an interface that "can be installed on a smart device (tablet, cell phone) or be presented as web-based"†. Their single most valuable element for this section is the priority-ordered physician view, carried forward since Section 3.5.4.

The honest position is therefore this: the clinician dashboard specified below has no implemented precedent in this corpus. One source implemented an interface without a clinician; the other designed a clinician interface without implementing it. Section 5.4.7 restates this as a limitation, and the dashboard is not presented as following established practice.

5.4.2 The Design Argument That Governs the Patient Application

Two findings in the corpus, placed together, determine where effort should go. Neither source makes the comparison.

alessa2019<empty citation> reviewed 186 hypertension self-management applications. 58.1% performed a single function. Of those supporting blood-pressure tracking,

[8] [**To verify**] The cardiologist-density figures attributed to **segun2025<empty citation>** in Section 5.4.4c must be checked against that paper directly rather than against the link compilation, since that compilation is the document whose attributions are in question. If the figures are not in the paper, the argument of Section 5.4.4c stands on clinician scarcity generally and the specific ratios should be removed.

73 of 84 required manual entry. Only 3.2% reported any health-professional input in development, and 70% shared data with third parties. Their behaviour-change-technique count has a median of 9; the reported mean of 18.4 cannot be reconciled with their stated range of 6–17, since a mean cannot exceed a maximum, and is not used here (Section 2.3.1).

zhou2024<empty citation> meta-analysed 20 randomised trials with 4 168 participants and report a systolic reduction of −5.78 mmHg (95% CI −7.97 to −3.59) with 85.2% retention. Their subgroup analyses are the operative result: interventions employing eleven or more behaviour-change techniques achieved −9.68 mmHg against −2.88 for ten or fewer, and multi-component interventions achieved −6.50 against −1.54 for single-component.

Set side by side, the median deployed application sits at nine techniques, which places it in the stratum that achieved −2.88 — below the count at which a clinically meaningful effect appears. The market does not merely underperform; it clusters immediately beneath the point at which the evidence says benefit begins. That is the gap this design must close, and Section 5.4.3d shows it closes it by construction rather than by content.

Two cautions attach and are stated here rather than appended. **zhou2024<empty citation>** rate the systolic estimate as low certainty, 75% of included trials were at high risk of bias, and only 30% were theory-based; the diastolic estimate of −3.28 is moderate certainty. The dose–response in technique count is a subgroup finding from a low-certainty body of evidence, so it is adopted here as a design target and not as a predicted effect size.

A third finding constrains what the application should promise. **mihevc2025<empty citation>** achieved a systolic reduction of −6.9 mmHg (95% CI −11 to −2.7, $p = .001$) and an HbA1c reduction of −0.5% (−0.8 to −0.2, $p = .002$) — and report that no behaviour change occurred. The mechanism was medication titration by the clinician in response to transmitted readings. Section 2.3.3 drew the conclusion and it governs here: the patient application's primary function is reliable capture and prompt escalation, not behaviour modification. **mihevc2025<empty citation>** also lost 10.9% of participants to inability to use the equipment, which makes entry burden an attrition mechanism rather than a usability nicety.

5.4.3 Patient Mobile Application

5.4.3a Platform

efendi2025<empty citation> built a native Android application. A cross-platform toolkit offers a single codebase and a single widget set.

The usual argument for cross-platform development, covering a second mobile platform, is weak in this deployment, where one platform's share is overwhelming and a build for the other would serve a small minority. The real reasons to prefer it are development speed for a single researcher and consistent rendering across heterogeneous low-end hardware. The clinician dashboard should not be built in the same framework: it is a dense tabular

interface used in a desktop browser, and a plain web application serves it better than a cross-platform widget toolkit.

One check must be performed before committing, and it cannot be performed from the corpus: the minimum platform version of **efendi2025<empty citation>** may already fall below the floor supported by current toolchains.[9]

5.4.3b Manual entry: three safety requirements

Manual entry is the dominant acquisition mode in the reviewed market — 73 of 84 applications — and Section 5.2.7 adopted Option C, under which manual entry carries blood pressure and glycaemia while instrumentation carries covariates only. The entry form is therefore the system's principal data path and not a fallback, which is a reversal of how such forms are usually treated.

The registration form of **efendi2025<empty citation>** is a reasonable starting point, but three requirements appear neither in it nor in any source, and all three are safety-critical.

The unit must be fixed per patient, displayed, and never selectable per entry. Glucose in mg/dL and in mmol/L differ by a factor of 18.0, so 7 mmol/L and 126 mg/dL are the same quantity. A value of 7 entered under the wrong unit is not a noisy reading; it is a reading that will cross or fail to cross every threshold in the system, and it is perfectly plausible under either unit, so no range check can catch it. Section 5.3.4d places a unit-matching predicate in the database rules, which catches a mismatch server-side; the interface must make the error unreachable client-side, by binding the unit to the patient record at enrolment and rendering it as fixed text beside the input.

The form must capture duplicate readings, not one. Section 3.2.5 established the guideline convention of duplicate measurement with averaging. A form offering a single field silently produces a different quantity from the one on which the thresholds are defined. Two fields with the average computed and displayed, together with `reading_index` recorded per Table 5.30, is the minimum conforming design.

Context is mandatory, not optional. Section 4.3.3 and the record schema require measurement context — at rest, post-exercise, post-meal, pre-meal — because the threshold applied depends on it. A context field the patient may skip yields records that cannot be labelled, which removes them from the training set entirely rather than merely degrading them.

Two further behaviours follow from earlier sections. Plausibility feedback at entry uses the ranges of Section 4.1.2 and *queries rather than rejects*: a hard rejection discards the

[9] [**To verify**] Obtain the current minimum API levels for the chosen cross-platform framework and for the on-device inference runtime, and state the API level this work actually targets. If the floor of **efendi2025<empty citation>** is unreachable, Table 5.41 must be revised upward and the implication for device cost stated.

genuine extreme value, which is precisely the value the system exists to catch, so the correct response to an implausible entry is a confirmation prompt whose confirmation is itself recorded. And the timestamp defaults to the present and may be edited backwards only, since Section 5.3.4d rejects future timestamps at write time — a client that permits a future timestamp produces a write the server will refuse, which the patient experiences as data loss.

5.4.3c Trends and logs

The heart-rate graph of **efendi2025<empty citation>** updates every five minutes. That cadence belongs to a live monitor and is the wrong unit for this system, where the protocol of Section 4.1.1 produces approximately two blood-pressure occasions per week.

The chart displays the fourteen-day window of Section 4.3 with the applicable threshold band drawn behind the series. Drawing the band is what makes the chart answer a clinical question rather than merely display numbers: the patient sees position relative to target, which is the discrepancy-between-current-state-and-goal technique counted in Table 5.40. The window is already available locally through offline persistence (Section 5.3.3), so the chart renders without a network round trip — which matters because a chart that blanks when connectivity drops teaches the patient that the application is unreliable.

The log is the same data in tabular form, showing both timestamps of Section 4.1.2, measurement time and record time, because a patient who entered a backdated reading should be able to see that the system recorded it as backdated.

What the patient is not shown. efendi2025<empty citation> display their classifier's output to the monitored person directly. This system does not, and the departure is deliberate: Section 5.4.4d establishes that even a clinician should not be shown a raw score from a model fitted at this sample size, and the patient has no means at all of discounting it. The patient sees the measured values, their position relative to the band, and — on amber or red — the instruction permitted by Section 3.5.5e. No probability, no score, and no label of their future state.

The threshold band cannot be drawn, and no target can be displayed, until the guideline editions recorded as outstanding in Table 4.43 are supplied. This is the most concrete consequence yet of that gap: the central visual element of the patient interface is unbuildable without them.

5.4.3d Self-management advice, and how the technique target is met

Section 3.5.5e is the binding constraint and is not negotiable: the application may tell the patient that a reading is high and that they should contact their health centre; it may not convey a dose change, a medication change, or an instruction to stop a treatment. The finding of **alessa2019<empty citation>** that 3.2% of applications had any professional

input, and that three claimed cuffless blood-pressure measurement without supporting evidence, is what unconstrained advice content produces.

The reconciliation between the eleven-technique target of **zhou2024<empty citation>** and the absence of behaviour change in **mihevc2025<empty citation>** is the design insight of this section. Many of the highest-value behaviour-change techniques are structural properties of a telemonitoring system rather than items of advice content. A system that measures, charts, reminds, and routes a clinician's response accumulates techniques by existing. The mapping below is therefore how the target is reached without a library of tips.

Table 5.40: Behaviour-change techniques delivered structurally. Ten of the twelve are architecture rather than content.

Technique	**Delivered by**
Self-monitoring of outcome	the measurement protocol itself (Section 4.1.1)
Feedback on outcome	the trend chart with threshold band (Section 5.4.3c)
Prompts and cues	scheduled reminder from `checkup_period_days` (Section 5.3.5b)
Goal setting, outcome	the target band displayed
Discrepancy between current state and goal	the reading plotted against that band
Review of outcome goals	the clinician verdict relayed back (Section 5.3.5)
Action planning	a fixed when-and-where measurement schedule
Habit formation	the same schedule, repeated
Social support, practical	nurse-assisted measurement (Table 5.5)
Credible source	advice attributed to the clinician and the guideline version
Information about health consequences	threshold explanation at the band
Behavioural contract	the enrolment protocol agreement

This mapping is a construction of this work. It has not been coded against the taxonomy by a trained coder, which is the standard the intervention-reporting literature expects, and the taxonomy itself is not in the supplied corpus. The count is therefore reported as a design intention subject to formal coding, not as an achieved score.[10]

5.4.3e Notifications

The tier mapping of Table 5.36 governs delivery. Green is silent and visible in the application; amber prompts the verification profile of Section 2.3.3; red raises an immediate high-priority notification together with a local on-device warning regardless of connectivity, per Section 5.1.2c — which is one of the two functions local inference exists to serve (Section 5.4.5d).

[10] [**To verify**] Obtain and cite the behaviour-change technique taxonomy, and have Table 5.40 coded by a second reader against it. Until then no numeric technique count should appear in an abstract or a results chapter, because a self-assigned count against an uncited taxonomy is not a measurement.

yazdani2018<empty citation> add the reminder in the other direction: "if no data is sent within the time limits set by the physician, a SMS reminder will be sent"†. That is the patient-facing half of the non-transmission sweep of Section 5.3.5b, and SMS rather than push is the correct channel for it, since the condition being detected is precisely that the application is not communicating — a push notification to a silent client is a message sent down the channel known to be broken.

One interaction with Section 5.3.6c must be respected. Measurement prompts must be jittered across a window rather than fired at a common hour, because a synchronised prompt synchronises the subsequent connections and restores the concurrent-connection peak the design exists to avoid.

5.4.3f Minimum device specification

The specification of **efendi2025<empty citation>** is the only one in the corpus and is adopted as a floor, subject to the toolchain check of Section 5.4.3a.

Table 5.41: Minimum device specification, after **efendi2025<empty citation>**, with one requirement relaxed.

Requirement	Value
Operating system	as **efendi2025<empty citation>** specify, subject to Section 5.4.3a
Free storage	1 GB
RAM	2 GB
Processor	1.2 GHz quad-core
Connectivity	*intermittent, any generation* — not 4G

The connectivity requirement is the one that must not be carried over. Section 5.2.5e established a transmitted volume of 231 kB per month for thirty patients under duty cycling, or 7.7 kB per patient. At the effective throughput of a second-generation mobile connection, the entire cohort's monthly traffic is of the order of a minute and a half of transfer time, and a single patient's is a few seconds. Specifying a fourth-generation connection as a floor, as **efendi2025<empty citation>** do for a continuously streaming monitor, would exclude exactly the resource-constrained settings the research framing document targets — and would do so to serve a volume that a far older network carries without difficulty. The requirement here is intermittent connectivity of any generation, which is what Sections 3.3.3b and 5.4.5 exist to accommodate.

5.4.4 Clinician Dashboard

5.4.4a A queue, not a patient list

The unit of clinical work is an unreviewed event, not a patient. Section 5.3.3 therefore stores the queue under a composed key, and Section 5.3.3c showed that reading

that path with a key ordering returns the list already sorted by priority and then by age within priority. The dashboard performs no sorting: the ordering specified by **yazdani2018<empty citation>** is a property of the data, not a feature of the view.

The justification is the latency budget of Section 5.1.3. Decomposing end-to-end delay showed that t_5, clinician review, exceeds every other term by orders of magnitude. **bediang2021<empty citation>** supply the measurement: their intervention accelerated access to examination from 33.3% to 98.9% (a relative risk of 2.97) and modified the diagnosis in 46.2% of cases, yet the delay to expert opinion was 10.3 against 11.2 days ($p = 0.82$), statistically indistinguishable, against a protocol requiring 72 hours. Automating acquisition while leaving review untouched produced no improvement in the term that dominates.

The dashboard is therefore the system's principal latency intervention, and the queue-age escalation sweep of Section 5.3.5b is what prevents the tail of that delay distribution from becoming this system's ordinary behaviour for some patients — which, as Section 5.1.3 established from the dispersion of those measurements, is what a right-skewed delay distribution produces if nothing bounds it.

5.4.4b Three tiers, not two

The research framing document specifies ranking by risk tier, normal against alert. This should be three tiers, for two reasons drawn from the corpus.

mihevc2025<empty citation> — the only trial in the corpus reporting both a protocol and a significant effect — used green, amber and red, and Section 2.3.3 built the two-stage verification profile on the amber tier. A binary scheme must collapse amber into one of the other two. Folded into normal, the verification profile never triggers and borderline readings go unexamined. Folded into alert, every borderline reading enters the queue and the arithmetic of Section 3.4.4b applies: at a 2% event rate and 90% sensitivity, a precision of 80% requires a specificity of 99.5%, which no model in this corpus approaches, so the queue fills with false alerts and ceases to be read.

The second reason is the experience of **efendi2025<empty citation>** themselves. Their binary classifier labelled a heart rate of 70.1 as stable, and their two stated reference bands for the stable class disagree with each other, so the label cannot be checked against a single stated criterion.[11]

The design point survives either reading. A two-state display offers no position in which to express uncertainty, so a classifier whose boundaries are incoherent emits a confident

[11] **[To verify] efendi2025<empty citation>** state the stable heart-rate band differently in two tables, and the lower bound differs between those statements. Record both bands verbatim with their table numbers, and state whether 70.1 falls inside or outside each. The argument here requires only that the two bands disagree, which is established; the stronger claim that the label contradicts a stated band requires knowing which lower bound applies.

label rather than a flag for review — and a confident wrong label is harder for a clinician to discount than an acknowledged borderline.

5.4.4c What a queue row carries

Clinician attention is the scarcest resource in this deployment. **segun2025<empty citation>** report that a substantial proportion of sub-Saharan African countries have no cardiologist at all, and specialist-to-population ratios of the order of one per several hundred thousand. Every element on the dashboard must therefore reduce the time to a triage decision, which means the row must be triageable without opening the patient record.

Table 5.42: Fields carried by a queue row.

Field	Why
Tier	the triage decision itself
Trigger source: rule, model, or non-transmission	Section 5.4.4d
The triggering value, with unit and context	the clinical fact
Days since last review	the escalation condition (Section 5.3.5b)
Direction of the fourteen-day trend	distinguishes an isolated excursion from a deterioration
Patient pseudonym and age band	identification without identity (Section 4.1.5c)
Model and threshold versions	Table 5.30

5.4.4d What must never be displayed

Two prohibitions, both derived in earlier chapters, and both easy to violate by building the obvious interface.

Never display a raw score from a class-weighted model. Section 4.4.5b derived the correction $s = s'/(s' + w(1 - s'))$. At a weight of $w = 20$ a raw output of $s' = 0.5$ corresponds to a true probability of $1/21 \approx 0.048$, so a clinician shown 0.5 would be reading a number about ten and a half times too large; at the conventional setting $w = (1 - \pi)/\pi = 49$ the same raw output corresponds to exactly the base rate of 0.02. Only the prior-corrected, calibrated value may be displayed. Given the independent sample size recorded in Table 4.43, even the corrected value carries false precision at two decimal places, and the tier together with the contributing reason communicates more honestly than any decimal.

Always display which mechanism fired. Section 4.4.6d set $\text{tier}_{\text{final}} = \max(\text{tier}_{\text{rule}}, \text{tier}_{\text{model}})$. A rule trigger is a threshold crossing traceable to a published guideline and is defensible on its own terms. A model trigger, from the sample size of Table 4.43, is weak evidence. Rendering both in the same colour with no attribution tells the clinician that two very different grades of evidence are the same thing, which is the specific failure that makes a

decision-support system distrusted after its first false alarm. The trigger source is therefore a primary field and not metadata.

5.4.4e Verdict capture is the label pipeline

Sections 4.1.2e and 4.4.2b established that the physician's adjudication is the only source of clinically validated labels in this design. The verdict control is therefore not administrative overhead; it is the training-data acquisition mechanism for the second contribution.

The consequence is a hard interface requirement: if recording a verdict is tedious, the label store stays empty and that contribution fails. The control must resolve an event in two or three interactions, with options mapping directly onto the label definition — true alert, false alert, indeterminate — plus the action taken, and an optional free-text comment that is never required. The requirement of **yazdani2018<empty citation>** that "all events... including physician and CDSS prescriptions, are stored as patient record"† is satisfied by writing `clinician_verdict` into the events tree, which Section 5.3.5 makes the trigger for closing the event and appending to the label store.

That the two-or-three-interaction figure is an argument rather than a measurement is stated in Section 5.4.7.

5.4.4f Roles

yazdani2018<empty citation> identify physician, healthcare team, patient and system administrator. Tables 5.5 and 5.32 constrain them: the physician holds a queue and may write a verdict and modify therapy; the nurse may measure and view assigned patients but holds no queue and may not modify therapy; the administrator manages enrolment and configuration but reads no clinical values.

The dashboard renders from the token claim of Section 5.3.4b, and the nurse view is not a greyed-out physician view but a different screen — a measurement and assistance worklist, matching what Table 5.5 assigns that role. A disabled-control version of the physician screen communicates that the nurse is a deficient physician, which is both wrong about the task-sharing arrangement of Section 3.1.3a and an invitation to work around the restriction.

5.4.4g It must replace a task, not add one

dodoo2022<empty citation> identified the decisive adoption barrier across 66 studies: "healthcare professionals view telemedicine as an additional task"†. Section 2.2.2 documented what the dashboard is competing against — paper registers, incomplete records, and readings recited from a card at a consultation.

The design conclusion is that the dashboard must become the place where home readings are reviewed, in place of the register, and not a second system consulted after

it. Concretely: the queue row must carry enough that no paper is required to triage; the patient view must show the same fourteen-day window the patient sees, so that the consultation and the dashboard do not disagree in front of the patient; and the verdict must be the single act that both closes the alert and records the clinical note. A dashboard requiring the clinician to transcribe anything into another system has added a task, and the evidence of **dodoo2022<empty citation>** predicts it will be abandoned.

5.4.5 Edge Inference

No source in the corpus implements this. **efendi2025<empty citation>** compute the prediction in the cloud and display it, and the decision support of **yazdani2018<empty citation>** is entirely server-side. What follows is a design proposal with no tested precedent here.

5.4.5a What actually has to be exported

The research framing document describes exporting the trained model to the mobile application. The model is the easy part and the smallest part. A score reproduced on the device requires the whole path from Sections 4.3 and 4.4 to be reproduced identically.

Table 5.43: What on-device scoring requires, by difficulty.

Component	Difficulty on device
The fourteen-day window	already solved by offline persistence (Section 5.3.3)
Feature construction (Section 4.3)	must match the cloud exactly
Per-patient normalisation constants	must be the training-time constants, shipped and versioned (Section 4.2.8b)
Masking and the decay $\gamma_t = \exp(-\kappa_c \, \Delta t)$	deterministic, portable
The model	trivially small (Section 5.4.5e)
Calibration and prior correction (Section 4.4.5)	mandatory, or the local score is an order of magnitude wrong
Thresholds τ_{amber}, τ_{red}	from the model registry (Table 5.30)

The difficulty is fidelity, not capacity. A feature pipeline reimplemented for the device against one written in the training language will diverge — in how a window boundary is treated, how a tie is broken, how a missing value is masked, which normalisation constants are used. A device producing silently different scores from the cloud is worse than a device producing none, because the discrepancy is invisible to everyone, including the clinician who sees one number and the researcher who computes another.

The only defensible arrangements are a single shared implementation compiled for both targets, or an exported pipeline artefact that both sides execute, in either case with conformance tests asserting tolerance-bounded agreement on a fixed test set as a release gate. The feature-set hash already recorded in Table 5.30 is what makes such a gate checkable at runtime rather than only at build time.

5.4.5b The rule engine is the part worth exporting

This follows directly from the composition of Section 4.4.6d.

The rule engine is a short list of comparisons of a value against a threshold, selected by channel and context. It is deterministic, has no numerical drift, requires no normalisation constants, and is a few hundred bytes of configuration. The model requires everything in Table 5.43.

The rule engine therefore delivers nearly all of the offline value at a small fraction of the risk. In an outage, a device carrying only the rule engine degrades to guideline conformity — which is precisely the floor Section 4.4.6d was constructed to guarantee — whereas a device carrying a mis-ported model pipeline degrades to an unknown function, and an unknown function is not a degradation mode that can be reasoned about or defended.

The recommendation is to ship the rule engine in the first version, treat local model scoring as a second stage gated on the conformance tests of Section 5.4.5a passing, and present it in that order.

5.4.5c Version stamping and reconciliation

An offline device may run a stale configuration for days. Every locally produced tier must therefore be stamped with the threshold version and, where a model was used, the model version and feature-set hash.

On reconnection the cloud re-scores the same window under the authoritative versions and records any discrepancy — exactly the mechanism Section 5.3.5 applies to the rule tier computed at the Edge, applied here to the device. This converts configuration drift from a silent failure into a measurable one, and the discrepancy rate becomes a reportable validation result in its own right.

5.4.5d What a local alert may actually do

Narrower than it first appears, and the narrowness is a safety property rather than a limitation.

The clinician is not on the device, and Section 3.5.5e forbids the application from issuing therapeutic instructions. A locally computed red tier may therefore do exactly two things: warn the patient to seek care, and queue the event for transmission at the earliest opportunity, with the local warning raised regardless of connectivity per Section 5.1.2c. It may not titrate, advise a dose, or stand in for review.

Under the transport design of Section 5.2.5c the queued event transmits at least-once with application-layer deduplication when connectivity returns, and the Last Will mechanism of Section 5.2.5 means the backend already knows the device went silent — so a red tier computed during an outage reaches the clinician through two independent routes, the queued event and the absence alarm.

5.4.5e Size is not the constraint

Worth stating because it is routinely assumed to be. A gradient-boosted ensemble of the size contemplated in Section 4.4.3c — depth two to three, with early stopping — is tens of kilobytes. The recurrent model of Section 4.4.3d has, at sixteen units over ten input channels, exactly

$$P = 4\Big[u(f+u)+u\Big] = 4\Big[16(10+16)+16\Big] = 1\,728 \tag{5.7}$$

parameters, which is under 7 kB at single precision and under 2 kB quantised. Against the storage and memory floor of Table 5.41 the model is roughly five orders of magnitude below the available budget.

The binding constraints are the fidelity requirement of Section 5.4.5a and the toolchain floor of Section 5.4.3a — not memory, storage or compute.

5.4.6 Specification

5.4.7 Limits of What This Section Establishes

Five limits apply.

No usability evaluation of this design exists. efendi2025<empty citation> evaluated their own application on their own cohort with their own instrument, and the dimension that scored lowest was security, at 4.06 of 5 — by their own measurement, the weakest aspect of the only implemented interface in the corpus. No instrument results exist for the design above, and the two-or-three-interaction requirement of Section 5.4.4e is an argument, not a measurement. A usability study is the obvious validation activity and is proposed in **??**.

There is no validated clinician interface to design against. Section 5.4.1 established the gap. The dashboard is reasoned from the latency budget, the alert arithmetic and the adoption literature; it is not copied from a working system.

Edge inference is unprecedented in this corpus, which is why Section 5.4.5b recommends shipping only the rule engine first.

The technique count is unverified. The mapping of Table 5.40 is a construction of this work, uncoded, against a taxonomy not in the corpus.

Current toolchain minimum platform versions are unknown, which determines whether the floor of **efendi2025<empty citation>** is reachable (Section 5.4.3a).

The outstanding documents now block a visible interface element rather than an internal convention. The guideline editions recorded in Table 4.43 are required in order to draw the threshold band and to display any target: without them the central visual of the patient application cannot be built, which makes them a dependency of the implementation and not only of the labelling. The Cameroonian instrument required by Section 5.1.6b now governs

Table 5.44: Interface requirements established in Section 5.4.

Requirement	Basis
Patient application cross-platform; dashboard a plain web application	Sections 5.4.3a and 5.4.4a
Unit fixed per patient at enrolment, displayed, never selected per entry	Section 5.4.3b; validated server-side in Section 5.3.4d
Entry form captures duplicate readings with computed average and `reading_index`	Section 3.2.5 and Table 5.30
Context field mandatory	Section 4.3.3
Implausible values queried and confirmed, never rejected	Section 4.1.2
Timestamp defaults to now, editable backwards only	Section 5.3.4d
Trend chart is the fourteen-day window with threshold band drawn	Sections 4.3 and 5.4.3c
Log displays both timestamps	Section 4.1.2
Patient shown values, band position and permitted instruction — never a score or label	Sections 3.5.5e, 5.4.3c and 5.4.4d
Advice content may never convey a dose or medication change	Section 3.5.5e
Technique count targeted at eleven or more, met structurally; formally coded before any claim	**zhou2024<empty citation>**; Section 5.4.3d
Non-transmission reminder by SMS, not push	Section 5.3.5b; **yazdani2018<empty citation>**
Measurement prompts jittered across a window	Sections 5.3.6c and 5.4.3e
Connectivity floor is intermittent, any generation — not 4G	Sections 5.2.5e and 5.4.3f
Dashboard is a queue of events, read pre-ordered, no client-side sort	Section 5.3.3c; **yazdani2018<empty citation>**
Three tiers, not two	**mihevc2025<empty citation>**; Sections 2.3.3 and 3.4.4b
Row triageable without opening the patient record	Section 5.4.4c
Never display a raw weighted-model score	Sections 4.4.5b and 5.4.4d
Trigger source displayed as a primary field	Sections 4.4.6d and 5.4.4d
Model and threshold versions shown on every event	Table 5.30
Verdict resolvable in two or three interactions; comment never required	Sections 4.1.2e and 5.4.4e
Nurse view a distinct worklist, not a restricted physician view	Table 5.5 and Section 5.4.4f
Dashboard must replace the register, not supplement it	**dodoo2022<empty citation>**; Section 5.4.4g
Rule engine exported first; local model scoring gated on conformance tests	Sections 4.4.6d and 5.4.5b
Shared or artefact-based feature pipeline, with agreement tests as a release gate	Section 5.4.5a
Local scores version-stamped; cloud re-scores and records the discrepancy	Sections 5.3.5 and 5.4.5c
Local red tier may only warn and queue	Sections 3.5.5e and 5.1.2c

two further items — what the patient may be shown about their own record, and the consent screen at enrolment — in addition to the residency decision of Section 5.3.8c and the consent document of Section 5.1.6d. The finding of **alessa2019<empty citation>** that 70% of reviewed applications shared data with third parties is the reason that instrument, and not a European regulation, must be identified before the consent screen is written.